A Structural Exhaustion Proof of the Erdős–Gyárfás Conjecture on Power-of-Two Cycles

Guillem Duran-Ballester
guillem@fragile.tech

August 20, 2026

Abstract

We prove the Erdős–Gyárfás conjecture that every finite simple graph of minimum degree at least three contains a cycle whose length is a power of two. Assuming a minimal counterexample, we reformulate the forbidden-cycle condition in terms of edge-rooted Mersenne returns and combine a boundaried replacement principle with the P_{13}-free theorem of Hegde, Sandeep, and Shashank. A maximal packing of induced P_{13} windows reduces the problem to a structured remainder, which is exhausted through two-step obstruction, compression, and surplus-routing alternatives. The remaining configurations are ruled out by explicit deficiency and entropy estimates together with a finite local analysis. All constants are stated explicitly, and the finite enumerations used in the proof are reproduced with code in the appendix.

Contents

0 Architecture of the proof

This section gives a proof-level overview before the local constructions are introduced. We describe the argument in standard graph-theoretic terms and refer to the formal definitions whenever the manuscript uses specialized terminology. Fix throughout a lexicographically minimal counterexample G. The main reductions are

minimal counterexample
$\Rightarrow$ no edge-rooted return path has Mersenne length
$\Rightarrow$ no proper boundaried subgraph has a smaller equivalent replacement
$\Rightarrow$ a maximal packing of induced P_{13}'s and a P_{13}-free remainder R
$\Rightarrow$ a near-cubic graph, or a forbidden cycle or high-degree local configuration
$\Rightarrow$ independent local cycle tests, or a dependence excluded by replacement
$\Rightarrow$ an entropy bound, or a remainder with small normalized degree deficit
$\Rightarrow$ a connected piece of negative charge
$\Rightarrow$ a subcubic boundary configuration or a high-degree fan
$\Rightarrow$ a terminal boundary-incidence configuration
$\Rightarrow$ a contradiction.

The four phases below identify the results proving these implications. The proof-dependency diagrams in Section 1 give the corresponding node-by-node expansion.

Phase 1: edge-rooted return lengths and minimal replacement. The first step expresses the desired cycle through the length of a path returning to an oriented edge. By Lemma 2.2, a power-of-two cycle exists precisely when some oriented edge has a simple return of length in

$$\mathcal{M} = \{2^k - 1 : k \geq 2\}.$$

Thus the counterexample condition is expressed as

$$R_e(G) \cap \mathcal{M} = \varnothing \qquad \text{for every oriented edge } e.$$

To use minimality locally, a connected subgraph is retained together with its boundary vertices and the operation of gluing it to an outside graph (Definitions 3.81 and 3.82). Two such pieces are compared only when their boundary degrees agree (Definition 3.83) and when every compatible outside graph gives the same answer to the power-of-two-cycle question. The manuscript calls the resulting exact boundary response *target-complete*; see Definition 3.85. Context universality and the replacement lemma show that a proper piece of G cannot have a smaller piece with the same boundary response and degree data (Lemmas 3.95 and 3.96 and Corollary 3.98). This is the form of minimality used later whenever local cycle tests become dependent.

Phase 2: packing induced P_{13} paths and bounding their density. The external input is the Hegde–Sandeep–Shashank theorem Theorem 5.1. Since G has minimum degree at least three and contains no power-of-two cycle, it contains an induced P_{13}. Choose a maximal vertex-disjoint family $\mathcal{P}$ of induced P_{13}'s. We call its members *windows*, solely as shorthand for these packed paths. Removing their vertices leaves a graph R whose components are P_{13}-free by maximality.

If $P = v_0 \dots v_{12}$ is a packed path and $x \notin V(P)$, its attachment to P is the subset of positions i for which xv_i is an edge. Pairs of positions at forbidden distances immediately create cycles of length four or eight; the remaining attachment sets and the analogous tests involving a short outside path form the finite enumeration in Lemmas 6.1 and 7.1. The exact degree count across all packed paths is

$$e(R, W) + 2e_{\times}(W) = 15p_{13} + \sigma_W,$$

with the notation and identity given in Definition 8.4 and Lemma 8.6.

Before normalizing the edge count, the proof separates the case in which the total degree surplus $\sigma(G) = \sum_v (d_G(v) - 3)$ is large. The exhaustive alternative Proposition 3.80 gives either a power-of-two cycle, a smaller boundary-equivalent replacement or support dependence, a high-degree fan passed to the local analysis, or $\sigma(G) = O(\sqrt{n})$. The last condition is called the *near-cubic spine* in Definition 4.1; it implies

$$B_{\text{skel}} = \frac{3}{2} n \log_2 n + o(n \log n),$$

for the number of labelled graphs under consideration.

No independence between distinct windows is assumed. At a fixed scale, either their conditional restrictions multiply, or their failure to do so produces a minimal connected overlap (Lemma 7.33). The overlap can be uncrossed into a serial chain of alternative corridor paths (Lemma 7.35). Sums of their length increments then contain long arithmetic progressions (Lemma 7.36); the arithmetic alternative in Lemma 7.37 yields either a power-of-two cycle or a periodic boundary response already excluded by contextual distinction, replacement, or the local structural alternatives. Thus the restrictions are additive exactly in the case where they are counted independently; see Lemmas 7.38 and 7.39. This proves the cap

$$\theta = \frac{p_{13}}{n} \leq \theta_{\text{win}} + o(1) = 0.01270017798\ldots + o(1),$$

and the sharper high-entropy cap 0.01198542083... in Proposition 7.42. In particular, R contains a fixed positive proportion of the vertices, while the displayed incidence identity bounds the positive degree deficiency that the packed paths can supply to it.

Phase 3: independent local cycle tests and the entropy alternative. Each length-two path in R supplies a local test for completing a power-of-two cycle through the rest of the graph. The obstruction rank $r_\Omega(R)$ is, by definition, the largest independent family of these tests; see Definition 10.1. A rank loss cannot simply be discarded. On a proper subgraph it makes the boundary response distinguishable by an outside graph, permits a smaller boundary-equivalent replacement, or enlarges the support of the dependence. These alternatives contradict the return-length condition, Corollary 3.98, or the proper-support dependence lemmas. A dependence using all of G is excluded by retaining the exact labelled cycle-response data; see Definitions 3.88 and 3.91 and Lemma 10.9. Consequently the surviving case has

$$r_\Omega(R) \geq W_2(R) - o(W_2(R)),$$

as stated in Lemma 10.10.

For the relevant family of possible remainders, entropy means the logarithm of the number of labelled possibilities per vertex (Definition 11.1). The exhaustive split in Proposition 11.6 has three cases. High entropy gives the sharper packing bound above. In the repetitive low-entropy case, one local type occurs often enough to supply a linear independent family of cycle tests, whose forced cost exceeds the remaining count. The nonrepetitive low-entropy case is retained for the degree calculation. In that last case the window and remainder estimates give

$$\Delta_{\text{net}}(R) = \frac{\partial_+(R) - \sigma_R}{|R|} \leq \tau_{\text{win}} + o(1) < \frac{1}{4}.$$

Here $\partial_+(R)$ is the sum of the positive boundary-degree deficits and σ_R is the degree surplus assigned to R.

Phase 4: localizing the degree deficit and excluding connected pieces. The graph R is decomposed into connected pieces together with the surplus units assigned to them. Such a piece is called *admissible* when it has the inherited P_{13}-free, boundary-degree, contextual, and minimality properties listed in Definition 13.2. Its net charge is

$$\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \frac{1}{4}|V(X)|$$

as defined in Definition 13.3. By Proposition 13.7, the global inequality $\Delta_{\text{net}}(R) < 1/4$ forces a connected admissible piece X with

$$\mathcal{N}(X) < 0.$$

There are two local cases. First suppose that X contains no vertex of degree greater than three in G (a *Type A support* in Definition 14.2). A boundary vertex w with internal degree at most two has deficit $q(w) = 3 - d_X(w)$. The canonical trace assigns each internal degree-three vertex to such a boundary vertex; the number assigned to w is its *receiver load*, defined in Definition 14.3. If some load is at least $4q(w)$, the eight exhaustive outcomes of Definition 14.16 give a forbidden cycle, an outside context that distinguishes two boundary responses, a smaller equivalent replacement, a support dependence, a high-degree fan, or the final minimal boundary data. If no load is saturated, the elementary $3, 7, 11$ discharging estimate gives nonnegative charge; see Lemma 14.45.

Now suppose that X contains a high-degree vertex (the *Type B case*). High-degree vertices are independent, and their degree-three neighbours form the fans specified in Definitions 14.17 and 14.18. The attachment restrictions either construct a power-of-two cycle or bound the deficit of an individual fan. Positive fan deficits are paid by distinct noncentral incidences, as proved in Lemma 14.79. When incidences belonging to different fans overlap, the minimal overlap retains the same outside-context and replacement properties; the exact alternatives are stated in Definition 14.90. The total contribution of failed or overlapping fan configurations is at most a constant times $\sigma(G)$ by Proposition 14.100, and hence is $O(\sqrt{n})$. It cannot account for the linear negative charge required in the surviving case.

After these exclusions, the only remaining object is a minimal set of at most two private boundary incidences that preserves the power-of-two-cycle response under every compatible outside graph. We call it here the *terminal boundary-incidence obstruction*; its formal manuscript name and precise minimality conditions are given in Definition 14.60. The local theorem Theorem 14.61 proves that it cannot occur.

How the alternatives close. The counting, rank, and degree arguments are separate checks on the same minimal counterexample. The packing count bounds how many induced paths can occur; the rank argument prevents dependent local cycle tests from being counted as independent information; and the charge argument localizes any remaining degree deficit. Every exceptional case is passed to a stated structural lemma, and the finite enumerations used there are recorded in the appendices. Thus the overview above is an exhaustive sequence of alternatives, not an assumption that the local restrictions are probabilistically independent.

1 Introduction and overview of the proof

The Erdős–Gyárfás power-of-two cycle problem asks whether every finite simple graph of minimum degree at least three contains a cycle whose length is a power of two. It belongs to the broader study of which cycle lengths a graph of bounded minimum degree must realize [2, 3], and the spanning-tree and leaf structure of minimum-degree-three graphs invoked below is classical [4]. We prove the following.

Theorem 1.1 (Main closure)**.** *Let G be a finite simple graph with minimum degree at least three. Suppose that G contains no cycle whose length is a power of two, and choose a lexicographically minimal such G. Then no such G exists.*

Proof. Assume, for contradiction, that such a lexicographically minimal counterexample G exists. By Lemma 2.2, a Mersenne edge-rooted return is equivalent to a power-of-two cycle, so no such return occurs in G. Since G has no power-of-two cycle, the Hegde–Sandeep–Shashank theorem Theorem 5.1 and Corollary 5.2 place G on the P_{13}-packing spine.

Before the normalized near-cubic budget is used, the sparse surplus-token branch is exhausted by Proposition 3.80. If a sparse surplus exit occurs, that sparse exit closes in the sparse branch. If a role-homogeneous same-token bottleneck produces decorated Type B fan data, that data is routed to the Type B fan ledger and is not a terminal non-near-cubic leaf. The remaining possibility is that G satisfies the near-cubic spine estimate of Definition 4.1. Thus every surviving minimal counterexample reaches the derived near-cubic entropy/net-charge spine.

The rank branch is exhausted by Lemma 10.4, Lemmas 10.6 and 10.9, and Lemma 10.10: a rank drop is routed to the target-defect, compression, or support-dependence closures, while the complementary branch has full obstruction rank. On that branch, Proposition 12.2 closes the small-budget entropy alternative and Proposition 11.6 routes the remaining case to the large-budget net-charge analysis. At that point every survivor has the near-cubic estimate, so the rank/full-budget analysis sends it to the residual families treated by Theorem 15.1.

Theorem 15.1 excludes all large-budget residuals except the Type A target-defect/route-8 ledger and the Type B bridge residual. The Type B bridge residual has sublinear total negative mass by Proposition 14.100, so it cannot carry the linear large-budget deficit. The remaining Type A ledger is closed by Theorem 15.48: target-defect entries are canonical exit-(4) peel data and finite descent removes them, while the terminal route-8 two-support entry is excluded by Proposition 14.62 and Theorem 14.61. This contradiction proves that no large-budget residual survives, and therefore no lexicographically minimal counterexample exists. □

Remark 1.2 (Status of the routed leaves)**.** *The normalized entropy and fan-mass arguments are invoked only after the sparse surplus-token branch has been discharged. Definition 4.1 names the estimate available on the surviving branch, and Lemma 4.4 computes the labelled skeleton budget once that estimate is available. The estimate is obtained, or the branch is already discharged, before the near-cubic budget is invoked: Proposition 3.80 proves that every survivor of the sparse surplus exits uses the full active-demand family, an $O(n)$-deficit baseline demand, and the capacity-token ledger, and then has only three possible outcomes: the near-cubic spine estimate, a sparse surplus exit, or decorated Type B fan data routed to the Type B fan ledger. The free/blocked pair split of Definition 3.33 and the no-overcounting identity Lemma 3.34, together with the token supply and no-overcounting identities Lemmas 3.38 and 3.39, give the exact high-load alternative of Lemma 3.57. The token-fibre graph accounting of Definitions 3.40 and 3.41 and Lemma 3.47 converts any large token load into either a same-token matching or a same-token star, and Lemma 3.43 refines it to a role-homogeneous same-token pattern in the window-incidence, remainder-surplus, or primitive blocker-support token class. Bounded role-fibre load routes to the near-cubic spine by Theorem 3.56. If bounded load fails, the exact numerical frontier is Theorem 3.75: it gives the forced blocked-pair demand $N_*(G)$, splits it over the six subtype supplies*

$$e(R, W), \qquad 2e_\times(W), \qquad \sigma_R, \qquad n, \qquad 2m, \qquad \sigma(G),$$

and gives the lower bound

$$\psi\left(\frac{N_*(G)}{36(4n + 15p_{13} + 3\sigma(G))}\right)$$

for the largest role-homogeneous pattern in the overloaded subtype. The exact same-token bottleneck routing of Definition 3.76, Lemma 3.77, and Theorem 3.78 then discharges that pattern: it gives a sparse surplus exit, ordinary Type B fan data, or fixed homogeneous caps and hence the near-cubic estimate. The exact window-join load of Corollary 13.5 records the global budget available for window-incidence load: without a negative admissible support, the graph must put a linear surplus excess on packed P_{13}-windows. The rank-dependence part is routed by Lemma 3.59: non-independent pair responses force a boundary-profile blocker, target-response blocker, target-defective quotient, target-complete compression, or global exact-profile contradiction. This is the surplus-token branch used before the near-cubic entropy spine. The route-8 Type A residual is reduced by Proposition 14.59 to the terminal two-support package of Definition 14.60, and that package is *excluded by Lemmas 14.54 and 14.55 and Theorem 14.61. The Type B bridge residual recorded in Definition 14.90 is not a terminal leaf: Proposition 14.100 bounds its total negative net charge by the global surplus budget. The Type A target-defect/route-8 ledger is closed by Theorem 15.48, which combines the the demand ledger, finite exit-*(4) *descent, and the terminal two-support route-8 exclusion. All other branches used in Theorem 15.1 are closed inside the paper or are explicitly routed to one of the displayed residual ledgers. In particular, the nonrepetitive low-entropy case in Proposition 11.6 is not a terminal leaf; it is routed to the large-budget net-charge analysis and therefore enters the same Type A route-8 and Type B bridge closure.*

The proof assumes such a counterexample exists and chooses one, G, lexicographically minimal by

$$(|V(G)|, |E(G)|),$$

and shows that any such graph is forced into the route-8 terminal package, which is ruled out by Theorem 14.61; this gives the closure stated in Theorem 1.1. The main route uses the P_{13}-free theorem of Hegde, Sandeep, and Shashank (Theorem 5.1) as a black box. The dense-window and route-8 bounded-block bookkeeping is recorded in the appendix as auxiliary resilience data.

Conventions. Throughout, G is a finite simple graph with $n = |V(G)|$ and $m = |E(G)|$. We write

$$\mathcal{D} = \{2^k : k \geq 2\} = \{4, 8, 16, \dots\} \qquad \text{and} \qquad \mathcal{M} = \{2^k - 1 : k \geq 2\} = \{3, 7, 15, \dots\}$$

for the power-of-two lengths and their predecessors. The *target* is a power-of-two cycle; a *target-X* notion means X taken with respect to the predicate "contains a power-of-two cycle". All asymptotic statements ($o(\cdot)$, $O(\cdot)$) are as $n \to \infty$, and "for large n" abbreviates "for all sufficiently large n".

Standing notation and constants

The recurring graph parameters are

$$\begin{aligned} \beta &= m - n + 1 && \text{(cycle rank)}, \\ \lambda &= 2n - 3 - m && \text{(sparsity rank)}, \\ \sigma(G) &= 2m - 3n && \text{(surplus above cubic)}. \end{aligned}$$

together with the maximum disjoint induced-P_{13} packing p_{13}, its density $\theta = p_{13}/n$, and the P_{13}-free remainder R. For a subgraph X we use the positive deficiency $\partial_+(X)$, ambient surplus $\sigma(X)$, length-two wedge count $W_2(X)$, obstruction rank $r_\Omega(X)$, per-vertex remainder entropy $\eta(R)$, and net charge $\mathcal{N}(X)$, each defined where it first appears. The explicit numerical constants are collected here for reference; each is established at the cited place and reproduced by the code in Section A.

Constant	Value	Meaning	Established in
c_Ω	$2.28922315244\ldots$	cost of one independent two-step obstruction test	Lemma 6.2
c_{13}	$118.108581006\ldots$	multi-scale P_{13}-window constant	Lemma 7.1, Section A
θ_{win}	$0.01270017798\ldots$	window-only P_{13} density cap on the derived spine branch	Proposition 7.42
θ_{high}	$0.01198542083\ldots$	sharper high-entropy P_{13} density cap	Proposition 7.42
ω_{win}	$2.54365026308\ldots$	wedge density of the remainder on the derived spine branch	Lemma 9.1
ω	$2.57407357888\ldots$	sharper high-entropy wedge density	Lemma 9.1
$K_{\mathrm{win}} = c_\Omega \omega_{\mathrm{win}}$	$5.82298307395\ldots$	window-only forced obstruction cost per remainder vertex, under the rank closure	Corollary 10.11
$K = c_\Omega \omega$	$5.89262883286\ldots$	sharper high-entropy forced obstruction cost per remainder vertex	Corollary 10.11
τ_{win}	$0.22817486846\ldots$	net-deficiency cap on the derived spine branch	Section 13
$\Theta(n)$	finite-n threshold	packing threshold closing the entropy cap	Definition 12.1

The proof is organized as a case analysis. Raw obstruction tests are not charged directly as independent entropy. Instead, the rank-forcing argument first proves that any surviving residual has full obstruction rank. This forced obstruction cost is then used only in the range where the remaining non-obstruction budget is strictly smaller than that cost. If the remaining budget is not small, the analysis reduces to an explicitly described large-budget residual and then to a local net-charge obstruction.

The proof proceeds in the following steps.

(1) Absence of power-of-two cycles is equivalent to absence of edge-rooted Mersenne returns.

(2) Minimality gives no proper internal 3-core and forbids target-complete replacements of proper boundaried pieces.

(3) The P_{13}-free theorem forces an induced P_{13}.

(4) The finite P_{13} obstruction algebra gives the constants

$$c_\Omega = \log_2(543958/111286) = 2.28922315244\ldots$$

and

$$c_{13} = 118.108581006\ldots.$$

(5) The sparse surplus-token branch is exhausted before the normalized skeleton budget is used: it either closes a sparse surplus exit, routes decorated Type B fan data to the Type B ledger, or supplies the near-cubic spine estimate.

(6) On the surviving spine branch, the independent-P_{13} window accounting gives a first upper bound on the packing density $\theta = p_{13}/n$.

(7) Removing a maximal disjoint induced-P_{13} packing leaves a large componentwise P_{13}-free remainder R.

(8) The rank-forcing argument proves that any rank drop among raw piece-obstruction tests gives target defect, replacement, target-complete proper-support compression, or whole-graph dependence. Therefore a surviving residual has full obstruction rank,

$$r_\Omega(R) \geq W_2(R) - o(W_2(R)).$$

(9) Full obstruction rank imposes the forced linear cost

$$c_\Omega W_2(R) \geq 5.82298307395\ldots|R| - o(|R|)$$

from the window-only constants, with the sharper high-entropy value $5.89262883286\ldots|R| - o(|R|)$ when the sharper deficiency cap is available.

(10) If the remaining non-obstruction budget is smaller than this forced cost, the entropy cap closes the branch.

(11) If the remaining budget is not small, the derived spine window-only cap still gives a large P_{13}-free remainder and the surplus-adjusted positive deficiency satisfies

$$\frac{\partial_+(R) - \sigma_R}{|R|} \leq \tau_{\text{win}} + o(1) < \frac{1}{4}.$$

(12) Thus any surviving large-budget residual must contain a connected admissible support X with negative net charge

$$\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \frac{1}{4}|V(X)| < 0.$$

Such an X is either a bounded low-deficiency P_{13}-free boundaried piece or a high-degree fan-safe surplus obstruction. The local exclusions close Type A outside route 8 and Type B outside the bridge residual. The final large-budget squeeze then extracts all route-8 Type A deficit, charges the remaining Type B bridge mass to the global surplus budget, applies the boundary-incidence ledger to reduce the surviving route-8 branch to a two-support entry, and invokes Theorem 14.61.

The proof-dependency diagram below records the branch structure in full detail. It is intentionally redundant: each displayed node corresponds to a definition, lemma, proposition, corollary, theorem, or local obstruction class in the body of the paper. One principal case split is

$$\text{rank drop} \Rightarrow \text{target defect/compression/support dependence},$$

while the complementary branch forces full obstruction rank. The final spine is the large-budget net-charge branch up to the terminal two-support route-8 exclusion; Type B fan-certificate residual centers and B2 failures are controlled there by the fan-mass estimate.

Proof-dependency diagram

The proof-dependency diagram is split across twelve numbered parts. Each diagram node has a unique bracketed integer, and the integers are consecutive along each expanded branch. Continuation arrows refer to later nodes by number rather than repeating the same node number. Round nodes are closed branches.

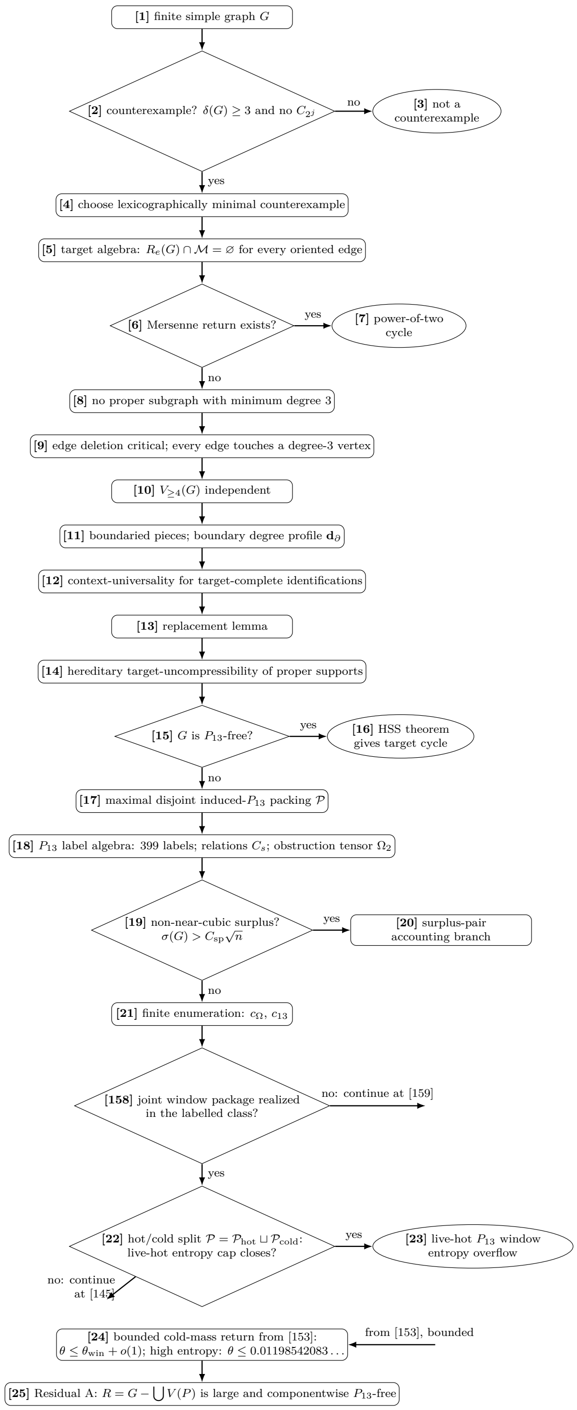

Figure 1: Proof-dependency diagram, Part I. Rectangles are assertions or residual states, diamonds are exhaustive branch tests, ellipses are terminal outcomes, and edge labels state which branch is followed. The bracketed numbers are unique diagram-node numbers used by the dependency table. Nodes [19] and [20] record the non-near-cubic surplus accounting branch used before the normalized spine budget; node [20] is the continuation expanded in Fig. 10. Node [158] is the exact finite form of the realization sentence of Lemma 7.1 and Proposition 7.42: it asks whether the joint window package of the fixed maximal packing is realized by the labelled skeleton class; its no-branch is the dense-packing residual expanded in Fig. 12. Node [22] creates the hot/cold dichotomy by splitting $\mathcal{P}$ into $\mathcal{P}_{\mathrm{hot}}$ and $\mathcal{P}_{\mathrm{cold}}$. Its no-branch continues at [145] in Fig. 11. The bounded arm of [153] returns to [24] with the density cap, and only that live residual continues to [25] and Fig. 2.

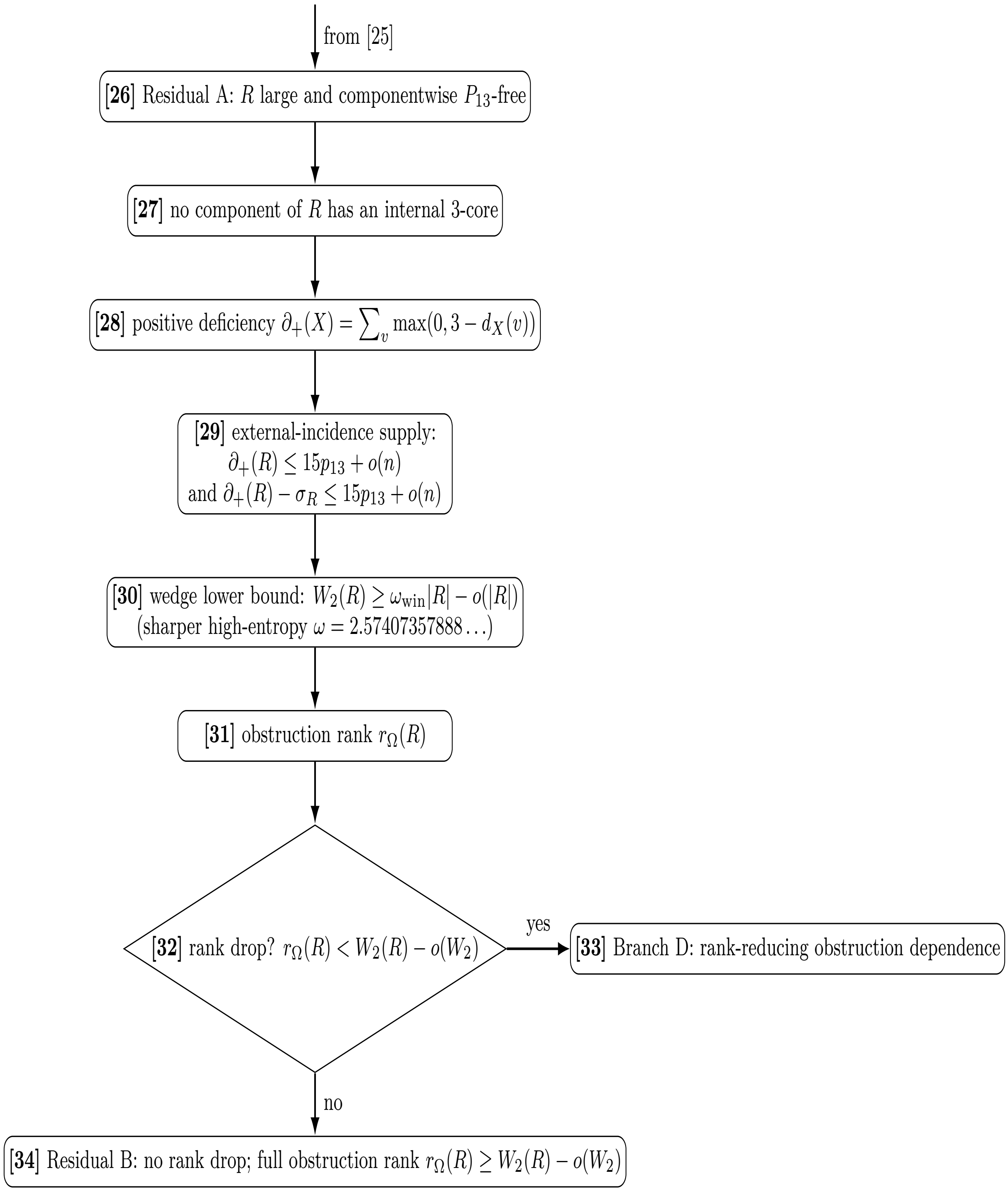

Figure 2: Proof-dependency diagram, Part II. Rectangles, diamonds, ellipses, bracketed node numbers, and edge labels use the convention of Fig. 1. At node [29], $\sigma_R = \sigma(R)$ denotes the ambient surplus on the remainder; the positive-deficiency bound $\partial_+(R) \le 15p_{13}+o(n)$ is the incidence-supply input, and the $\partial_+(R) - \sigma_R$ bound is its surplus-adjusted consequence. The diagram produces two continuations: the rank-drop branch, closed in Part III, and the full-rank residual, continued in Part IV.

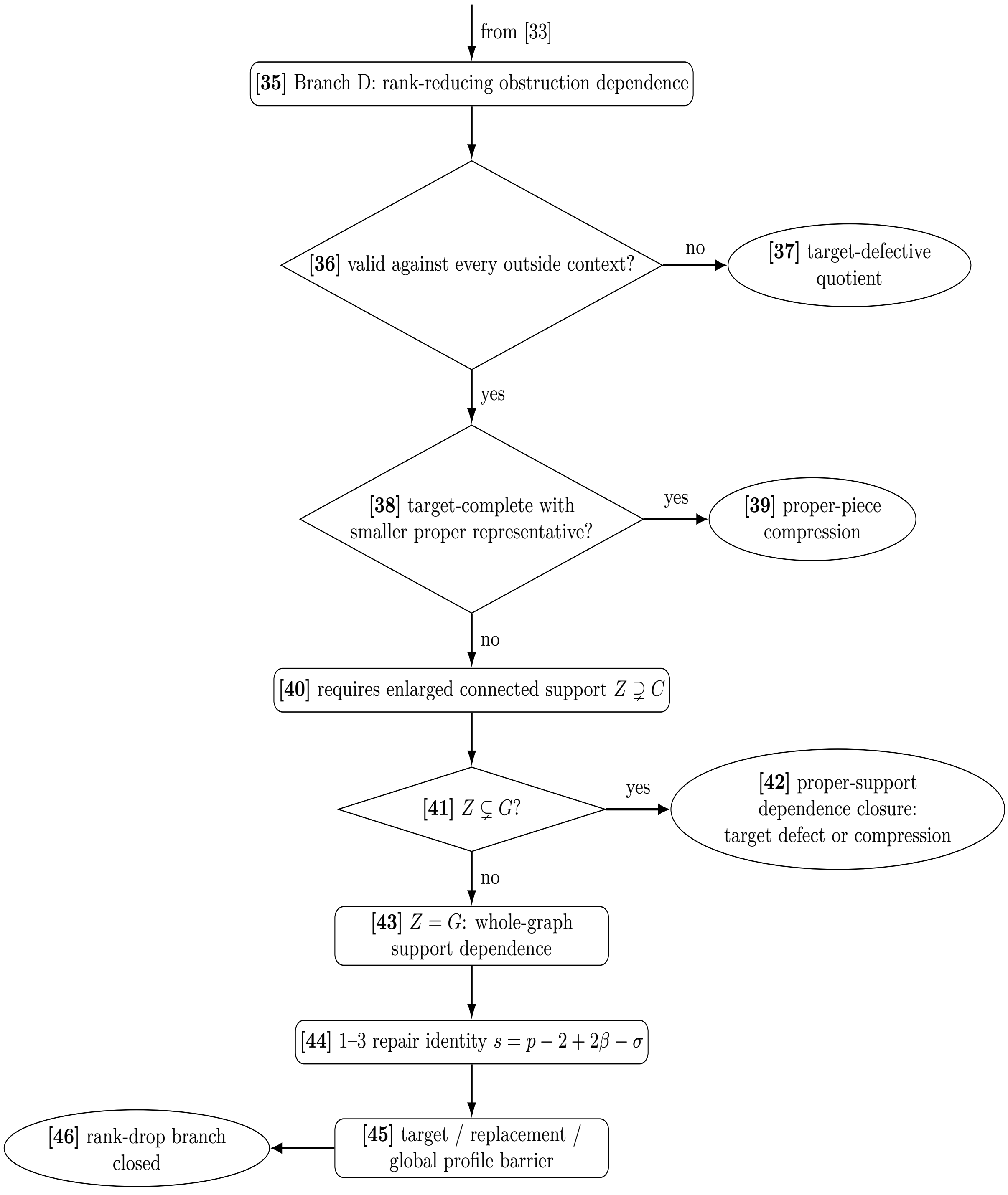

Figure 3: Proof-dependency diagram, Part III. Rectangles, diamonds, ellipses, bracketed node numbers, and edge labels use the convention of Fig. 1. Every rank-drop branch terminates in a closed round node.

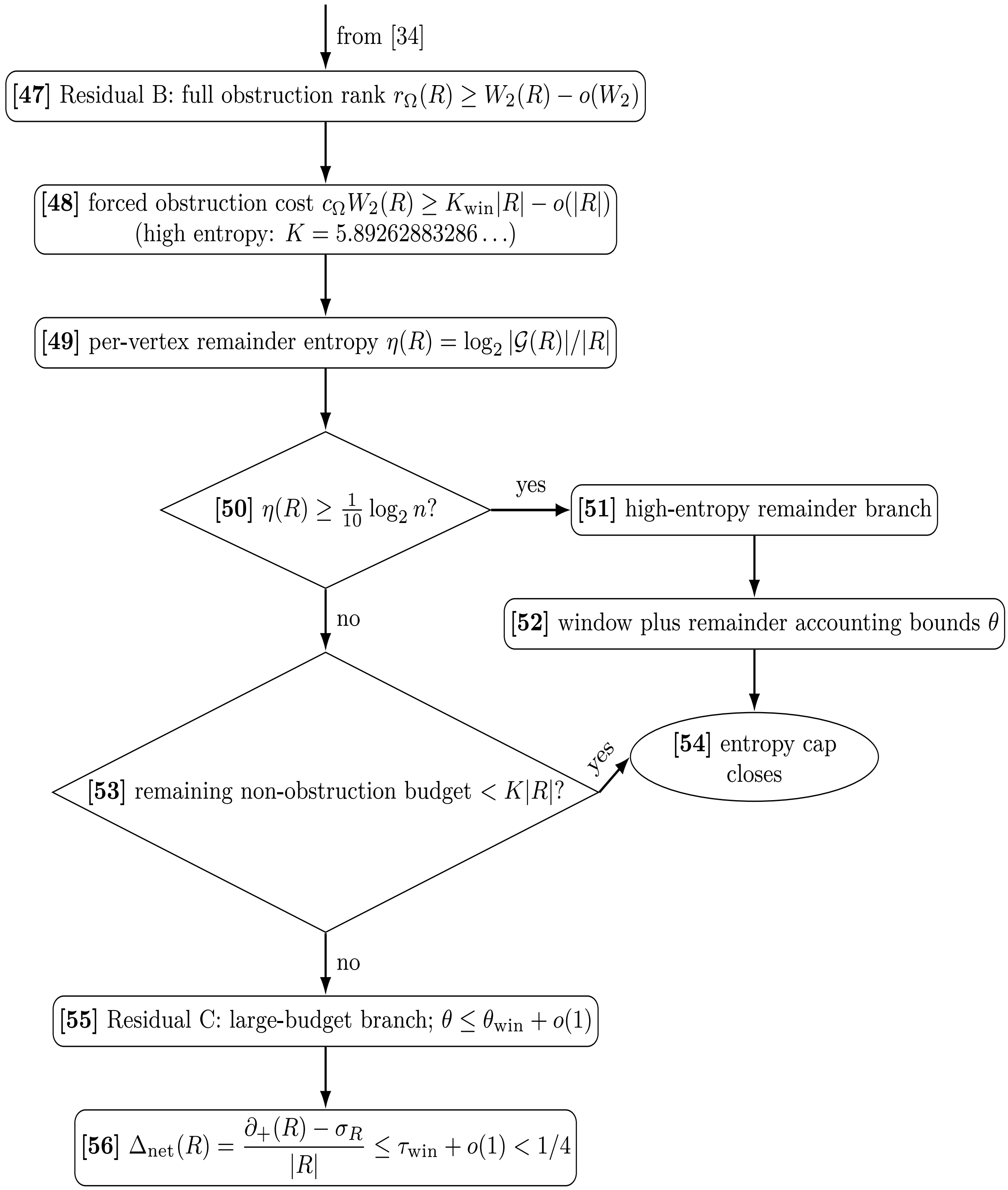

Figure 4: Proof-dependency diagram, Part IV. Rectangles, diamonds, ellipses, bracketed node numbers, and edge labels use the convention of Fig. 1. The surviving continuation node is the large-budget net-deficiency residual.

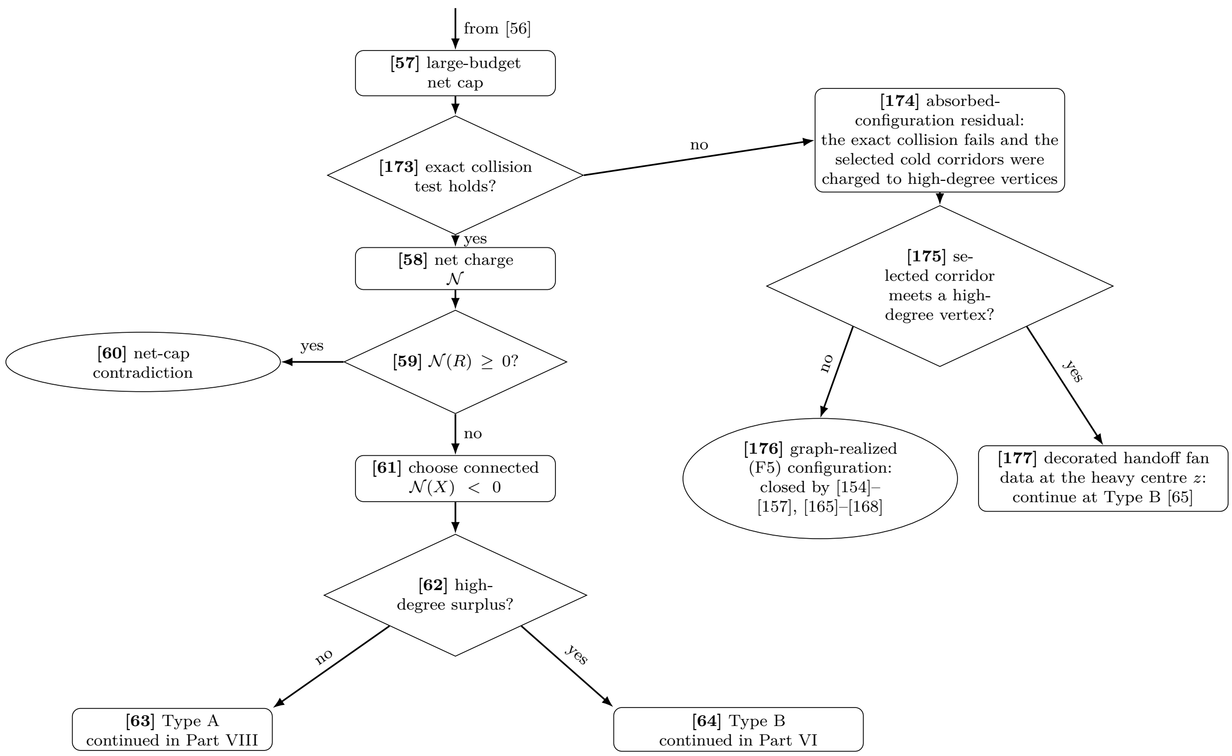

Figure 5: Proof-dependency diagram, Part V. Conventions are as in Fig. 1. This panel performs the net-charge split [57]–[64] and expands the no-edge of [173] in place. Node [173] is the exact net-cap collision test of Lemma 7.28; its failure gives the absorbed-configuration residual [174]. Node [175] restores the discarded charge by Lemma 7.29: a selected corridor with no high-degree vertex is a genuine (F5) configuration and closes at [176], while the other branch supplies decorated handoff fan data and enters Type B [65] at [177]. Node [63] continues in Fig. 8, and nodes [64] and [177] continue in Fig. 6.

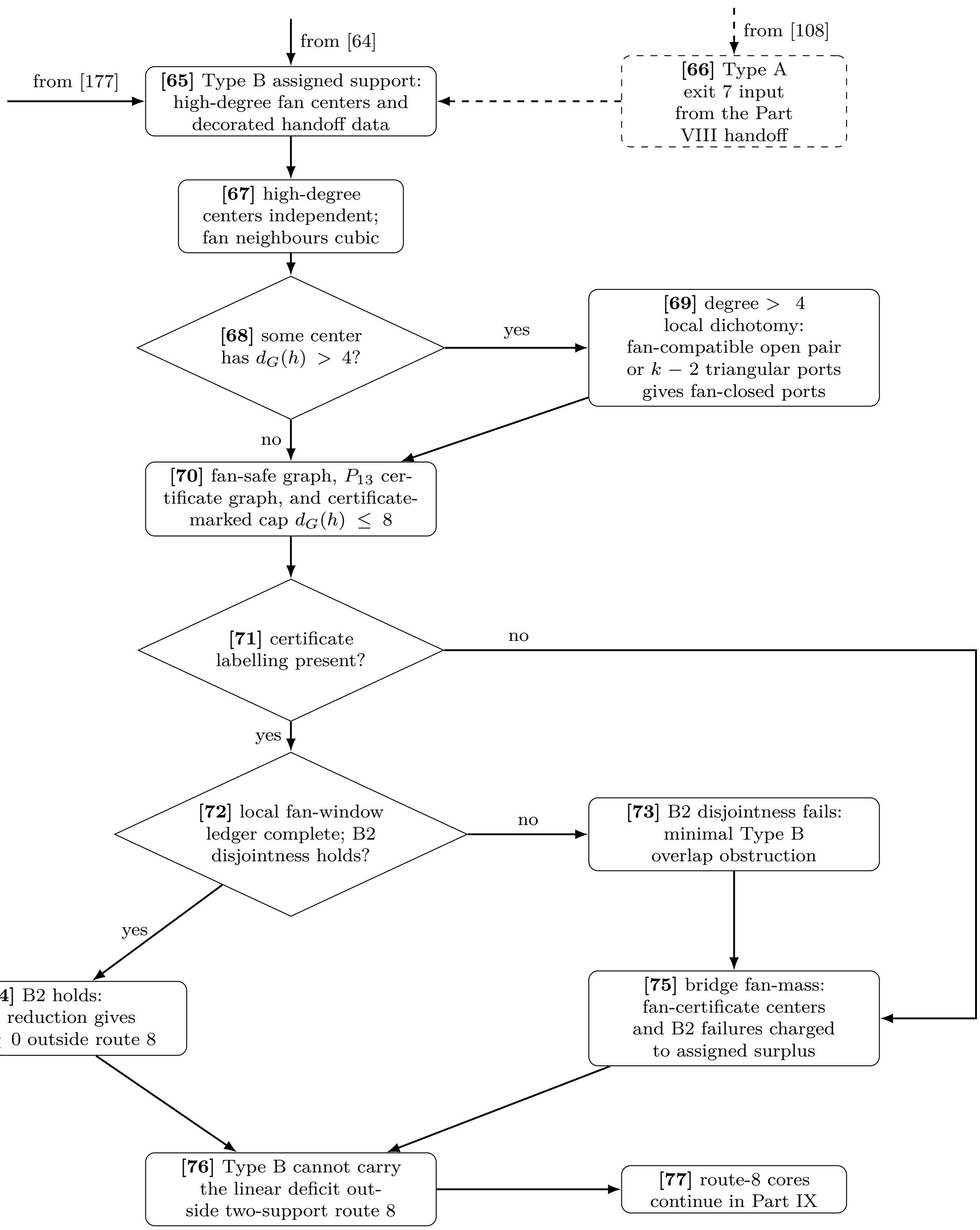

Figure 6: Proof-dependency diagram, Part VI. The Type B continuation starts from node [64], receives the incoming Type A handoff through the dashed input [66], splits at [68], and applies [69]–[76] before the route-8 continuation [77]. The degree-4 no-branch of [68] is expanded in Fig. 7.

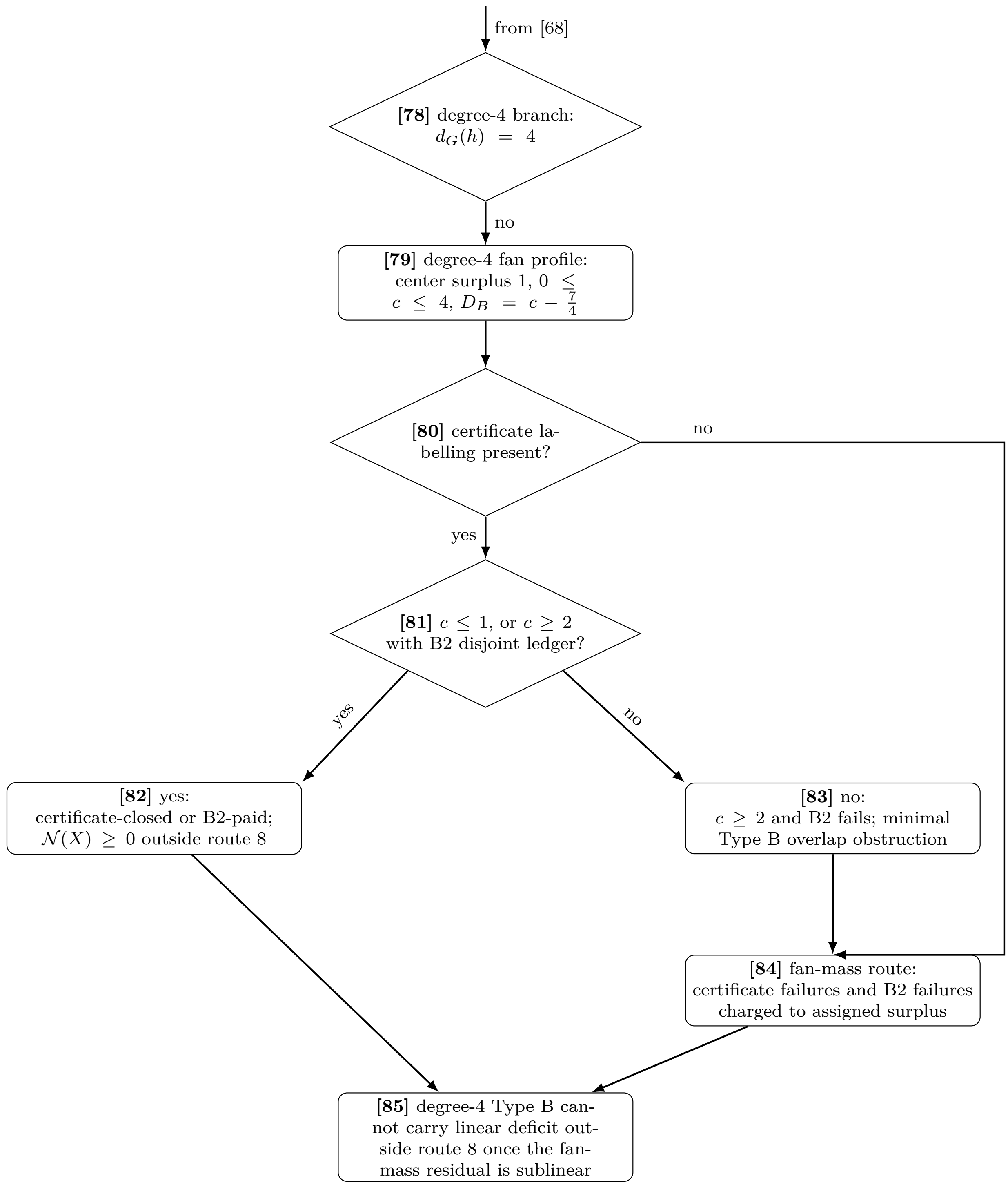

Figure 7: Proof-dependency diagram, Part VII. This panel expands the degree-4 no-branch of [68]. The local closed cases are $c \leq 1$; the difficult degree-4 cases are $c = 2, 3, 4$, where closure depends on the B1 incidence payment and the B2 disjoint ledger at [81], or else routes to the residual fan-mass node [84].

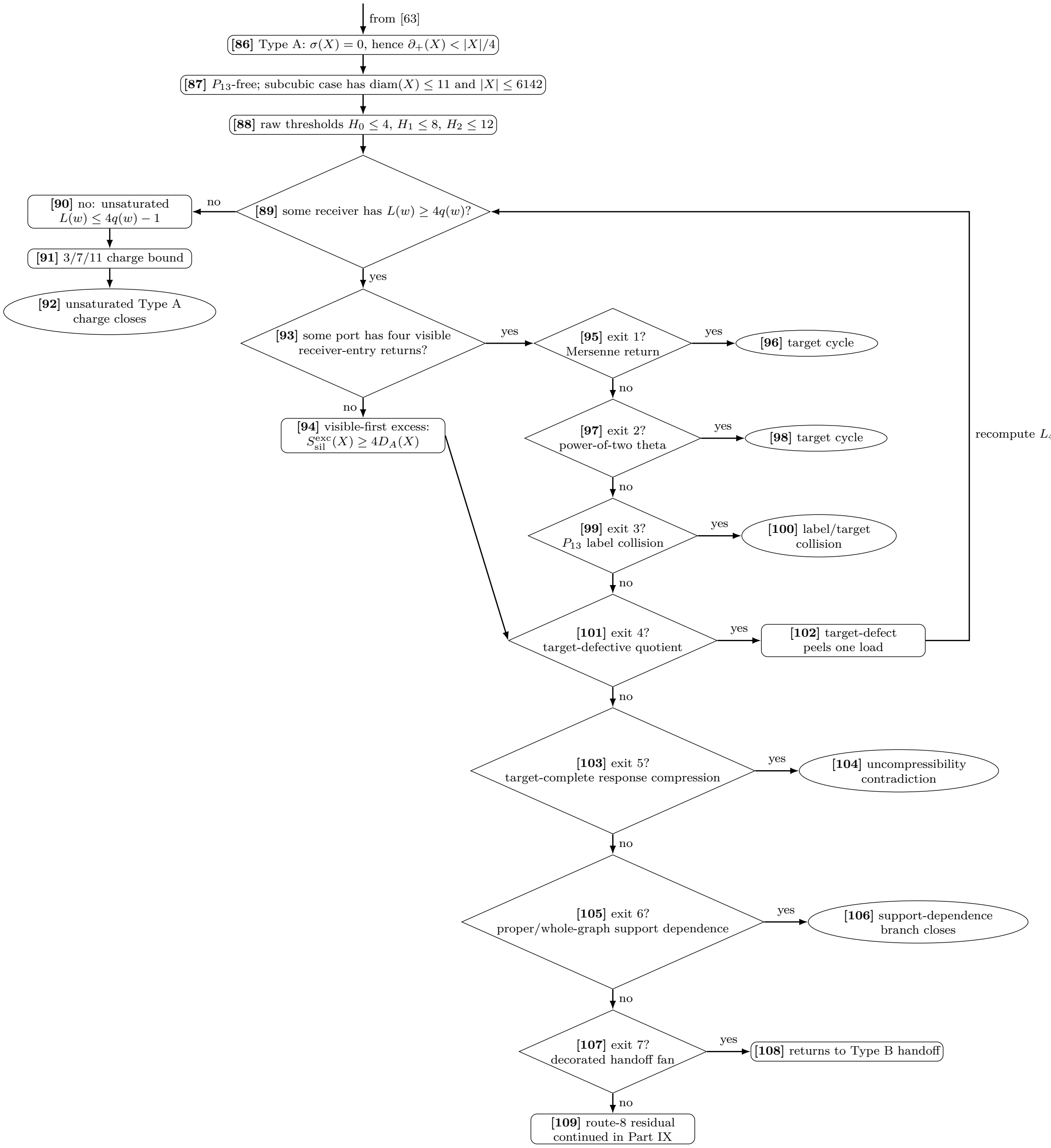

Figure 8: Proof-dependency diagram, Part VIII. Rectangles, diamonds, ellipses, bracketed node numbers, and edge labels use the convention of Fig. 1. Type A expands from node [63]. The 3/7/11 calculation applies only after the saturated branch [89] has been exhausted. If [89] is yes, exits 1–7 are tested explicitly: exits 1–3, 5, and 6 close at [96], [98], [100], [104], and [106], exit 4 peels one target-defective routed load at [102] and returns to [89] with the residual load L_4, while exit 7 hands off to Type B [65] through node [108]. On the no-branch of [93], the visible-first excess calculation gives $S^{\mathrm{exc}}_{\mathrm{sil}}(X) \ge 4D_A(X)$. Exit (8), the route-8 residual [109], is expanded in Fig. 9.

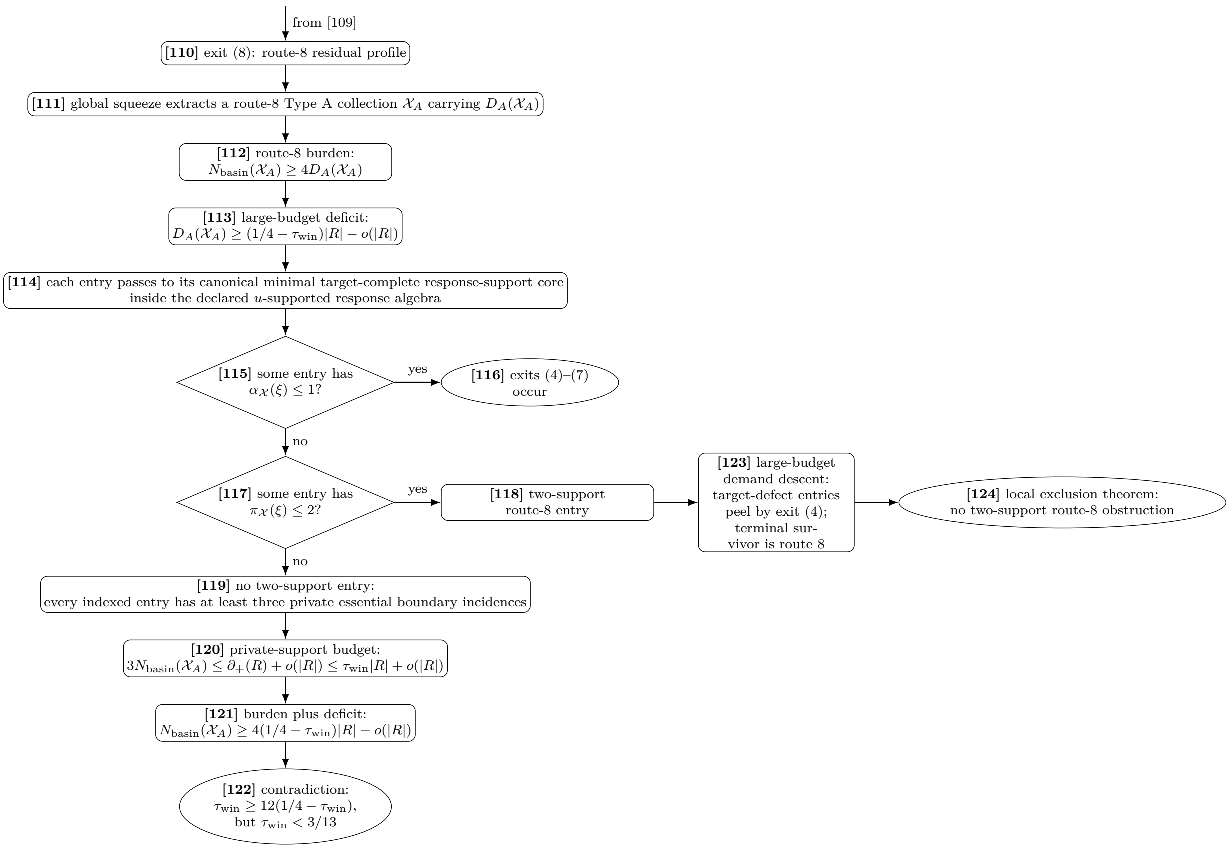

Figure 9: Proof-dependency diagram, Part IX. Rectangles, diamonds, ellipses, bracketed node numbers, and edge labels use the convention of Fig. 1. The figure expands exit (8) from Fig. 8 and records how it feeds the full demand closure. On the derived spine branch, the route-8 response-support reduction closes the no-two-support branch: the global squeeze extracts $\mathcal{X}_A$ carrying D_A, the route-8 burden gives $N_{\text{basin}} \geq 4D_A$, zero/one-essential-core entries trigger exits (4)–(7), and absence of a two-support entry gives $3N_{\text{basin}} \leq \tau_{\text{win}}|R| + o(|R|)$, contradicting $3N_{\text{basin}} \geq 12(1/4 - \tau_{\text{win}})|R| - o(|R|)$ and $\tau_{\text{win}} < 3/13$. A two-support entry enters node [123], the large-budget demand descent: target-defect entries are canonical exit-(4) peel data and decrease Λ_4, while a terminal route-8 entry is sent to node [124], where the local exclusion theorem excludes it.

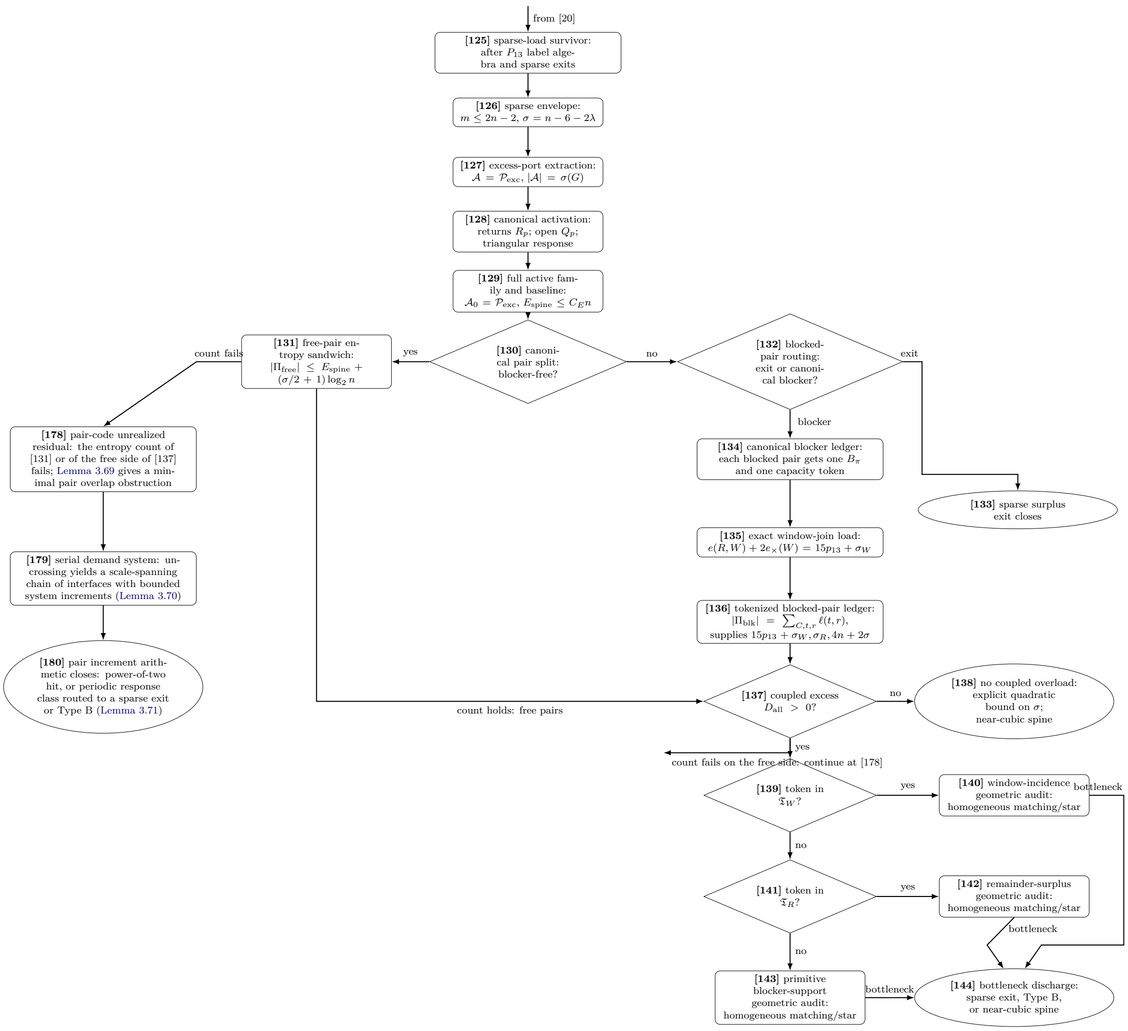

Figure 10: Proof-dependency diagram, Part X. This panel expands the sparse non-near-cubic branch [20]. Nodes [126]–[129] build the full active surplus family. Node [130] splits canonical pairs. Free pairs are charged by the entropy sandwich [131], while blocked pairs pass through the blocker route [132]–[136]; the role fibres in [136] partition $\Pi_{\rm blk}$, so this ledger has no double counting. Node [137] is the coupled single-graph test of Corollaries 3.53 and 3.54 and Proposition 3.55: all three token classes use the same $n, p_{13}, \sigma_W, \sigma_R$. If $D_{\rm all} = 0$, the explicit quadratic bound routes to [138]. If $D_{\rm all} > 0$, the role-fibre excess enters the window-incidence, remainder-surplus, or primitive blocker-support geometric audit [140], [142], [143]. Node [144] applies Theorem 3.78: the bottleneck realizes a sparse surplus exit, produces Type B fan data, or is bounded by fixed homogeneous caps and routes to [138]. The entropy counts of [131] and of the free side of [137] are branch tests, exactly as at [158]: when the count holds the branch continues as drawn, and when it fails the residual [178] is closed by structural accounting instead of independent bits — Lemma 3.69 extracts a minimal overlap obstruction among the pair-response supports, Lemma 3.70 uncrosses it into a serial demand system [179], and Lemma 3.71 closes that system by the doubling-orbit arithmetic of Lemma 7.15: a power-of-two hit, or a periodic response class routed to a sparse surplus exit or to Type B fan data [180].

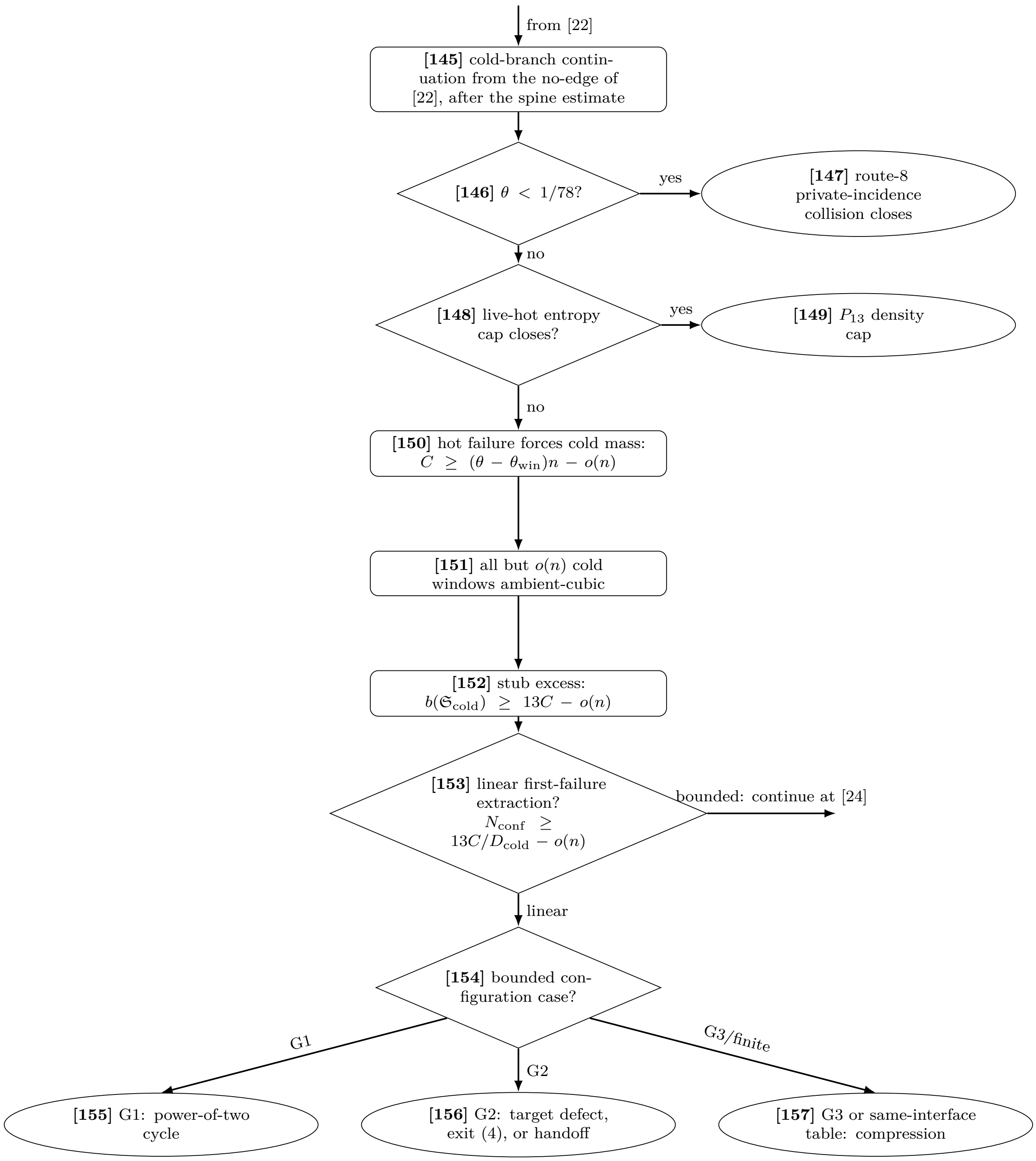

Figure 11: Proof-dependency diagram, Part XI. This panel continues the no-edge of the hot/cold split at node [22]. If $\theta < 1/78$, the existing route-8 response-support inequality closes at [147]. Otherwise the live-hot entropy comparison either closes at [149], or its failure forces the unpaid cold family to have linear size [150]. On the near-cubic spine, only $o(n)$ cold windows are non-cubic [151]; each ambient-cubic cold window has 15 external stubs, and after two corridor ends are absorbed it contributes 13 branch-excess units [152]. Node [153] makes the exact finite split hidden by the asymptotic notation: its bounded arm supplies the density cap and returns to [24], while its linear arm extracts a positive disjoint configuration family and continues to [154]. The bounded-configuration trichotomy [154] has no terminal residual: G1 gives a power-of-two cycle [155], G2 routes to target defect, exit (4), or the already existing handoff ledgers [156], and G3 together with the finite same-interface table gives a target-complete compression [157].

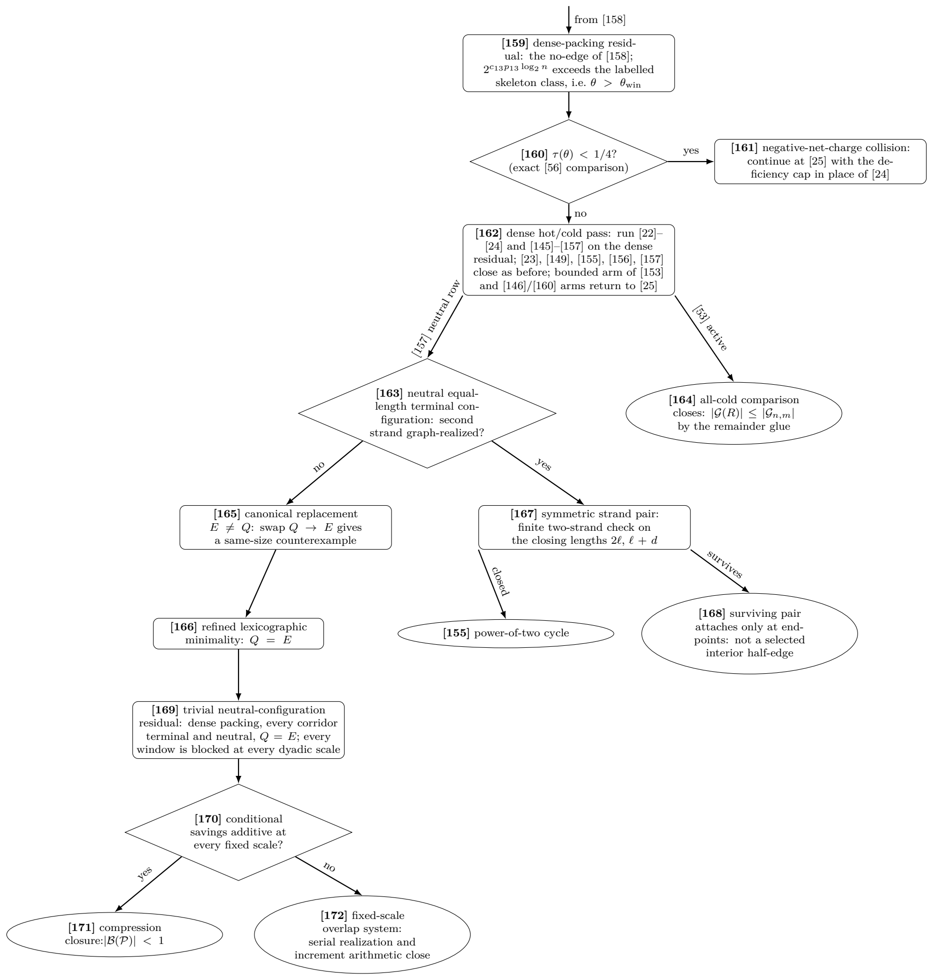

Figure 12: Proof-dependency diagram, Part XII. The dense residual [159] passes through the exact deficiency test [160] and the dense hot/cold pass [162]. The all-cold arm closes at [164]; the neutral arm [163] splits into the replacement route [165]–[166] and the two-strand route [167], [155], [168]. The continuation of [166] is included here: the trivial neutral residual [169] reaches either compression [171] through additivity [170], or the serial-system arithmetic closure [172].

Detailed dependency table

The table records every named object in the proof-dependency diagram. The *Diagram node(s)* column uses the bracketed node numbers from the twelve numbered parts. The Type A continuation node [63] is expanded in Fig. 8, the Type B continuation nodes [64] and [177] are expanded in Fig. 6, the route-8 residual node [109] is expanded in Fig. 9, the surplus-pair accounting branch [20] is expanded in Fig. 10, and the hot/cold window interface is expanded in Fig. 11. The dense-packing residual and its trivial-neutral continuation form the consecutive run [159]–[172] in Fig. 12; the exact-collision and absorbed-configuration branch [173]–[177] is displayed at its origin in Fig. 5. The item numbers remain theorem-level bookkeeping and are not diagram-node identifiers.

Item	Diagram node(s)	Node / theorem	Formal content	Failure route	Label
0	[1]–[3]	Counterexample setup	$\delta(G) \geq 3$ and no C_{2j}	If false, theorem holds for G	Definition 2.4
1	[4]	Minimality	choose minimal $(\|V\|, \|E\|)$ counterexample	smaller counterexample impossible	Assumption
2	[5]–[7]	Edge-rooted target equivalence	no C_{2j} iff all $R_e(G) \cap \mathcal{M} = \varnothing$	Mersenne return gives target cycle	Lemma 2.2
3	[8]	No proper core	every proper subgraph has $\delta \leq 2$	proper smaller counterexample	Lemma 3.1
4	[9], [10], [67]	High-degree independence	$V_{\geq 4}$ independent	high-high edge deletion	Lemma 3.2
5	[11]	Boundaried graphs	pieces are T-boundaried supports	replacement formalism	Definition 3.81
6	[11], [36], [37]	Boundary degree profiles	quotients fibrewise over $\mathbf{d}_\partial$	otherwise target-defective	Lemma 3.94
7	[12], [36]	Context universality	target-complete identities work against every context	otherwise quotient defect	Lemma 3.95
8	[13]	Replacement	smaller, at most as obstructing representative gives smaller counterexample	minimality contradiction	Lemma 3.96
9	[14], [38], [39]	Hereditary uncompressibility	no proper target-complete compression	replacement contradiction	Corollary 3.98
10	[31]	Target rank	independent coordinates realize 2^k states	entropy only counts independent coordinates	Definition 3.99
11	—	Skeleton budget	labelled edge sets dominate all canonical data	no extra auxiliary degrees of freedom	Lemma 4.2
12	—	Near-cubic budget	$B_{\text{skel}} = \frac{3}{2} n \log_2 n + o(n \log n)$	budget scale	Lemma 4.4
13	[15], [16]	P_{13}-free theorem	P_{13}-free $+\delta \geq 3$ gives target cycle	forces induced P_{13}	Theorem 5.1
14	[18]	P_{13} label algebra	399 labels and relations C_s	finite algebra basis	Lemma 6.1
15	[21]	Two-step obstruction enumeration	$543958, 432672, 111286$ and c_Ω	verified finite computation	Lemma 6.2
16	[21]	Multi-scale window package	$c_{13} = 118.108581006\ldots$ and $(c_{13} - o(1))p_{13} \log_2 n$ independently target-testable window coordinates	controls P_{13} packing density	Lemma 7.1; Section A
17	[22]–[24]	P_{13} density bound	after surplus-token routing supplies Definition 4.1, $\theta \leq \theta_{\text{win}} + o(1)$; high entropy gives $\theta \leq 0.01198542083\ldots + o(1)$	window entropy overflow / window-only cap	Lemma 7.1, Proposition 7.42
18	[145]–[157]	Hot/cold window branch	if $\theta < 1/78$, route 8 closes; otherwise hot failure forces $C \geq (\theta - \theta_{\text{win}})n - o(n)$, $b(\mathfrak{S}_{\text{cold}}) \geq 13C - o(n)$, and $N_{\text{conf}} \geq 13C/D_{\text{cold}} - o(n)$; bounded configurations route to hit, defect, handoff, route 8, same-interface table, or compression	no terminal cold residual after the spine estimate	Definitions 7.2, 7.3, 7.5, 7.8, 7.9 and 7.13, Lemmas 7.4, 7.6, 7.7, 7.10 to 7.12, 7.14 and 7.15, and Theorem 7.16
19	[17], [25]–[27]	Remainder R	large, componentwise P_{13}-free, and with empty internal 3-core	maximality of packing and the HSS theorem	Lemma 8.2

20	[28]	Positive deficiency	$\partial_+(X) = \sum_v \max(0, 3 - d_X(v))$	high degree cannot cancel deficient vertices	Definition 8.3
21	[29]	External-incidence supply	$\partial_+(R) \leq 15p_{13} + o(n)$ and hence $\partial_+(R) - \sigma_R \leq 15p_{13} + o(n)$	few windows give small supply	Lemma 8.5
22	[30]	Wedge lower bound	componentwise $W_2(C) \geq 3\|V(C)\| - 2\partial_+(C)$; with $\partial_+(R) \leq (\tau_{\text{win}} + o(1))\|R\|$, this gives $W_2(R) \geq \omega_{\text{win}}\|R\| - o(\|R\|)$; high entropy gives $\omega\|R\| - o(\|R\|)$	degree-count proof	Lemma 9.1
23	[31]	Obstruction rank	$r_\Omega(R)$ counts independent obstruction tests via functional admissible rank quotients of exact response profiles	raw tests are not charged directly; closed empty-boundary data stay label-injective	Definitions 3.88, 3.91, 3.92 and 10.1
24	[32], [33], [35]–[37]	Localization of obstruction dependence	rank drop gives a finite functional target-dependence and routes to target defect/admissible compression/support	dependence cannot be silent locally	Lemmas 10.3 and 10.4
25	[35]–[37]	Separated wedge testers	identical separated wedge balls are context-universal or defective	supports full-rank forcing	Lemma 10.5
26	[40]–[42]	Proper support enlargement	enlarged $Z \subsetneq G$ gives target defect or target-complete proper-support compression	context defect or replacement contradiction	Lemma 10.6
27	[44]	Repair identity	$s = p - 2 + 2\beta_Z - \sigma_Z$	whole-graph dependence rank barrier	Lemma 10.8
28	[43], [45]	No silent whole-graph dependence	$Z = G$ gives target-defect, smaller closed counterexample, or exact label-injective global data	no global silent rank reduction	Definitions 3.88 and 3.91 and Lemma 10.9
29	[34], [46], [47]	Full obstruction rank	$r_\Omega(R) \geq W_2(R) - o(W_2)$	if rank drops, contradiction	Lemma 10.10
30	[48]	Forced obstruction cost	$c_\Omega W_2(R) \geq K_{\text{win}}\|R\| - o(\|R\|)$; high entropy gives $K\|R\| - o(\|R\|)$	finite-size cost	Corollary 10.11
31	[49]	Per-vertex remainder entropy	$\eta(R) = \log_2 \|\mathcal{G}(R)\|/\|R\|$	separates high-/low-entropy remainder regimes	Definition 11.1
32	[49], [50]	Dominant local type	low-entropy remainder has dominant rooted type	otherwise too many labelled remainder states	Lemma 11.2
33	[51], [52]	Translates independent	separated dominant wedges are independently target-testable	otherwise rank-forcing applies	Lemma 11.4
34	[50]–[54]	Two-budget routing	high-entropy branch uses window entropy; repetitive low-entropy branch gives positive obstruction rank; nonrepetitive low-entropy branch is routed forward	prevents entropy overcounting	Proposition 11.6
35	[53], [54]	Finite threshold $\Theta(n)$	high-θ branch closed by entropy cap	small-budget closure	Definition 12.1, Proposition 12.2
36	[55]–[57]	Net-deficiency cap	large-budget branch has $\Delta_{\text{net}} \leq \tau_{\text{win}} + o(1) < 1/4$	very low net deficiency required	Section 13
37	[58]	Net charge	$\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \|X\|/4$	failure localizes to negative support	Definition 13.3
38	[59]–[62]	Net charge localization	negative global net charge gives connected negative support	Type A/Type B split	Lemma 13.4, Proposition 13.7

39	[63], [86]–[109]	Type A obstruction	bounded low-deficiency P_{13}-free boundaried piece split into closed saturated exits, target-defect peeling, Type B handoff, unsaturated discharging, and the route-8 trace-support branch	raw 4/8/12 thresholds, exits 1–3, 5, and 6 close, exit 4 peels exactly one target-defective routed load, exit 7 hands off to Type B through connector-configuration routing, cubic-switch absorption, and high-degree handoff, unsaturated 3/7/11 discharging, route-8 burden $S^{\mathrm{exc}}_{\mathrm{sil}}(X) \geq 4D_A(X)$, declared u-supported response algebra, canonical minimal target-complete response-support core, cut parity, zero/one essential-support collapse, private-support reduction to a two-support entry, support-deletion witnesses, and the terminal exclusion theorem	Definition 14.3, Definition 14.11, Definition 14.16, Definition 14.33, Lemma 14.23, Lemma 14.24, Lemma 14.25, Lemma 14.34, Lemma 14.35, Lemma 14.36, Lemma 14.37, Lemma 14.39, Lemma 14.27, Lemma 14.40, Lemma 14.43, Lemma 14.45, Definition 14.15, Definition 14.46, Definition 14.48, Lemma 14.47, Lemma 14.56, Lemma 14.58, Definition 14.51, Lemma 14.52, Lemma 14.53, Lemma 14.54, Lemma 14.55, Definition 14.60, Theorem 14.61, Proposition 14.62

40	[64]–[85], [108]	Type B obstruction	degree-4 fan-safe surplus support, degree-greater-than-four heavy-center side branch, and decorated handoff fan data	degree-four centers give either a fan-compatible open pair or two triangular ports; heavy centers reduce to a fan-compatible open pair or three triangular ports; in either case two fan-closed ports give a positive-deficit Type B fan; triangular shoulders have only central/cross/outside completion landings; certificate-marked fans have the local cap and local certificate/B1 ledger; fan-certificate residual centers go to fan-mass; B2 failures form minimal overlap obstructions; on the derived spine branch all Type B bridge residual mass is bounded by the global surplus budget	Definition 14.19, Lemma 14.26, Definition 14.18, Definition 3.7, Definition 3.14, Lemma 3.8, Definition 3.9, Lemma 3.10, Corollary 3.13, Lemma 3.11, Corollary 3.12, Lemma 3.15, Lemma 3.16, Lemma 3.17, Lemma 3.18, Definition 14.69, Lemma 14.70, Proposition 14.80, Corollary 14.81, Corollary 14.82, Definition 14.68, Definition 14.72, Definition 14.73, Definition 14.74, Lemma 14.75, Lemma 14.76, Lemma 14.77, Lemma 14.78, Lemma 14.79, Definition 14.83, Definition 14.84, Definition 14.85, Lemma 14.87, Lemma 14.88, Proposition 14.89, Definition 14.90, Proposition 14.92, Definition 14.71, Lemma 14.93, Definition 14.95, Lemma 14.97, Lemma 14.98, Lemma 14.99, Proposition 14.100

41	[76], [85]	Branch-closure theorem	After surplus-token routing has supplied the near-cubic spine estimate, pure Type A outside route 8 closes locally and Type B bridge residual centers and B2 failures are sublinear globally	outside explicit Type A route 8 and Type B bridge residuals, the entropy-admissible branch has no survivor	Theorem 15.1
42	[123]	Large-budget demand descent	The remaining linear Type A deficit is placed in the unified target-defect/route-8 ledger; every target-defect two-support entry is canonical exit-(4) peel data, and finite descent removes it	the terminal large-budget branch can survive only through a two-support route-8 entry, which is then routed to node [124]	Definition 15.2, Definition 15.12, Definition 15.45, Lemma 15.46, Lemma 15.44, Lemma 15.47, Theorem 15.48
43	[110]–[124]	Route-8 exit closure	after target-defect descent, the route-8 burden gives $N_{\text{basin}} \geq 4D_A$; canonical response-support cores close zero/one-essential-core entries; absence of a two-support entry forces incompatible private-support upper and lower bounds; a two-support entry supplies a terminal obstruction with declared deletion witnesses	node [124] rules out the terminal two-support obstruction because the two-support condition places each declared deletion quotient in the canonical exit-(4) family, while exit (4) is absent in a true route-8 residual	Definition 14.46, Definition 14.48, Definition 14.50, Lemma 14.47, Lemma 14.56, Lemma 14.58, Proposition 14.59, Definition 14.51, Lemma 14.52, Lemma 14.53, Lemma 14.54, Lemma 14.55, Definition 14.60, Theorem 14.61, Proposition 14.62, Theorem 15.48
44	—	Enumeration appendix	reproduces all finite constants including c_{13}	constants are checkable	Section A
51	[158]–[168]	Dense-packing residual	the realization sentence of Lemma 7.1 is decided at [158]; its no-branch [159] is closed for $\tau(\theta) < 1/4$ by the deficiency routing [160]–[161] and otherwise runs the hot/cold pass [162]; the neutral equal-length terminal configuration [163] is a symmetry, split into the canonical-replacement case closed by refined minimality [165]–[166] and the symmetric strand pair closed by the two-strand finite check [167]; the surviving symmetric pairs attach only at window endpoints and are never selected interior half-edges [168]; the all-cold comparison [164] closes by the remainder glue injection	closed	Definition 7.17, Lemma 7.18, Lemma 7.19, Definition 7.20, Lemma 7.21, Lemma 7.22, Lemma 7.23, Lemma 7.24, Definition 7.25, Lemma 7.26

52	[173]–[177]	Small-order repair	node [56]'s collision is decided exactly on the object at [173]; its failure forces linearly many cold windows on the bounded arm of [153] whose corridors were charged as the configuration-extraction loss; [175] restores that charge: subcubic supports are genuine (F5) configurations [176], supports through a high-degree vertex are decorated handoff fan data entering Type B at [65] [177]	no sufficient-order condition remains	Lemma 7.28, Lemma 7.29
53	[169]–[172]	Trivial neutral configuration: compression or system-arithmetic closure	the residual left by [166] is exposed barrier by barrier; additive conditional fibres give the blocked-class compression [171], while failure at one scale gives a minimal local overlap obstruction, a serial simple-cycle system, and the increment-periodicity closure [172]	closed	Definition 7.31, Definition 7.32, Lemma 7.33, Definition 7.34, Lemma 7.35, Lemma 7.36, Lemma 7.37, Lemma 7.38, Lemma 7.39
54	[178]–[180]	Pair-code unrealized residual	the entropy counts of [131] and of the free side of [137] are branch tests; their failure supplies a minimal pair overlap obstruction [178], uncrossed into a serial demand system [179] and closed by the pair increment arithmetic [180] (power-of-two hit, or periodic response class routed to a sparse exit or Type B)	closed	Definition 3.68, Lemma 3.69, Lemma 3.70, Lemma 3.71, Lemma 3.72

45	[19], [20]	Non-near-cubic surplus branch	every survivor of the sparse surplus exits has the full active-demand family and, after an $O(n)$-deficit baseline demand is fixed, satisfies the exact capacity-token budget; the geometric same-token bottleneck lemma gives fixed homogeneous caps unless the load realizes a sparse surplus exit or ordinary Type B fan data, and the caps give $\sigma(G) = O(\sqrt{n})$	derives the near-cubic spine before the normalized entropy budget is used, except for branches that are already routed to sparse exits or to the Type B fan ledger; blocker-free pair responses are charged by the exact cubic-baseline sandwich and the spine lower-bound deficits, while blocked pairs are assigned to capacity tokens; the exact token-fibre budget gives the load forced by any attempted violation, and Lemma 3.77 turns a large homogeneous fibre into a geometric first-separator/fan route	Lemma 3.3, Lemma 3.25, Definition 3.26, Definition 3.29, Lemma 3.30, Definition 3.31, Definition 3.33, Lemma 3.34, Definition 3.35, Lemma 3.36, Definition 3.37, Lemma 3.38, Lemma 3.39, Definition 3.40, Definition 3.41, Definition 3.76, Lemma 3.47, Lemma 3.43, Lemma 3.77, Theorem 3.78, Corollary 3.44, Definition 3.62, Lemma 3.63, Lemma 3.64, Proposition 3.66, Proposition 3.67, Theorem 3.56, Definition 8.4, Lemma 8.6, Corollary 13.5, Proposition 3.80
46	[125]–[129]	Sparse accounting setup	a sparse-load survivor yields the full active surplus family and an $O(n)$-deficit baseline demand	[129] fixes the common baseline used by the later sandwich	Definition 3.26, Definition 3.29, Lemma 3.30, Definition 3.62
47	[130]–[132]	Free/blocked pair split	pair-response coordinates split into blocker-free and blocked pairs; free pairs are charged by the entropy sandwich	blocked-pair debt is the only branch passed to the token ledger	Definition 3.58, Lemma 3.59, Proposition 3.67
48	[132]–[136]	Blocker and token ledger	canonical blockers, primitive blocker supports, and capacity tokens count blocked pairs exactly once	[136] is the exact token-fibre partition and no-overcounting closure	Definition 3.31, Definition 3.32, Definition 3.33, Lemma 3.34, Definition 3.35, Lemma 3.36, Definition 3.37, Lemma 3.38, Lemma 3.39
49	[135], [136]	Window-join load	the exact packed-window join identity supplies $15p_{13} + \sigma_W$ and the surplus-aware window-stub load	supplies the window/remainder capacity side of the token ledger	Definition 8.4, Lemma 8.6, Corollary 13.5, Remark 3.79
50	[137]–[143]	Coupled high-load split	the single-graph high-load test splits the forced homogeneous pattern into window-incidence, remainder-surplus, and primitive-blocker-support token classes	[139]–[143] are the three token-class audits	Definition 3.40, Definition 3.41, Definition 3.42, Lemma 3.50, Lemma 3.47, Theorem 3.51, Theorem 3.56
51	[140], [142]–[144]	Geometric bottleneck discharge	a large same-token fibre gives a homogeneous matching or star, then realizes a sparse exit, Type B fan data, or fixed homogeneous caps	[144] sends the branch to a sparse exit, Type B, or [138]	Definition 3.76, Lemma 3.77, Theorem 3.78, Corollary 3.44

The constraint ledger

The reduction tracks a fixed list of *invariants*: numbered structural constraints that a minimal counterexample G must satisfy, together with the entropy or deficiency *budget* each one spends and the labeled results that introduce or consume it. Throughout, G is the lexicographically minimal counterexample, so $\delta(G) \geq 3$ and $R_e(G) \cap \mathcal{M} = \varnothing$ for every oriented edge. Budgets quoted in the skeleton normalization $B_{\text{skel}} = \frac{3}{2} n \log_2 n + o(n \log n)$ are used only after Proposition 3.80 has either routed the non-near-cubic sparse-surplus branch to a closed ledger or supplied the near-cubic spine estimate of Definition 4.1. We write $\theta = p_{13}/n$, with densities taken per $|R|$ or per n as noted. We recall the recurring parameters $\beta = m - n + 1$ (cycle rank), $\lambda = 2n - 3 - m$ (sparsity rank), $\sigma = 2m - 3n$ (surplus above cubic), and, for a boundaried piece C, its deficiency $d(C) = \sum_v (3 - d_C(v))$. The ledger is organized into blocks; the reverse (result $\to$ invariant) view is given in the next subsection.

External inputs (black boxes). The argument imports four external results; only the first enters the final proof as a load-bearing black box. The non-near-cubic surplus branch is internal: it uses the free/blocked pair split, the capacity-token ledger, and the same-token bottleneck routing of Lemma 3.77.

Input	Statement used	Constraint it imposes on G	Used by
HSS P_{13}-free theorem (Theorem 5.1)	P_{13}-free + $\delta \geq 3 \Rightarrow$ a power-of-two cycle	G must contain an induced P_{13}; after removing a maximal disjoint P_{13} family, every remainder component is P_{13}-free and boundary-deficient	Corollary 5.2, Lemma 8.5, invariants 20–24, entire remainder analysis
Bondy–Vince / Gao–Ma	Few sub-cubic vertices force two cycle lengths differing by 1 or 2; bounded-block cap $\lvert X\rvert < 5b(X)^2$	Large almost-cubic boundaried pieces must export close cycle lengths or pay quadratic boundary	Appendix resilience bookkeeping
Kleitman–West max-leaf	$\delta \geq 3 \Rightarrow$ spanning tree with $\geq n/4 + 2$ leaves	Cotree fundamental cycles must avoid Mersenne tree distances	Appendix tree–cotree profile bookkeeping
Degree-3-critical / 1–3 tree literature	Extremal graphs with no proper $\delta \geq 3$ subgraph have 1–3 tree / leaf-path structure	Guides the repair-network identity	Appendix marked length-rank bookkeeping

Block I: minimality and target algebra (invariants 1–8).

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
1	Mersenne-return target algebra	Target [5]	No power-of-two cycle $\iff$ $R_e(G) \cap \mathcal{M} = \varnothing$ for every directed edge	— (defines the target predicate)	Lemma 2.2 (states it); Definition 2.5; consumed by *every* quotient/piece/profile result: Lemma 3.94, Lemma 3.95, Lemma 3.96, Corollary 3.98, invariants 30–38
2	Minimal proper-subgraph condition	Core [8]	Every proper subgraph has $\delta \leq 2$	—	Lemma 3.1 (states it); underlies Lemma 3.5, Lemma 3.96
3	Edge-deletion criticality	Delete [9]	Every edge touches a degree-3 vertex	—	Lemma 3.2 (states it)

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
4	High-degree independence	High-degree [10]	$V_{\geq 4}$ is independent	—	Lemma 3.2 (corollary); used by Lemma 14.93 (fan aggregation), invariant 16
5	No passive degree-2 storage	Counterex. [2]	Final G has no degree-2 vertex; frontier degree-2 must be compensated	—	Motivates 1–3 networks; Lemma 8.5, repair-network auxiliaries
6	Replacement dominance	Replace [13]	If X' is smaller with $\mathsf{H}(X') \subseteq \mathsf{H}(X)$, replacing X by X' gives a smaller counterexample	—	Lemma 3.96 (states it); Lemma 3.94, Lemma 3.95 feed it; Corollary 3.98 derives from it
7	Lumpability defect	Univ. [12]	A quotient is valid only if it preserves target-hitting under every outside context	—	Lemma 3.95 (states it); used in final spine step 7; Lemma 6.1 (defect at path-lengths $1, 2, 3$)
8	Hereditary target-uncompressibility	Uncomp. [14]	No proper boundaried piece admits a nontrivial target-complete compression	—	Corollary 3.98 (states it); **the contradiction lever** in final spine step 7; consumed by Lemma 14.45, Lemma 14.93, invariant 29

Block II: degree, rank, sparsity (invariants 9–19).

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
9	Minimum edge count	Counterex. [2]	$m \geq \lceil 3n/2 \rceil$	—	Lemma 3.5; near-cubic specialization Lemma 4.4
10	Sparse upper envelope	Core [8]	$m \leq 2n - 2$	—	Lemmas 3.3 and 4.3; Laman identity (invariant 12)
11	Cycle-rank constraint	Minimality [4]	$\beta = m - n + 1 \geq n/2 + 1$ (up to parity)	—	Lemma 3.5 (states it); feeds 12, 13
12	Laman slack identity	Rank ids [4]	$\beta + \lambda = n - 2$	—	Definition 11.1, two-budget split; standing notation; Lemma 4.4
13	Surplus–rank relation	Rank ids [4]	$\sigma = 2\beta - n - 2$, $\lambda = n/2 - 3 - \sigma/2$	—	Near-cubic accounting; Lemma 13.4 (via σ)
14	High-degree surplus bound	Near-cubic spine [21] (the no-edge of the surplus test); surplus branch [125]–[144] supplies the same estimate for its survivors	$\sigma = O(\sqrt{n})$ for survivors after sparse exits and Type B handoff data are routed, because Theorem 3.78 supplies fixed role-homogeneous same-token caps for all three token classes	σ-budget: $O(\sqrt{n})$	Definition 4.1, Definition 3.40, Definition 3.41, Definition 3.76, Lemma 3.47, Lemma 3.43, Lemma 3.77, Theorem 3.78, Corollary 3.44, Proposition 3.80; Proposition 13.7
15	Near-cubic edge count	Near-cubic spine [21] (the no-edge of the surplus test); surplus branch [125]–[144] supplies the same estimate for its survivors	$m = \frac{3}{2}n + O(\sqrt{n})$, $\bar{d} = 3 + O(n^{-1/2})$ for the same survivors	sets B_{skel} coefficient $\frac{3}{2}$	Definition 4.1, Theorem 3.78, Proposition 3.80, Lemma 4.4, Lemma 4.2, all entropy inequalities
16	Certificate-marked fan degree	High-degree [67]	certificate-marked Type B fans satisfy $d_G(h) \leq 8$; fan-certificate residual centers are charged by $\sum_h (d_G(h) - 3)$	label-packing cap plus σ-budget	Lemmas 14.66 and 14.93 and Proposition 14.100; related to invariant 4
17	Spectral feasibility mask	Spectral [70]	$A(H) \leq M$, $\delta(H) \geq 3 \;\Rightarrow \rho(M) \geq 3$	—	Auxiliary (spectral masks, backup)
18	Spectral equality branch	Spectral [70]	$\rho(M) = 3$ closing $\Rightarrow H$ cubic	—	Auxiliary

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
19	Spectral surplus branch	Spectral [70]	$\rho(M) > 3 \Rightarrow$ surplus used as dense/compensation/critical/entropy	—	Auxiliary (organizes repair recursion)

Block III: induced P_{13} packing and forced remainder (invariants 20–24).

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
20	Low P_{13} density	Density [24]	after Proposition 3.80 supplies Definition 4.1, $p_{13} \le \theta_{\mathrm{win}} n + o(n)$; high entropy gives $p_{13} \le 0.01198542083 \cdot n + o(n)$	window-only entropy vs B_{skel}; sharper high-entropy cap	Lemma 7.1, Proposition 7.42; Eq. (1); Proposition 12.2; final spine steps 1–2
21	Forced remainder	Residual A [25]	$\lvert R\rvert \ge 0.83489768623 \cdot n - o(n)$	$= (1-13\theta)n$	Lemma 8.5, Lemma 9.1, invariant 28; all boundaried-piece analysis
22	P_{13}-free piece condition	Residual A [25]	Components of R are P_{13}-free, have empty internal 3-core, and are boundary-deficient	—	Lemmas 8.2 and 8.5; Lemma 14.45, Lemma 14.93
23	Window stub capacity	Stub accounting [29]; exact identity on the surplus branch [135]	Exact split $e(R,W) + 2e_\times(W) = 15p_{13} + \sigma_W$; on the derived spine branch this gives $e(R,W) \le 15p_{13} + o(n)$	15 per window plus explicit window surplus	Lemma 8.6, Lemma 8.7, Lemma 8.5; invariant 24
24	Remainder deficiency density	Stub accounting [29]; surplus-branch form [136]	$\partial_+(R) \le 15p_{13} + O(\sqrt{n})$ on the derived spine branch; before that estimate is supplied, $\partial_+(R) - \sigma_R \le 15p_{13} + \sigma_W - \sigma_R$ and no-negative-support forces $\sigma_W - \sigma_R \ge (n - 73p_{13})/4$	15θ aggregate plus window-join load	Lemma 8.5, Lemma 8.7, Corollary 13.5; final spine step 3; Proposition 13.7; **net-charge collision** (vs Type A/B output)

Block IV: obstruction constraints (invariants 25–29).

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
25	Legal P_{13} labels	Label algebra [18]	399 legal labels; the zero-defect quotient through path-lengths $1,2,3$ is the identity	finite (399)	Lemma 6.1 (states it); Lemma 6.2; Remark 14.67; invariant 7; Lemma 14.45, Lemma 14.93
26	Two-step obstruction density	Obstruction enumeration [21]	$432672/543958 = 0.795414\ldots$ of locally safe wedges are obstructing	finite enumeration	Lemma 6.2 (states it); invariant 28; Section A
27	Flatness entropy cost	Cost [48]	$c_\Omega = 2.28922315244\ldots$ bits per independent obstruction coordinate	bits/coord	Corollary 10.11, Remark 10.12, Proposition 11.6, Eq. (2)
28	Raw piece-obstruction supply	Wedge lower bound [30]	after surplus-token routing supplies Definition 4.1, $W_2(R) \ge 2.54365026308 \cdot \lvert R\rvert$; high entropy gives $2.57407357888 \cdot \lvert R\rvert$	lower bound (demand)	Lemma 9.1 (states it); Lemma 10.10, Corollary 10.11; final spine step 4

#	Invariant	Node (first tracked)	Constraint on G	Budget	Lemmas/theorems using it
29	Obstruction-rank compression ratio	Full-rank squeeze [47]	$r_\Omega/W_2 \le 0.23758 + o(1)$ forces a rank-defect route unless the full-rank squeeze applies	budget cap $0.611561 \cdot \lvert R\rvert$ independent tests	Lemma 10.10, Proposition 11.6, Theorem 15.1; final spine steps 5–7; collides with invariant 8 outside explicit residuals

Block V: odd-divisor / cycle-space constraints (invariants 30–38). These refine the target predicate (invariant 1) on the cycle space; all are consumed by the target-algebra layer and the replacement/uncompressibility machinery.

#	Invariant	Node (first tracked)	Constraint on G	Used by
30	Two-path criterion	Target algebra [5]	Two disjoint paths summing to 2^k $\Rightarrow$ forbidden power-of-two cycle	Lemma 2.2; Lemma 14.45 (return spectrum)
31	Theta closure	Target algebra [5]	All branch-pair sums in a theta avoid powers of two	Piece obstruction profiles; Lemma 3.96
32	Ear closure	Target algebra [5]	Ear + base-path length avoids powers of two	Hazard profiles; Lemma 3.95
33	Symmetric difference	Context universality [12]	Overlapping cycles: primitive power-of-two output kills	Lemma 3.95
34	Rank-one power-of-two target support	Obstruction rank [31], developed at [40]	All power-of-two signals must have support rank ≥ 2	Lemma 10.9, Lemma 10.6
35	Composite cancellation	Obstruction rank [31], developed at [44]; imported by the cold branch at [145]	Power-of-two cancellation must stay nonprimitive, non-realized	Lemma 10.8
36	Odd divisor present	Target algebra [5]	Every cycle length has an odd prime divisor (no odd divisor means a power of two)	Target predicate refinement
37	No common odd divisor	Divisor transport [35]	No single odd prime divides all cycle lengths	Divisor-transport analysis
38	Polarized divisor transport	Obstruction rank [31], developed at [35]; imported by the cold branch at [145]	Overlap formula $q_p(E) = q_p(C) + q_p(D) - 2t$ flat and non-killing	Lemma 10.5, support-dependence lemmas

Block VI: obstruction-rank routing and two-budget machinery. These manuscript results operate the obstruction invariants (25–29) without being separate ledger invariants; the result $\to$ invariant direction is displayed for completeness.

Result	Role	Invariants consumed
Lemma 10.4	Localizes where obstruction dependence can occur (3 cases)	26, 28
Lemma 10.5	Separated wedges tested by disjoint contexts	28, 38

Result	Role	Invariants consumed
Lemma 10.6	Proper nonlocal supports forbidden	34
Lemma 10.8	Enlarged nonlocal supports = repair networks	35
Lemma 10.9	No silent whole-graph dependence by closed exact profiles	34, 35; Definitions 3.88 and 3.91
Lemma 10.10	Forces full obstruction rank $r_\Omega \geq W_2 - o(\cdot)$	28, 29
Corollary 10.11	Obstruction rank $\to$ entropy tax on the derived spine branch	27, 28, Definition 4.1
Lemma 11.2	Dominant local type in low-entropy branch	15, 26
Lemma 11.4	Dominant type $\Rightarrow$ independent obstruction translates	26, 29
Proposition 11.6	Two-budget routing (entropy, repetitive low entropy, nonrepetitive low entropy) on the derived spine branch	20, 27, 29, Definition 4.1
Proposition 12.2	Closes high-θ branch via entropy cap on the derived spine branch	20, 27, Definition 4.1
Remark 12.4	Closure survives at $K = 0$ (obstruction machinery dispensable)	8, 24, 29

Block VII: large-budget residual and local exclusion (the spine's terminus).

Result	Role	Invariants consumed
Definition 13.1	Canonical support decomposition	21, 22
Definition 13.2	Admissible support (carries cumulative constraints)	1, 6, 8, 22, 23, 24
Lemma 13.4	Exact net-charge localization over the connected components of R	13, 14, 24
Proposition 13.7	Large-budget residual is negative net charge	14, 24, Lemma 13.4
Definition 14.3, Lemma 14.4	Define receivers, missing-port counts $q(w)$, and routed load $L(w)$	23, 24
Definition 14.5	Defines anchored returns, receiver-entry returns, visible loads, and silent loads at a completion port	1, 23, 30
Lemma 14.7, Lemma 14.8	First entries of anchored returns are receivers; degree-two receiver entries satisfy the stated port budget	23, 24
Lemma 14.9	Every completion port has an anchored return	1, 3, 30; Lemma 3.4
Definition 14.10, Lemma 14.12, Definition 14.15	Define the reduced route-8 trace-basin data, the declared u-supported response algebra, and connector-band constraints	1, 22, 23, 25, 30, 31
Definition 14.16	Enumerates the eight saturated receiver exits	1, 6, 8, 22, 25, 30, 31, 34, 35
Definition 14.33, Lemma 14.34, Lemma 14.38	Defines exit-(4) peeling and proves the exact one-load, 1/4-charge update	6, 8; Definition 14.16
Lemma 14.35	Routes four visible unpeeled receiver-entry returns to one of exits 1–7	1, 8, 25, 30, 31; Definition 14.33

Result	Role	Invariants consumed
Lemma 14.36	Routes the no-four-visible-return case to exits 4–8, with exit 4 carrying an unpeeled excess load	6, 8, 34, 35; Definition 14.33, Definition 14.15
Lemma 14.37	Combines the visible and silent residual routing lemmas after prior exit-(4) peeling	Lemma 14.35, Lemma 14.36
Definition 14.11	Defines connector configurations, first separators, and finite continuation classes	6, 8, 30, 31
Lemma 14.23	Routes nonabsorbed four-coordinate continuation data to exits 4–6 or a surviving first separator	6, 8, 30, 31; Definition 14.11
Lemma 14.24, Lemma 14.25	Turns surviving first separators into decorated Type B handoff envelopes	4, 6, 8; Definition 14.11
Lemma 14.39	Closes exits 1–3, 5, and 6, peels exit 4, and transfers exit 7 to Type B	1, 6, 8, 25, 30, 31, 34, 35; Lemma 14.38, Lemma 14.25
Lemma 14.27	Computes raw receiver thresholds $H_0 \leq 4$, $H_1 \leq 8$, $H_2 \leq 12$	4, 24
Lemma 14.28, Lemma 14.32, Lemma 14.30, Lemma 14.31	Split the saturated 4/8/12 receiver branch into routes 1–8 and quantify the reduced silent trace-basin branch	1, 8, 22, 25, 30, 31, 34, 35
Lemma 14.40	Removes closed exits 1–3, 5, and 6 and all exit-4 peeled target-defective loads before Type A discharging, hands exit 7 to Type B, and leaves reduced route 8 as a residual	Lemma 14.28, Lemma 14.32, Lemma 14.35, Lemma 14.36, Lemma 14.37, Lemma 14.39, Definition 14.15
Definition 14.46	Defines the Type A route-8 deficit D_A	14, 24; Definition 14.15, Lemma 14.31
Lemma 14.47	Gives the indexed basin lower bound $N_{\text{basin}} \geq 4D_A$ and the large-budget lower bound for D_A	14, 24; Definition 14.46
Definition 14.48, Definition 14.50	Defines route-8 entries, essential response-support cores, private response supports, and true route-8 residual entries	8, 23, 30, 31
Lemma 14.56	Forces mixed target events in a response-support core to cross at least two supports	1, 8, 30, 31; Definition 14.48
Lemma 14.58	Uses the cut-parity constraint to close zero/one essential-support entries by exits 4–7	1, 8, 30, 31; Lemma 14.56
Proposition 14.59	Spends the route-8 basin burden against the boundary-incidence ledger and reduces every surviving large-budget route-8 branch to a two-support entry	14, 23, 24; Corollary 8.8, Lemma 14.47, Lemma 14.58
Definition 14.51, Lemma 14.52, Lemma 14.53	Records deletion witnesses forced by inclusion-minimality of the essential response-support core and records their declared and response supports	6, 8, 23, 24; Definition 14.48

Result	Role	Invariants consumed
Lemma 14.54, Lemma 14.55	Proves that a two-support declared deletion quotient is canonical exit (4), and that it realizes exit (4)	1, 6, 8, 30, 31; Lemma 14.53
Definition 14.60, Theorem 14.61, Proposition 14.62	Packages the terminal two-support route-8 obstruction, proves the local exclusion theorem, and derives full route-8 closure	Proposition 14.59, Lemma 14.54, Lemma 14.55
Lemma 14.43	Unsaturated receivers give the 3/7/11 charge inequality	4, 24
Lemma 14.45	Dense $\sigma = 0$ supports outside target-defect peeling, route 8, and the Type B handoff satisfy $\partial_+ \geq \lvert X\rvert/4$; a negative one has target-defect, route-8, or Type B handoff data	1, 4, 8, 22, 23, 24, 25, 30, 31; Lemma 14.40, Lemma 14.43
Definition 14.68, Definition 14.69, Definition 14.72	Define fan-window incidences, fan-closed ports, and the hybrid window/non-window credit ledger for one Type B fan	23, 24, 25
Definition 14.73, Definition 14.74, Lemma 14.75, Lemma 14.76	Direct fan-window and two-window cycles are eliminated structurally before incidence credit is counted	1, 23, 30, 31
Lemma 14.77, Lemma 14.78, Lemma 14.79	Every certificate-marked positive-deficit survivor has deficit $D_B = c - \frac{11-k}{4}$ paid locally by half-credit on disjoint window and non-window incidences before the global B2 ledger is invoked	23, 24, 25
Proposition 14.80, Corollary 14.81, Corollary 14.82	Compatible-pair, degree-four triangular, and heavy triangular local routes enter the local Type B fan ledger	4, 8, 25, 30, 31
Definition 14.83, Definition 14.84, Definition 14.85, Lemma 14.87, Lemma 14.88, Proposition 14.89, Definition 14.90, Proposition 14.92	For supports with no fan-certificate residual center, local Type B closure is reduced to the B2 refined support ledger after local B1 is proved; disjoint-incidence failure is a minimal overlap obstruction inheriting global constraints	1, 4, 8, 22, 23, 24, 25, 34, 35
Definition 14.19, Lemma 14.26	Records Type A exit-(7) data as an admissible decorated Type B handoff envelope	1, 4, 8, 22, 25; Lemma 14.25
Definition 14.18, Remark 14.67	Defines certificate-marked Type B fans and records the fan-degree label-packing cap	16, 25
Definition 3.7, Definition 3.14, Lemma 3.8, Definition 3.9, Lemma 3.10, Corollary 3.13, Lemma 3.11, Corollary 3.12	Splits degree-four and degree-greater-than-four centers into fan-compatible open pairs or triangular-port alternatives	4, 8, 16, 25
Lemma 3.15, Lemma 3.16	Shows triangular shoulders have completion edges and that relevant returns enter through shoulders	1, 4, 8, 25, 30, 31
Lemma 3.17, Lemma 3.18	Exhausts first landings for triangular shoulders and supplies the second port in the cross-shoulder case	1, 4, 8, 25, 30, 31
Lemma 14.70	Routes compatible open pairs into two fan-closed ports	4, 8, 25; Definition 3.9

Result	Role	Invariants consumed
Definition 14.71, Lemma 14.93	Excludes certificate-marked high-degree supports with no fan-certificate residual center under B2; positive-deficit fans are paid locally and B2 ledger failures are isolated	4, 8, 16, 24, 25; Lemma 14.78, Lemma 14.79
Definition 14.95, Lemma 14.97, Lemma 14.98, Lemma 14.99, Proposition 14.100	Fan-certificate residual centers, grouped decorated envelopes, and B2 failures are charged to high-degree fan-center surplus units, with any route-8 non-window core extracted into the Type A deficit ledger; on the derived spine branch the remaining bridge mass is $O(\sigma(G)) = o(\|R\|)$	4, 14, 16, 24, 35, Definition 4.1, Lemma 14.79, Definition 14.90
Theorem 15.1	Near-cubic entropy-admissible branch closure outside explicit Type A route 8 and Type B bridge residuals	20, 27, 28, 29, 35, 39, Definition 4.1
Theorem 15.48	Closes the remaining Type A target-defect/route-8 ledger: target-defect two-support entries are peeled by finite exit-(4) descent, and the terminal two-support route-8 entry is impossible	20, 27, 28, 29, 35, 39, Definition 4.1, Proposition 14.100, Definition 15.45, Lemma 15.46, Lemma 15.44, Lemma 15.47, Proposition 14.59
Theorem 1.1	Main closure through sparse-surplus routing, the near-cubic entropy spine, and the terminal two-support route-8 exclusion (using HSS)	all spine invariants: 1–8, 15, 20–24, 25–29, 35–41, 43–44; Proposition 3.80, Definition 4.1

Spine dependency summary (the load path through route 8).

Theorem 5.1 plus Definition 4.1 $\Rightarrow$ inv 20 (low P_{13} density, via Lemma 7.1 and Proposition 7.42)
$\Rightarrow$ inv 21 (forced remainder $\geq 84.4\%$)
$\Rightarrow$ inv 23, 24 (stub capacity 15/window $\Rightarrow$ deficiency $\leq \tau_{\text{win}} = 0.2282\ldots$)
$\Rightarrow$ inv 28 (raw obstruction supply $\geq 2.543|R|$; high entropy gives $\geq 2.574|R|$) [DEMAND]
$\Rightarrow$ inv 29 (budget caps independent tests $\leq 0.611|R|$) [SUPPLY]
$\Rightarrow$ $> 76\%$ correlation forced
$\Rightarrow$ inv 7/8 (lumpability defect *or* target-complete compression)
$\Rightarrow$ Corollary 3.98: contradiction outside route 8, with Type B fan-certificate residual centers and B2 failures made sublinear by Proposition 14.100

Auxiliary path. The auxiliary invariants (spectral masks 17–19, tree–cotree / Kleitman–West, marked length-rank, HIST branch) are not used as load-bearing inputs in Theorem 1.1. They record possible repair directions if one later replaces a quantitative estimate in invariants 27–29. Likewise, Corollaries 3.46, 3.48, 3.52 and 3.61 are recorded for the surplus-token repair path; the proof of Theorem 1.1 uses the sharper routed statements Theorems 3.56, 3.75 and 3.78.

Per-lemma invariant requirements

The following per-result checklist lists, for each labeled object, the invariants it *requires as input*, a plain-language description, its role, and its position in the proof flow. A lemma may be verified in isolation by confirming that every invariant in its **Requires** cell is already established at an earlier-or-equal stage; "earlier" is measured along the proof-dependency diagram, so each required invariant is tracked at the diagram node the row names or at a node upstream of it, and never only on a sibling branch. Two position columns are given so the reader may cross-check against either presentation: a coarse **Stage** label and the fine-grained **diagram-node** identifier of the proof-dependency diagram.

Stage legend. **R** root/setup · **MIN** minimality · **REP** replacement layer · **ENT** entropy/skeleton budget · **REM** P_{13}-free remainder · **CURV** obstruction supply/rank · **2BUD** two-budget routing · **LB** large-budget residual · **EXC** local exclusion (terminus).

Remark 1.3 (Two distinct integer scales)**.** *Ledger-invariant numbers and diagram-node numbers are separate namespaces. The tables below say "inv" for ledger invariants and use bracketed integers only for unique diagram nodes.*

Setup and target algebra.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 2.2	R	[5]–[7] `(target/ret/cycle)`	inv 1	A graph has a 2^k cycle iff some edge has a return path of length $2^k - 1$ (a Mersenne number).	Converts the geometric target into an arithmetic one; the foundation.
Definition 2.5	R	[5] `(target)`	inv 1	Defines "target-safe" *contextually* (a cycle may exit and re-enter).	Makes safety a property of embedding — the cumulative constraint.

Minimality and replacement.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 3.1	MIN	[8] `(core)`	inv 2, minimality	Every proper subgraph has a vertex of degree ≤ 2.	Forces 2-degeneracy; no smaller $\delta \geq 3$ piece.
Lemma 3.2	MIN	[9] `(delete)`, [10] `(hiind)`	inv 3	Every edge touches a degree-3 vertex; $V_{\geq 4}$ independent.	Buffers high-degree surplus through cubic vertices.
Lemma 3.5	MIN	[4] `(min)` $\to$ rank identities	inv 9, 11	Cycle rank $\beta \geq n/2 + 1$ from $\delta \geq 3$.	Quantifies forced cycle structure.
Lemma 3.94	REP	[11] `(bd)`	inv 1	Distinct boundary degree profiles cannot be quotient-merged.	Protects boundary bookkeeping under replacement.
Lemma 3.95	REP	[12] `(univ)`	inv 1, 7	A target-complete identification stays valid in every outside context.	Makes hazard profiles context-independent.
Lemma 3.96	REP	[13] `(replace)`	inv 1, 6, 7; Lemma 3.94, Lemma 3.95	Swap a sub-piece for a smaller no-more-hazardous one $\Rightarrow$ smaller graph.	Engine of minimality.
Corollary 3.98	REP	[14] `(uncomp)`	inv 8; Lemma 3.96	No proper boundaried piece can be target-compressed.	**The contradiction lever.**

Entropy / skeleton budget.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 3.100	ENT	[18] `(p13alg)` $\to$ [22]	inv 1	Independent target-testable coordinates each consume skeleton entropy.	Sets the bit "currency."
Lemma 4.2	ENT	[21] `(enum)`	inv 15	The labelled skeleton count dominates auxiliary data.	Justifies one common budget B_{skel} once the edge-count branch is fixed.
Lemma 4.3	ENT	[21] `(enum)`	inv 10	Budget valid across the edge-count range.	Robustness to edge-count variation.
Lemma 4.4	ENT	[21] `(enum)`	Definition 4.1	Near-cubic budget $B_{\mathrm{skel}} = \frac{3}{2} n \log_2 n + o(n \log n)$.	Pins the $\frac{3}{2}$ constant on the branch where the surplus-token routing has supplied the spine estimate.
Lemma 3.30	ENT/ SURP	[125]–[128] `(active setup)`	inv 10, 14, 15; sparse surplus exits absent	A sparse-load survivor has the full active surplus family.	Sets up the active-demand side of the surplus branch.
Definition 3.62	ENT/ SURP	[129] `(baseline)`	inv 14, 15; Lemma 3.30	Fixes the $O(n)$-deficit baseline demand used by the surplus accounting branch.	Supplies the common demand baseline for the later sandwich.
Lemma 3.63, Lemma 3.64	ENT/ SURP	[131] `(free pairs)`	inv 10, 14, 15; Definition 3.62	Computes the cubic baseline budget and the incremental edge-count room above cubic.	Supplies the numerical room used by the surplus-pair entropy sandwich.
Proposition 3.66	ENT/ SURP	[131] `(free pairs)`	inv 10, 14, 15; Lemma 3.63, Lemma 3.64	Charges blocker-free surplus-pair responses to the exact cubic-baseline sandwich.	Handles the free-pair side.
Proposition 3.67	ENT/ SURP	[132] `(blocked?)`	inv 10, 14, 15; Proposition 3.66	Extends the entropy sandwich to the blocked case, leaving only canonical blocker debt.	Sends blocked pairs to the blocker ledger.
Definition 3.73	ENT/ SURP	[129] `(baseline)`	inv 14, 15; Definition 3.62	Records the lower-bound deficit packages.	Supplies the deficit side of the surplus estimates.
Corollary 3.74	ENT/ SURP	[138] `(nearclose)`	inv 14, 15; Definition 3.73	Converts the lower-bound packages into surplus estimates used when the branch routes to the near-cubic spine.	Supplies the near-cubic cap side of the surplus sandwich.
Definition 3.33	ENT/ SURP	[132]–[134] `(canonical blockers)`	inv 10, 14, 15; Proposition 3.67	Defines the canonical blocker assigned to each blocked response.	Makes the blocked-pair assignment single-valued.
Lemma 3.34	ENT/ SURP	[134] `(blocker ledger)`	inv 10, 14, 15; Definition 3.33	Proves the canonical blocker ledger counts each blocked response once.	Prevents overcounting before capacity tokens are introduced.
Definition 3.35, Lemma 3.36	ENT/ SURP	[134] `(primitive blocker support)`	inv 14, 15; Definition 3.33	Primitive sparse blocker supports supply the primitive-blocker-support token class.	Supplies the primitive-blocker-support side of the token ledger.
Definition 3.37	ENT/ SURP	[134]–[136] `(capacity tokens)`	inv 10, 14, 15; Lemma 3.34	Defines the capacity-token fibres attached to the blocked-pair ledger.	Names the token side of the blocked-pair accounting.
Lemma 3.38	ENT/ SURP	[136] `(token supply)`	inv 10, 14, 15; Definition 3.37	Records the window, remainder-surplus, and primitive-blocker-support token supplies.	Supplies the three capacity classes for the coupled high-load test.
Lemma 3.39	ENT/ SURP	[136] `(token partition)`	inv 10, 14, 15; Definition 3.37	Proves the token fibres partition the blocked-pair set exactly.	Prevents double counting on the token side.
Definition 8.4	ENT/ SURP	[135] `(join)`	inv 14; Definition 3.37	Defines the window/remainder surplus split.	Sets the notation for the packed-window join identity.
Lemma 8.6, Lemma 8.7	ENT/ SURP	[135] `(join)`	inv 14; Definition 8.4	Gives the exact window-incidence supply $15p_{13} + \sigma_W$ and the surplus-aware stub load.	Supplies the window side of the token capacities.
Corollary 13.5	ENT/ SURP	[136] `(supply)`	inv 14; Lemma 8.6, Lemma 8.7	Globalizes the join load into the coupled token budget.	Supplies the window/remainder side of the token capacities.
Definition 3.40	ENT/ SURP	[136] `(token fibres)`	inv 14, 15; Definition 3.37	Defines same-token fibres.	Names the fibres used by the high-load split.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Definition 3.41	ENT/ SURP	[139]–[143] `(token roles)`	inv 14, 15; Definition 3.40	Defines the window-incidence, remainder-surplus, and primitive-blocker-support roles.	Names the three token classes used in the high-load split.
Corollary 3.45, Lemma 3.57	ENT/ SURP	[137] `(high-load test)`	inv 10, 14, 15; Lemma 3.38, Lemma 3.39	Gives the coupled single-graph high-load alternative.	Determines whether the branch routes to [138] or to a role class.
Theorem 3.56	ENT/ SURP	[137]–[143] `(role split)`	inv 10, 14, 15; Corollary 3.45, Lemma 3.57	Splits the forced load into the window-incidence, remainder-surplus, and primitive-blocker-support token classes.	Records the sharp load alternative before geometric routing.
Definition 3.76	ENT/ SURP	[140], [142], [143] `(role audits)`	inv 4, 8, 14, 15; Definition 3.41	Defines the routing configurations used in the three role audits.	Names the geometric data audited by the bottleneck lemma.
Lemma 3.47	ENT/ SURP	[140], [142], [143] `(role audits)`	inv 4, 8, 14, 15; Definition 3.76	A large same-token load contains a matching or star in the active role class.	Produces the combinatorial pattern audited by homogeneous extraction.
Lemma 3.43	ENT/ SURP	[140], [142], [143] `(role audits)`	inv 4, 8, 14, 15; Lemma 3.47	The matching or star contains a homogeneous role subpattern.	Produces the geometric pattern audited by the bottleneck lemma.
Lemma 3.77, Theorem 3.78	ENT/ SURP	[144] `(bottleneck)`	inv 4, 8, 14, 15; sparse surplus exits and Type B routing where applicable	The homogeneous bottleneck realizes a sparse surplus exit, produces Type B fan data, or falls under fixed homogeneous caps.	Discharges the geometric audit.
Corollary 3.44	ENT/ SURP	[138] `(nearclose)`	inv 14, 15; Theorem 3.78	Fixed homogeneous caps route the remaining branch to the near-cubic spine.	Supplies the cap-to-spine route.
Proposition 3.80	ENT/ SURP	[144] `(bottleneck)`	inv 14, 15; Corollary 3.44	Combines the sparse-load discharge with the cap-to-spine route.	Closes the non-near-cubic surplus branch or derives the near-cubic spine.
Lemma 4.7	ENT	[22] `(many)`	inv 1; Lemma 4.2	#realizable states $\leq$ #labelled graphs.	The counting contradiction step.
Definition 7.17	ENT	[158] `(realized)`, [159] `(dense)`	Lemma 7.1, Lemma 4.2	Decides whether the joint window package is realized by the labelled class; the no-branch is the dense-packing residual.	Turns the realization sentence into a branch test.
Lemma 7.18	ENT/ CURV	[160] `(tau)`, [161] `(netcharge)`	Proposition 13.7, Lemma 13.4	For $\tau(\theta) < 1/4$ the dense residual enters [25] with the deficiency cap and is closed by [56]–[64].	Closes the moderately dense regime by net charge.
Lemma 7.19	ENT/ EXC	[162] `(coldpass)`	Lemma 7.10, Lemma 7.12	The hot/cold pass executes on the dense residual with all its closures.	Reduces the dense regime to [163] and [164].
Definition 7.20, Lemma 7.21	EXC	[163] `(neutral)`	Lemma 7.14, Definition 7.8	The neutral equal-length terminal configuration is a symmetry; split into canonical replacement and genuine second strand.	Routes the neutral row to [165]/[167].
Lemma 7.22	EXC	[165] `(swap)`, [166] `(refined)`	node [4] minimality; Definition 3.89	Exchanging the corridor for its canonical representative gives a same-size counterexample smaller in the refined order.	Closes the canonical-replacement case.
Lemma 7.23	CURV	[167] `(twostrand)`, [155]	Definition 7.9 (F1); `Graph/TwoStrandEnumeration.lean`	A symmetric strand pair closes the cycles 2ℓ and $\ell + d$; one has power-of-two length when either value is a power of two.	Closes the symmetric case with a power-of-two closing length.
Lemma 7.24	CURV/ EXC	[168] `(twosurvivor)`	Lemma 7.6, Lemma 7.23	Interior window vertices carry one stub; genuine symmetric pairs attach only at endpoints and are never selected interior half-edges.	Closes the surviving symmetric pairs; the extraction is charged on interior stubs (9 per window).
Definition 7.25, Lemma 7.26	ENT	[164] `(allcold)`	Definition 11.1, Lemma 4.2	The remainder class with the inherited edge count injects into the labelled skeleton class, so the all-cold comparison cannot be active.	Closes [164].

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 7.28	CURV/NET	[173] `(exact)`, [174] `(absorbed)`	Lemma 8.5, Proposition 13.7, node [22]	Node [56]'s collision decided on the object; its failure gives linearly many cold windows on the bounded arm of [153].	Removes the sufficient-order condition.
Lemma 7.29	EXC/TYPB	[175] `(meets)`, [176] `(genuine)`, [177] `(fan)`	inv 4 (node [10]); Lemma 7.10, Lemma 14.25, Definition 14.19	A selected corridor is a genuine (F5) configuration or decorated handoff fan data at a heavy centre; the configuration-extraction loss is charged to Type B.	Closes the absorbed-configuration residual.
Definition 3.68, Lemma 3.69	ENT/SURP	[178] `(pairunreal)`	Proposition 3.67, Lemma 3.59	Failure of the pair count gives a minimal connected pair overlap obstruction.	Turns the sandwich into a branch test.
Lemma 3.70	EXC/SURP	[179] `(pairserial)`	Lemma 7.35, Lemma 3.77, Definition 7.9	Uncrossing closes by exits (b)/(c), Type B, or yields a serial demand system with bounded increments.	Serial realization of the pair obstruction.
Lemma 3.71, Lemma 3.72	ENT/EXC	[180] `(pairclose)`	Lemma 7.36, Lemma 7.15, Lemma 3.77	Doubling-orbit hit or periodic response class routed to a sparse exit or Type B.	Closes the pair-code unrealized residual.
Definition 7.31, Lemma 7.39	ENT	[169] `(trivial)`, [171] `(compress)`	Lemma 7.1, Lemma 4.4, Lemma 7.38	The blocked class compresses by $(c_{13} - o(1))p \log_2 n$ bits, hence is empty for $\theta > \theta_{\mathrm{win}}$.	The exact entropy step.
Definitions 7.32 and 7.34 and Lemmas 7.33 and 7.35 to 7.38	ENT/EXC	[170] `(additive)`, [172] `(arithmetic)`	Section A, Lemma 7.1, Lemma 7.10, Lemma 7.15; G2/G3 and declared Type A/Type B interfaces	Conditional savings add unless their failure supplies a minimal same-scale overlap system; local uncrossing makes its cycle spectrum a realizable serial sumset, whose doubling-orbit alternative is a power-of-two hit or an already routed periodic response class.	Closes the nonadditive fixed-scale branch at [172].
Lemma 7.1	ENT	[21] `(enum)` $\to$ [22] `(many)`	inv 1, 25; Definition 4.1; Section A	Converts the finite 91-barrier computation into $(c_{13} - o(1))p_{13} \log_2 n$ independent window coordinates.	Names the load-bearing window package used by the density cap.
Definitions 7.2, 7.3, 7.5, 7.8, 7.9 and 7.13, Lemmas 7.4, 7.6, 7.7, 7.10 to 7.12, 7.14 and 7.15, and Theorem 7.16	ENT/EXC	[145]–[157] `(cold)`	inv 1, 8, 14, 15, 35, 38; Definition 4.1; Lemma 7.1; Theorem 15.48	Quantifies the hot/cold branch: $\theta < 1/78$ is route-8 closed; hot failure gives $C \geq (\theta - \theta_{\mathrm{win}})n - o(n)$, $b \geq 13C - o(n)$, and $N_{\mathrm{conf}} \geq 13C/D_{\mathrm{cold}} - o(n)$; bounded configurations route by hit, defect, same-interface table, handoff, or compression.	Records the terminal cold-window closure used by the downstream large-budget route.
Proposition 7.42	ENT	[24] `(theta)`	Definition 4.1; Lemma 7.1; Lemma 3.100	Independent induced-P_{13} density $\theta \leq \theta_{\mathrm{win}}$ after the surplus-token branch supplies the spine estimate, with sharper high-entropy cap $\theta \leq 0.011985$.	Kills the high-P_{13} branch on the derived spine branch; forces a large remainder.

P_{13} packing and forced remainder.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Corollary 5.2	REM	[15] `(p13free)`	Theorem 5.1	A counterexample contains an induced P_{13}.	Entry to the window/remainder split.
Lemma 8.2	REM	[27] `(no3)`	maximality of $\mathcal{P}$; Theorem 5.1	Every component of the fixed remainder R is P_{13}-free and has empty internal 3-core.	Normalizes the selected remainder for all later componentwise routing.
Lemma 6.1	REM	[18] `(p13alg)`	inv 7, 25	399 legal P_{13} labels; zero-defect quotient through lengths $1, 2, 3$ is the identity.	The finite window-safety algebra.
Lemma 6.2	REM/CURV	[21] `(enum)`	inv 25, 26	432672/543958 of safe wedges are obstructing.	Finite count feeding obstruction supply.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 8.5	REM	[26] `(R)`, [29] `(stub)`	Theorem 5.1, inv 21, 22	Remainder ($\geq 83.4\%$) gets ≤ 15 incidences/window; $\partial_+(R) \leq 15p_{13} + o(n)$ and $\partial_+(R) - \sigma_R \leq 15p_{13} + o(n)$.	**Supply ceiling** of the final collision.

Obstruction supply, rank, two-budget.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 9.1	CURV	[30] `(wedge)`	inv 21, 28; Corollary 8.8	Remainder has $\geq 2.543 \cdot \lvert R\rvert$ raw two-step obstruction tests on the derived spine branch, with the sharper high-entropy value $\geq 2.574 \cdot \lvert R\rvert$.	**Demand floor** of the final collision.
Lemma 10.3	CURV	[31] `(rank)`, [32] `(drop)`	Definitions 3.92 and 10.1	Extracts a finite functional target-dependence from a rank drop.	Makes the rank-drop interface explicit.
Lemma 10.4	CURV	[32] `(drop)`, [33] `(depopen)`	inv 26, 28; Lemma 10.3	Localizes the three ways obstruction dependence arises.	Organizes the forced-rank case analysis.
Lemma 10.5	CURV	[35] `(dep)`	inv 28, 38	Separated wedges tested by disjoint contexts.	Ensures obstruction tests are independent.
Lemma 10.6	CURV	[40] `(Z)`, [41] `(properZ)`	inv 34	Proper nonlocal supports are forbidden.	Blocks the "spread it out" escape.
Lemma 10.8	CURV	[44] `(rankwall)`	inv 35	Enlarged nonlocal supports = repair networks.	Converts the other escape into killed structure.
Lemma 10.9	CURV	[43] `(global)`	inv 34, 35; Definitions 3.88 and 3.91	Whole-graph dependence is target-defective, a smaller closed counterexample, or exact on raw obstruction labels.	Closes the whole-graph-dependence loophole.
Lemma 10.10	CURV	[47] `(fullrank)`	inv 28; Lemma 10.9	Obstruction rank $\geq W_2 - o(\lvert R\rvert)$.	Forces near-maximal demand.
Corollary 10.11	CURV	[48] `(cost)`	inv 27, 28; Lemma 10.10	Obstruction rank $\to$ entropy tax (c_Ω bits each).	Puts demand into budget currency.
Lemma 11.2	2BUD	[49] `(eta)`, [50] `(expensive)`	inv 15, 26	A dominant local vertex-type appears in the low-entropy branch.	Handles the cheap-entropy regime.
Lemma 11.4	2BUD	[51] `(expbranch)`	inv 26, 29; Lemma 11.2	The dominant type forces a positive linear family of independent obstruction translates.	Supplies a rank input in the repetitive low-entropy branch.
Proposition 11.6	2BUD	[52] `(windowacct)`	inv 20, 27, 29	Two-budget routing: high entropy closes by window count; repetitive low entropy gives a rank input; nonrepetitive low entropy routes forward.	Prevents an invalid entropy overcount and preserves the net-charge route.
Proposition 12.2	2BUD	[53] `(smallbudget)`	inv 20, 27	The high-θ branch is closed by the entropy cap.	Closes one side of the dichotomy.
Remark 12.4	2BUD	[54] `(entcap)`	inv 8, 24, 29	Closure already holds at $K = 0$ (obstruction machinery dispensable).	Over-determination / graceful degradation.

Large-budget residual and local exclusion (terminus).

Result	Stage	Diagram node(s)	Requires	What it does	Role
Definition 13.1	LB	[55] `(thetaN)`	inv 21, 22	Canonical decomposition of a support into boundaried pieces.	Sets up per-cell net-charge bookkeeping.
Definition 13.2	LB	[55] `(thetaN)`	inv 1, 6, 8, 22, 23, 24	Admissible support carrying all cumulative constraints.	The object the exclusion lemmas act on.
Lemma 13.4	LB	[56] `(netcap)`, [58] `(Ndef)`	inv 13, 14, 24	Net charge localizes exactly over the connected components of R.	Cell-wise $\mathcal{N} \geq 0$ sums to a global deficiency bound.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Proposition 13.7	LB	[59] `(globalnet)`	inv 14, 24; Lemma 13.4	The large-budget residual has negative net charge.	Selects the supports to kill.
Definition 14.3, Lemma 14.4	EXC	[88] `(thresholds)`	inv 23, 24	Defines receivers, missing-port count $q(w)$, routed load $L(w)$, and canonical load routing.	Supplies the bookkeeping variables for Type A.
Definition 14.5	EXC	[93] `(visible)`	Definition 14.3	Defines anchored returns, receiver-entry returns, visible loads, and silent loads for one completion port.	Supplies the visible/silent split tested at node [93].
Lemma 14.7, Lemma 14.8	EXC	[93] `(visible)`	inv 23; Definition 14.5	Proves that the first entry of an anchored return is a receiver and records the degree-two entry budget.	Keeps receiver-entry returns inside the Type A receiver bookkeeping.
Lemma 14.9	EXC	[89] `(saturated)`, [93] `(visible)`, [94] `(silent)`, [109] `(route8)`	inv 1, 30; Lemma 3.4	Every completion port has an anchored return.	Makes every saturated port test nonvacuous.
Definition 14.10, Lemma 14.12, Definition 14.15	EXC	[109] `(route8)`	inv 1, 22, 23, 25, 30, 31	Defines route-8 residual data, target-complete-minimal trace basins, the declared u-supported response algebra, and connector bands forced by all internal channels.	Records the reduced silent trace-basin branch.
Definition 14.16	EXC	[95]–[109]	inv 1, 6, 8, 22, 25, 30, 31, 34, 35	Enumerates the eight alternatives tested after node [89] is saturated.	Provides the common exit language for nodes [95]–[109].
Definition 14.33	EXC	[101] `(exit4)`	inv 6, 8; Definition 14.16	Defines the peeling set for exit (4).	Names the loads that can be removed by the exit-(4) loop.
Lemma 14.34	EXC	[102] `(peel)`	inv 6, 8; Definition 14.33	Proves the exact 1/4 charge update for one peeled load.	Supplies the numerical update for the loop.
Lemma 14.38	EXC	[102] `(peel)`, [89] `(loop)`	inv 6, 8; Lemma 14.34	Proves that one supported unpeeled load is removed at exit (4).	Makes the [102] return to [89] a finite strictly decreasing loop in $L_4(w)$.
Lemma 14.35	EXC	[93] `(visible)`, [95]–[108]	inv 1, 8, 25, 30, 31; Definition 14.33	Four visible unpeeled receiver-entry returns realize one of exits (1)–(7), with exit (4) supporting an unpeeled load.	Routes the yes-branch of [93].
Lemma 14.36	EXC	[94] `(silent)`, [101]–[109]	inv 6, 8, 34, 35; Definition 14.33, Definition 14.15	If no port has four visible unpeeled returns, the residual excess set realizes one of exits (4)–(8), with exit (4) supporting an unpeeled excess load.	Routes the no-branch of [93].
Lemma 14.37	EXC	[89] `(saturated)`, [93] `(visible)`, [94] `(silent)`, [101] `(exit4)`, [102] `(peel)`, [109] `(route8)`	Lemma 14.35, Lemma 14.36	Combines the visible and silent residual routing lemmas after any previous peeling.	Iterates saturated routing until a closed exit, Type B handoff, route 8, or unsaturation occurs.
Definition 14.11	EXC	[107] `(exit7)`	inv 6, 8, 30, 31	Defines connector configurations, first separators, and the finite continuation classes.	Supplies the vocabulary for the exit-(7) split.
Lemma 14.23	EXC	[107] `(exit7)`; feeds [101], [103], [105] when absorbed	inv 6, 8, 30, 31; Definition 14.11	A nonabsorbed four-coordinate family either realizes exits (4)–(6) or has a surviving first separator.	Routes the absorbed subcases back to the already listed saturated exits.
Lemma 14.24	EXC	[107] `(exit7)`	inv 4, 6, 8; Definition 14.11	Cubic switches are absorbed unless a surviving first separator remains.	Removes the cubic part of exit (7).
Lemma 14.25	EXC	[108] `(decorB)`	inv 4, 6, 8; Lemma 14.24	A surviving separator has ambient degree at least four and produces a decorated Type B handoff envelope.	Converts exit (7) into the Type B input [108].

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 14.39	EXC	[95]–[108]	inv 1, 6, 8, 25, 30, 31, 34, 35; Lemma 14.38, Lemma 14.25	Closes exits (1)–(3), (5), and (6), peels one load at exit (4), and transfers exit (7) to Type B.	Records the outcome of the explicit saturated-exit tests.
Lemma 14.27	EXC	[88] `(thresholds)`	inv 4, 24	Computes $H_0 \leq 4$, $H_1 \leq 8$, $H_2 \leq 12$.	Identifies the saturated branch-switch values.
Lemma 14.28	EXC	[93] `(visible)`	inv 1, 8, 25, 30, 31	Four visible receiver-entry returns force target, quotient, support dependence, or decorated handoff fan data.	Closes the visible saturated branch.
Lemma 14.32	EXC	[94] `(silent)`	inv 6, 8, 34, 35; Definition 14.19, Definition 14.15	Silent/mixed excess forces compression, support dependence, decorated handoff fan data, or indexed target-complete-minimal trace basins.	Closes the silent branch outside route 8.
Lemma 14.30, Lemma 14.31	EXC	[109] `(route8)`	inv 6, 8, 34, 35; Lemma 14.32	Visible-first excess gives $S^{\mathrm{exc}}_{\mathrm{sil}}(X) \geq 4D_A(X)$ and records the reduced route-8 residual.	Quantifies the silent route-8 residual.
Lemma 14.40	EXC	[89] `(saturated)`, [109] `(route8)`	Lemma 14.28, Lemma 14.32, Lemma 14.35, Lemma 14.36, Lemma 14.37, Lemma 14.39	Removes closed exits 1–3, 5, and 6 and exit-4 peeled loads from the 4/8/12 branch, hands exit 7 to Type B, and leaves reduced route 8 as a residual.	Leaves unsaturated Type A plus the target-defect exit, the Type B handoff, and the quantified route-8 branch.
Definition 14.46	EXC	[111] `(deficit)`	inv 14, 24; Definition 14.15, Lemma 14.31	Defines the Type A route-8 deficit D_A extracted from the reduced silent residual.	Names the route-8 part of the final demand/route-8 closure.
Lemma 14.47	EXC	[112] `(basins)`, [113] `(burden)`	inv 14, 24; Definition 14.46	Proves the indexed basin lower bound $N_{\mathrm{basin}} \geq 4D_A$ and the large-budget lower bound for D_A.	Gives the lower side of the route-8 response-support count.
Definition 14.48	EXC	[110] `(route8)`	inv 8, 23, 30, 31; Definition 14.15	Defines indexed route-8 entries, active boundary incidences, essential response-support cores, and private response supports.	Supplies the response-support vocabulary.
Definition 14.50	EXC	[114] `(core)`	inv 8, 23, 30, 31; Definition 14.48	Defines true route-8 residual entries after exits (1)–(7) are absent.	States the residual condition used after [114].
Lemma 14.56	EXC	[114] `(core)`	inv 1, 8, 30, 31; Definition 14.48	A mixed target event in the response-support core crosses at least two supports.	Gives the cut-parity constraint on essential response-support cores.
Lemma 14.58	EXC	[115] `(smallcore)`, [116] `(closedexits)`	inv 1, 8, 30, 31; Lemma 14.56	Zero/one-essential-support entries realize exits (4)–(7).	Removes the small-core branch at [115].
Proposition 14.59	EXC	[117]–[123]	inv 14, 23, 24; Corollary 8.8, Lemma 14.47, Lemma 14.58	If no two-support entry exists, three private response supports per basin violate the boundary-incidence budget; hence a two-support entry exists.	Closes route 8 outside the terminal two-support entry.
Definition 14.51, Lemma 14.52, Lemma 14.53	EXC	[118] `(two-support)`	inv 6, 8, 23, 24; Definition 14.48	Inclusion-minimality of the essential response-support core gives a declared deletion witness for every essential response support.	Supplies the witness data used at [124].
Lemma 14.54	EXC	[118] `(two-support)`	inv 1, 6, 8, 30, 31; Lemma 14.53	In a two-support entry, each declared deletion quotient lies in the canonical exit-(4) family.	Converts the terminal response-support data into canonical exit-(4) data.
Lemma 14.55	EXC	[124] `(exclusion)`	inv 1, 6, 8, 30, 31; Lemma 14.54	Each canonical deletion quotient realizes exit (4).	Produces the forbidden absent exit in the terminal residual.
Definition 14.60, Theorem 14.61, Proposition 14.62	EXC	[124] `(exclusion)`	Proposition 14.59, Lemma 14.54, Lemma 14.55	Packages the terminal two-support obstruction, proves it empty, and derives route-8 closure.	Closes the terminal route-8 interface.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Lemma 14.43	EXC	[90] `(unsat)`	Lemma 14.4, Lemma 14.27	Unsaturated receivers give the 3/7/11 charge inequality.	Supplies the Type A charge bound.
Lemma 14.45	EXC	[86], [87]–[109]	inv 1, 4, 8, 22, 23, 24, 25, 30, 31; Lemma 14.40, Lemma 14.43	Dense $\sigma = 0$ supports outside target-defect peeling, route 8, and the Type B handoff satisfy $\partial_+(X) \geq \frac{1}{4}\lvert V(X)\rvert$; a negative support has target-defect, route-8, or Type B handoff data.	Kills pure Type A after the split.
Remark 14.67	EXC	[70] `(fsafe)`, [71] `(certq)`	inv 25	Certificate-marked fan-degree cap is a structural label-packing bound (≤ 8/window); fan-certificate residual centers are routed to Type B residual mass.	Supplies the Type B constant.
Definition 14.68, Definition 14.69, Definition 14.72	EXC	[72] `(typeBfan)`	inv 23, 24, 25	Defines fan-window incidences, fan-closed ports, and the hybrid window/non-window credit ledger used by one Type B fan.	Supplies the accounting objects used at the local B1 node.
Definition 14.73, Definition 14.74, Lemma 14.75, Lemma 14.76	EXC	[72] `(typeBfan)`	inv 1, 23, 30, 31; Definition 14.68	Constructs the direct cycles forced by same-window and two-window closed incidences.	Removes direct fan-window and two-window cycle configurations before the incidence payment.
Lemma 14.77, Lemma 14.78, Lemma 14.79	EXC	[72] `(typeBfan)`	inv 23, 24, 25; Definition 14.72	Every certificate-marked positive-deficit residual uses $2c$ distinct non-h incidences, whose half-credit pays D_B with slack.	Quantifies and closes the local B1 Type B fan branch before the global B2 ledger argument.
Proposition 14.80	EXC	[72] `(typeBfan)`	inv 4, 8, 25; Definition 14.69, Lemma 14.79	Two fan-closed ports give $D_B \geq (k-3)/4 > 0$ and enter the local Type B fan ledger.	Closes a fan-closed local Type B center.
Corollary 14.81	EXC	[69] `(localgt)`, [72] `(typeBfan)`	inv 4, 8, 25; Lemma 14.70, Proposition 14.80	A compatible open pair produces the two fan-closed ports required by the local Type B fan ledger.	Routes compatible-pair data to node [72].
Corollary 14.82	EXC	[78]–[81], [72] `(typeBfan)`	inv 4, 8, 25; Lemma 3.17, Lemma 3.18, Proposition 14.80	Degree-four triangular and heavy triangular routes produce the fan-closed ports required at node [72]; the heavy triangular route gives $D_B \geq (5k-19)/4 > 0$.	Routes triangular local data to the local Type B fan ledger.
Definition 14.83, Definition 14.84, Definition 14.85, Lemma 14.87, Lemma 14.88, Proposition 14.89, Definition 14.90, Proposition 14.92	EXC	[73]	inv 1, 4, 8, 22, 23, 24, 25, 34, 35; Lemma 14.78	Defines Type B incidence ledgers for supports with no fan-certificate residual center, proves B2 failures produce minimal overlap obstructions, and records the global-to-local constraints inherited by such obstructions.	Reduces local Type B closure to the B2 refined support ledger after local B1 is proved; disjoint-incidence failure is a globally admissible overlap obstruction.
Definition 14.19	EXC	[108] `(decorB)`	inv 1, 4, 8, 22, 25; Lemma 14.25	Defines the decorated fan envelope produced by Type A exit (7).	Records the incoming Type B support from node [108].
Lemma 14.26	EXC	[65] `(typeB)`	inv 1, 4, 8, 22, 25; Definition 14.19	Proves the decorated handoff is admissible Type B fan-envelope data.	Makes the node [108] handoff usable at Type B node [65].
Definition 14.18	EXC	[70] `(fsafe)`	inv 16, 25	Defines certificate-marked Type B fans.	Supplies the local certificate object for Type B fans.
Remark 14.67	EXC	[71] `(certq)`	inv 16, 25; Definition 14.18	Records the fan-degree label-packing cap.	Supplies the local certificate cap for Type B fans.
Definition 3.7	EXC	[78] `(degree4)`	inv 4, 8, 16, 25	Defines triangular ports at high-degree centers.	Names the triangular local configuration.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Definition 3.14	EXC	[79] `(profile)`	inv 4, 8, 16, 25; Definition 3.7	Defines triangular fan cores.	Supplies the profile object for triangular routing.
Lemma 3.8	EXC	[68] `(degsplit)`, [69] `(localgt)`	inv 4, 8, 16, 25	Puts the high-degree neighbourhood into the normal form used by the local split.	Starts the Type B degree split.
Definition 3.9, Lemma 3.10	EXC	[69] `(localgt)`	inv 4, 8, 16, 25; Lemma 3.8	Defines fan-compatible open ports and proves compatibility for same-center open pairs.	Supplies the fan-compatible branch of the local split.
Corollary 3.13	EXC	[78] `(degree4)`	inv 4, 8, 16, 25; Lemma 3.8	Activates the degree-four local branch.	Separates the degree-four case before triangular routing.
Lemma 3.11	EXC	[69] `(localgt)`	inv 4, 8, 16, 25; Definition 3.7, Definition 3.9	Splits heavy centers into a fan-compatible open pair or a triangular-port alternative.	Supplies the heavy-center local split.
Corollary 3.12	EXC	[79] `(profile)`	inv 4, 8, 16, 25; Lemma 3.11	Records the local dichotomy in the profile used downstream.	Supplies the local Type B case split before the incidence ledger.
Lemma 3.15	EXC	[78] `(degree4)`	inv 1, 4, 8, 25, 30, 31; Definition 3.14	Triangular shoulders have completion edges.	Sets up the triangular first-landing analysis.
Lemma 3.16	EXC	[79] `(profile)`	inv 1, 4, 8, 25, 30, 31; Lemma 3.15	Every relevant return enters through a shoulder.	Routes triangular returns to the landing analysis.
Lemma 3.17	EXC	[80] `(landing)`	inv 1, 4, 8, 25, 30, 31; Lemma 3.16	The first landing is central, cross-triangular, or outside.	Routes triangular ports toward fan-closed data.
Lemma 3.18	EXC	[81] `(cross)`	inv 1, 4, 8, 25, 30, 31; Lemma 3.17	Cross-shoulder multiplicity supplies the required second port.	Completes the triangular route to fan-closed data.
Lemma 14.70	EXC	[69] `(localgt)`, [72] `(typeBfan)`	inv 4, 8, 25; Definition 3.9	A compatible open pair supplies two fan-closed ports at the same center.	Routes compatible-pair subcases into the Type B incidence payment.
Definition 14.7	EXC	[71]–[73], [82]–[85]	inv 4, 8, 16, 24, 25	Defines the certificate-marked multiclosed Type B residuals, including the B2-failure residual classes.	Names the Type B residuals that survive the local B1 ledger.
Lemma 14.93	EXC	[71]–[73], [82]–[85]	inv 4, 8, 16, 24, 25; Definition 14.71, Lemma 14.78, Lemma 14.79	Applies the refined Type B ledger: certificate-closed fans have nonnegative charge, positive-deficit fans are paid by B1, and B2 failures are isolated as residuals.	Kills Type B outside route 8 and outside the Type B bridge residual.
Definition 14.95	EXC	[75] `(fanmass)`	inv 4, 14, 16, 24, 35; Definition 4.1	Defines the Type B residual mass for fan-certificate residual centers, grouped decorated envelopes, and B2 failures.	Names the global bridge-mass quantity.
Lemma 14.97	EXC	[75] `(fanmass)`	inv 4, 14, 16, 24, 35; Definition 14.95	Extracts a route-8 non-window core from grouped decorated envelopes into D_A.	Keeps decorated-envelope mass separate from Type A route-8 deficit.
Lemma 14.99	EXC	[84] `(fanmass)`	inv 4, 14, 16, 24, 35; Definition 14.95	Extracts a route-8 non-window core from Type B bridge residual data into D_A.	Keeps Type B bridge mass separate from Type A route-8 deficit.
Lemma 14.98, Proposition 14.100	EXC	[84] `(fanmass)`	inv 4, 14, 16, 24, 35; Definition 4.1; Lemma 14.79, Definition 14.90	Charges residual centers and B2 failures to high-degree fan-center surplus and obtains $M_B \leq 16\sigma(G) = o(\|R\|)$.	Prevents Type B bridge residuals from carrying the linear large-budget deficit on the derived spine branch.
Theorem 15.1	EXC	[76] `(branch)`, [85] `(typeB closed)`, [123] `(demand closed)`	inv 20, 27, 28, 29, 34, 35	Closes the entropy-admissible branch outside explicit Type A target-defect/route-8 and Type B bridge residuals.	Combines the local exclusion theorems before the final demand descent.

Result	Stage	Diagram node(s)	Requires	What it does	Role
Theorem 15.48	EXC	[123] `(demand closed)`	inv 20, 27, 28, 29, 34, 35; Proposition 14.100, Definition 15.45, Lemma 15.46, Lemma 15.44, Lemma 15.47, Proposition 14.59	Closes the unified Type A target-defect/route-8 ledger by finite exit-(4) descent and the terminal two-support route-8 exclusion.	Sends only the terminal route-8 survivor to node [124], where it is impossible.
Theorem 1.1	(root goal)	[2] `(bad)` → terminal	Theorem 5.1 + surplus-token routing + all spine invariants	Every minimal counterexample either closes in the sparse-surplus branch, routes to Type B fan data, or reaches the near-cubic spine and then the terminal two-support route-8 exclusion, which is empty.	The main closure theorem.

Invariant → lemmas reverse index (impact analysis). If invariant k needs revision, these results are affected; diagram-node numbers are shown in parentheses.

Inv.	Invariant	First node	Affected results	Impact note
1	Mersenne-return target algebra	Target [5]	Lemma 2.2 ([5]–[7]), Definition 2.5 ([5]), Lemma 3.94 ([11]), Lemma 3.95 ([12]), Lemma 3.96 ([13]), Lemma 3.100 ([18]), Lemma 4.7 ([22]), Definition 13.2 ([55]), Lemma 14.8, Lemma 14.9, Lemma 14.12 ([109]), Definition 14.16, Lemma 14.39, Lemma 14.28 ([93]), Lemma 14.75 ([72]), Lemma 14.44, Lemma 14.45 ([86]), Definition 14.51, Theorem 14.61 ([124])	Pervasive target predicate.
2	Minimal proper-subgraph condition	Core [8]	Lemma 3.1 ([8]); downstream use in Lemma 3.5, Lemma 3.96, Lemma 10.6	Excludes smaller $\delta \geq 3$ cores.
3	Edge-deletion criticality	Delete [9]	Lemma 3.2 ([9]); its high-degree corollary feeds inv 4	Edge-local minimality.
4	High-degree independence	High-degree [10]	Lemma 3.2 ([10]), Lemma 14.27 ([88]), Lemma 14.43 ([90]), Lemma 14.45 ([86]), Lemma 14.93 ([65]), inv 16	Controls surplus aggregation.
5	No passive degree-2 storage	Counterex. [2]	Lemma 8.5 ([29]), Lemma 10.8 ([44]), Lemma 10.9 ([43]), auxiliary repair-network closures	Prevents unrecorded subdivision storage.
6	Replacement dominance	Replace [13]	Lemma 3.94 ([11]), Lemma 3.96 ([13]), Definition 13.2 ([55]), Lemma 14.32 ([94]), Lemma 14.39, Definition 14.51, Definition 14.60 ([124])	Orders boundary profiles.
7	Lumpability defect	Univ. [12]	Lemma 3.95 ([12]), Lemma 3.96 ([13]), Lemma 6.1 ([18]); final spine step through inv 8	Detects invalid quotients.
8	Hereditary target-uncompressibility	Uncomp. [14]	Corollary 3.98 ([14]), Definition 13.2 ([55]), Remark 12.4 ([54]), Definition 14.15 ([109]), Definition 14.16, Lemma 14.39, Lemma 14.28 ([93]), Lemma 14.32 ([94]), Lemma 14.40 ([89]), Lemma 14.44, Lemma 14.45 ([86]), Lemma 14.93 ([65]), Definition 14.51, Definition 14.60, Theorem 14.61 ([124]), Theorem 15.1 ([76], [123])	Main contradiction lever.
9	Minimum edge count	Counterex. [2]	Lemma 3.5 ([4]), Lemma 4.4 ([21])	Lower edge bound.
10	Sparse upper envelope	Core [8]	Lemma 3.3, Lemma 4.3 ([21]); feeds the Laman identity inv 12	Upper edge budget.
11	Cycle-rank constraint	Minimality [4]	Lemma 3.5 ([4]); feeds inv 12 and inv 13	Cycle-rank lower bound.
12	Laman slack identity	Rank ids [4]	Lemma 4.4 ([21]), Definition 11.1, Proposition 11.6 ([52])	Links rank and sparsity.
13	Surplus–rank relation	Rank ids [4]	Lemma 13.4 ([56], [58]); net-charge notation throughout the large-budget branch	Translates surplus into charge.

Inv.	Invariant	First node	Affected results	Impact note
14	High-degree surplus bound	Near-cubic spine [21] (the no-edge of the surplus test); surplus branch [125]–[144] supplies the same estimate for its survivors	Definition 4.1, Proposition 3.66, Proposition 3.67, Corollary 3.74, Lemma 3.30, Definition 3.33, Lemma 3.34, Definition 3.35, Lemma 3.36, Definition 3.37, Lemma 3.38, Lemma 3.39, Definition 3.40, Definition 3.41, Definition 3.76, Lemma 3.47, Lemma 3.43, Lemma 3.77, Theorem 3.78, Corollary 3.44, Corollary 3.45, Lemma 3.57, Theorem 3.56, Definition 8.4, Lemma 8.6, Corollary 13.5, Proposition 3.80, Lemma 13.4 ([56], [58]), Proposition 13.7 ([59]), Proposition 14.59, Definition 14.60 ([124])	Keeps surplus sublinear on the derived spine branch, or derives the exact high-load alternative from the free-pair entropy sandwich plus the capacity-token ledger; blocked pairs are counted by token fibres, the P_{13} join identity quantifies window concentration, and any large homogeneous same-token fibre is a geometric bottleneck routed to sparse exits or Type B.
15	Near-cubic edge count	Near-cubic spine [21] (the no-edge of the surplus test); surplus branch [125]–[144] supplies the same estimate for its survivors	Definition 4.1, Proposition 3.66, Corollary 3.74, Theorem 3.78, Proposition 3.80, Lemma 4.4 ([21]), Lemma 4.2 ([21]), Lemma 11.2 ([49], [50]); entropy estimates using B_{skel}	Fixes the budget coefficient after the near-cubic bound is supplied by the sparse surplus branch.
16	Certificate-marked fan degree	High-degree [67]	Lemma 14.93 ([65]), Remark 14.67 ([70], [71]), Lemma 14.66; related to inv 4	Auxiliary fan-label cap; fan-certificate residual centers are charged through surplus mass.
17	Spectral feasibility mask	Spectral [70]	Lemma 14.93 ([70]), Lemma 14.66, Remark B.5	Auxiliary spectral mask.
18	Spectral equality branch	Spectral [70]	Lemma 14.93 ([70]), Lemma 14.66, Remark B.5	Cubic equality subcase.
19	Spectral surplus branch	Spectral [70]	Lemma 14.93 ([70]), Lemma 14.66, Remark B.5	Surplus spectral subcase.
20	Low P_{13} density	Density [24]	Lemma 7.1 ([21]–[22]), Proposition 7.42 ([24]), Proposition 11.6 ([52]), Proposition 12.2 ([53]), Theorem 15.1 ([76], [123]), Lemma B.2	Window-density cap.
21	Forced remainder	Residual A [25]	Lemma 8.5 ([26], [29]), Lemma 9.1 ([30]), Definition 13.1 ([55]), Lemma B.2; all boundaried-piece analysis	Supplies the large remainder.
22	P_{13}-free piece condition	Residual A [25]	Corollary 5.2 ([15]), Lemma 8.5 ([29]), Definition 13.1 ([55]), Definition 13.2 ([55]), Definition 14.19 ([108]), Definition 14.15 ([109]), Definition 14.16, Lemma 14.28 ([93]), Lemma 14.45 ([86]), Lemma 14.88	Local P_{13}-free structure inherited by overlap supports.
23	Window stub capacity	Stub accounting [29]; exact identity on the surplus branch [135]	Lemma 8.5 ([29]), Definition 13.2 ([55]), Definition 14.3, Definition 14.5, Lemma 14.8, Definition 14.10, Definition 14.15 ([109]), Definition 14.48, Definition 14.60 ([124]), Definition 14.68, Definition 14.72, Lemma 14.77 ([72]), Lemma 14.78, Lemma 14.79, Definition 14.83, Definition 14.84, Lemma 14.88, Definition 14.90, Proposition 14.92, Lemma 14.45 ([86]); feeds inv 24	Outside-incidence supply; positive-deficit Type B residuals use hybrid half-credit on window and non-window incidences, and B2 is the required disjoint ledger.

Inv.	Invariant	First node	Affected results	Impact note
24	Remainder deficiency density	Stub accounting [29]; surplus-branch form [136]	Lemma 8.5 ([29]), Definition 13.2 ([55]), Lemma 13.4 ([56], [58]), Proposition 13.7 ([59]), Remark 12.4 ([54]), Lemma 14.27 ([88]), Lemma 14.43 ([90]), Proposition 14.59, Definition 14.60 ([124]), Lemma 14.77 ([72]), Definition 14.83, Definition 14.84, Definition 14.85, Definition 14.90, Proposition 14.92, Lemma 14.44, Lemma 14.45 ([86])	Supply ceiling; Type B residual deficit is $D_B = c - \frac{11-k}{4}$ and bridge failure is represented by a minimal overlap obstruction.
25	Legal P_{13} labels	Label algebra [18]	Lemma 6.1 ([18]), Lemma 6.2 ([21]), Remark 14.67 ([70], [71]), Definition 14.15 ([109]), Definition 14.16, Lemma 14.39, Lemma 14.28 ([93]), Definition 14.73 ([72]), Definition 14.74, Lemma 14.75 ([72]), Lemma 14.76, Lemma 14.88, Lemma 14.44, Lemma 14.45 ([86]), Lemma 14.93 ([65]), Lemma 14.66	Finite label algebra and structural fan-window endpoint bookkeeping.
26	Two-step obstruction density	Obstruction enumeration [21]	Lemma 6.2 ([21]), Lemma 10.4 ([31], [32]), Lemma 11.2 ([49], [50]), Lemma 11.4 ([51]), Section A	Obstructing wedge count.
27	Flatness entropy cost	Cost [48]	Corollary 10.11 ([48]), Remark 10.12, Proposition 11.6 ([52]), Proposition 12.2 ([53]), Theorem 15.1 ([76], [123])	Bits per independent test.
28	Raw piece-obstruction supply	Wedge lower bound [30]	Lemma 9.1 ([30]), Lemma 10.4 ([31], [32]), Lemma 10.5 ([35]), Lemma 10.10 ([47]), Corollary 10.11 ([48]), Theorem 15.1 ([76], [123])	Demand floor.
29	Obstruction-rank compression ratio	Full-rank squeeze [47]	Lemma 10.10 ([47]), Lemma 11.4 ([51]), Proposition 11.6 ([52]), Remark 12.4 ([54]), Theorem 15.1 ([76], [123]); collides with inv 8	Independent-test cap.
30	Two-path criterion	Target algebra [5]	Lemma 2.2 ([5]–[7]), Lemma 14.12 ([109]), Definition 14.16, Lemma 14.39, Lemma 14.28 ([93]), Definition 14.51, Definition 14.60, Theorem 14.61 ([124]), Lemma 14.44, Lemma 14.45 ([86])	Pair-of-path target test.
31	Theta closure	Target algebra [5]	Lemma 3.95 ([12]), Lemma 3.96 ([13]), Definition 14.15 ([109]), Lemma 14.28 ([93]), Definition 14.51, Definition 14.60 ([124]); piece obstruction profiles	Theta sums avoid targets.
32	Ear closure	Target algebra [5]	Lemma 3.95 ([12]), Lemma 3.96 ([13]); target-response profiles	Ear completions avoid targets.
33	Symmetric difference	Context univ. [12]	Lemma 3.95 ([12]), Lemma 3.96 ([13]), Lemma 10.9 ([43])	Overlap-cycle target test.
34	Rank-one power-of-two target support	Obstruction rank [31], developed at [40]	Lemma 10.6 ([40], [41]), Lemma 10.9 ([43]), Lemma 14.28 ([93]), Lemma 14.32 ([94]), Lemma 14.39, Definition 14.60, Theorem 14.61 ([124]), Lemma 10.10 ([47]), Lemma B.3	Blocks rank-one support enlargement.
35	Composite cancellation	Obstruction rank [31], developed at [44]; imported by the cold branch at [145]	Lemma 10.8 ([44]), Lemma 10.9 ([43]), Lemma 14.32 ([94]), Lemma 14.39, Definition 14.60, Theorem 14.61 ([124]), Lemma B.3	Controls cancellation repairs.
36	Odd response support nonempty	Target algebra [5]	Lemma 2.2 ([5]–[7]), Lemma 3.95 ([12]), Lemma 10.9 ([43])	Power-of-two response-support test.
37	No common response support	Response-support transport [35]	Lemma 10.5 ([35]), Lemma 10.6 ([40], [41]), Lemma 10.9 ([43])	Response-support transport obstruction.
38	Polarized response-support transport	Obstruction rank [31], developed at [35]; imported by the cold branch at [145]	Lemma 10.5 ([35]), Lemma 10.4 ([31], [32]), Lemma 10.6 ([40], [41]), Lemma 10.9 ([43])	Separates response-support testers.

Fast gap-check usage. To check the proof, confirm for each result that every invariant in its **Requires** cell is established at an earlier-or-equal node; a required invariant introduced at a strictly later node is a forward-reference gap. For the contradiction-producing results Lemma 14.45, Lemma 14.93, and Theorem 15.1, confirm the chain inv 8 ← Corollary 3.98 ← Lemma 3.96 is intact and not circular with the result invoking it. The two-sided squeeze lives in exactly two cells: **demand** is Lemma 9.1 (invariant 28) and **supply** is Lemma 8.5 / Proposition 11.6 (invariants 24, 29); on the derived spine branch, any repair to an obstruction estimate must keep demand > supply, i.e. $2.543 > 0.611$ per $|R|$ (sharper high-entropy demand 2.574), or in the net-charge form $0.25 > 0.2282$.

2 Target cycles as Mersenne returns

Definition 2.1 (Mersenne return set)**.** *Let G be a finite simple graph and let $e = uv$ be an oriented edge. Define*

$$R_e(G) = \{r : \textit{there is a simple path of length } r \textit{ from } v \textit{ to } u \textit{ in } G - e\}.$$

Let

$$\mathcal{M} = \{2^j - 1 : j \geq 2\} = \{3, 7, 15, 31, \ldots\}.$$

Lemma 2.2 (Edge-rooted target equivalence)**.** *A graph G contains a cycle of length 2^j if and only if there is an oriented edge e such that $2^j - 1 \in R_e(G)$. Hence G has no power-of-two cycle if and only if*

$$R_e(G) \cap \mathcal{M} = \varnothing \qquad \textit{for every oriented edge } e.$$

Proof. Suppose C is a cycle of length 2^j. Choose an oriented edge $e = uv$ on C. Removing e from C leaves a simple path of length $2^j - 1$ from v to u, and therefore $2^j - 1 \in R_e(G)$. Conversely, if $2^j - 1 \in R_e(G)$, then e together with the corresponding simple return path gives a simple cycle of length 2^j. □

Lemma 2.3 (Two-path criterion)**.** *Let P and Q be internally vertex-disjoint simple paths with the same endpoints. Then $P \cup Q$ is a simple cycle of length $|P| + |Q|$. In particular, if $|P| + |Q| \in \mathcal{D}$, then G contains a power-of-two cycle.*

Proof. The two paths meet only in their common endpoints, so traversing P from one endpoint to the other and then traversing Q back gives a cycle with no repeated internal vertex. Its edge set is the disjoint union of the edge sets of P and Q, hence its length is $|P| + |Q|$. □

Definition 2.4 (Counterexample)**.** *A graph G is a* counterexample *if*

$$\delta(G) \geq 3 \qquad \textit{and} \qquad R_e(G) \cap \mathcal{M} = \varnothing \quad \forall e \in \vec{E}(G).$$

Definition 2.5 (Target-safe)**.** *A subgraph $X \subseteq G$ is* target-safe *if G contains no power-of-two cycle meeting $V(X)$. This is the contextual notion: such a cycle may run partly through $G - X$, so target-safety of X is a property of how X is embedded in G, strictly stronger than the absence of a power-of-two cycle internal to X. For $X = G$ it is exactly the no-power-of-two-cycle clause of the counterexample condition.*

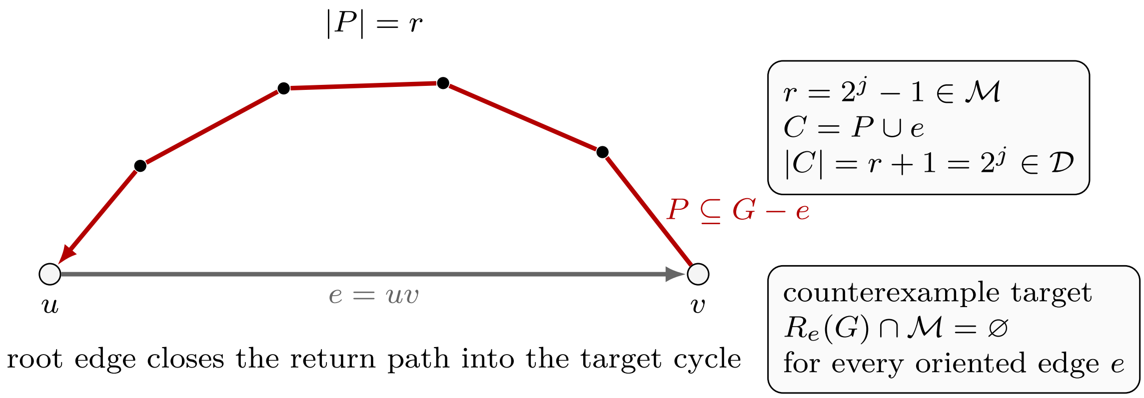

Figure 13: Visual definition of the edge-rooted target algebra. The white circled vertices are the endpoints u and v, the grey arrow is the oriented root edge $e = uv$, and the black dots are internal vertices of the return path. The red path P lies in $G - e$, runs from v back to u, and has length $r = |P|$. The box labelled $r = 2^j - 1 \in \mathcal{M}$ records the target case: adjoining the root edge gives the simple cycle $C = P \cup e$ of length $|C| = r + 1 = 2^j \in \mathcal{D}$. The counterexample box records the equivalent local target condition $R_e(G) \cap \mathcal{M} = \varnothing$ for every oriented edge e.

For contradiction, throughout the paper we assume that a counterexample exists and choose one, denoted G, lexicographically minimal by

$$(|V(G)|, |E(G)|).$$

Set

$$n = |V(G)|, \qquad m = |E(G)|.$$

Framework branch table. The target-cycle section supplies the first branch table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Mersenne return	Some oriented edge e has $R_e(G) \cap \mathcal{M} \neq \varnothing$.	No residual demand remains.	The root edge and return path.	One oriented edge.	Closed by Lemma 2.2.
Two-path closure	Two internally disjoint paths have power-of-two total length.	No residual demand remains.	The two path lengths.	One unordered endpoint pair.	Closed by Lemma 2.3.
Counterexample branch	All oriented edges avoid Mersenne returns.	The target-safe condition.	The counterexample definition.	Every oriented edge of G.	This is the standing branch of Definition 2.4.

3 Minimality and replacement

Lemma 3.1 (No proper core). *Every proper subgraph $H \subsetneq G$ satisfies $\delta(H) \leq 2$.*

Proof. If $H \subsetneq G$ had $\delta(H) \geq 3$, then H would also have no power-of-two cycle, because every cycle of H is a cycle of G. Thus H would be a smaller counterexample, contradicting the minimality of G. $\square$

Lemma 3.2 (Deletion criticality)**.** *Every edge of G has at least one endpoint of degree 3. In particular, the set $V_{\geq 4}(G)$ is independent.*

Proof. Suppose for contradiction that some edge $xy \in E(G)$ has $d_G(x), d_G(y) \geq 4$. Deleting xy lowers the degrees of x and y by one, to at least 3, and leaves every other degree unchanged, so $\delta(G - xy) \geq 3$. Moreover every cycle of $G - xy$ is a cycle of G, so deleting xy creates no new cycle and in particular no new power-of-two cycle; hence $R_e(G - xy) \subseteq R_e(G)$ for every oriented edge e, and $G - xy$ has no power-of-two cycle. Thus $G - xy$ is a counterexample with the same vertex set and one fewer edge, contradicting the edge-minimality of G. Therefore no edge joins two vertices of degree at least 4; equivalently, every edge has an endpoint of degree exactly 3, and the set $V_{\geq 4}(G)$ of vertices of degree at least 4 is independent. $\square$

Lemma 3.3 (Sparse upper envelope)**.** *The minimal counterexample satisfies*

$$m \leq 2n - 2.$$

Proof. First G has at least one vertex of degree 3. Otherwise every vertex would have degree at least 4, and any edge of G would have two endpoints of degree at least 4, contradicting Lemma 3.2. Fix a degree-3 vertex v.

Every subgraph of $G - v$ is a proper subgraph of G. By Lemma 3.1, each such subgraph has a vertex of degree at most 2. Hence $G - v$ is 2-degenerate. A 2-degenerate graph on $N \geq 2$ vertices has at most $2N - 3$ edges: remove vertices in a degeneracy order, count at most 2 deleted edges at each step until two vertices remain, and then count at most one final edge.

Applying this with $N = n - 1$ gives

$$e(G - v) \leq 2(n - 1) - 3.$$

Since $d_G(v) = 3$,

$$m = e(G - v) + d_G(v) \leq 2(n - 1) - 3 + 3 = 2n - 2.$$

$\square$

Lemma 3.4 (Bridgelessness of the minimal counterexample)**.** *The graph G has no bridge. Consequently every edge of G lies on a cycle; equivalently, $R_e(G) \neq \varnothing$ for every oriented edge $e \in \vec{E}(G)$.*

Proof. Suppose for contradiction that $e = uv$ is a bridge. Let A and B be the components of $G - e$ containing u and v, respectively. Form G' from $G - e$ by identifying u and v to a single vertex z.

The graph G' is simple. Indeed, after deleting e, the neighbours of u lie in $A - \{u\}$ and the neighbours of v lie in $B - \{v\}$, which are disjoint. Thus identifying u and v creates no parallel edge, and deleting e before the identification creates no loop.

The minimum degree remains at least 3. Every vertex other than z has the same degree as in G. The identified vertex has

$$d_{G'}(z) = d_G(u) + d_G(v) - 2 \geq 4,$$

since $d_G(u), d_G(v) \geq 3$ and the bridge uv is counted once at each endpoint before deletion.

Next, G' has no power-of-two cycle. Any cycle of G' that uses z and has vertices on both sides would contain, after deleting z, a path from a vertex of $A - \{u\}$ to a vertex of $B - \{v\}$ in $G' - z$. This is impossible because $A - \{u\}$ and $B - \{v\}$ are disconnected in $G' - z$. Hence every cycle of

G' lies wholly in the image of A or wholly in the image of B, and is therefore already a cycle of G. Since G has no power-of-two cycle, neither does G'.

Thus G' is a counterexample with $|V(G')| = |V(G)| - 1$, contradicting the vertex-minimality of G. Therefore G has no bridge. The final assertion follows because an edge is a bridge if and only if it lies on no cycle, and a cycle through an oriented edge $e = ab$ is exactly a return path from b to a in $G - e$ together with e. □

Lemma 3.5 (Cycle-rank constraint). *The cycle rank*

$$\beta(G) = m - n + 1$$

satisfies

$$\beta(G) \geq \frac{n}{2} + 1.$$

Proof. Since $\delta(G) \geq 3$, we have $2m = \sum_v d(v) \geq 3n$. Therefore $m - n + 1 \geq n/2 + 1$. □

Definition 3.6 (Surplus ports and excess selectors). *Let*

$$H = V_{\geq 4}(G).$$

By Lemma 3.2, H is independent, so every neighbour of a vertex in H has degree exactly 3.

A surplus port *is an ordered edge*

$$p = (h, x)$$

with $h \in H$ and $x \in N_G(h)$. We write

$$c(p) = h, \qquad x(p) = x$$

for its high centre and cubic endpoint. Since $d_G(x) = 3$, there are two vertices, denoted a_p, b_p, such that

$$N_G(x(p)) = \{c(p), a_p, b_p\}.$$

The unordered pair

$$s(p) = \{a_p, b_p\}$$

is the shoulder pair *of p, and the unordered pair*

$$e_p^* = a_p b_p$$

is its shoulder chord.

An excess selector *is a choice, for each $h \in H$, of a set*

$$\mathcal{P}_{\text{exc}}(h) \subseteq \{(h, x) : x \in N_G(h)\}$$

of size $d_G(h) - 3$. We set

$$\mathcal{P}_{\text{exc}} = \bigcup_{h \in H} \mathcal{P}_{\text{exc}}(h).$$

Thus

$$|\mathcal{P}_{\text{exc}}| = \sum_{h \in H} (d_G(h) - 3) = \sigma(G).$$

The three unselected incident ports at a high vertex are its base ports*; the selected ports are exactly the units of degree surplus that one would like to remove in order to make the graph cubic.*

Definition 3.7 (High centers and port types)**.** *A vertex $h \in V(G)$ is a* high center *if $d_G(h) \geq 4$, and a* heavy center *if $d_G(h) \geq 5$. Let $p = (h, x)$ be a surplus port at a high center. The port p is* triangular *if its shoulder chord lies in G:*

$$e_p^* \in E(G).$$

Equivalently, if $N_G(x) = \{h, a_p, b_p\}$, then $x a_p b_p x$ is a triangle not using h. The port is open *if*

$$e_p^* \notin E(G).$$

The word open records only the absence of the shoulder chord in G; no suppressed graph is associated to the port.

Lemma 3.8 (High-neighbourhood normal form)**.** *Let h be a high center. Then the following hold.*

(a) Every vertex in $N_G(h)$ has degree 3.

(b) The graph induced by $N_G(h)$ is a matching.

(c) If $x, y \in N_G(h)$ are nonadjacent, then x and y have no common neighbour outside $\{h\}$.

Proof. For (a), apply Lemma 3.2 to the edge hx. Since $d_G(h) \geq 4$, the degree-3 endpoint of hx is x.

For (b), suppose that $G[N_G(h)]$ contained two adjacent edges xy and yz. Then

$$hxyzh$$

is a simple cycle of length 4, contradicting the assumption that G contains no power-of-two cycle. Thus every vertex of $G[N_G(h)]$ has degree at most 1, so $G[N_G(h)]$ is a matching.

For (c), suppose that nonadjacent $x, y \in N_G(h)$ had a common neighbour $z \neq h$. Then

$$hxzyh$$

is a simple cycle of length 4, again impossible. Hence no such z exists. □

Definition 3.9 (Fan-compatible open ports)**.** *Let h be a high center, and let $p = (h, x)$ and $q = (h, y)$ be distinct open ports at h. The pair $\{p, q\}$ is* fan-compatible *if*

$$x \notin s(q), \qquad y \notin s(p), \qquad s(p) \cap s(q) = \varnothing.$$

Equivalently, the two shoulder pairs are disjoint and neither port vertex is a shoulder endpoint for the other port; in this case the four non-h incidences can be assigned as local Type B fan incidences.

Lemma 3.10 (Same-center open ports are fan-compatible)**.** *Let h be a high center, and let $p = (h, x)$ and $q = (h, y)$ be distinct open ports at h. If $xy \notin E(G)$, then $\{p, q\}$ is fan-compatible.*

Proof. Since $xy \notin E(G)$, the endpoint x is not in the shoulder pair $s(q)$, and y is not in the shoulder pair $s(p)$. Suppose that $s(p) \cap s(q)$ contained a vertex z. Then $z \neq h$, and the two nonadjacent neighbours $x, y \in N_G(h)$ would have the common neighbour z outside $\{h\}$. This contradicts Lemma 3.8. Hence the shoulder pairs are disjoint, and the pair is fan-compatible by definition. □

Lemma 3.11 (Heavy-center triangular alternative)**.** *Let h be a heavy center of degree k. If no two open ports at h form a fan-compatible open pair, then at least $k - 2$ ports at h are triangular. In particular, at least three ports at h are triangular.*

Proof. Let $U \subseteq N_G(h)$ be the set of vertices x such that the port (h, x) is open. If two vertices $x, y \in U$ were nonadjacent, then Lemma 3.10 would make the two ports (h, x) and (h, y) fan-compatible, contrary to the hypothesis. Therefore every two vertices in U are adjacent.

By Lemma 3.8, $G[N_G(h)]$ is a matching. A matching has no clique of size 3, so $|U| \leq 2$. All ports at h whose cubic endpoint is not in U are not open, hence triangular by Definition 3.7. Therefore the number of triangular ports at h is at least

$$k - |U| \geq k - 2.$$

Since $k \geq 5$, this is at least 3. □

Corollary 3.12 (Heavy-center local dichotomy)**.** *Let h be a heavy center. Then one of the following alternatives holds.*

(i) There are two open ports at h that are fan-compatible.

(ii) At least $d_G(h) - 2$ ports at h are triangular; in particular, h has three triangular ports.

Proof. If the first alternative fails, then Lemma 3.11 gives the second alternative. □

Corollary 3.13 (Degree-four local activation)**.** *Let h be a high center with $d_G(h) = 4$. Then one of the following alternatives holds.*

(i) There are two open ports at h that are fan-compatible.

(ii) At least two ports at h are triangular.

In case (i)*, any admissible Type B fan-window profile that records the two ports and assigns their four non-h incidences routes through Corollary 14.81. In case* (ii)*, any admissible Type B fan-window profile that records two triangular ports and assigns their four triangle incidences routes through Proposition 14.80; in the certificate-marked B1 survivor it has*

$$D_B(\mathfrak{F}_h) \geq 2 - \frac{11 - 4}{4} = \frac{1}{4} > 0.$$

Proof. Let $U \subseteq N_G(h)$ be the set of vertices x such that the port (h, x) is open. If two vertices $x, y \in U$ are nonadjacent, then Lemma 3.10 gives a fan-compatible open pair, which is alternative (i). Suppose no such pair exists. Then every two vertices of U are adjacent. By Lemma 3.8, the graph induced by $N_G(h)$ is a matching, so U has size at most 2. Since $d_G(h) = 4$, at least $4 - |U| \geq 2$ ports at h are not open. By Definition 3.7, those ports are triangular. This is alternative (ii).

The routing assertion in case (i) is exactly Corollary 14.81, applied with $k = 4$. In case (ii), choose two triangular ports. For each of them the two triangle edges are precisely the two non-h incidences of the port vertex. Once those incidences are assigned to the fan envelope, both ports are fan-closed by Definition 14.69. Applying Proposition 14.80 with $r = 2$ and $k = 4$ gives the displayed deficit bound and the listed Type B alternatives. □

Definition 3.14 (Triangular fan core and shoulder completions)**.** *Let h be a heavy center and let*

$$\mathcal{T}_h = \{p_i = (h, x_i) : i \in I\}$$

be a nonempty set of triangular ports at h. For each $i \in I$, write

$$N_G(x_i) = \{h, a_i, b_i\}, \qquad S_i = \{a_i, b_i\},$$

so that $x_i a_i b_i x_i$ is a triangle. The triangular fan core *generated by $\mathcal{T}_h$ is*

$$F_h(\mathcal{T}_h) = G\Big[\{h\} \cup \{x_i : i \in I\} \cup \bigcup_{i \in I} S_i\Big].$$

The vertices in S_i are the shoulders *of p_i.*

Let $s \in S_i$, and let s'_i denote the other vertex of S_i. A shoulder-completion incidence *at s is an edge $sy \in E(G)$ with*

$$y \notin \{x_i, s'_i\}.$$

It is central *when $y = h$,* cross-triangular *when*

$$y \in \bigcup_{j \in I,\ j \neq i} S_j,$$

and outside *when*

$$y \notin V(F_h(\mathcal{T}_h)) \cup N_G(h).$$

An outside incidence is the first edge by which the displayed triangular fan core attaches to the rest of the graph.

Lemma 3.15 (Shoulder-completion bookkeeping)**.** *Let $p_i = (h, x_i)$ be a triangular port at a heavy center h, and let $S_i = \{a_i, b_i\}$. Then the following hold.*

(a) Each shoulder $s \in S_i$ has at least one shoulder-completion incidence.

(b) At most one shoulder in S_i lies in $N_G(h)$.

(c) If $s \in S_i \cap N_G(h)$, then $d_G(s) = 3$, its unique shoulder-completion incidence is the central edge sh, and s has no outside incidence.

(d) No shoulder-completion incidence has endpoint in $N_G(h) \setminus \{h\}$.

Proof. Fix $s \in S_i$, and let s'_i be the other shoulder. The two edges sx_i and ss'_i are present. Since $\delta(G) \geq 3$, s has at least one incident edge different from these two; this proves (a).

If both a_i and b_i lay in $N_G(h)$, then the vertex $x_i \in N_G(h)$ would be adjacent inside $G[N_G(h)]$ to both a_i and b_i. This contradicts the matching conclusion in Lemma 3.8. Hence (b) holds.

Assume $s \in N_G(h)$. By Lemma 3.8, $d_G(s) = 3$. Its three neighbours are already h, x_i, and s'_i. Thus the only shoulder-completion incidence at s is the central edge sh, proving (c).

Finally suppose that sy is a shoulder-completion incidence with $y \in N_G(h) \setminus \{h\}$. If y were adjacent to x_i, then $y \in S_i$, because the only neighbours of x_i are h, a_i, b_i; this is excluded by the definition of shoulder-completion incidence. Hence x_i and y are nonadjacent vertices of $N_G(h)$. They have the common neighbour $s \neq h$, contradicting Lemma 3.8. This proves (d). □

Lemma 3.16 (Port returns enter through shoulders)**.** *Let $p_i = (h, x_i)$ be a triangular port at a heavy center h. Then there is a shoulder $s \in S_i$ and a simple s–h path Q in $G - x_i$. Moreover, the cycle obtained from Q by adding the two edges*

$$hx_i, \qquad x_i s$$

has length $|E(Q)| + 2$, and therefore

$$|E(Q)| \notin \{2^j - 2 : j \geq 2\}.$$

If Q is not the single central edge sh, then after deleting a possible initial use of the shoulder edge $a_i b_i$, the path Q uses a noncentral shoulder-completion incidence of one of the two shoulders of p_i.

Proof. By Lemma 3.4, the edge hx_i lies on a cycle C. Deleting hx_i from C gives a simple x_i–h path in $G - hx_i$. The first edge of this path is x_is for some $s \in S_i$. Removing that first edge leaves a simple s–h path Q in $G - x_i$.

The edges of Q, together with x_is and hx_i, reconstruct the cycle C. Since G has no power-of-two cycle, $|E(Q)| + 2$ is not a power of two, which gives the displayed exclusion.

If Q has length 1, then $Q = sh$ and the return is central. Suppose $|E(Q)| \geq 2$. If the first edge of Q is not ss'_i, where s'_i is the other shoulder, then it is a noncentral shoulder-completion incidence at s. If the first edge is ss'_i, delete that initial edge. The next edge cannot return to s, because Q is simple, and cannot go to x_i, because $Q \subseteq G - x_i$. Hence the next edge is either the central edge s'_ih or a noncentral shoulder-completion incidence at s'_i. This is the asserted alternative. □

Lemma 3.17 (First landing exhaustion for triangular shoulders)**.** *Let $\mathcal{T}_h$ be a set of triangular ports at a heavy center h. Every shoulder-completion incidence in the triangular fan core $F_h(\mathcal{T}_h)$ is exactly one of the following:*

(i) a central incidence to h;

(ii) a cross-triangular incidence to a shoulder of another port in $\mathcal{T}_h$;

(iii) an outside incidence with endpoint in

$$V(G) \setminus \big(V(F_h(\mathcal{T}_h)) \cup N_G(h)\big).$$

No completion incidence lands at another neighbour of h, and no completion incidence lands at another port vertex x_j.

Proof. Let sy be a shoulder-completion incidence at a shoulder $s \in S_i$. If $y = h$, it is central. If $y \in N_G(h) \setminus \{h\}$, then Lemma 3.15 forbids the incidence; in particular this excludes $y = x_j$ for every port vertex x_j, because each $x_j \in N_G(h)$.

It remains that $y \notin N_G(h)$. If $y \in V(F_h(\mathcal{T}_h))$, then y cannot be h, cannot be a port vertex x_j, and cannot be x_i or the other shoulder of p_i by the definition of completion incidence. Thus y is a shoulder of some other triangular port, and the incidence is cross-triangular. If $y \notin V(F_h(\mathcal{T}_h))$, then it is an outside incidence by definition. These cases are disjoint and exhaustive. □

Lemma 3.18 (Cross-shoulder multiplicity)**.** *Let p_i and p_j be distinct triangular ports at the same heavy center h. If there are two distinct cross-triangular incidences between S_i and S_j, then either G contains a 4-cycle or at least one shoulder in $S_i \cup S_j$ has ambient degree at least 4. Consequently, after routing high shoulders to the Type B surplus branch, the cross-triangular incidences between any two shoulder pairs form a matching of size at most one.*

Proof. Let the two cross-triangular edges be e_1 and e_2. If they share a shoulder endpoint s, then s is incident with the two triangle edges inside its own port and with both e_1 and e_2. Hence $d_G(s) \geq 4$.

Assume therefore that e_1 and e_2 are vertex-disjoint. Write $S_i = \{a_i, b_i\}$ and $S_j = \{a_j, b_j\}$. After relabelling the two shoulders inside each pair, the cross edges are a_ia_j and b_ib_j. Then

$$a_ia_jb_jb_ia_i$$

is a simple cycle of length 4, using the cross edge a_ia_j, the triangular shoulder edge a_jb_j, the cross edge b_jb_i, and the triangular shoulder edge b_ia_i. This contradicts target-safety unless the high-shoulder alternative already occurred in the shared-endpoint case.

If all shoulders have ambient degree 3, the shared-endpoint case is impossible and the vertex-disjoint case gives a forbidden 4-cycle. Thus at most one cross-triangular edge can join S_i to S_j, and the remaining incidences form a matching. □

Remark 3.19 (Degree-greater-than-four residual leaf)**.** *Corollary 3.12 isolates the degree-greater-than-four branch into two local configurations. In Leaf A, a fan-compatible open pair is recorded as two fan-closed ports by Lemma 14.70; Corollary 14.81 then gives a positive-deficit Type B fan whenever the fan is certificate-marked and survives the direct target, compression, and support-dependence alternatives. If one of those direct alternatives occurs, the branch is already closed; if the fan is unmarked or the local entries cannot be globally disjoint, the branch is recorded as a fan-certificate residual center or a B2-overlap obstruction. In Leaf B, the triangular branch is routed by Definition 3.14 and Lemmas 3.15, 3.17 and 3.18 and by the Type B fan ledger in Corollary 14.82: the guaranteed triangular ports are cubic-closed fan neighbours, their shoulder-completion incidences have only the central, cross-triangular, and outside first landings listed above, and the cross-triangular multiplicity left after high-shoulder routing is a matching survivor. The common primitive in both leaves is fan-closedness: two assigned non-h incidences at a cubic neighbour of h contribute one cubic-closed fan neighbour. Thus any heavy center of degree $k \geq 5$ supplies at least two fan-closed ports after the local dichotomy, and Proposition 14.80 gives*

$$D_B \geq 2 - \frac{11-k}{4} = \frac{k-3}{4} > 0.$$

Thus the active degree-greater-than-four closure is the Type B fan-closed ledger.

Remark 3.20 (Interface with the surplus-token branch)**.** *Corollary 3.12 separates the degree-greater-than-four branch from the degree-4 branch. The degree-greater-than-four branch is routed through fan-closed Type B data by Corollaries 14.81 and 14.82. This local dichotomy is one input to the sparse surplus-token branch, not a replacement for it. Excess ports that do not close through the direct target, compression, support dependence, or fan-closed Type B alternatives are counted by the active surplus family $\mathcal{P}_{\mathrm{exc}}$. The non-near-cubic branch then applies the free/blocked pair split and the capacity-token ledger of Definition 3.37. Consequently any attempted survivor above the near-cubic range first produces one of the three role-homogeneous same-token audit classes of Definition 3.41 and Corollary 3.45: window-incidence, remainder-surplus, or primitive blocker-support load, with a fixed blocker type and token subtype. The same-token bottleneck routing of Lemma 3.77 then either realizes a sparse surplus exit, produces ordinary Type B fan data, or leaves fixed homogeneous caps and hence the near-cubic estimate.*

Definition 3.21 (Open-port suppression)**.** *Let $p = (h, x)$ be an open surplus port, so*

$$N_G(x) = \{h, a_p, b_p\}, \qquad a_p b_p \notin E(G).$$

The suppression *of p is the graph*

$$G/p := G - x + a_p b_p.$$

A finite family $\mathcal{Q}$ of open surplus ports is suppressible *if the following four conditions hold.*

(a) The non-center supports

$$T(p) = \{x(p), a_p, b_p\} \qquad (p \in \mathcal{Q})$$

are pairwise disjoint.

(b) No selected center is contained in any non-center support:

$$\{c(p) : p \in \mathcal{Q}\} \cap \bigcup_{p \in \mathcal{Q}} T(p) = \varnothing.$$

(c) The shoulder chords $e_p^* = a_p b_p$, $p \in \mathcal{Q}$, are pairwise distinct and none is an edge of G.

(d) For every high center h,

$$|\{p \in \mathcal{Q} : c(p) = h\}| \le d_G(h) - 3.$$

For a suppressible family $\mathcal{Q}$, define

$$G/\mathcal{Q} = G - \{x(p) : p \in \mathcal{Q}\} + \{a_p b_p : p \in \mathcal{Q}\}.$$

Thus $G/\{p\} = G/p$.

Lemma 3.22 (Suppression preserves simplicity and minimum degree)**.** *If $\mathcal{Q}$ is a suppressible family of open surplus ports, then $G/\mathcal{Q}$ is finite and simple, and every vertex of $G/\mathcal{Q}$ has degree at least 3.*

Proof. Only the vertices $x(p)$, $p \in \mathcal{Q}$, are deleted, so the resulting graph is finite. No added edge is a loop, because $a_p \neq b_p$. No added edge duplicates an old edge, and no two added edges duplicate each other, by the definition of a suppressible family. Hence $G/\mathcal{Q}$ is simple.

Let v be a vertex that remains after the suppression. If $v = c(p)$ for some selected ports, then the only deleted neighbours of v are the port vertices $x(p)$ with center v: condition (b) prevents v from lying in any other selected non-center support. Thus

$$d_{G/\mathcal{Q}}(v) = d_G(v) - |\{p \in \mathcal{Q} : c(p) = v\}| \ge d_G(v) - (d_G(v) - 3) = 3$$

by condition (d).

If v is a shoulder endpoint for a selected port p, then condition (a) makes this selected port unique. The vertex v loses the edge $vx(p)$ and gains exactly the shoulder chord e_p^*. It is not deleted and is not a selected center by condition (b). Hence its degree is unchanged. Every other remaining vertex loses no incident edge and gains no incident edge, or only gains an added chord. Since $\delta(G) \ge 3$, all remaining vertices have degree at least 3. □

Lemma 3.23 (Single-port suppression witness)**.** *Let G be the lexicographically minimal counterexample, and let $p = (h, x)$ be an open surplus port. Then $G - x$ contains a simple a_p–b_p path of length $2^j - 1$ for some $j \ge 2$.*

Proof. The singleton $\{p\}$ is suppressible. By Lemma 3.22, G/p is a finite simple graph with minimum degree at least 3. It has one fewer vertex than G. By the minimality of G, the graph G/p contains a cycle C of length 2^j for some $j \ge 2$.

The cycle C must use the added shoulder chord $a_p b_p$. Otherwise all edges of C are old edges of G not incident with the deleted vertex x, and C is already a power-of-two cycle in G, impossible. Deleting the chord $a_p b_p$ from C leaves a simple a_p–b_p path in $G - x$, and its length is $2^j - 1$. □

Lemma 3.24 (Suppressed-family critical cycle)**.** *Let G be the lexicographically minimal counterexample, and let $\varnothing \neq \mathcal{Q}$ be a suppressible family of open surplus ports. Then $G/\mathcal{Q}$ contains a power-of-two cycle using at least one added shoulder chord. If a power-of-two cycle C in $G/\mathcal{Q}$ has length 2^j and uses exactly the added chords indexed by $\mathcal{S} \subseteq \mathcal{Q}$, then $\mathcal{S} \neq \varnothing$, and replacing each added chord $a_p b_p$ with the two-edge path*

$$a_p x(p) b_p$$

gives a simple cycle of G of length $2^j + |\mathcal{S}|$. Consequently

$$2^j + |\mathcal{S}|$$

is not a power of two.

Proof. By Lemma 3.22, $G/\mathcal{Q}$ is finite, simple, and has minimum degree at least 3. Since $\mathcal{Q} \neq \varnothing$, it has fewer vertices than G. Minimality gives a cycle C in $G/\mathcal{Q}$ whose length is a power of two. If C used no added shoulder chord, then C would be a power-of-two cycle already present in G, a contradiction. Hence the set $\mathcal{S}$ of added chords used by C is nonempty.

For each $p \in \mathcal{S}$, replace the edge $a_p b_p$ on C by the path $a_p x(p) b_p$. The supports $T(p)$ are pairwise disjoint, and none of the vertices $x(p)$ is present in $G/\mathcal{Q}$. Therefore these replacements are internally disjoint and introduce no repeated vertex. The resulting closed walk is a simple cycle in G. Each replacement adds exactly one edge, so the new length is $2^j + |\mathcal{S}|$. Since G has no power-of-two cycle, this integer is not a power of two. ☐

Lemma 3.25 (Sparse slack identity)**.** *For the standing parameters*

$$\lambda = 2n - 3 - m, \qquad \sigma(G) = 2m - 3n,$$

one has

$$\sigma(G) = n - 6 - 2\lambda \qquad \text{and} \qquad m = \frac{3}{2}n + \frac{1}{2}\sigma(G).$$

Thus every additional unit of high-degree surplus consumes exactly one half-unit of the sparse slack λ.

Proof. Substituting $m = 2n - 3 - \lambda$ gives

$$2m - 3n = 2(2n - 3 - \lambda) - 3n = n - 6 - 2\lambda.$$

The second identity is the same equality solved for m. ☐

Definition 3.26 (Sparse surplus exits)**.** *A* sparse surplus exit *is one of the following conclusions, each of which is defined without invoking the near-cubic Type A/Type B branch machinery:*

- *(a) a direct power-of-two contradiction, equivalently a power-of-two cycle or a Mersenne return in the sense of Lemma 2.2;*
- *(b) a target-defective quotient, as defined by Lemma 3.95;*
- *(c) a nontrivial target-complete compression of a proper boundaried piece, forbidden by Lemma 3.96 and Corollary 3.98;*
- *(d) a proper or whole-graph support-dependence coordinate governed by Lemmas 10.6 and 10.9;*
- *(e) an open-port suppression cycle whose chord set violates the arithmetic conclusion of Lemma 3.24.*

A graph survives the sparse surplus exits *when none of these conclusions occurs for any selected surplus demand, any selected pair of surplus demands, or any baseline spine coordinate used in the entropy sandwich of Definition 3.62. The label name* `def:named-surplus-exits` *is retained for references in the dependency diagram; the exits in this branch are precisely the sparse exits listed above.*

Lemma 3.27 (Excess-port extraction)**.** *Let G be the lexicographically minimal counterexample, and let $\mathcal{P}_{\mathrm{exc}}$ be the excess selector of Definition 3.6. Then*

$$|\mathcal{P}_{\mathrm{exc}}| = \sigma(G).$$

Moreover, for every $p = (h, x) \in \mathcal{P}_{\mathrm{exc}}$, the vertex h has degree at least 4, the vertex x has degree 3, and

$$N_G(x) = \{h, a_p, b_p\}$$

for two distinct vertices $a_p, b_p \neq h$.

Proof. By definition, each high-degree vertex h contributes exactly $d_G(h) - 3$ selected incident ports to $\mathcal{P}_{\mathrm{exc}}$. The high-degree vertices are precisely the vertices contributing positive terms to $\sigma(G) = \sum_v (d_G(v) - 3)$, because every vertex of G has degree at least 3. Hence

$$|\mathcal{P}_{\mathrm{exc}}| = \sum_{h \in V_{\geq 4}(G)} (d_G(h) - 3) = \sigma(G).$$

If $p = (h, x)$ is a selected port, then $h \in V_{\geq 4}(G)$. By Lemma 3.2, every edge has at least one degree-3 endpoint and $V_{\geq 4}(G)$ is independent. Since h is not degree 3, the other endpoint x must have degree 3. Thus x has exactly two neighbours different from h, denoted a_p and b_p. ☐

Lemma 3.28 (Sparse port activation)**.** *For every selected port $p = (h, x) \in \mathcal{P}_{\mathrm{exc}}$, there is a canonical finite response support $\Gamma(p)$ with the following properties.*

(a) The port support is $T(p) = \{x, a_p, b_p\}$.

(b) There is a simple x–h path $R_p \subseteq G - hx$. Its first edge after x is either xa_p or xb_p.

(c) If p is open, then $\Gamma(p)$ contains the lexicographically first path $Q_p \subseteq G - x$ supplied by Lemma 3.23; this path joins a_p to b_p and has length $2^{j(p)} - 1$ for some $j(p) \geq 2$.

(d) If p is triangular, then $\Gamma(p)$ contains the triangle

$$x a_p b_p x$$

and the lexicographically first return path R_p from (b).

The construction of $\Gamma(p)$ uses only G, the ordered vertex labels, and the port p; hence it is canonical.

Proof. By Lemma 3.4, the edge hx is not a bridge. Therefore h and x remain connected in $G - hx$, and there is a simple x–h path in $G - hx$. Choose the lexicographically first such path and call it R_p. Since $d_G(x) = 3$ and $N_G(x) = \{h, a_p, b_p\}$, the first edge of R_p after x is one of xa_p, xb_p.

If p is open, Lemma 3.23 supplies a simple a_p–b_p path $Q_p \subseteq G - x$ of length $2^{j(p)} - 1$; choose the lexicographically first such path. If p is triangular, then $a_p b_p \in E(G)$, so $x a_p b_p x$ is a triangle. In that case record this triangle together with R_p. These rules determine a finite subgraph $\Gamma(p)$ canonically from G and p. ☐

Definition 3.29 (Active surplus demands)**.** *Let G be a lexicographically minimal counterexample reaching node* [18]*. An* active surplus demand *is a selected surplus port*

$$p = (h, x) \in \mathcal{P}_{\mathrm{exc}}$$

equipped with the canonical data $T(p)$, R_p, and $\Gamma(p)$ from Lemma 3.28, and not already removed by a sparse surplus exit of Definition 3.26. The response support $\Gamma(p)$ is used only as a support for target-response data; no disjointness between distinct response supports is assumed unless it is explicitly stated.

Lemma 3.30 (Surviving active family is the full excess-port family)**.** *If G survives the sparse surplus exits, then*

$$\mathcal{A}_0 := \mathcal{P}_{\mathrm{exc}}$$

is a finite family of active surplus demands and

$$|\mathcal{A}_0| = \sigma(G).$$

Proof. By Lemma 3.27, every selected port $p \in \mathcal{P}_{\rm exc}$ has the canonical data $T(p)$, R_p, and $\Gamma(p)$ supplied by Lemma 3.28. Since G survives the sparse surplus exits, no selected surplus demand is removed by an exit of Definition 3.26. Thus every $p \in \mathcal{P}_{\rm exc}$ is active in the sense of Definition 3.29. The cardinality is exactly $|\mathcal{P}_{\rm exc}| = \sigma(G)$, again by Lemma 3.27. □

Definition 3.31 (Sparse surplus blockers)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. A* sparse surplus blocker *for an unordered pair $\{p, q\} \subseteq \mathcal{A}_0$ is one of the following concrete objects.*

(a) a vertex or edge-incidence contained in both declared demand supports

$$(T(p) \cup \Gamma(p)) \cap (T(q) \cup \Gamma(q));$$

(b) a common vertex or common edge-incidence of the two canonical return paths R_p and R_q;

(c) a common shoulder endpoint or shared cubic buffer vertex among $\{a_p, b_p, x(p), a_q, b_q, x(q)\}$;

(d) a boundary-degree-profile coordinate which prevents a quotient or replacement from staying in a single fibre, in the sense of Lemma 3.94;

(e) a target-response coordinate witnessing a target-defective quotient, target-complete compression, or support-dependence event, in the sense of Lemmas 3.95, 10.6 and 10.9 and Corollary 3.98;

(f) an arithmetic chord-set obstruction from a suppressed open-port family, namely the concrete set $\mathcal{S}$ of added shoulder chords in Lemma 3.24.

The blocker must be an object explicitly listed above. A pair is not declared blocked merely because the proof has not found a closure; its failed compatibility must be represented by a finite-capacity object of one of these six types.

Definition 3.32 (Canonical sparse-blocker order)**.** *Fix the vertex order used in the lexicographic choices above. Sparse surplus blockers are ordered as follows. First order by the type number (a)–(f) in Definition 3.31. Within a type, order vertices by the fixed vertex order, edge-incidences lexicographically by their ordered endpoint pair and incident end, boundary-profile and target-response coordinates by the lexicographic order of their declared labels and supports, and chord sets by the lexicographic order of their increasing list of added shoulder chords. This gives a total order on every finite set of blockers arising from a fixed pair of active surplus demands.*

Definition 3.33 (Canonical blocker ledger)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands and put*

$$\Pi(\mathcal{A}_0) = \binom{\mathcal{A}_0}{2}.$$

For $\pi = \{p, q\} \in \Pi(\mathcal{A}_0)$, let $\mathsf{Blk}(\pi)$ be the finite set of sparse surplus blockers for π in the sense of Definition 3.31. Define

$$\Pi_{\rm blk} := \{\pi \in \Pi(\mathcal{A}_0) : \mathsf{Blk}(\pi) \neq \varnothing\}, \qquad \Pi_{\rm free} := \Pi(\mathcal{A}_0) \setminus \Pi_{\rm blk}.$$

For every $\pi \in \Pi_{\rm blk}$, the canonical blocker *is*

$$\Phi_{\rm can}(\pi) := \min_{\prec} \mathsf{Blk}(\pi),$$

where $\prec$ is the order of Definition 3.32. The canonical blocker set and incidence ledger are

$$\mathcal{B}_{\mathrm{can}} := \Phi_{\mathrm{can}}(\Pi_{\mathrm{blk}}), \qquad \mathfrak{I}_{\mathrm{can}} := \{(\pi, \Phi_{\mathrm{can}}(\pi)) : \pi \in \Pi_{\mathrm{blk}}\}.$$

For $B \in \mathcal{B}_{\mathrm{can}}$, write

$$\mu(B) := |\Phi_{\mathrm{can}}^{-1}(B)|.$$

Lemma 3.34 (Canonical blocker ledger has no overcounting)**.** *For every finite family $\mathcal{A}_0$ of active surplus demands,*

$$|\Pi_{\mathrm{blk}}| = |\mathfrak{I}_{\mathrm{can}}| = \sum_{B \in \mathcal{B}_{\mathrm{can}}} \mu(B).$$

Consequently, if $\mu(B) \leq M$ for every $B \in \mathcal{B}_{\mathrm{can}}$ and $|\mathcal{B}_{\mathrm{can}}| \leq C_B n$, then

$$|\Pi_{\mathrm{blk}}| \leq M C_B n.$$

Proof. The map

$$\pi \longmapsto (\pi, \Phi_{\mathrm{can}}(\pi))$$

is a bijection from Π_{blk} to $\mathfrak{I}_{\mathrm{can}}$, so $|\Pi_{\mathrm{blk}}| = |\mathfrak{I}_{\mathrm{can}}|$. The fibres $\Phi_{\mathrm{can}}^{-1}(B)$, $B \in \mathcal{B}_{\mathrm{can}}$, form a disjoint partition of Π_{blk}. Hence

$$|\Pi_{\mathrm{blk}}| = \sum_{B \in \mathcal{B}_{\mathrm{can}}} |\Phi_{\mathrm{can}}^{-1}(B)| = \sum_{B \in \mathcal{B}_{\mathrm{can}}} \mu(B).$$

The displayed upper bound follows by applying $\mu(B) \leq M$ to each summand and then using $|\mathcal{B}_{\mathrm{can}}| \leq C_B n$. $\square$

Definition 3.35 (Primitive blocker support)**.** *An* edge-incidence *of G is an ordered pair (e, v) with $e \in E(G)$ and $v \in e$. The primitive sparse blocker-support universe is*

$$\mathfrak{U}_{\mathrm{sp}}(G) := V(G) \sqcup I_E(G) \sqcup \mathcal{P}_{\mathrm{exc}}, \qquad I_E(G) := \{(e, v) : e \in E(G),\, v \in e\}.$$

Each sparse surplus blocker B has a primitive blocker support

$$\kappa(B) \in \mathfrak{U}_{\mathrm{sp}}(G)$$

defined as follows.

- *(a) If B is a vertex, then $\kappa(B) = B$.*
- *(b) If B is an edge-incidence, then $\kappa(B) = B$.*
- *(c) If B is a common shoulder endpoint or cubic buffer vertex, then $\kappa(B) = B$.*
- *(d) If B is a boundary-degree-profile coordinate or target-response coordinate, then $\kappa(B)$ is the least vertex, edge-incidence, or selected surplus port contained in the declared support of that coordinate. Paths and windows contribute their least contained vertex or edge-incidence. This support exists because every sparse pair-response coordinate is supported on $X_\pi \supseteq T(p) \cup T(q)$, and the declared signature of Definition 3.87 generates quotient coordinates only from finite unions of declared supports.*
- *(e) If B is a suppressed-family chord-set obstruction $\mathcal{S}$, then $\kappa(B)$ is the least selected surplus port whose shoulder chord belongs to $\mathcal{S}$. The set $\mathcal{S}$ is nonempty by Lemma 3.24.*

All minima use the fixed vertex order, the induced lexicographic order on edge-incidences, and the selected-port order used in Definition 3.32.

Lemma 3.36 (Primitive blocker-support supply is linear)**.** *For every lexicographically minimal counterexample reaching the sparse-surplus branch,*

$$|\mathfrak{U}_{\mathrm{sp}}(G)| \le 6n.$$

Proof.

$$|\mathfrak{U}_{\mathrm{sp}}(G)| = |V(G)| + |I_E(G)| + |\mathcal{P}_{\mathrm{exc}}| = n + 2m + \sigma(G).$$

By Lemma 3.3, $m \le 2n-2$, so $2m \le 4n-4$. Also

$$\sigma(G) = 2m - 3n \le n - 4 \le n,$$

again using $m \le 2n-2$. Hence

$$|\mathfrak{U}_{\mathrm{sp}}(G)| \le n + (4n-4) + n \le 6n.$$

□

Definition 3.37 (Capacity tokens for blocked surplus pairs)**.** *The* capacity-token universe *is the disjoint union*

$$\mathfrak{T}_{\mathrm{cap}} = \mathfrak{T}_{\mathrm{prim}} \sqcup \mathfrak{T}_R \sqcup \mathfrak{T}_W,$$

defined as follows.

$$\mathfrak{T}_{\mathrm{prim}} := \mathfrak{U}_{\mathrm{sp}}(G).$$

The remainder-surplus tokens are

$$\mathfrak{T}_R := \{(v,j) : v \in R,\ 1 \le j \le d_G(v) - 3\}.$$

Thus $|\mathfrak{T}_R| = \sigma_R$. The window-join tokens are

$$\mathfrak{T}_W := \{(e, RW) : e \in E(R,W)\} \sqcup \{(e,a),(e,b) : e = ab,\ a,b \textit{ lie in distinct packed } P_{13} \textit{ windows}\}.$$

Thus each window–remainder edge contributes one token and each cross-window edge contributes one token at each of its two window ends.

For a sparse surplus blocker B, let $\mathrm{supp}(B)$ *be its declared support: the singleton vertex for a vertex blocker, the marked edge for an edge-incidence blocker, the declared support of a boundary-profile or target-response coordinate, and the union of the shoulder chords for a chord-set blocker. Order the unordered pairs in Π_{blk} lexicographically by their two active surplus demands, using the selected-port order induced by the fixed vertex order. This is the* canonical pair order*. For $\pi \in \Pi_{\mathrm{blk}}$, put $B_\pi = \Phi_{\mathrm{can}}(\pi)$. The* capacity token $\Theta_{\mathrm{cap}}(\pi)$ *is defined by the following canonical rule.*

- *(a) If* $\mathrm{supp}(B_\pi)$ *contains a window–remainder edge, choose the least such edge and take the corresponding token $(e, RW) \in \mathfrak{T}_W$.*
- *(b) Otherwise, if* $\mathrm{supp}(B_\pi)$ *contains a cross-window edge, choose the least such edge $e = ab$ and then the least of its two endpoint tokens in $\mathfrak{T}_W$.*

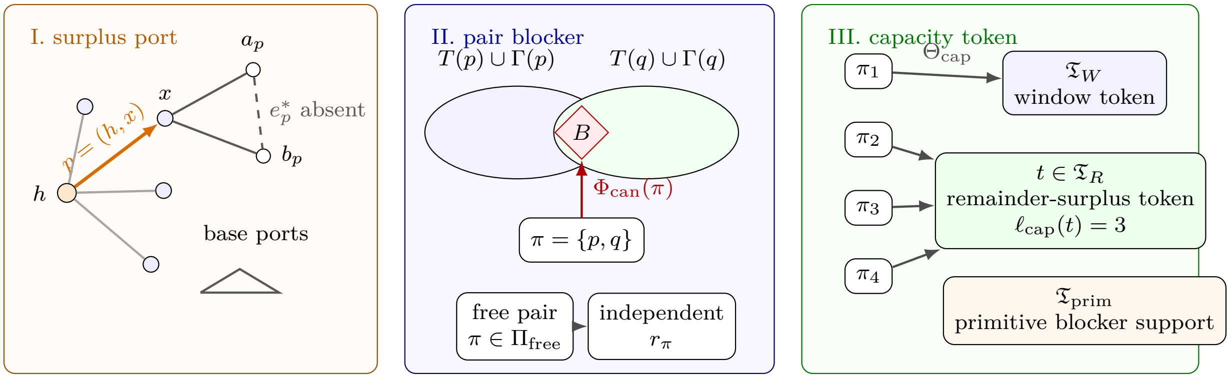

Figure 14: Visual definition of the sparse surplus port, blocker, and capacity-token ledger. Panel I shows a high center h, a selected surplus port $p = (h, x)$, and the cubic endpoint x with its two shoulder vertices a_p, b_p. The dashed shoulder chord $e_p^* = a_p b_p$ illustrates an open port; the small triangle below illustrates the triangular case $e_p^* \in E(G)$. The selected ports in $\mathcal{P}_{\text{exc}}(h)$ are the $d_G(h) - 3$ surplus units at h, while the unselected incidences are base ports. Panel II shows two active demands p, q, their declared supports $T(p) \cup \Gamma(p)$ and $T(q) \cup \Gamma(q)$, and a blocker B for the pair $\pi = \{p, q\}$. A blocked pair is assigned its canonical blocker $\Phi_{\text{can}}(\pi)$; a free pair lies in Π_{free} and contributes an independent pair-response coordinate r_π. Panel III shows the capacity-token map Θ_{cap}. Each blocked pair is assigned exactly one token in the disjoint union $\mathfrak{T}_{\text{cap}} = \mathfrak{T}_W \sqcup \mathfrak{T}_R \sqcup \mathfrak{T}_{\text{prim}}$, where $\mathfrak{T}_W$ records window incidences, $\mathfrak{T}_R$ records surplus units in the remainder, and $\mathfrak{T}_{\text{prim}}$ records primitive blocker supports. Multiple blocked pairs may land on the same token t; their number is the token load $\ell_{\text{cap}}(t) = |\Theta_{\text{cap}}^{-1}(t)|$.

(c) *Otherwise, if* $\operatorname{supp}(B_\pi)$ *contains a vertex* $v \in R$ *with* $d_G(v) > 3$*, choose the least such vertex* $v = v(\pi)$*. Let* $\mathcal{P}_v$ *be the set of all* $\pi' \in \Pi_{\text{blk}}$ *for which the first two cases fail and the least high-degree remainder vertex in* $\operatorname{supp}(B_{\pi'})$ *is* v*. Order* $\mathcal{P}_v$ *by the canonical pair order and let*

$$\operatorname{rk}_v(\pi) \in \{0, \ldots, |\mathcal{P}_v| - 1\}$$

be the rank of π *in this order. Set*

$$j(\pi) = 1 + \big(\operatorname{rk}_v(\pi) \bmod (d_G(v) - 3)\big)$$

and take the token $(v, j(\pi)) \in \mathfrak{T}_R$*.*

(d) *If none of the preceding cases applies, set*

$$\Theta_{\text{cap}}(\pi) := \kappa(B_\pi) \in \mathfrak{T}_{\text{prim}}.$$

In cases (a)–(c), $\Theta_{\text{cap}}(\pi)$ *is the token selected there. For* $t \in \mathfrak{T}_{\text{cap}}$*, its* token load *is*

$$\ell_{\text{cap}}(t) := |\Theta_{\text{cap}}^{-1}(t)|.$$

Lemma 3.38 (Capacity-token supply)**.** *The capacity-token universe satisfies*

$$|\mathfrak{T}_{\text{cap}}| \leq 8n + \sigma(G).$$

More precisely,

$$|\mathfrak{T}_{\text{cap}}| = |\mathfrak{U}_{\text{sp}}(G)| + \sigma_R + e(R, W) + 2e_\times(W) = |\mathfrak{U}_{\text{sp}}(G)| + 15p_{13} + \sigma(G).$$

Proof. By construction,

$$|\mathfrak{T}_{\mathrm{cap}}| = |\mathfrak{T}_{\mathrm{prim}}| + |\mathfrak{T}_R| + |\mathfrak{T}_W| = |\mathfrak{U}_{\mathrm{sp}}(G)| + \sigma_R + e(R, W) + 2e_\times(W).$$

The exact window-join identity Lemma 8.6 gives

$$e(R, W) + 2e_\times(W) = 15p_{13} + \sigma_W.$$

Since $\sigma(G) = \sigma_R + \sigma_W$,

$$|\mathfrak{T}_{\mathrm{cap}}| = |\mathfrak{U}_{\mathrm{sp}}(G)| + 15p_{13} + \sigma(G).$$

By Lemma 3.36, $|\mathfrak{U}_{\mathrm{sp}}(G)| \le 6n$. Also $13p_{13} \le n$, because the packed P_{13} windows are vertex-disjoint. Hence $15p_{13} \le (15/13)n < 2n$, and therefore

$$|\mathfrak{T}_{\mathrm{cap}}| \le 8n + \sigma(G).$$

□

Lemma 3.39 (Token ledger has no overcounting)**.** *For every finite family $\mathcal{A}_0$ of active surplus demands,*

$$|\Pi_{\mathrm{blk}}| = \sum_{t \in \mathfrak{T}_{\mathrm{cap}}} \ell_{\mathrm{cap}}(t).$$

Consequently, if $\ell_{\mathrm{cap}}(t) \le M_0$ for every $t \in \mathfrak{T}_{\mathrm{cap}}$, then

$$|\Pi_{\mathrm{blk}}| \le M_0(8n + \sigma(G)).$$

Proof. The map $\Theta_{\mathrm{cap}} : \Pi_{\mathrm{blk}} \to \mathfrak{T}_{\mathrm{cap}}$ assigns exactly one token to each blocked pair. The fibres $\Theta_{\mathrm{cap}}^{-1}(t)$, $t \in \mathfrak{T}_{\mathrm{cap}}$, form a disjoint partition of Π_{blk}. Therefore

$$|\Pi_{\mathrm{blk}}| = \sum_{t \in \mathfrak{T}_{\mathrm{cap}}} |\Theta_{\mathrm{cap}}^{-1}(t)| = \sum_{t \in \mathfrak{T}_{\mathrm{cap}}} \ell_{\mathrm{cap}}(t).$$

If every token load is at most M_0, then

$$|\Pi_{\mathrm{blk}}| \le M_0 |\mathfrak{T}_{\mathrm{cap}}| \le M_0(8n + \sigma(G))$$

by Lemma 3.38. □

Definition 3.40 (Token-fibre graphs and same-token patterns)**.** *For $t \in \mathfrak{T}_{\mathrm{cap}}$, the* token-fibre graph *H_t is the simple graph with vertex set $\mathcal{A}_0$ and edge set*

$$E(H_t) = \Pi_t = \Theta_{\mathrm{cap}}^{-1}(t).$$

Thus

$$e(H_t) = \ell_{\mathrm{cap}}(t).$$

A same-token K-matching *at t is a matching of size K in H_t. A* same-token K-star *at t is a set of K edges of H_t incident with one common active demand. The token class of the pattern is the token class containing t: window-incidence, remainder-surplus, or primitive blocker support.*

Definition 3.41 (Same-token blocker roles)**.** *Let $t \in \mathfrak{T}_{\mathrm{cap}}$, and let $\pi \in \Theta_{\mathrm{cap}}^{-1}(t)$. Put $B_\pi = \Phi_{\mathrm{can}}(\pi)$. The* same-token role *of π at t is the following finite label:*

$$\rho_t(\pi) = (\mathrm{type}(B_\pi), \mathrm{class}(t), \mathrm{sub}(t)).$$

Here $\mathrm{type}(B_\pi) \in \{a, b, c, d, e, f\}$ is the blocker type in Definition 3.31. The token class $\mathrm{class}(t)$ is one of W, R, P, according as

$$t \in \mathfrak{T}_W, \qquad t \in \mathfrak{T}_R, \qquad t \in \mathfrak{T}_{\mathrm{prim}}.$$

The subtype is defined by

$$\mathrm{sub}(t) = \begin{cases} RW, & t = (e, RW) \in \mathfrak{T}_W, \\ WW, & t = (e, a) \in \mathfrak{T}_W \text{ is a cross-window endpoint token}, \\ R, & t \in \mathfrak{T}_R, \\ V, & t \in V(G) \subseteq \mathfrak{T}_{\mathrm{prim}}, \\ I, & t \in I_E(G) \subseteq \mathfrak{T}_{\mathrm{prim}}, \\ P, & t \in \mathcal{P}_{\mathrm{exc}} \subseteq \mathfrak{T}_{\mathrm{prim}}. \end{cases}$$

Let $\mathfrak{R}_{\mathrm{st}}$ be the set of all such labels. Then

$$|\mathfrak{R}_{\mathrm{st}}| \le 6(2 + 1 + 3) = 36.$$

We write $Q_{\mathrm{st}} := 36$ for this uniform role bound. For $r \in \mathfrak{R}_{\mathrm{st}}$, define the role fibre *at t by*

$$\Pi_{t,r} := \{\pi \in \Theta_{\mathrm{cap}}^{-1}(t) : \rho_t(\pi) = r\}, \qquad \ell(t, r) := |\Pi_{t,r}|.$$

The role-fibre graph $H_{t,r}$ is the spanning subgraph of H_t with edge set $\Pi_{t,r}$. The sets $\Pi_{t,r}$, $r \in \mathfrak{R}_{\mathrm{st}}$, partition $\Theta_{\mathrm{cap}}^{-1}(t)$, and therefore

$$\ell_{\mathrm{cap}}(t) = \sum_{r \in \mathfrak{R}_{\mathrm{st}}} \ell(t, r).$$

A same-token matching or same-token star is role-homogeneous *if all of its edges π have the same label $\rho_t(\pi)$.*

Definition 3.42 (Token-load threshold and homogeneous cap charge)**.** *For $x \ge 0$, define*

$$\psi(x) := \begin{cases} 0, & x = 0, \\ \left\lceil \dfrac{1 + \sqrt{1 + 8x}}{4} \right\rceil, & x > 0. \end{cases}$$

Equivalently, $\psi(x)$ is the least integer $k \ge 0$ such that

$$x \le k(2k - 1).$$

For an integer $L \ge 1$, define the role-homogeneous token cap charge

$$\mathrm{Cap}_{\mathrm{hom}}(L) := Q_{\mathrm{st}}(L - 1)(2L - 3).$$

Thus $\mathrm{Cap}_{\mathrm{hom}}(L)$ is the uniform token load allowed by charging each of the at most Q_{st} role fibres separately when no role-homogeneous same-token L-matching or L-star occurs at that token.

Lemma 3.43 (Homogeneous extraction from a same-token overload)**.** *If a token $t \in \mathfrak{T}_{\mathrm{cap}}$ supports a same-token K-matching, then t supports a role-homogeneous same-token matching of size at least $\lceil K/Q_{\mathrm{st}} \rceil$. If t supports a same-token K-star, then t supports a role-homogeneous same-token star of size at least $\lceil K/Q_{\mathrm{st}} \rceil$.*

Proof. The edges of H_t are coloured by the role map $\rho_t : E(H_t) \to \mathfrak{R}_{\mathrm{st}}$. By Definition 3.41, at most $Q_{\mathrm{st}} = 36$ colours occur. If M is a same-token K-matching, then the sets

$$M_r := \{\pi \in M : \rho_t(\pi) = r\}, \qquad r \in \mathfrak{R}_{\mathrm{st}},$$

partition M. Hence one colour class has cardinality at least $\lceil K/Q_{\mathrm{st}} \rceil$. That colour class is still a matching, because it is a subset of M, and it is role-homogeneous by construction.

The star case is identical. If S is a same-token K-star with centre p, then each colour class $S_r \subseteq S$ is still a star with the same centre p. One colour class has size at least $\lceil K/Q_{\mathrm{st}} \rceil$. □

Corollary 3.44 (Homogeneous same-token caps close the surplus branch)**.** *Assume G survives the sparse surplus exits, admits a baseline spine demand with $E_{\mathrm{spine}}(n) \le C_E n$, and has full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$. Suppose there are fixed integers L_W, L_R, L_P such that:*

- *(a) no token in $\mathfrak{T}_W$ supports a role-homogeneous same-token L_W-matching or role-homogeneous same-token L_W-star;*
- *(b) no token in $\mathfrak{T}_R$ supports a role-homogeneous same-token L_R-matching or role-homogeneous same-token L_R-star;*
- *(c) no token in $\mathfrak{T}_{\mathrm{prim}}$ supports a role-homogeneous same-token L_P-matching or role-homogeneous same-token L_P-star.*

Then

$$\sigma(G) = O(\sqrt{n}),$$

where the implicit constant depends only on C_E, L_W, L_R, L_P.

Proof. Let

$$M_0 := \max\{\mathrm{Cap}_{\mathrm{hom}}(L_W),\ \mathrm{Cap}_{\mathrm{hom}}(L_R),\ \mathrm{Cap}_{\mathrm{hom}}(L_P)\}.$$

We show that every token has load at most M_0. Fix $t \in \mathfrak{T}_W$. For each role r, the graph $H_{t,r}$ has no L_W-matching and no L_W-star; otherwise that matching or star would be a role-homogeneous same-token L_W-pattern at t, contrary to (a). By Lemma 3.47,

$$\ell(t,r) \le (L_W - 1)(2L_W - 3)$$

for every r. Summing the role-fibre partition of Definition 3.41 gives

$$\ell_{\mathrm{cap}}(t) = \sum_{r \in \mathfrak{R}_{\mathrm{st}}} \ell(t,r) \le Q_{\mathrm{st}}(L_W - 1)(2L_W - 3) = \mathrm{Cap}_{\mathrm{hom}}(L_W) \le M_0.$$

The argument for a token in $\mathfrak{T}_W$ used only the role-fibre partition and the corresponding homogeneous cap. Replacing L_W by L_R or L_P gives the same bound for tokens in $\mathfrak{T}_R$ and $\mathfrak{T}_{\mathrm{prim}}$. The hypotheses of Theorem 3.56 now hold with $c_1 = 1$, because $|\mathcal{A}_0| = \sigma(G)$ by Lemma 3.30. That theorem gives $\sigma(G) = O(\sqrt{n})$. □

Corollary 3.45 (Forced homogeneous same-token scale)**.** *Assume G survives the sparse surplus exits, admits a baseline spine demand with deficit $E_{\mathrm{spine}}(n)$, and has full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$. Let $\Lambda(G)$ be the quantity in Corollary 3.49. Then some capacity token supports a role-homogeneous same-token matching or role-homogeneous same-token star of size at least*

$$\psi\left(\frac{\Lambda(G)}{Q_{\mathrm{st}}}\right).$$

In particular, if $E_{\mathrm{spine}}(n) = O(n)$ and $\sigma(G)/\sqrt{n} \to \infty$, then the forced role-homogeneous same-token matching-or-star size tends to infinity.

Proof. Lemma 3.57 gives a token t with

$$\ell_{\mathrm{cap}}(t) \geq \Lambda(G).$$

The role fibres at t partition $\Theta_{\mathrm{cap}}^{-1}(t)$, so

$$\sum_{r \in \mathfrak{R}_{\mathrm{st}}} \ell(t, r) = \ell_{\mathrm{cap}}(t).$$

Since at most Q_{st} roles occur, some role r satisfies

$$\ell(t, r) \geq \frac{\Lambda(G)}{Q_{\mathrm{st}}}.$$

Apply Lemma 3.47 to the role-fibre graph $H_{t,r}$. If $s = \max\{\nu(H_{t,r}), \Delta(H_{t,r})\}$, then

$$\ell(t, r) = e(H_{t,r}) \leq s(2s - 1).$$

By the definition of ψ,

$$s \geq \psi\left(\frac{\Lambda(G)}{Q_{\mathrm{st}}}\right).$$

Thus $H_{t,r}$ contains a matching or star of that size. Because all edges of $H_{t,r}$ have the same role r, this is a role-homogeneous same-token matching or star. The final assertion follows because Q_{st} is fixed. ◻

Corollary 3.46 (Quantitative homogeneous overload lower bound)**.** *Assume G survives the sparse surplus exits, has full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$, and admits a baseline spine demand with deficit $E_{\mathrm{spine}}(n) \leq C_E n$. Let $K_{\mathrm{hom}}(G)$ be the largest integer K for which some capacity token supports a role-homogeneous same-token K-matching or a role-homogeneous same-token K-star. Then*

$$K_{\mathrm{hom}}(G) \geq \psi\left(\frac{\max\left\{0,\ \binom{\sigma(G)}{2} - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n\right\}}{Q_{\mathrm{st}}(8n + \sigma(G))}\right).$$

Consequently:

(a) If $\sigma(G) \geq A\sqrt{n}$, $A^2 \geq 6(C_E + 1)$, and n is sufficiently large in terms of A and C_E, then

$$K_{\mathrm{hom}}(G) \geq \left\lceil \frac{A}{12\sqrt{Q_{\mathrm{st}}}} \right\rceil.$$

(b) If $\sigma(G) \geq \alpha n$ for some fixed $\alpha > 0$, and n is sufficiently large in terms of α and C_E, then

$$K_{\mathrm{hom}}(G) \geq \left\lceil \frac{\alpha}{12\sqrt{Q_{\mathrm{st}}}}\sqrt{n} \right\rceil.$$

Proof. The displayed bound is Corollary 3.45 with $E_{\mathrm{spine}}(n) \leq C_E n$ substituted into $\Lambda(G)$.

For (a), assume $\sigma(G) \geq A\sqrt{n}$. Since $\sigma(G) \leq n$ by Lemma 3.3, the denominator $8n + \sigma(G)$ is at most $9n$. For fixed A, the term $(\sigma(G)/2 + 1)\log_2 n$ is $o(n)$. For all sufficiently large n,

$$\binom{\sigma(G)}{2} - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n \geq \left(\frac{A^2}{3} - C_E - 1\right) n \geq \frac{A^2}{6} n,$$

where the last inequality uses $A^2 \geq 6(C_E + 1)$. Hence

$$\frac{\binom{\sigma(G)}{2} - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n}{Q_{\mathrm{st}}(8n + \sigma(G))} \geq \frac{A^2}{54Q_{\mathrm{st}}}.$$

If $k < A/(12\sqrt{Q_{\mathrm{st}}})$, then

$$k(2k-1) < 2k^2 < \frac{A^2}{72Q_{\mathrm{st}}} < \frac{A^2}{54Q_{\mathrm{st}}}.$$

By the defining minimality of ψ, this gives (a).

For (b), assume $\sigma(G) \geq \alpha n$. For all sufficiently large n,

$$\binom{\sigma(G)}{2} - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n \geq \frac{\alpha^2}{3} n^2.$$

Again $8n + \sigma(G) \leq 9n$. Therefore

$$\frac{\binom{\sigma(G)}{2} - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n}{Q_{\mathrm{st}}(8n + \sigma(G))} \geq \frac{\alpha^2}{27Q_{\mathrm{st}}} n.$$

If $k < \alpha\sqrt{n}/(12\sqrt{Q_{\mathrm{st}}})$, then

$$k(2k-1) < 2k^2 < \frac{\alpha^2}{72Q_{\mathrm{st}}} n < \frac{\alpha^2}{27Q_{\mathrm{st}}} n.$$

The defining minimality of ψ gives (b). ∎

Lemma 3.47 (Matching–star accounting)**.** *Let H be a finite simple graph with $e(H)$ edges, matching number $\nu(H)$, and maximum degree $\Delta(H)$. Then*

$$e(H) \leq \nu(H)(2\Delta(H) - 1).$$

Consequently, if $K \geq 1$ and H contains neither a matching of size K nor a star of size K, then

$$e(H) \leq (K-1)(2K-3).$$

Equivalently, if $e(H) > (K-1)(2K-3)$, then H contains a matching of size K or a star of size K.

Proof. Choose a maximum matching M in H. Since M is maximal, every edge of H has at least one endpoint in the set U of vertices covered by M. Thus U is a vertex cover of H, and $|U| = 2\nu(H)$. Each vertex of U is incident with at most $\Delta(H)$ edges of H. Hence

$$\sum_{u \in U} d_H(u) \leq 2\nu(H)\Delta(H).$$

The $\nu(H)$ matching edges of M have both endpoints in U, and are therefore counted twice in the degree sum. Every other edge is counted at least once, because U is a vertex cover. Hence

$$e(H) \leq \sum_{u \in U} d_H(u) - \nu(H) \leq \nu(H)(2\Delta(H) - 1).$$

If H contains no matching of size K, then $\nu(H) \leq K - 1$. If it contains no star of size K, then $\Delta(H) \leq K - 1$. Substituting these two bounds gives $e(H) \leq (K-1)(2K-3)$. The final statement is the contrapositive. $\square$

Corollary 3.48 (Same-token pattern caps close the surplus branch)**.** *Assume G survives the sparse surplus exits, admits a baseline spine demand with $E_{\text{spine}}(n) \leq C_E n$, and has full active family $\mathcal{A}_0 = \mathcal{P}_{\text{exc}}$. Suppose there are fixed integers K_W, K_R, K_P such that:*

(a) no token in $\mathfrak{T}_W$ supports a same-token K_W-matching or a same-token K_W-star;

(b) no token in $\mathfrak{T}_R$ supports a same-token K_R-matching or a same-token K_R-star;

(c) no token in $\mathfrak{T}_{\text{prim}}$ supports a same-token K_P-matching or a same-token K_P-star.

Then

$$\sigma(G) = O(\sqrt{n}),$$

where the implicit constant depends only on C_E, K_W, K_R, K_P.

Proof. Put

$$M_0 := \max\{(K_W - 1)(2K_W - 3),\, (K_R - 1)(2K_R - 3),\, (K_P - 1)(2K_P - 3)\}.$$

By Lemma 3.47, every token load satisfies

$$\ell_{\text{cap}}(t) \leq M_0.$$

The hypotheses of Theorem 3.56 now hold with $c_1 = 1$, because $|\mathcal{A}_0| = \sigma(G)$ by Lemma 3.30. That theorem gives $\sigma(G) = O(\sqrt{n})$. $\square$

Corollary 3.49 (Forced same-token overload scale)**.** *Assume G survives the sparse surplus exits, admits a baseline spine demand with deficit $E_{\text{spine}}(n)$, and has full active family $\mathcal{A}_0 = \mathcal{P}_{\text{exc}}$. Put*

$$\Lambda(G) := \frac{\max\left\{0,\, \binom{\sigma(G)}{2} - E_{\text{spine}}(n) - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n\right\}}{8n + \sigma(G)}.$$

Then some capacity token supports a same-token matching or a same-token star of size at least

$$\psi(\Lambda(G)).$$

In particular, if $E_{\text{spine}}(n) = O(n)$ and $\sigma(G)/\sqrt{n} \to \infty$, then the forced same-token matching-or-star size tends to infinity.

Proof. Lemma 3.57 gives a token t with

$$\ell_{\mathrm{cap}}(t) \geq \Lambda(G).$$

Let $r = \max\{\nu(H_t), \Delta(H_t)\}$. By Lemma 3.47,

$$\ell_{\mathrm{cap}}(t) = e(H_t) \leq r(2r-1).$$

Since $\ell_{\mathrm{cap}}(t) \geq \Lambda(G)$, the definition of ψ gives

$$r \geq \psi(\Lambda(G)).$$

Thus H_t contains either a matching or a star of size $\psi(\Lambda(G))$, which is precisely a same-token matching or same-token star at t. If $E_{\mathrm{spine}}(n) = O(n)$ and $\sigma(G)/\sqrt{n} \to \infty$, then the numerator in $\Lambda(G)$ is $\frac{1}{2}\sigma(G)^2 - o(\sigma(G)^2)$, while $8n + \sigma(G) \leq 9n$ by $\sigma(G) \leq n$, which follows from Lemma 3.3. Hence $\Lambda(G) \to \infty$, and so the displayed same-token pattern size tends to infinity. □

Lemma 3.50 (Exact surplus-pair charge partition)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. For $C \in \{W, R, P\}$, use the token classes*

$$\mathfrak{T}_W, \qquad \mathfrak{T}_R, \qquad \mathfrak{T}_P := \mathfrak{T}_{\mathrm{prim}}.$$

Then the following disjoint union is exact:

$$\binom{\mathcal{A}_0}{2} = \Pi_{\mathrm{free}} \sqcup \bigsqcup_{C \in \{W,R,P\}} \bigsqcup_{t \in \mathfrak{T}_C} \bigsqcup_{r \in \mathfrak{R}_{\mathrm{st}}} \Pi_{t,r}.$$

Consequently,

$$\binom{|\mathcal{A}_0|}{2} = |\Pi_{\mathrm{free}}| + \sum_{C \in \{W,R,P\}} \sum_{t \in \mathfrak{T}_C} \sum_{r \in \mathfrak{R}_{\mathrm{st}}} \ell(t,r).$$

Every unordered pair of active demands appears in exactly one summand on the right-hand side.

Proof. By Definition 3.33, the pair set $\binom{\mathcal{A}_0}{2}$ is the disjoint union

$$\Pi_{\mathrm{free}} \sqcup \Pi_{\mathrm{blk}}.$$

If $\pi \in \Pi_{\mathrm{blk}}$, then $\Theta_{\mathrm{cap}}(\pi)$ is defined and is a single element of the disjoint token universe

$$\mathfrak{T}_{\mathrm{cap}} = \mathfrak{T}_W \sqcup \mathfrak{T}_R \sqcup \mathfrak{T}_{\mathrm{prim}}.$$

Thus π lies in exactly one token class C and exactly one token $t \in \mathfrak{T}_C$. For that fixed t, the role map ρ_t assigns exactly one role $r \in \mathfrak{R}_{\mathrm{st}}$, so $\pi \in \Pi_{t,r}$. This proves that every blocked pair appears in at least one displayed role-fibre summand.

The appearance is unique. The token classes are disjoint, Θ_{cap} is a function, and ρ_t is a function once t is fixed. Hence a blocked pair cannot belong to two distinct triples (C, t, r). Conversely, every $\Pi_{t,r}$ is a subset of $\Theta_{\mathrm{cap}}^{-1}(t) \subseteq \Pi_{\mathrm{blk}}$, so no free pair appears in any role-fibre summand. Therefore the displayed union is disjoint and exhaustive. Taking cardinalities gives the second formula. □

Theorem 3.51 (Sharp classwise homogeneous token budget)**.** *Assume G survives the sparse surplus exits, admits a baseline spine demand with deficit $E_{\mathrm{spine}}(n)$, and has full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$. Put*

$$N_*(G) := \max\left\{0,\ \binom{\sigma(G)}{2} - E_{\mathrm{spine}}(n) - \left(\frac{1}{2}\sigma(G)+1\right)\log_2 n\right\}.$$

For $C \in \{W, R, P\}$, set

$$\mathfrak{T}_C = \begin{cases} \mathfrak{T}_W, & C = W, \\ \mathfrak{T}_R, & C = R, \\ \mathfrak{T}_{\mathrm{prim}}, & C = P, \end{cases} \qquad \Pi_C := \{\pi \in \Pi_{\mathrm{blk}} : \Theta_{\mathrm{cap}}(\pi) \in \mathfrak{T}_C\},$$

and write $B_C := |\Pi_C|$, $S_C := |\mathfrak{T}_C|$. Then:

(a) The blocked-pair demand satisfies the exact class split

$$B_W + B_R + B_P = |\Pi_{\mathrm{blk}}| \geq N_*(G).$$

(b) The class supplies are

$$S_W = 15p_{13} + \sigma_W, \qquad S_R = \sigma_R, \qquad S_P = |\mathfrak{U}_{\mathrm{sp}}(G)| \leq 6n,$$

and hence

$$S_W + S_R + S_P \leq 8n + \sigma(G).$$

(c) If no token in $\mathfrak{T}_C$ supports a role-homogeneous same-token L_C-matching or role-homogeneous same-token L_C-star, then

$$B_C \leq \mathrm{Cap}_{\mathrm{hom}}(L_C) S_C = Q_{\mathrm{st}}(L_C - 1)(2L_C - 3) S_C.$$

(d) If the three homogeneous caps L_W, L_R, L_P hold in the three token classes, then the necessary classwise budget inequality is

$$N_*(G) \leq \mathrm{Cap}_{\mathrm{hom}}(L_W)(15p_{13} + \sigma_W) + \mathrm{Cap}_{\mathrm{hom}}(L_R)\sigma_R + \mathrm{Cap}_{\mathrm{hom}}(L_P)|\mathfrak{U}_{\mathrm{sp}}(G)|.$$

In particular,

$$N_*(G) \leq \mathrm{Cap}_{\mathrm{hom}}(L_W)(15p_{13} + \sigma_W) + \mathrm{Cap}_{\mathrm{hom}}(L_R)\sigma_R + 6\,\mathrm{Cap}_{\mathrm{hom}}(L_P)n.$$

(e) If $N_*(G) > 0$, then for some nonempty token class C there is a token $t \in \mathfrak{T}_C$ supporting a role-homogeneous same-token matching or role-homogeneous same-token star of size at least

$$\psi\left(\frac{B_C}{Q_{\mathrm{st}} S_C}\right),$$

and in particular at least

$$\psi\left(\frac{N_*(G)}{Q_{\mathrm{st}}(8n + \sigma(G))}\right).$$

Proof. For (a), the free-pair entropy sandwich with blockers gives

$$|\Pi_{\mathrm{free}}| \le E_{\mathrm{spine}}(n) + \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n,$$

because $|\mathcal{A}_0| = \sigma(G)$. Since

$$|\Pi_{\mathrm{free}}| + |\Pi_{\mathrm{blk}}| = \binom{\sigma(G)}{2},$$

we obtain $|\Pi_{\mathrm{blk}}| \ge N_*(G)$. The three sets Π_W, Π_R, Π_P partition Π_{blk}, because $\mathfrak{T}_{\mathrm{cap}}$ is the disjoint union $\mathfrak{T}_W \sqcup \mathfrak{T}_R \sqcup \mathfrak{T}_{\mathrm{prim}}$ and Θ_{cap} assigns exactly one token to each blocked pair. Hence

$$B_W + B_R + B_P = |\Pi_{\mathrm{blk}}| \ge N_*(G).$$

Part (b) is exactly Lemma 3.38, with $\mathfrak{T}_{\mathrm{prim}} = \mathfrak{U}_{\mathrm{sp}}(G)$ and Lemma 3.36 applied to S_P.

For (c), fix a class C and a token $t \in \mathfrak{T}_C$. For every role r, the role-fibre graph $H_{t,r}$ has no matching of size L_C and no star of size L_C; otherwise that matching or star would be a role-homogeneous same-token L_C-pattern at t. By Lemma 3.47,

$$\ell(t,r) = e(H_{t,r}) \le (L_C - 1)(2L_C - 3).$$

The role fibres partition the token fibre, so

$$\ell_{\mathrm{cap}}(t) = \sum_{r \in \mathfrak{R}_{\mathrm{st}}} \ell(t,r) \le Q_{\mathrm{st}}(L_C - 1)(2L_C - 3) = \mathrm{Cap}_{\mathrm{hom}}(L_C).$$

Summing the exact token fibres in class C gives

$$B_C = \sum_{t \in \mathfrak{T}_C} \ell_{\mathrm{cap}}(t) \le \mathrm{Cap}_{\mathrm{hom}}(L_C) S_C.$$

Part (d) follows by summing the three estimates in (c) and using (a) and (b).

For (e), choose a class C for which B_C/S_C is maximal among nonempty classes with $S_C > 0$. Since the nonempty classes partition Π_{blk},

$$\frac{B_C}{S_C} \ge \frac{B_W + B_R + B_P}{S_W + S_R + S_P} \ge \frac{N_*(G)}{8n + \sigma(G)}.$$

Some token $t \in \mathfrak{T}_C$ then has load

$$\ell_{\mathrm{cap}}(t) \ge B_C/S_C.$$

The role fibres at t partition the token fibre, so some role r has

$$\ell(t,r) \ge \frac{B_C}{Q_{\mathrm{st}} S_C}.$$

Apply Lemma 3.47 to $H_{t,r}$. If $s = \max\{\nu(H_{t,r}), \Delta(H_{t,r})\}$, then

$$\ell(t,r) = e(H_{t,r}) \le s(2s - 1).$$

Therefore

$$s \ge \psi\left(\frac{B_C}{Q_{\mathrm{st}} S_C}\right).$$

The graph $H_{t,r}$ contains a matching or star of that size, and all its edges have the same role r. This gives the first displayed homogeneous lower bound. The second follows from

$$\frac{B_C}{S_C} \ge \frac{N_*(G)}{8n + \sigma(G)}$$

and the monotonicity of ψ. ☐

Corollary 3.52 (Quantified homogeneous class overload)**.** *Assume the hypotheses of Theorem 3.51, and fix proposed homogeneous caps L_W, L_R, L_P. Define the class capacities*

$$A_W := \mathrm{Cap}_{\mathrm{hom}}(L_W)(15p_{13} + \sigma_W), \qquad A_R := \mathrm{Cap}_{\mathrm{hom}}(L_R)\sigma_R, \qquad A_P := \mathrm{Cap}_{\mathrm{hom}}(L_P)|\mathfrak{U}_{\mathrm{sp}}(G)|.$$

Put

$$D_{\mathrm{exc}} := (N_*(G) - A_W - A_R - A_P)_+ \,,$$

and, for $C \in \{W, R, P\}$, put

$$\Omega_C := (B_C - A_C)_+.$$

Then:

(a) The total class overload satisfies

$$\Omega_W + \Omega_R + \Omega_P \geq D_{\mathrm{exc}}.$$

(b) Hence at least one class C satisfies

$$\Omega_C \geq \frac{1}{3} D_{\mathrm{exc}}.$$

(c) If $\Omega_C > 0$, then $S_C > 0$, and class C contains a role-homogeneous same-token matching or role-homogeneous same-token star of size at least

$$\psi\left((L_C - 1)(2L_C - 3) + \frac{\Omega_C}{Q_{\mathrm{st}} S_C}\right).$$

(d) If $D_{\mathrm{exc}} > 0$, and the two classes different from C do not overload, i.e. $B_{C'} \leq A_{C'}$ for $C' \neq C$, then class C overloads by at least the whole deficit:

$$\Omega_C \geq D_{\mathrm{exc}},$$

and class C contains a role-homogeneous same-token matching or role-homogeneous same-token star of size at least

$$\psi\left((L_C - 1)(2L_C - 3) + \frac{D_{\mathrm{exc}}}{Q_{\mathrm{st}} S_C}\right).$$

(e) The three pure-overload tests have the following sharp class-by-class lower bounds. Define

$$F_W := \big(N_*(G) - A_R - A_P\big)_+, \qquad F_R := \big(N_*(G) - A_W - A_P\big)_+, \qquad F_P := \big(N_*(G) - A_W - A_R\big)_+.$$

If $B_R \leq A_R$ and $B_P \leq A_P$, then

$$B_W \geq F_W, \qquad \Omega_W \geq (F_W - A_W)_+ = D_{\mathrm{exc}}.$$

If $S_W > 0$, the forced window-incidence homogeneous pattern has size at least

$$\psi\left(\frac{F_W}{Q_{\mathrm{st}} S_W}\right).$$

If $B_W \leq A_W$ and $B_P \leq A_P$, then

$$B_R \geq F_R, \qquad \Omega_R \geq (F_R - A_R)_+ = D_{\mathrm{exc}}.$$

If $S_R > 0$, the forced remainder-surplus homogeneous pattern has size at least

$$\psi\left(\frac{F_R}{Q_{\mathrm{st}} S_R}\right).$$

If $B_W \le A_W$ and $B_R \le A_R$, then

$$B_P \ge F_P, \qquad \Omega_P \ge (F_P - A_P)_+ = D_{\mathrm{exc}}.$$

If $S_P > 0$, the forced primitive blocker-support homogeneous pattern has size at least

$$\psi\left(\frac{F_P}{Q_{\mathrm{st}} S_P}\right).$$

The estimate is sharp for the scalar class-load ledger: the load vector

$$B_C = A_C + D_{\mathrm{exc}}, \qquad B_{C'} = A_{C'} \quad (C' \ne C)$$

has total load $A_W + A_R + A_P + D_{\mathrm{exc}}$ and overloads only class C.

Proof. For (a), use the elementary inequality

$$\sum_C (B_C - A_C)_+ \ge \left(\sum_C B_C - \sum_C A_C\right)_+.$$

By Theorem 3.51,

$$\sum_C B_C = B_W + B_R + B_P \ge N_*(G),$$

and therefore

$$\Omega_W + \Omega_R + \Omega_P \ge (N_*(G) - A_W - A_R - A_P)_+ = D_{\mathrm{exc}}.$$

Part (b) is the pigeonhole principle applied to the three nonnegative numbers $\Omega_W, \Omega_R, \Omega_P$.

For (c), if $\Omega_C > 0$, then $B_C > A_C$, so $B_C > 0$, and hence $S_C > 0$. By Theorem 3.51, class C contains a role-homogeneous same-token matching or star of size at least

$$\psi\left(\frac{B_C}{Q_{\mathrm{st}} S_C}\right).$$

Since $B_C = A_C + \Omega_C$ and

$$A_C = \mathrm{Cap}_{\mathrm{hom}}(L_C) S_C = Q_{\mathrm{st}}(L_C - 1)(2L_C - 3) S_C,$$

we have

$$\frac{B_C}{Q_{\mathrm{st}} S_C} = (L_C - 1)(2L_C - 3) + \frac{\Omega_C}{Q_{\mathrm{st}} S_C}.$$

This proves (c).

For (d), assume $B_{C'} \le A_{C'}$ for the two classes $C' \ne C$. Then

$$B_C \ge N_*(G) - \sum_{C' \ne C} B_{C'} \ge N_*(G) - \sum_{C' \ne C} A_{C'} = A_C + (N_*(G) - A_W - A_R - A_P).$$

If $D_{\mathrm{exc}} > 0$, this gives $B_C \ge A_C + D_{\mathrm{exc}}$, hence $\Omega_C \ge D_{\mathrm{exc}}$. Substituting this bound into (c) gives the displayed pattern-size lower bound. The final sentence is a direct substitution into the scalar identities defining the class-load ledger.

For (e), first assume $B_R \le A_R$ and $B_P \le A_P$. Then

$$B_W \ge N_*(G) - B_R - B_P \ge N_*(G) - A_R - A_P.$$

Since $B_W \ge 0$, this gives $B_W \ge F_W$. Subtracting A_W and taking positive parts gives

$$\Omega_W \ge (F_W - A_W)_+ = \big(N_*(G) - A_W - A_R - A_P\big)_+ = D_{\mathrm{exc}}.$$

If $S_W > 0$, then Theorem 3.51 gives a window-incidence role-homogeneous same-token matching or star of size at least

$$\psi\left(\frac{B_W}{Q_{\mathrm{st}} S_W}\right) \ge \psi\left(\frac{F_W}{Q_{\mathrm{st}} S_W}\right),$$

using monotonicity of ψ. The R- and P-estimates are the same argument with the class names permuted. □

Corollary 3.53 (Coupled single-graph overload budget)**.** *Assume the hypotheses of Theorem 3.51, and fix proposed homogeneous caps L_W, L_R, L_P. Put*

$$b_W := (L_W - 1)(2L_W - 3), \qquad b_R := (L_R - 1)(2L_R - 3), \qquad b_P := (L_P - 1)(2L_P - 3).$$

Since $Q_{\mathrm{st}} = 36$, the exact coupled capacity of the three classes in the single graph G is

$$A_{\mathrm{all}} = 36\Big[b_W(15p_{13} + \sigma_W) + b_R\sigma_R + b_P(4n + 2\sigma(G))\Big].$$

Equivalently, using $\sigma(G) = \sigma_W + \sigma_R$,

$$A_{\mathrm{all}} = 36\Big[4b_P n + 15 b_W p_{13} + (b_W + 2b_P)\sigma_W + (b_R + 2b_P)\sigma_R\Big].$$

Define the coupled excess

$$D_{\mathrm{all}} := \big(N_*(G) - A_{\mathrm{all}}\big)_+.$$

Then:

(a) The total role-fibre overload in the single graph is at least

$$D_{\mathrm{all}}.$$

(b) The total number of role-fibre slots available to absorb this excess is at most

$$36(S_W + S_R + S_P) = 36(4n + 15p_{13} + 3\sigma(G)).$$

(c) If $D_{\mathrm{all}} > 0$, then for some class $C \in \{W, R, P\}$ there is a role fibre in that class whose load is at least

$$b_C + \frac{D_{\mathrm{all}}}{36(4n + 15p_{13} + 3\sigma(G))}.$$

Consequently G contains a role-homogeneous same-token matching or role-homogeneous same-token star of size at least

$$\psi\left(b_C + \frac{D_{\mathrm{all}}}{36(4n + 15p_{13} + 3\sigma(G))}\right)$$

for that same class C.

Proof. The exact primitive-token supply is

$$S_P = |\mathfrak{U}_{\mathrm{sp}}(G)| = n + 2m + \sigma(G).$$

Since $\sigma(G) = 2m - 3n$, we have $m = (3n + \sigma(G))/2$, and therefore

$$S_P = n + 2m + \sigma(G) = 4n + 2\sigma(G).$$

Together with $S_W = 15p_{13} + \sigma_W$ and $S_R = \sigma_R$, this gives the displayed formula for A_{all}; the second displayed form is obtained by substituting $\sigma(G) = \sigma_W + \sigma_R$.

For each role fibre in class C, the cap threshold before a homogeneous L_C-matching or L_C-star is b_C. Thus the total role-fibre capacity over all three classes is

$$36b_W S_W + 36b_R S_R + 36b_P S_P = A_{\mathrm{all}}.$$

Since the exact class theorem gives a blocked-pair demand at least $N_*(G)$, the excess above this coupled capacity is at least D_{all}. This proves (a).

The number of role-fibre slots is at most $36(S_W + S_R + S_P)$. Using the exact supplies,

$$S_W + S_R + S_P = (15p_{13} + \sigma_W) + \sigma_R + (4n + 2\sigma(G)) = 4n + 15p_{13} + 3\sigma(G),$$

which proves (b).

For a token $t \in \mathfrak{T}_C$ and a role $r \in \mathfrak{R}_{\mathrm{st}}$, write

$$E_{t,r} := \big(\ell(t,r) - b_C\big)_+.$$

The elementary inequality $\sum_i (x_i - y_i)_+ \ge (\sum_i x_i - \sum_i y_i)_+$, applied to all triples (C,t,r), gives

$$\sum_{C\in\{W,R,P\}} \sum_{t\in\mathfrak{T}_C} \sum_{r\in\mathfrak{R}_{\mathrm{st}}} E_{t,r} \ge \left(\sum_C B_C - A_{\mathrm{all}}\right)_+ \ge D_{\mathrm{all}}.$$

If $D_{\mathrm{all}} > 0$, distribute these explicitly defined role-fibre excesses over the at most

$$36\big(4n + 15p_{13} + 3\sigma(G)\big)$$

role-fibre slots. Some role fibre in some class C has excess at least the average displayed in (c), so its load is at least

$$b_C + \frac{D_{\mathrm{all}}}{36\big(4n + 15p_{13} + 3\sigma(G)\big)}.$$

Applying the matching–star bound of Lemma 3.47 to that role-fibre graph gives the stated homogeneous matching-or-star lower bound by the definition of ψ. □

Corollary 3.54 (Numerical normalized single-graph budget)**.** *Assume the hypotheses of Corollary 3.53. Set*

$$\theta := \frac{p_{13}}{n}, \qquad w := \frac{\sigma_W}{n}, \qquad r := \frac{\sigma_R}{n}, \qquad s := \frac{\sigma(G)}{n} = w + r.$$

Then $0 \le \theta \le 1/13$ and $0 \le s \le 1$, and the coupled capacity is exactly

$$A_{\mathrm{all}} = n\,\Gamma_L(\theta, w, r),$$

where

$$\Gamma_L(\theta, w, r) := 144b_P + 540b_W\theta + 36(b_W + 2b_P)w + 36(b_R + 2b_P)r.$$

Equivalently, the three token classes contribute the following single-graph budget:

class	S_C/n	*role-fibre cap*	A_C/n
W	$15\theta + w$	b_W	$36b_W(15\theta + w)$
R	r	b_R	$36b_Rr$
P	$4 + 2s$	b_P	$36b_P(4 + 2s)$

The exact coupled excess is

$$D_{\text{all}} = \left[\frac{\sigma(G)(\sigma(G)-1)}{2} - \left(\frac{\sigma(G)}{2} + 1\right)\log_2 n - E_{\text{spine}}(n) - n\,\Gamma_L(\theta, w, r)\right]_+.$$

If $E_{\text{spine}}(n) \leq C_E n$*, then*

$$D_{\text{all}} \geq \left[\frac{\sigma(G)(\sigma(G)-1)}{2} - \left(\frac{\sigma(G)}{2} + 1\right)\log_2 n - n(C_E + \Gamma_L(\theta, w, r))\right]_+.$$

Moreover the total role-fibre slot count is at most

$$n\,\Delta(\theta, s), \qquad \Delta(\theta, s) := 144 + 540\theta + 108s \leq \frac{3816}{13} = 293.5384615\ldots.$$

Thus, if $D_{\text{all}} > 0$*, some role fibre in one of the three classes has load at least*

$$b_C + \frac{D_{\text{all}}}{n\,\Delta(\theta, s)} \geq b_C + \frac{13D_{\text{all}}}{3816\,n},$$

and hence supports a role-homogeneous same-token matching or star of size at least

$$\psi\left(b_C + \frac{13D_{\text{all}}}{3816\,n}\right).$$

For reference, the numerical cap charges are

L	$b(L) = (L-1)(2L-3)$	$36b(L)$
2	1	36
3	6	216
4	15	540
5	28	1008
6	45	1620
7	66	2376
8	91	3276

If $L_W = L_R = L_P = L$*, the worst-case linear capacity coefficient is*

$$\Gamma_L(\theta, w, r) = 36b(L)(4 + 15\theta + 3s) \leq \frac{3816}{13}\,b(L).$$

Proof. The identities for A_{all} and the three class contributions are the formula of Corollary 3.53 divided by n, using $p_{13} = \theta n$, $\sigma_W = wn$, $\sigma_R = rn$, and $\sigma(G) = sn$. The inequality $\theta \leq 1/13$ follows from the vertex-disjointness of the packed P_{13} windows, and $s \leq 1$ follows from $\sigma(G) = 2m - 3n \leq n$ in Lemma 3.3.

Since $D_{\mathrm{all}} = (N_*(G) - A_{\mathrm{all}})_+$ and $A_{\mathrm{all}} \geq 0$, the identity $([X]_+ - A)_+ = [X - A]_+$ gives the displayed exact formula for D_{all}. Replacing $E_{\mathrm{spine}}(n)$ by the upper bound $C_E n$ gives the next inequality.

The slot count in Corollary 3.53 is

$$36(4n + 15p_{13} + 3\sigma(G)) = n(144 + 540\theta + 108s) = n\Delta(\theta, s).$$

Using $\theta \leq 1/13$ and $s \leq 1$ gives

$$\Delta(\theta, s) \leq 144 + \frac{540}{13} + 108 = \frac{3816}{13}.$$

Substitution into the role-fibre overload bound of Corollary 3.53 proves the displayed load and matching–star estimates. The numerical table is obtained by substituting $L = 2, \ldots, 8$ into $b(L) = (L-1)(2L-3)$. $\square$

Proposition 3.55 (Single-graph sparse-load routing)**.** *Assume G reaches node* [125]*, satisfies the structural constraints recorded at nodes* [126]*–*[136]*, has $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$, and admits a baseline spine demand with $E_{\mathrm{spine}}(n) \leq C_E n$. Fix proposed homogeneous caps L_W, L_R, L_P, and let b_W, b_R, b_P be as in Corollary 3.53. Put*

$$\Gamma_{\max} := 144 b_P + \frac{540}{13} b_W + 36 \max\{b_W + 2b_P,\ b_R + 2b_P\}.$$

Define

$$R_L(n) := \frac{1 + \log_2 n + \sqrt{(1 + \log_2 n)^2 + 8(C_E + \Gamma_{\max})n + 8\log_2 n}}{2}.$$

Then the sparse-load survivor has the following exhaustive alternatives.

(a) If no token in $\mathfrak{T}_W, \mathfrak{T}_R, \mathfrak{T}_{\mathrm{prim}}$ supports the corresponding role-homogeneous L_W, L_R, L_P matching or star, then

$$\sigma(G) \leq R_L(n).$$

In particular, for all sufficiently large n, this alternative routes to node [138]*, the near-cubic spine.*

(b) If $\sigma(G) > R_L(n)$, then $D_{\mathrm{all}} > 0$. Hence, by Corollary 3.54, G contains a role-homogeneous same-token matching or star in one of the three token classes. According to the class of the token, G is routed to node [140], [142]*, or* [143]*.*

(c) Therefore no graph can remain at node [137]*. An alleged survivor which is neither routed to* [138] *nor to one of* [140], [142], [143] *must violate at least one of the structural inputs used in the derivation: the sparse envelope $m \leq 2n - 2$ at* [126]*, the full active family and baseline demand at* [129]*, the exact window-join identity at* [135]*, or the exact role-fibre partition at* [136]*.*

Proof. Assume first that the three homogeneous caps hold. Then each role fibre in class C has load at most b_C, by Lemma 3.47; hence the total blocked-pair load is at most A_{all}. Since Theorem 3.51 gives blocked-pair demand at least $N_*(G)$, we have $N_*(G) \leq A_{\mathrm{all}}$. Thus

$$\binom{\sigma(G)}{2} \leq E_{\mathrm{spine}}(n) + \left(\frac{\sigma(G)}{2} + 1\right) \log_2 n + A_{\mathrm{all}}.$$

By Corollary 3.54,

$$A_{\text{all}} = n\Gamma_L(\theta, w, r), \qquad \Gamma_L(\theta, w, r) = 144b_P + 540b_W\theta + 36(b_W + 2b_P)w + 36(b_R + 2b_P)r.$$

The structural constraints at [126] and the disjointness of the packed P_{13} windows give

$$0 \le \theta \le \frac{1}{13}, \qquad w + r = \frac{\sigma(G)}{n} \le 1.$$

Therefore

$$\Gamma_L(\theta, w, r) \le 144b_P + \frac{540}{13}b_W + 36\max\{b_W + 2b_P,\ b_R + 2b_P\} = \Gamma_{\max}.$$

Using $E_{\text{spine}}(n) \le C_E n$ gives

$$\frac{\sigma(G)(\sigma(G)-1)}{2} \le (C_E + \Gamma_{\max})n + \left(\frac{\sigma(G)}{2} + 1\right)\log_2 n.$$

After multiplying by 2 and moving the $\sigma(G)$-terms to the left,

$$\sigma(G)^2 - (1 + \log_2 n)\sigma(G) - 2(C_E + \Gamma_{\max})n - 2\log_2 n \le 0.$$

The positive root of the corresponding quadratic is precisely $R_L(n)$, so $\sigma(G) \le R_L(n)$. This proves (a). Since C_E and $\Gamma_{\max}$ are fixed once the caps are fixed, $R_L(n) = O(\sqrt{n} + \log n) = O(\sqrt{n})$; for large n, this is the near-cubic route [138].

For (b), take the contrapositive of (a). If $\sigma(G) > R_L(n)$, the three homogeneous caps cannot all hold. Equivalently the coupled cap inequality is violated. Quantitatively, the displayed quadratic inequality is false, so

$$\binom{\sigma(G)}{2} - E_{\text{spine}}(n) - \left(\frac{\sigma(G)}{2} + 1\right)\log_2 n > A_{\text{all}},$$

and hence $D_{\text{all}} > 0$. Corollary 3.54 then produces a role fibre in class W, R, or P with load above its cap; Lemma 3.47 turns that load into the corresponding role-homogeneous matching or star. The class labels are exactly the diagram branches [140], [142], and [143].

Part (c) is the logical union of (a) and (b). Every inequality used above is one of the stated structural inputs: $m \le 2n - 2$ gives $\sigma(G)/n \le 1$, the full active family and baseline demand give the lower demand $N_*(G)$, the window-join identity gives the W-token supply, and the role-fibre partition gives the exact blocked-pair sum. If a graph avoids all four routed outcomes, at least one of those inputs must fail. ☐

Theorem 3.56 (Tokenized surplus accounting closure)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands such that*

$$|\mathcal{A}_0| \ge c_1\sigma(G)$$

for a fixed $c_1 > 0$. Suppose G survives the sparse surplus exits and there is a baseline spine demand with $E_{\text{spine}}(n) \le C_E n$. Suppose also that the capacity-token ledger satisfies

$$\ell_{\text{cap}}(t) \le M_0 \qquad (t \in \mathfrak{T}_{\text{cap}})$$

for a fixed constant M_0. Then

$$\sigma(G) = O(\sqrt{n}),$$

where the implicit constant depends only on c_1, C_E, and M_0.

Proof. The pair set is the disjoint union

$$\binom{\mathcal{A}_0}{2} = \Pi_{\mathrm{free}} \sqcup \Pi_{\mathrm{blk}}.$$

By Proposition 3.67,

$$|\Pi_{\mathrm{free}}| \le C_E n + \left(\frac{1}{2}\sigma(G) + 1\right) \log_2 n.$$

By Lemma 3.39,

$$|\Pi_{\mathrm{blk}}| \le M_0(8n + \sigma(G)).$$

Thus

$$\binom{|\mathcal{A}_0|}{2} \le (C_E + 8M_0)n + M_0\sigma(G) + \left(\frac{1}{2}\sigma(G) + 1\right) \log_2 n.$$

For all sufficiently large $\sigma(G)$,

$$\binom{|\mathcal{A}_0|}{2} \ge \frac{c_1^2}{4}\sigma(G)^2.$$

Combining the two displays gives constants C_1, C_2, depending only on c_1, C_E, M_0, such that

$$\sigma(G)^2 \le C_1 n + C_2\sigma(G) \log_2 n + C_2\sigma(G) + C_2 \log_2 n.$$

If $\sigma(G) \le 4C_2 \log_2 n + 4C_2$, then $\sigma(G) = O(\log n) = O(\sqrt{n})$. Otherwise the terms linear in $\sigma(G)$ and $\sigma(G)\log_2 n$ are absorbed into $\frac{1}{2}\sigma(G)^2$, after increasing constants. The remaining inequality gives $\sigma(G)^2 \le C_3 n$, and hence $\sigma(G) = O(\sqrt{n})$. □

Lemma 3.57 (Exact capacity-token load alternative)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands, and put*

$$s := |\mathcal{A}_0|, \qquad L_{\max} := \max_{t \in \mathfrak{T}_{\mathrm{cap}}} \ell_{\mathrm{cap}}(t).$$

Suppose G survives the sparse surplus exits and admits a baseline spine demand with deficit $E_{\mathrm{spine}}(n)$. Then

$$\binom{s}{2} \le E_{\mathrm{spine}}(n) + \left(\frac{1}{2}\sigma(G) + 1\right) \log_2 n + L_{\max}|\mathfrak{T}_{\mathrm{cap}}|.$$

Equivalently,

$$L_{\max} \ge \frac{\max\left\{0,\ \binom{s}{2} - E_{\mathrm{spine}}(n) - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n\right\}}{|\mathfrak{T}_{\mathrm{cap}}|}.$$

Using Lemma 3.38, this implies the coarser but denominator-free bound

$$L_{\max} \ge \frac{\max\left\{0,\ \binom{s}{2} - E_{\mathrm{spine}}(n) - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n\right\}}{8n + \sigma(G)}.$$

In particular, for the full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$, if $E_{\mathrm{spine}}(n) \le C_E n$ and $\sigma(G) \ge 2$, then

$$L_{\max} \ge \frac{\max\left\{0,\ \frac{1}{4}\sigma(G)^2 - C_E n - \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n\right\}}{8n + \sigma(G)}.$$

Proof. The canonical ledger gives the disjoint partition

$$\binom{\mathcal{A}_0}{2} = \Pi_{\mathrm{free}} \sqcup \Pi_{\mathrm{blk}}.$$

By Proposition 3.67,

$$|\Pi_{\mathrm{free}}| \le E_{\mathrm{spine}}(n) + \left(\frac{1}{2}\sigma(G) + 1\right) \log_2 n.$$

By Lemma 3.39,

$$|\Pi_{\mathrm{blk}}| = \sum_{t \in \mathfrak{T}_{\mathrm{cap}}} \ell_{\mathrm{cap}}(t) \le L_{\max} |\mathfrak{T}_{\mathrm{cap}}|.$$

Adding the two bounds proves the first displayed inequality. Rearranging gives the exact lower bound for $L_{\max}$; if the numerator is negative, the trivial lower bound $L_{\max} \ge 0$ is stronger, which accounts for the maximum with 0. The denominator-free version follows from $|\mathfrak{T}_{\mathrm{cap}}| \le 8n + \sigma(G)$. For the full active family, Lemma 3.30 gives $s = \sigma(G)$, and $\binom{\sigma(G)}{2} \ge \sigma(G)^2/4$ for $\sigma(G) \ge 2$. ☐

Definition 3.58 (Sparse surplus-pair response coordinates)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. For an unordered pair $\pi = \{p, q\} \subseteq \mathcal{A}_0$, let X_π be the lexicographically first connected subgraph of G with the minimum possible number of vertices that contains*

$$T(p) \cup \Gamma(p) \cup T(q) \cup \Gamma(q).$$

Let

$$\partial X_\pi = \{v \in V(X_\pi) : v \text{ is incident with an edge of } G - X_\pi\}.$$

The sparse pair-response coordinate *r_π is the declared coordinate of $\rho^{\mathrm{ex}}_{\partial X_\pi}(X_\pi)$ whose label is π, whose support is X_π, and whose value is the exact target-response data of the two labelled demands p and q inside X_π. This coordinate is part of the declared signature by clauses (D2), (D7), and (D8) of Definition 3.87.*

For a set $\Pi \subseteq \binom{\mathcal{A}_0}{2}$, let X_Π be the lexicographically first connected subgraph of G with the minimum possible number of vertices that contains every X_π with $\pi \in \Pi$. Each r_π is then viewed as a labelled coordinate of $\rho^{\mathrm{ex}}_{\partial X_\Pi}(X_\Pi)$ by restriction to its declared support.

Lemma 3.59 (Sparse pair dependence forces a blocker or an exit)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands and let $\Pi \subseteq \binom{\mathcal{A}_0}{2}$. Suppose the coordinate family*

$$\mathcal{R}_\Pi = \{r_\pi : \pi \in \Pi\}$$

does not survive every admissible rank quotient of $\rho^{\mathrm{ex}}_{\partial X_\Pi}(X_\Pi)$. Then either G has a sparse surplus exit in the sense of Definition 3.26, or some $\pi \in \Pi$ has a sparse surplus blocker of type (d) or (e) in Definition 3.31.

Proof. Choose a maximal subfamily $\mathcal{I} \subseteq \mathcal{R}_\Pi$ that survives every admissible rank quotient. Since $\mathcal{R}_\Pi$ does not survive, there is a coordinate $r_\pi \in \mathcal{R}_\Pi \setminus \mathcal{I}$. The finite circuit argument of Lemma 10.3, applied to the declared coordinates $\mathcal{R}_\Pi$, gives a subfamily $\mathcal{B} \subseteq \mathcal{I}$ and an admissible quotient

$$q : \rho^{\mathrm{ex}}_{\partial Z}(Z) \to Q$$

on an inclusion-minimal connected determination support $Z \subseteq X_\Pi$ which determines r_π from the coordinates in $\mathcal{B}$.

If the determination attempts to identify states with different boundary degree profiles, then Lemma 3.94 forbids a target-complete identification. The offending boundary-degree entry is a blocker of type (d). Thus assume the boundary degree profile is preserved.

If the determination is not target-complete against all ∂Z-boundaried contexts, then by Lemma 3.95 it is target-defective. This is a sparse surplus exit of type (b), and the distinguishing target-response coordinate is also a blocker of type (e).

Assume, therefore, that the quotient is target-complete. If $Z \subsetneq G$, then the admissibility rule of Definition 3.91 supplies a strictly smaller proper representative Z' in the sense of Definition 3.89. The five defining properties of a proper representative are exactly the five hypotheses of the replacement lemma Lemma 3.96 for the support Z. Hence replacing Z by Z' would give a smaller counterexample, contradicting the minimality of G. Equivalently, the dependence is a forbidden target-complete compression, which is a sparse surplus exit of type (c) and a blocker of type (e).

It remains to consider $Z = G$. Then the quotient is a closed exact-profile quotient. By the closed clause of Definition 3.91, a rank-reducing closed quotient is admissible only if it has a strictly smaller admissible closed representative H. By Definition 3.90, H is finite, simple, strictly smaller than G, and $\delta(H) \geq 3$. Also $\mathsf{H}_\varnothing(H) \subseteq \mathsf{H}_\varnothing(G)$. If H contained a power-of-two cycle, the empty closed context witnessing that power-of-two cycle would lie in $\mathsf{H}_\varnothing(H)$. It does not lie in $\mathsf{H}_\varnothing(G)$, because G is a counterexample with no power-of-two cycle. Hence H has no power-of-two cycle, so H is a strictly smaller counterexample, contradicting minimality. If no smaller closed representative exists, the quotient is label-injective on the declared coordinates under discussion and therefore cannot reduce the rank of $\mathcal{R}_\Pi$, contrary to the choice of q. Thus the whole-graph case also produces a sparse surplus exit of type (d), represented concretely by the target-response coordinate of the failed global quotient. ☐

Proposition 3.60 (Sparse pair independence/compression dichotomy)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. If G survives the sparse surplus exits and no pair in $\binom{\mathcal{A}_0}{2}$ has a sparse surplus blocker, then the full family*

$$\{r_{\{p,q\}} : \{p,q\} \in \binom{\mathcal{A}_0}{2}\}$$

is independently target-testable.

Proof. If the displayed family were not independently target-testable, it would fail to survive some admissible rank quotient. Applying Lemma 3.59 with $\Pi = \binom{\mathcal{A}_0}{2}$ gives either a sparse surplus exit or a sparse surplus blocker for one of the pairs. Both alternatives are excluded by hypothesis. Hence no admissible quotient reduces the family, which is exactly independent target-testability in the sense of Definition 3.99. ☐

Corollary 3.61 (Entropy saturation of an unblocked surplus family)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. If G survives the sparse surplus exits and no pair in $\binom{\mathcal{A}_0}{2}$ has a sparse surplus blocker, then*

$$\binom{|\mathcal{A}_0|}{2} \leq \log_2 \binom{\binom{n}{2}}{m}.$$

In particular, using Lemma 3.3,

$$\binom{|\mathcal{A}_0|}{2} = O(n \log n).$$

Proof. By Proposition 3.60, all pair-response coordinates indexed by $\binom{\mathcal{A}_0}{2}$ are independently target-testable. Lemmas 3.100 and 4.2 therefore give

$$2^{\binom{|\mathcal{A}_0|}{2}} \le \binom{\binom{n}{2}}{m},$$

which is the first displayed inequality after taking logarithms.

By Lemma 3.3, $m \le 2n-2$. With $N = \binom{n}{2}$, the standard estimate

$$\binom{N}{m} \le \left(\frac{eN}{m}\right)^m$$

and the lower bound $m \ge 3n/2$ from $\delta(G) \ge 3$ give

$$\log_2 \binom{N}{m} \le (2n-2)\log_2\left(\frac{eN}{3n/2}\right) = O(n\log n).$$

Combining this with the first inequality proves the second. □

Definition 3.62 (Baseline spine demand)**.** *Put*

$$N = \binom{n}{2}, \qquad m_0 = \left\lceil \frac{3}{2}n \right\rceil, \qquad B_0(n) = \log_2\binom{N}{m_0}.$$

A family $\mathcal{I}_{\mathrm{spine}}$ of declared target coordinates is a baseline spine demand with deficit $E_{\mathrm{spine}}(n)$ *if $\mathcal{I}_{\mathrm{spine}}$ is independently target-testable and*

$$|\mathcal{I}_{\mathrm{spine}}| \ge B_0(n) - E_{\mathrm{spine}}(n).$$

In applications, $\mathcal{I}_{\mathrm{spine}}$ is the union of the independent target coordinates already forced by the near-cubic spine before sparse surplus-pair coordinates are added. The deficit $E_{\mathrm{spine}}(n)$ measures the unused room in the near-cubic skeleton budget after the window, remainder, obstruction, and local-residual demands have been imposed.

Lemma 3.63 (Exact cubic baseline budget)**.** *With*

$$N = \binom{n}{2}, \qquad m_0 = \left\lceil \frac{3}{2}n \right\rceil, \qquad B_0(n) = \log_2\binom{N}{m_0},$$

one has

$$B_0(n) = \frac{3}{2}n\log_2 n + O(n).$$

Proof. For the upper bound, use

$$\binom{N}{m_0} \le \left(\frac{eN}{m_0}\right)^{m_0}.$$

Since $N = \frac{1}{2}n^2 + O(n)$ and $m_0 = \frac{3}{2}n + O(1)$, the ratio eN/m_0 is $\Theta(n)$, and therefore

$$\log_2\binom{N}{m_0} \le \left(\frac{3}{2}n + O(1)\right)(\log_2 n + O(1)) = \frac{3}{2}n\log_2 n + O(n).$$

For the lower bound, the standard product estimate

$$\binom{N}{k} = \prod_{i=0}^{k-1} \frac{N-i}{k-i} \geq \left(\frac{N}{k}\right)^k \qquad (1 \leq k \leq N)$$

applied with $k = m_0$ gives

$$\log_2 \binom{N}{m_0} \geq m_0 \log_2 \left(\frac{N}{m_0}\right) = \left(\frac{3}{2}n + O(1)\right)(\log_2 n + O(1)) = \frac{3}{2}n \log_2 n - O(n).$$

$\square$

Lemma 3.64 (Incremental edge-count room above cubic)**.** *Let* $m_0 = \lceil 3n/2 \rceil$, $N = \binom{n}{2}$, *and let*

$$m = m_0 + s$$

with $s \geq 0$ *and* $m \leq 2n - 2$*. Then*

$$\log_2 \binom{N}{m} - \log_2 \binom{N}{m_0} \leq s \log_2 n.$$

For the minimal counterexample,

$$s \leq \frac{1}{2}\sigma(G) + 1.$$

Proof. For $s = 0$ there is nothing to prove. If $s > 0$, then

$$\frac{\binom{N}{m_0+s}}{\binom{N}{m_0}} = \prod_{i=1}^{s} \frac{N - m_0 - i + 1}{m_0 + i} \leq \left(\frac{N}{m_0}\right)^s \leq n^s,$$

because $N \leq n^2/2$ and $m_0 \geq 1$. Taking logarithms gives the first claim. Finally,

$$\sigma(G) = 2m - 3n = 2(m_0 + s) - 3n = 2s + (2m_0 - 3n),$$

where $2m_0 - 3n \in \{0, 1\}$. Hence $s \leq \sigma(G)/2 + 1$. $\square$

Lemma 3.65 (Mixed sparse-spine dependence forces compression or a blocker)**.** *Let* $\mathcal{A}_0$ *be a finite family of active surplus demands, let* $\mathcal{I}_{\text{spine}}$ *be any family of declared baseline spine coordinates, and put*

$$\mathcal{R}_{\mathcal{A}_0} = \{r_{\{p,q\}} : \{p,q\} \in \binom{\mathcal{A}_0}{2}\}.$$

If the union

$$\mathcal{I}_{\text{spine}} \cup \mathcal{R}_{\mathcal{A}_0}$$

is not independently target-testable, then either G *has a sparse surplus exit, or some pair* $\{p,q\} \subseteq \mathcal{A}_0$ *has a sparse surplus blocker of type (d) or (e) in Definition 3.31.*

Proof. Apply the circuit extraction lemma Lemma 10.3 to the declared coordinate family $\mathcal{I}_{\text{spine}} \cup \mathcal{R}_{\mathcal{A}_0}$. Choose an inclusion-minimal determination certificate for the first coordinate whose label-injectivity fails after a maximal independent subfamily is fixed.

If the determination attempts to identify states with different boundary degree profiles, then Lemma 3.94 forbids a target-complete identification. If the determined coordinate is a pair coordinate $r_{\{p,q\}}$, the offending boundary-degree entry is a blocker of type (d); otherwise it is a sparse surplus exit of type (b), because a non-fibrewise quotient is target-defective.

Assume from now on that the boundary degree profile is preserved. If the quotient is not target-complete against all contexts, then Lemma 3.95 makes it target-defective. This is a sparse surplus exit of type (b). If the determined coordinate is $r_{\{p,q\}}$, the distinguishing target-response coordinate is also a blocker of type (e).

It remains that the quotient is target-complete. If its determination support is proper in G, then admissibility supplies a strictly smaller proper representative in the sense of Definition 3.89; by Lemma 3.96 and Corollary 3.98 this is impossible in the minimal counterexample. Thus the attempted correlation is a sparse surplus exit of type (c), and, for a pair coordinate, a blocker of type (e).

If the support is all of G, the closed exact-profile clause of Definition 3.91 applies. A non-injective closed quotient is admissible only when represented by a strictly smaller admissible closed graph H. Then H is finite, simple, satisfies $\delta(H) \geq 3$, and has closed profile contained in $\mathsf{H}_\varnothing(G)$. A power-of-two cycle in H would add the corresponding empty-context target event to $\mathsf{H}_\varnothing(H)$, impossible because $\mathsf{H}_\varnothing(G)$ contains no such event. Hence H is a strictly smaller counterexample, contradicting minimality. If no such representative exists, the quotient is label-injective on the declared coordinates under discussion and therefore is not a rank reduction. Hence every genuine mixed dependence produces either a sparse surplus exit or, when a pair coordinate is involved, a blocker of type (d) or (e). $\square$

Proposition 3.66 (Entropy sandwich for independent surplus pairs)**.** *Let $\mathcal{A}_0$ be a finite family of active surplus demands. Suppose G survives the sparse surplus exits and no pair in $\binom{\mathcal{A}_0}{2}$ has a sparse surplus blocker. Suppose also that there is a baseline spine demand $\mathcal{I}_{\text{spine}}$ with deficit $E_{\text{spine}}(n)$. Then*

$$\binom{|\mathcal{A}_0|}{2} \leq E_{\text{spine}}(n) + \left(\frac{1}{2}\sigma(G) + 1\right) \log_2 n.$$

Consequently, if $E_{\text{spine}}(n) = O(n)$ and $|\mathcal{A}_0| \geq c_1\sigma(G)$ for some fixed $c_1 > 0$, then

$$\sigma(G) = O(\sqrt{n}).$$

Proof. By Lemma 3.65, the union

$$\mathcal{I}_{\text{spine}} \cup \{r_{\{p,q\}} : \{p,q\} \in \binom{\mathcal{A}_0}{2}\}$$

is independently target-testable; otherwise a sparse exit or a sparse blocker would occur, contrary to the hypotheses. Therefore Lemmas 3.100 and 4.2 give

$$|\mathcal{I}_{\text{spine}}| + \binom{|\mathcal{A}_0|}{2} \leq \log_2 \binom{N}{m}.$$

Since $|\mathcal{I}_{\text{spine}}| \geq B_0(n) - E_{\text{spine}}(n)$,

$$\binom{|\mathcal{A}_0|}{2} \leq E_{\text{spine}}(n) + \log_2 \binom{N}{m} - \log_2 \binom{N}{m_0}.$$

The first displayed bound now follows from Lemma 3.64.

Assume $E_{\mathrm{spine}}(n) \le C_E n$ and $|\mathcal{A}_0| \ge c_1\sigma(G)$. The preceding inequality gives constants C_1, C_2, depending only on c_1 and C_E, such that

$$\sigma(G)^2 \le C_1 n + C_2\sigma(G)\log_2 n + C_2 \log_2 n.$$

If $\sigma(G) \le 2C_2 \log_2 n$, then $\sigma(G) = O(\log n) = O(\sqrt{n})$. Otherwise the middle term is at most $\frac{1}{2}\sigma(G)^2$ after increasing constants, and the inequality gives $\sigma(G)^2 \le C_3 n$. Thus $\sigma(G) = O(\sqrt{n})$ in all cases. ☐

Proposition 3.67 (Entropy sandwich with canonical blockers). *Let $\mathcal{A}_0$ be a finite family of active surplus demands, and let Π_{free} be the unblocked pair set from Definition 3.33. Suppose G survives the sparse surplus exits and that there is a baseline spine demand $\mathcal{I}_{\mathrm{spine}}$ with deficit $E_{\mathrm{spine}}(n)$. Then*

$$|\Pi_{\mathrm{free}}| \le E_{\mathrm{spine}}(n) + \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n.$$

Proof. Consider the declared coordinate family

$$\mathcal{I}_{\mathrm{spine}} \cup \{r_\pi : \pi \in \Pi_{\mathrm{free}}\}.$$

If this family is not independently target-testable, apply the finite circuit extraction argument of Lemma 10.3 to an inclusion-minimal determination support for a first dependent coordinate. The same alternatives as in Lemma 3.65 occur. A boundary-degree-profile failure gives a sparse surplus exit unless the dependent coordinate is a pair coordinate, in which case it gives a blocker of type (d). A failure of context-universality gives a sparse surplus exit unless the dependent coordinate is a pair coordinate, in which case it gives a blocker of type (e). A target-complete proper support contradicts Lemma 3.96 and Corollary 3.98; this is the sparse surplus exit of type (c), and for a pair coordinate also a blocker of type (e). A closed whole-graph quotient is either represented by a smaller admissible closed graph, contradicting minimality, or is label-injective on the declared coordinates and therefore not a rank reduction. Thus any dependence gives a sparse surplus exit or a blocker for some pair in Π_{free}. Both are impossible: the graph survives the sparse surplus exits, and pairs in Π_{free} have no sparse surplus blocker by definition. Hence the displayed coordinate family is independently target-testable.

By Lemmas 3.100 and 4.2,

$$|\mathcal{I}_{\mathrm{spine}}| + |\Pi_{\mathrm{free}}| \le \log_2 \binom{N}{m}.$$

Since $|\mathcal{I}_{\mathrm{spine}}| \ge B_0(n) - E_{\mathrm{spine}}(n)$, we have

$$|\Pi_{\mathrm{free}}| \le E_{\mathrm{spine}}(n) + \log_2 \binom{N}{m} - \log_2 \binom{N}{m_0}.$$

Lemma 3.64 gives

$$\log_2 \binom{N}{m} - \log_2 \binom{N}{m_0} \le \left(\frac{1}{2}\sigma(G) + 1\right)\log_2 n,$$

which proves the result. ☐

3.0.1 The pair-code unrealized residual: nodes [178]–[180]

The entropy count of Proposition 3.67 is, like the window realization at node [158], a branch test: on the residual at hand either the mixed family $\mathcal{I}_{\mathrm{spine}} \cup \{r_\pi\}$ realizes its code among the labelled skeletons, and the sandwich holds as displayed, or it does not. The pair coordinates r_π are functions of the graph whose supports X_π share the spine routes, so their independence is not a consequence of label-injectivity, and the second alternative is a genuine residual. It is closed by the same structural accounting as the trivial neutral configuration of nodes [169]–[172]: the failure of the count supplies an overlap obstruction, the obstruction is uncrossed into a serial system, and the increment arithmetic of Lemma 7.15 closes the system.

Definition 3.68 (Pair overlap system)**.** *Fix the active family $\mathcal{A}_0$, the pair set Π under discussion (Π_{free} at node [131], the free side of a capacity fibre at node [137]), the edges of G outside the union of the port returns $\bigcup_p T(p) \cup \Gamma(p)$, and the coordinates already exposed in the canonical encoding order. For $\pi \in \Pi$ the* response support *is X_π of Definition 3.58; two pair coordinates* overlap *when their response supports meet outside the port returns of their own demands. For a finite $\mathcal{U} \subseteq \Pi$ let $\mathcal{S}(\mathcal{U})$ be the set of joint response values on $\mathcal{U}$ realized by graphs in the current conditional fibre. A* pair overlap obstruction *is a nonempty $\mathcal{U}$ for which no ordering $\pi_1, \dots, \pi_t$ has $|\mathcal{S}(\pi_i \mid \pi_1, \dots, \pi_{i-1})| \le 1$ for all i, in every nonempty conditional fibre; that is, no exposure order makes each pair coordinate determined by the previously exposed data. It is* minimal *when every proper nonempty subfamily has such an ordering, and its* overlap support *is the union of its response supports; a disconnected union is treated componentwise.*

Lemma 3.69 (Failure of the pair count produces a local overlap obstruction)**.** *On the strict branch, if the mixed family $\mathcal{I}_{\mathrm{spine}} \cup \{r_\pi : \pi \in \Pi\}$ does not realize its code among the labelled skeletons, i.e. the count of Proposition 3.67 fails at node [131] or at the free side of node [137], then the current conditional fibre contains a minimal pair overlap obstruction, and its overlap support is connected.*

Proof. For a family of pair coordinates whose response supports are pairwise disjoint outside their own port returns, changing the response of one pair changes no edge, port return, or configuration occurring in another pair's support, and Lemma 3.59 applied to that family (which survives the sparse exits and has no blocker) makes it independently target-testable after conditioning on the others; exposing such coordinates in any order realizes the product code. Consequently a failure of the count is witnessed by a finite family for which no exposure order works; choose one minimal by cardinality. If its overlap support were disconnected, the components have disjoint declared supports and minimality supplies an admissible order inside each proper component; concatenating these orders would be admissible for the whole family, a contradiction. □

Lemma 3.70 (Uncrossing and serial realizability of a pair obstruction)**.** *Let $\mathcal{U}$ be a minimal pair overlap obstruction on the strict branch. Exactly one of the following occurs.*

- *(i) an uncrossing of two overlapping response supports realizes a power-of-two cycle;*
- *(ii) the uncrossed responses are distinguished by a compatible context, a target-defective quotient: sparse surplus exit of type (b);*
- *(iii) the responses are context-equivalent and one is a smaller proper representative: sparse surplus exit of type (c);*
- *(iv) the first nonserial intersection is a same-token routed bottleneck whose first separator is a high-degree vertex: the decorated Type B fan data of Lemma 3.77;*

(v) after deleting common subpaths, the overlap support is a serial demand system: ordered interfaces $x_0, \dots, x_s$ along the shared spine route, at each index a nonempty family of mutually internally disjoint corridor pieces, and two closing pieces (the two port returns) meeting the system only at x_0, x_s; every nonzero system increment is at most $D_{\mathrm{sp}} := 2M_{\mathrm{cold}} + 2\ell_{\mathrm{ret}}$, where ℓ_{ret} is the registered bound on a canonical port return; and every choice of one piece per index closes an actual simple cycle of G through the two demands.

On the strict branch, alternatives (i)–(iv) *are already closed: (i) by the counterexample condition, (ii) and (iii) by the sparse surplus survivor of node [125], and (iv) by the Type B continuation.*

Proof. Identical to Lemma 7.35 with the response supports X_π in place of completion supports and the port returns $T(p), T(q)$ in place of the window offsets. Choose the obstruction with the fewest support edges and then the fewest intersections; orient every support from its first demand. For two intersecting supports take their first and last common vertices; between them the union contains two internally disjoint strands, and the uncrossed pairing preserves the boundary-degree profile and all exposed coordinates. A power-of-two closure is (i); a distinguishing context is (ii) by Lemma 3.95; equal responses with different sizes give (iii) by Lemma 3.96 and Corollary 3.98; a branching intersection at a high-degree first separator is the routed bottleneck of Definition 3.76 and Lemma 3.77, whose first-separator reading is decorated Type B fan data, (iv), and at an ambient-cubic separator it is the cubic-switch absorption of Lemma 14.24. Otherwise the strands are equal-length neutral pieces, and node [166] identifies them, removing an intersection against secondary minimality unless it is a common subpath. The intersection graph is then a path, cut at its common subpaths into the ordered interfaces of a serial system. A cell longer than D_{sp} is read from both ends by cold cut-states of Definition 7.9; two states repeat before the readings meet, the intervening piece is a first-failure exchange closed by (F1)–(F5) exactly as in Lemma 7.35, so every cell has the stated bound. Distinct cells have disjoint interiors and each interface uses two distinct recorded incidences, so every choice closes a simple cycle through the two demands' port returns. If all realizable cycles avoided the tested lengths for a reason supported on a bounded end segment, the dependence would be a bounded (F5) row; hence the system is scale-spanning in the sense of Definition 7.34. ☐

Lemma 3.71 (Pair increment arithmetic and periodic response class; node [180)**.** *] On the strict branch, a scale-spanning serial demand system is impossible. With g the gcd of the frequent increments and $\mathcal{R}$ the realizable residues (Lemma 7.36 with D_{sp} in place of D_{sys} and the port-return offsets in place of the window offsets), either the doubling orbit modulo the odd part of g meets $\mathcal{R}$ and the central spectrum contains a power of two, or the residue map on the ordered interfaces is a periodic response class: a compatible context that sees it gives a target-defective quotient (sparse exit (b)); no context gives a target-complete quotient of a proper support (sparse exit (c)); and a response class reaching a same-token routed bottleneck is the periodic trace class that Lemma 3.77 routes to Type B fan data or to the sparse exits. For odd g the hit criterion is $\mathrm{ord}_g(2) > g - |\mathcal{R}|$; for $g = 2^a u$ it applies to u after the transient $k < a$.*

Proof. Word for word the proof of Lemma 7.37: by Lemma 7.36 every length of each displayed coset is realized by a simple cycle away from a bounded end segment; a congruence $2^k \equiv L + r \pmod{g}$ with $r \in \mathcal{R}$ is a power-of-two cycle through the two demands, contradicting the counterexample condition. Under avoidance the residue of the route length from x_0 is well defined and transported through every corridor; two equal-residue states distinguished by a context are a target-defective quotient (Lemma 3.95, sparse exit (b)); otherwise the quotient is target-complete and on a proper support gives a smaller representative (Lemma 3.96 and Corollary 3.98, sparse exit (c)); at a routed bottleneck the residue coordinate uses the same connector configurations as Definition 3.76, and Lemma 3.77 routes it. Both exits are refuted by the sparse surplus survivor of node [125], and the Type B outcome is the continuation of node [144]. □

Lemma 3.72 (Pair count or arithmetic closure; node [178]**.** *] On the strict branch, either the entropy count of Proposition 3.67 holds on the current residual (node [131], and the free side of node [137]), and the branch continues as drawn, or the residual carries a minimal pair overlap obstruction (Lemma 3.69), which Lemma 3.70 turns into a scale-spanning serial demand system, closed by Lemma 3.71. Hence the pair-code unrealized residual [178] has no survivor.*

Proof. Immediate from the three lemmas. □

Definition 3.73 (Spine lower-bound deficits)**.** *Let*

$$c_{13} = 118.108581006, \qquad K = 5.89262883286\ldots, \qquad \theta = \frac{p_{13}}{n},$$

and put

$$\theta_{\mathrm{he}} := \frac{1.4}{116.808581006} = 0.01198542083\ldots.$$

The near-cubic branch supplies the following independent-coordinate lower-bound packages, whenever their stated hypotheses hold. We record their deficits relative to the exact cubic baseline $B_0(n)$.

(a) Window-only package. *The packed P_{13}-windows contribute at least*

$$L_{\mathrm{win}} = c_{13}\theta n \log_2 n$$

independent target coordinates. By Lemma 3.63, its deficit satisfies

$$D_{\mathrm{win}} := B_0(n) - L_{\mathrm{win}} \le (1.5 - c_{13}\theta) n \log_2 n + O(n).$$

(b) High-remainder-entropy package. *In the high-entropy branch $\eta(R) \ge \frac{1}{10}\log_2 n$, the remainder contributes at least $\frac{1}{10}|R|\log_2 n$ additional independent coordinates. Since $|R| = n - 13p_{13} = (1 - 13\theta)n$, this package has lower bound*

$$L_{\mathrm{he}} = \left(\frac{1-13\theta}{10} + c_{13}\theta\right) n \log_2 n = (0.1 + 116.808581006\,\theta) n \log_2 n,$$

and deficit

$$D_{\mathrm{he}} := B_0(n) - L_{\mathrm{he}} \le (1.4 - 116.808581006\,\theta) n \log_2 n + O(n).$$

(c) High-entropy plus forced-obstruction package. *When the full-rank obstruction cost of Corollary 10.11 is also available in the high-entropy branch, the additional lower bound is*

$$K|R| - o(n) = K(1-13\theta)n - o(n).$$

The corresponding deficit satisfies

$$D_{\mathrm{he}+\Omega} \le (1.4 - 116.808581006\,\theta) n \log_2 n - K(1-13\theta)n + o(n) + O(n).$$

Each deficit is an admissible upper bound for $E_{\mathrm{spine}}(n)$ in Definition 3.62, for the corresponding lower-bound package.

Corollary 3.74 (Surplus estimates from spine lower bounds)**.** *Assume G survives the sparse surplus exits, no pair in $\binom{\mathcal{A}_0}{2}$ has a sparse surplus blocker, and*

$$|\mathcal{A}_0| \geq c_1\sigma(G)$$

for a fixed $c_1 > 0$. Then the following estimates are rigorous consequences of the lower-bound packages in Definition 3.73.

(a) From the window-only package,

$$\binom{|\mathcal{A}_0|}{2} \leq D_{\mathrm{win}} + \left(\frac{1}{2}\sigma(G)+1\right)\log_2 n,$$

hence

$$\sigma(G) = O\Big(\sqrt{(1.5-c_{13}\theta)_+\, n\log n + n} + \log n\Big).$$

In particular, if $1.5 - c_{13}\theta = O(1/\log n)$, then $\sigma(G) = O(\sqrt{n})$.

(b) In the high-remainder-entropy package,

$$\binom{|\mathcal{A}_0|}{2} \leq D_{\mathrm{he}} + \left(\frac{1}{2}\sigma(G)+1\right)\log_2 n,$$

hence

$$\sigma(G) = O\Big(\sqrt{(1.4-116.808581006\,\theta)_+\, n\log n + n} + \log n\Big).$$

In particular, if

$$1.4 - 116.808581006\,\theta = O(1/\log n),$$

then $\sigma(G) = O(\sqrt{n})$.

(c) In the high-entropy plus forced-obstruction package, if

$$(1.4 - 116.808581006\,\theta)n\log_2 n - K(1-13\theta)n = O(n),$$

then $\sigma(G) = O(\sqrt{n})$. If the same expression is $< -Cn$ for a sufficiently large fixed constant C, then the branch has no survivor before surplus-pair coordinates are added.

Proof. Each package in Definition 3.73 is a baseline spine demand with deficit bounded above by the corresponding D-quantity. Thus Proposition 3.66 gives the displayed pair inequalities.

We prove the first square-root estimate; the second is identical. From $|\mathcal{A}_0| \geq c_1\sigma(G)$, for all sufficiently large $\sigma(G)$ we have

$$\binom{|\mathcal{A}_0|}{2} \geq \frac{c_1^2}{4}\sigma(G)^2.$$

Combining with the window-only pair inequality gives

$$\sigma(G)^2 \leq C\left((1.5-c_{13}\theta)_+ n\log n + n + \sigma(G)\log n\right)$$

for a constant C depending only on c_1. If $\sigma(G) \leq 2C\log n$, then $\sigma(G) = O(\sqrt{n})$ for $n \geq 2$. Otherwise $C\sigma(G)\log n \leq \frac{1}{2}\sigma(G)^2$, after increasing C, and the term is absorbed into the left-hand side. This proves the window-only bound and its $O(\sqrt{n})$ consequence when $1.5-c_{13}\theta = O(1/\log n)$.

Replacing the window-only deficit term by D_{he} in the displayed inequality gives the stated high-entropy estimate.

For (c), the stated $O(n)$ hypothesis says precisely that $D_{\mathrm{he}+\Omega} = O(n)$, so Proposition 3.66 gives $\sigma(G) = O(\sqrt{n})$. If the displayed expression is less than $-Cn$ for C larger than the constants hidden in Lemma 3.63 and in the $o(n)$ obstruction term, then the baseline lower-bound package itself exceeds $B_0(n)$. This contradicts Lemmas 3.100 and 4.2. $\square$

Theorem 3.75 (Sharp surplus-overload audit)**.** *Let G reach node* [125]*, survive the sparse surplus exits, have full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$, and admit a baseline spine demand with deficit $E_{\mathrm{spine}}(n)$. Define $N_*(G)$ as in Theorem 3.51. Let*

$$x := e_\times(W)/n, \qquad \theta := p_{13}/n, \qquad w := \sigma_W/n, \qquad r := \sigma_R/n, \qquad s := \sigma(G)/n = w + r.$$

For each subtype $u \in \{RW, WW, R, V, I, P\}$, let S_u be the number of capacity tokens of subtype u, and let B_u be the number of blocked pairs assigned to such tokens. Then:

(a) The subtype supplies are exact:

subtype	S_u	S_u/n
RW	$e(R,W)$	$15\theta + w - 2x$
WW	$2e_\times(W)$	$2x$
R	σ_R	r
V	n	1
I	$2m$	$3 + s$
P	$\sigma(G)$	s

where $0 \le x \le (15\theta + w)/2$, $0 \le \theta \le 1/13$, and $0 \le s \le 1$.

(b) The blocked-pair demand has the exact subtype split

$$B_{RW} + B_{WW} + B_R + B_V + B_I + B_P = |\Pi_{\mathrm{blk}}| \ge N_*(G).$$

(c) The total token supply and total role-fibre slot supply are

$$\sum_u S_u = 4n + 15p_{13} + 3\sigma(G) = n(4 + 15\theta + 3s),$$

and

$$Q_{\mathrm{st}} \sum_u S_u = 36n(4 + 15\theta + 3s) \le \frac{3816}{13} n.$$

(d) If $S_u > 0$, then some token of subtype u supports a role-homogeneous same-token matching or star of size at least

$$\psi\left(\frac{B_u}{Q_{\mathrm{st}} S_u}\right).$$

Consequently, if $N_*(G) > 0$, some subtype u supports such a pattern of size at least

$$\psi\left(\frac{N_*(G)}{36(4n + 15p_{13} + 3\sigma(G))}\right) \ge \psi\left(\frac{13 N_*(G)}{3816 n}\right).$$

Proof. The exact window split follows from Definition 3.37: RW tokens are the window–remainder edges and WW tokens are the two endpoint tokens on each cross-window edge. Hence

$$S_{RW} = e(R, W), \qquad S_{WW} = 2e_\times(W).$$

By Lemma 8.6,

$$e(R, W) = 15p_{13} + \sigma_W - 2e_\times(W),$$

which gives the first two normalized rows. The bound on x is the nonnegativity of $e(R, W)$. The R-row is the definition of $\mathfrak{T}_R$:

$$S_R = \sum_{v \in R} (d_G(v) - 3) = \sigma_R.$$

For primitive tokens,

$$\mathfrak{T}_{\mathrm{prim}} = V(G) \sqcup I_E(G) \sqcup \mathcal{P}_{\mathrm{exc}},$$

so the subtype supplies are

$$S_V = n, \qquad S_I = 2m, \qquad S_P = |\mathcal{P}_{\mathrm{exc}}| = \sigma(G).$$

Since $\sigma(G) = 2m - 3n$, one has $2m = 3n + \sigma(G)$, giving $S_I/n = 3 + s$. The inequalities $0 \le \theta \le 1/13$ and $0 \le s \le 1$ follow respectively from the vertex-disjointness of the packed P_{13} windows and from Lemma 3.3.

The map Θ_{cap} assigns every blocked pair to exactly one capacity token by Definition 3.37. The six subtypes listed in Definition 3.41 partition the capacity-token universe. Thus the subtype fibres partition Π_{blk}. The lower bound $|\Pi_{\mathrm{blk}}| \ge N_*(G)$ is Theorem 3.51.

Summing the six supply rows gives

$$\sum_u S_u = e(R, W) + 2e_\times(W) + \sigma_R + n + 2m + \sigma(G).$$

Using Lemma 8.6, $\sigma(G) = \sigma_W + \sigma_R$, and $2m = 3n + \sigma(G)$, this is

$$15p_{13} + \sigma_W + \sigma_R + n + (3n + \sigma(G)) + \sigma(G) = 4n + 15p_{13} + 3\sigma(G).$$

Multiplication by $Q_{\mathrm{st}} = 36$ gives the role-fibre slot count. Since $\theta \le 1/13$ and $s \le 1$,

$$36(4 + 15\theta + 3s) \le 36\left(4 + \frac{15}{13} + 3\right) = \frac{3816}{13}.$$

Fix a subtype u with $S_u > 0$. Among the S_u tokens of this subtype, some token t has load at least B_u/S_u. Its role fibres partition its load into at most Q_{st} parts, so some role fibre has size at least $B_u/(Q_{\mathrm{st}} S_u)$. Applying Lemma 3.47 to that role-fibre graph gives a role-homogeneous same-token matching or star of size at least $\psi(B_u/(Q_{\mathrm{st}} S_u))$.

If $N_*(G) > 0$, choose a subtype with maximal ratio B_u/S_u among nonempty subtypes. The preceding partition gives

$$\frac{B_u}{S_u} \ge \frac{\sum_v B_v}{\sum_v S_v} \ge \frac{N_*(G)}{4n + 15p_{13} + 3\sigma(G)}.$$

The displayed global lower bound follows by monotonicity of ψ. $\square$

Definition 3.76 (Same-token routing configurations)**.** *Let $t \in \mathfrak{T}_{\mathrm{cap}}$, let $r \in \mathfrak{R}_{\mathrm{st}}$, and let $\pi = \{p, q\} \in \Pi_{t,r}$. The* same-token routing support

$$Z(\pi; t, r)$$

is the finite support generated by the declared signature Definition 3.87 from the following data: the capacity token t, the canonical blocker $\Phi_{\mathrm{can}}(\pi)$, the selected port supports $T(p), T(q)$, and the canonical response supports $\Gamma(p), \Gamma(q)$.

A routing configuration *of π at t is one of the declared connector configurations in $Z(\pi; t, r)$ which starts at the primitive blocker support of t and ends at one of the two selected port supports $T(p)$ or $T(q)$. If two routing configurations have a maximal common initial segment and then leave through two distinct next incidences, the last vertex of the common initial segment is their* first separator*. If no such vertex exists before both configurations enter the same declared selected-port support with the same endpoint label, the two configurations are* parallel*.*

The routing label *of a pair in $\Pi_{t,r}$ records the same-token role r, the subtype of t, the ordered endpoint of the pair under discussion, the local open/triangular status of the corresponding selected ports, the boundary-degree profile of the bounded port supports $T(p), T(q)$, the P_{13}-label entries appearing in the bounded part of the support, and the suppressed-chord flag when the blocker has type (f). These labels form a finite set; denote its cardinality by Q_{geom}.*

Lemma 3.77 (Same-token bottleneck routing)**.** *Assume G survives the sparse surplus exits of Definition 3.26. Let $t \in \mathfrak{T}_{\mathrm{cap}}$ and $r \in \mathfrak{R}_{\mathrm{st}}$. If a role-homogeneous same-token matching or star in the role-fibre graph $H_{t,r}$ has more than Q_{geom} edges, then one of the following occurs:*

(a) a sparse surplus exit occurs; or

(b) the support produces a decorated Type B handoff fan envelope, hence is routed to the Type B fan ledger.

Proof. Let $\mathcal{M}$ be the matching or star. Since there are only Q_{geom} routing labels, two distinct edges $\pi_1, \pi_2 \in \mathcal{M}$ have the same routing label. If $\mathcal{M}$ is a star, choose the two noncentral endpoints of π_1 and π_2; if $\mathcal{M}$ is a matching, choose an endpoint from each edge. In both cases we obtain two distinct selected surplus demands whose connector data have the same token, the same blocker type, the same token subtype, and the same boundary-degree fibre.

Consider the two routing configurations from the primitive blocker support of t toward those two selected demands. If the configurations are parallel, then the two declared response coordinates have the same bounded support data and lie in the same boundary-degree fibre. Identifying them is a nontrivial quotient of declared coordinates. If some outside context distinguishes the two responses, the quotient is target-defective by Lemma 3.95, which is a sparse surplus exit. If the identification is target-complete on a proper support, then Lemma 3.96 and Corollary 3.98 give the target-complete compression exit. If the identification becomes valid only after adjoining a larger connected support, then Lemmas 10.6 and 10.9 give the proper or whole-graph support-dependence exit. In the closed exact-profile case, Definition 3.91 permits a non-injective quotient only when there is a smaller admissible closed representative; that representative would be a smaller counterexample. Thus the parallel case is impossible in a survivor.

It remains that the two configurations have a first separator z. If $d_G(z) = 3$, the root incidence of the common prefix together with the two distinct next incidences exhausts all incidences at z. The finite switch support at z therefore has no unused ambient incidence through which the two coordinates can be distinguished without using the declared quotient data. There are then exactly three target-completeness possibilities for the switch quotient. If some compatible outside context distinguishes the two switched coordinates, the quotient is target-defective by Lemma 3.95; this is sparse surplus exit (b) in Definition 3.26. If every compatible context preserves the identification on the finite switch support, then the two same-interface coordinates give a nontrivial target-complete compression of a proper support, contradicting Lemma 3.96 and Corollary 3.98 and entering sparse surplus exit (c). If context-universality holds only after adjoining a larger connected support, the support has proper or whole-graph dependence; these are excluded by Lemmas 10.6 and 10.9 and enter sparse surplus exit (d). Thus a cubic first separator cannot survive, and any surviving first separator must satisfy

$$d_G(z) \geq 4.$$

The high-degree separator z, together with the separated connector tails from z to the two selected surplus demands, is exactly the decorated handoff data of Definition 14.19. The fan-safety condition for this envelope says that every pair of separated tails has no two-tail closure of power-of-two length, has compatible P_{13}-window labels, and lies in one boundary-degree fibre of the declared response algebra. If a two-tail closure has power-of-two length, then sparse surplus exit (a) of Definition 3.26 occurs. If the closure does not have power-of-two length but the two P_{13}-labels are incompatible, the P_{13}-label collision is the finite label exit already included in sparse surplus exit (a). If the labels are compatible but the boundary-degree fibre or target response is separated by an outside context, the quotient is target-defective, giving exit (b). If the same response is target-complete on a proper finite support, Lemma 3.96 and Corollary 3.98 give exit (c). If it becomes target-complete only after adjoining a larger connected support, Lemmas 10.6 and 10.9 give exit (d). These alternatives exhaust every failure of fan-safety. For a blocker of type (f), the only additional possibility is that the common support is the suppressed open-port chord set; if the chord arithmetic produces a power-of-two length, exit (e) of Definition 3.26 occurs, and otherwise the two open-port shoulders are assigned as fan incidences by Lemma 3.10 and Corollary 14.81 or, in the triangular alternative, by Corollary 14.82. Thus every surviving separated case enters the Type B fan ledger. □

Theorem 3.78 (Geometric closure of homogeneous same-token overloads)**.** *Assume G reaches node* [125]*, survives the sparse surplus exits, has full active family $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$, and admits a baseline spine demand with $E_{\mathrm{spine}}(n) \leq C_E n$. Put*

$$L_{\mathrm{geom}} := Q_{\mathrm{geom}} + 1.$$

Then every role-homogeneous same-token matching or star of size L_{geom} in the window-incidence, remainder-surplus, or primitive blocker-support token classes realizes either a sparse surplus exit or a decorated Type B handoff fan envelope.

Consequently, on the subbranch in which sparse surplus exits are absent and all decorated Type B handoff data have been routed into the Type B fan ledger, the three fixed homogeneous caps

$$L_W = L_R = L_P = L_{\mathrm{geom}}$$

hold. On that subbranch,

$$\sigma(G) = O(\sqrt{n}) \qquad \text{and} \qquad m = \frac{3}{2}n + O(\sqrt{n}).$$

Proof. The first assertion is exactly Lemma 3.77, applied to the role fibre containing the alleged matching or star.

If sparse exits are absent and decorated Type B handoff data have already been routed to the Type B fan ledger, the first assertion says that no capacity token in any of the three classes can

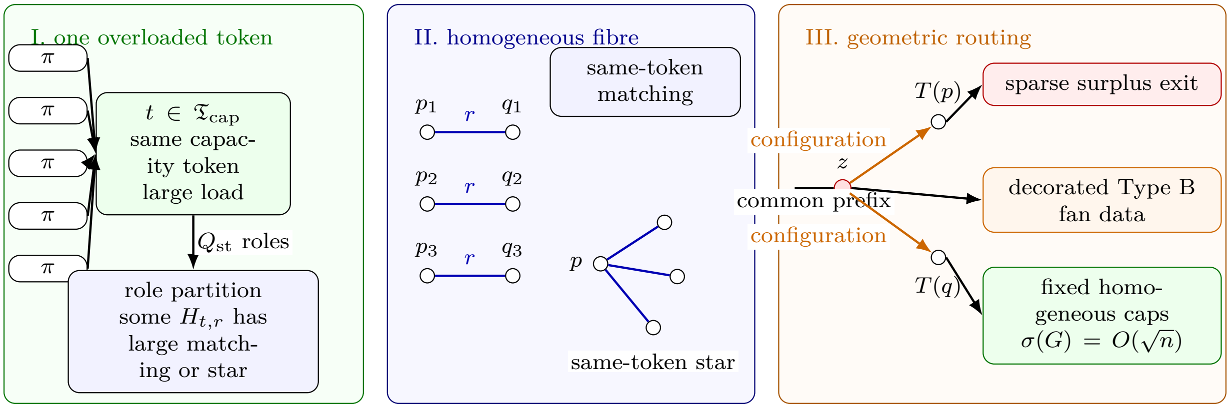

Figure 15: Visual definition of the homogeneous same-token bottleneck. Panel I shows blocked active surplus pairs assigned by the capacity-token map to one overloaded token $t \in \mathfrak{T}_{\mathrm{cap}}$. The role partition of the fibre over t produces a role r for which the same-token role graph $H_{t,r}$ contains a large matching or a large star. Panel II displays the two possible homogeneous patterns: every shown edge has the same token t and the same role r. Panel III shows the geometric routing of two selected demands from the same homogeneous pattern. Parallel routing configurations give one of the quotient, compression, or support-dependence exits already included among the sparse surplus exits. If the configurations first separate at a high-degree vertex z, the separated tails form decorated Type B handoff fan data. When neither a sparse surplus exit nor Type B fan data remains, all three token classes have fixed homogeneous caps, and Corollary 3.44 gives $\sigma(G) = O(\sqrt{n})$.

support a role-homogeneous same-token L_{geom}-matching or L_{geom}-star. Therefore the hypotheses of Corollary 3.44 hold with $L_W = L_R = L_P = L_{\mathrm{geom}}$. That corollary gives $\sigma(G) = O(\sqrt{n})$. Since

$$m = \frac{3}{2}n + \frac{1}{2}\sigma(G)$$

by Lemma 3.25, the edge-count estimate follows. □

Remark 3.79 (Exact surplus frontier)**.** *The sparse surplus bookkeeping has no disjointness loss. The active family is $\mathcal{A}_0 = \mathcal{P}_{\mathrm{exc}}$ by Lemma 3.30. Free pairs are charged by Proposition 3.67. Blocked pairs are assigned a canonical blocker and then one capacity token by Definition 3.37. The token fibres partition the blocked-pair set by Lemma 3.39. Therefore any attempted non-near-cubic surplus survivor must first pass through one of the six subtype loads in Theorem 3.75, grouped into the three diagram classes $W = \{RW, WW\}$, $R = \{R\}$, and $P = \{V, I, P\}$. A large load in any one of those fibres is not terminal: Theorem 3.78 routes it to a sparse surplus exit, to Type B fan data, or to the fixed homogeneous caps that imply the near-cubic spine.*

Proposition 3.80 (Non-near-cubic sparse-load routing)**.** *Let G be a lexicographically minimal counterexample that reaches node* [18]*. If G survives the sparse surplus exits, then the exact sparse-load audit has only the following outcomes:*

(a) G satisfies the near-cubic spine estimate $\sigma(G) = O(\sqrt{n})$;

(b) a sparse surplus exit occurs; or

(c) a role-homogeneous same-token bottleneck produces decorated Type B fan data and is routed to the Type B fan ledger.

In particular, the window-incidence, remainder-surplus, and primitive blocker-support same-token classes are not terminal non-near-cubic leaves.

Proof. If a sparse surplus exit occurs, there is nothing to route. Otherwise Lemma 3.30 gives $\mathcal{A}_0 = \mathcal{P}_{\text{exc}}$, and Proposition 3.67 supplies the free-pair entropy bound used in $N_*(G)$. The blocked-pair ledger and token ledger are exact by Lemmas 3.34 and 3.39.

Apply Theorem 3.78. If no homogeneous same-token pattern reaches size L_{geom}, the fixed caps $L_W = L_R = L_P = L_{\text{geom}}$ hold and Corollary 3.44 gives $\sigma(G) = O(\sqrt{n})$. If such a pattern exists, then Lemma 3.77 gives either a sparse surplus exit or decorated Type B handoff data. The subtype identity in Theorem 3.75 records where the load came from: RW and WW are window-incidence loads, R is a remainder-surplus load, and V, I, P are primitive blocker-support loads. In all cases the load is either closed, routed to Type B, or bounded by the fixed cap that yields the near-cubic estimate. $\square$

3.1 Boundaried pieces, target-complete quotients, and replacement

Definition 3.81 (Boundaried graph and gluing)**.** *Let T be a finite label set. A T-boundaried graph is a graph X together with an injective labelling of a subset $\partial X \subseteq V(X)$ by T. The vertices in ∂X are boundary vertices; all others are internal. If X and Y are T-boundaried graphs with disjoint internal vertex sets, their gluing is denoted*

$$X \oplus_T Y,$$

and is obtained by identifying equally labelled boundary vertices and taking the union of edges.

Definition 3.82 (Boundaried piece)**.** *A* boundaried piece *of G is a connected T-boundaried subgraph X occurring as one side of a decomposition $G = X \oplus_T Y$ along a boundary $T \subseteq V(G)$; the other side Y is its* context*. The piece is* proper *if $X \neq G$. The replacement and uncompressibility arguments of this section quantify over proper boundaried pieces.*

Definition 3.83 (Boundary degree profile)**.** *Let X be a T-boundaried graph. For $t \in T$, let v_t be the boundary vertex of X labelled by t. The boundary degree profile of X is the vector*

$$\mathbf{d}_\partial(X) = (d_X(v_t))_{t \in T}.$$

All target-response states in this paper are taken fibrewise over $\mathbf{d}_\partial$. In particular, two boundaried states with different boundary degree profiles are never eligible to be identified by a target-complete quotient.

Definition 3.84 (Obstruction profile)**.** *For a T-boundaried graph X, define its target obstruction profile*

$$\mathsf{H}_T(X) = \{Y \in \mathsf{Ctx}_T : X \oplus_T Y \text{ contains a power-of-two cycle}\},$$

where Ctx_T is the class of finite T-boundaried contexts that can be glued to X. If X' and X are T-boundaried graphs, write

$$X' \preceq_T X$$

if

$$\mathsf{H}_T(X') \subseteq \mathsf{H}_T(X).$$

Thus X' is at most as obstructing as X.

Definition 3.85 (Target-complete quotient)**.** *Let X be a T-boundaried piece, and let $\mathcal{R}(X)$ be a collection of local target-response coordinates of X, such as terminal trace lengths, edge-rooted Mersenne-return tests, two-step obstruction tests, boundary degree data, and other marked response data. A quotient of $\mathcal{R}(X)$ is* target-complete *if every identification made by the quotient preserves both:*

(a) the boundary degree profile $\mathbf{d}_\partial(X)$; and

(b) the target predicate after gluing X to every T-boundaried context.

Thus two coordinates may be identified only inside a fixed boundary-degree fibre and only when no context $Y \in \mathsf{Ctx}_T$ creates a power-of-two cycle from one coordinate and not from the other. An identification failing this context-universal test is target-defective.

Lemma 3.86 (Composition of target-complete quotients)**.** *Let*

$$\rho \longrightarrow Q \longrightarrow Q'$$

be boundary-degree-preserving response quotients tested against the same class of compatible boundaried contexts. If both displayed quotients are target-complete, then the composite quotient $\rho \to Q'$ is target-complete. Consequently, if $\rho \to Q$ is target-complete and the composite $\rho \to Q'$ is not target-complete, then $Q \to Q'$ is not target-complete. Moreover, any compatible context that distinguishes a realization of Q' from the state Q also distinguishes that realization from ρ after composing with the target-complete quotient $\rho \to Q$.

Proof. Fix a compatible outside context Y. Target-completeness of $Q \to Q'$ says that every realization of Q', glued to Y, has the same target predicate as the corresponding Q-state. Target-completeness of $\rho \to Q$ says that the Q-state, glued to Y, has the same target predicate as ρ. Hence every realization of Q' has the same target predicate as ρ after gluing to Y. Since Y was arbitrary, the composite is target-complete. The contrapositive gives the second assertion. The last assertion is the same transitivity statement applied to a fixed distinguishing context. □

Definition 3.87 (Declared response-coordinate signature)**.** *The response-coordinate signature used in the proof is fixed once and for all. For a T-boundaried support X, a declared coordinate is a tuple*

$$(k, \iota, \operatorname{supp}_X(r), \operatorname{val}_X(r)),$$

where k is its kind, ι is its label, $\operatorname{supp}_X(r)$ is a finite set of vertices, edges, boundary incidences, paths, or windows of X, and $\operatorname{val}_X(r)$ is the corresponding value in the embedded support. The declared coordinates are generated by the following clauses.

(D1) boundary-degree entries at labelled boundary vertices;

(D2) edge-rooted return data, completion-port data, first-entry receivers, connector lengths, receiver-entry channels, and the corresponding Mersenne return tests;

(D3) induced-P_{13} window labels, legal-label relations, packed-window incidences, and cross-window incidence data;

(D4) raw obstruction coordinates indexed by internal length-two wedges;

(D5) Type A trace data: canonical traces, trace-incidence coordinates, connector-band constraints, cross-port theta constraints, silent-basin response coordinates, response-support restrictions, and their labelled subcoordinates;

(D6) Type B fan and bridge data: fan centers, fan-safe pairs, certificate labels, closed fan-window pairs, hybrid incidence entries, candidate ledger entries, overlap demands, decorated handoff fan response coordinates, and decorated handoff-arm coordinates;

(D7) sparse surplus data: selected surplus ports, canonical port returns, open-port suppression paths, triangular-port response triangles, and sparse surplus-pair response coordinates;

(D8) finite products, labelled copies, restrictions, and quotient images of coordinates already generated in clauses (D1)–(D7), with support equal to the union of the supports of the entries used.

No other local response coordinate is available to a quotient. Whenever a later definition names a coordinate family, it names the subfamily generated by this signature with the displayed labels. In particular, a phrase such as "needed to evaluate a return" means that the relevant port, first-entry incidence, connector endpoint, channel, window, or boundary incidence is a member of the declared support specified by the corresponding clause above.

Definition 3.88 (Exact response profile)**.** *Let X be a T-boundaried support. Let $\mathfrak{S}_T(X)$ be the finite labelled family of coordinates generated on X by Definition 3.87. The* exact response profile *of X is*

$$\rho_T^{\mathrm{ex}}(X) = \big(\mathbf{d}_\partial(X), \mathsf{H}_T(X), \mathcal{R}_T^{\mathrm{ex}}(X)\big),$$

where $\mathcal{R}_T^{\mathrm{ex}}(X) = \mathfrak{S}_T(X)$, including the raw obstruction coordinates carried by X when they are present. The profile is exact in the following sense: two distinct coordinate labels remain distinct entries of $\mathcal{R}_T^{\mathrm{ex}}(X)$ even if their numerical values in the embedded graph coincide.

A quotient

$$q : \rho_T^{\mathrm{ex}}(X) \longrightarrow Q$$

is label-injective on a subfamily *$\mathcal{A} \subseteq \mathcal{R}_T^{\mathrm{ex}}(X)$ if every coordinate of $\mathcal{A}$ is retained by q and*

$$q(a) = q(b) \quad \text{with } a, b \in \mathcal{A} \qquad \Longrightarrow \qquad a = b.$$

It is rank-reducing *on $\mathcal{A}$ if it is not label-injective on $\mathcal{A}$.*

Definition 3.89 (Proper representative of an exact-profile quotient)**.** *Let $G = X \oplus_T Y$, where $X \subsetneq G$ is a proper T-boundaried support, and let*

$$q : \rho_T^{\mathrm{ex}}(X) \to Q$$

be a quotient of its exact profile. A T-boundaried graph X' properly represents *q if there is a quotient*

$$q_{X'} : \rho_T^{\mathrm{ex}}(X') \to Q$$

with the following properties.

(a) $\mathsf{H}_T(X') \subseteq \mathsf{H}_T(X)$.

(b) $\mathbf{d}_\partial(X') = \mathbf{d}_\partial(X)$.

(c) X' contains no internal power-of-two cycle.

(d) Every internal vertex of X' has degree at least 3.

(e) *Every labelled coordinate of Q is the image under $q_{X'}$ of a coordinate generated by Definition 3.87, and the quotient relations among the coordinates under discussion are the same as those imposed by q.*

The representative is strictly smaller *if X' is smaller than X in the lexicographic order used for the minimal counterexample.*

Definition 3.90 (Closed representative of an exact-profile quotient)**.** *Let*

$$q : \rho^{\mathrm{ex}}_{\varnothing}(G) \to Q$$

be a quotient of the closed exact profile. A finite simple closed graph H represents *q if there is a quotient*

$$q_H : \rho^{\mathrm{ex}}_{\varnothing}(H) \to Q$$

with the following properties.

(a) *The obstruction-profile component of q_H is no larger than that of G:*

$$\mathsf{H}_{\varnothing}(H) \subseteq \mathsf{H}_{\varnothing}(G).$$

(b) *The boundary-degree component is the empty boundary profile.*

(c) *Every labelled coordinate of Q is the image under q_H of a coordinate generated by the declared signature of Definition 3.87, and the quotient relations among raw obstruction coordinates recorded in Q are the same quotient relations as those imposed by q.*

The representative is strictly smaller *when H is smaller than G in the lexicographic minimality order. It is* admissible closed *when, in addition, $\delta(H) \geq 3$.*

Definition 3.91 (Admissible rank quotient)**.** *Let $\mathcal{A}$ be a family of declared response coordinates carried by a connected support $X \subseteq G$, and put*

$$T = \{v \in V(X) : v \text{ is incident with an edge of } G - X\}.$$

An admissible rank quotient *for $\mathcal{A}$ is a quotient*

$$q : \rho^{\mathrm{ex}}_{T}(X) \longrightarrow Q$$

which preserves the boundary degree profile and is target-complete against all T-boundaried contexts in the sense above.

If $X \subsetneq G$, then a rank-reducing admissible quotient must be represented by a strictly smaller proper representative in the sense of Definition 3.89. Equivalently, a target-complete proper-support correlation that has no smaller graph representative is not an admissible rank reduction; it is only an abstract identification of labels and does not reduce the graph-realizable target rank used in this proof.

If $T = \varnothing$ and $X = G$, the quotient is evaluated in the closed exact profile $\rho^{\mathrm{ex}}_{\varnothing}(G)$. In that closed case, absence of an outside context is not an identification rule: an admissible rank quotient must be label-injective on the declared coordinates under discussion unless it is represented by a strictly smaller admissible closed representative in the sense of Definition 3.90.

Thus the empty-boundary profile may record exact global information, but exact global information does not collapse two labelled response coordinates. A closed non-injective quotient can affect target rank only when it supplies a strictly smaller closed representative; such a representative would be a smaller counterexample if the target predicate is still false.

Definition 3.92 (Functional rank quotient). *Let $q : \rho_T^{\mathrm{ex}}(X) \to Q$ be an admissible rank quotient for a finite coordinate family $\mathcal{A}$. The quotient q is* functional on $\mathcal{A}$ *if the following rank axiom holds. Whenever $\mathcal{I} \subseteq \mathcal{A}$ and $a \in \mathcal{A} \setminus \mathcal{I}$ are such that q is label-injective on $\mathcal{I}$, but is not label-injective on $\mathcal{I} \cup \{a\}$, there is a finite subfamily $\mathcal{B} \subseteq \mathcal{I}$ and a function ϕ on the realized q-image vectors of $\mathcal{B}$ such that*

$$q(a) = \phi((q(b))_{b\in\mathcal{B}})$$

for every realization of the exact response profile under q. If the quotient sends a to a single quotient value, then $\mathcal{B} = \varnothing$. If the quotient identifies a with a coordinate $b \in \mathcal{I}$, then $\mathcal{B} = \{b\}$ and $\phi(q(b)) = q(b)$.

The admissible quotient system used to compute target-rank consists only of admissible rank quotients that are functional on the coordinate family under discussion.

Remark 3.93 (Rank coordinates and entropy charge). *The obstruction rank r_Ω is always computed from the declared coordinates of the exact response profile and from admissible rank quotients in the sense of Definitions 3.88, 3.91 and 3.92. Thus a rank coordinate contributes to the entropy count only when it remains a labelled, independently target-testable coordinate after all admissible quotients have been applied. This is the interface used by Lemma 3.100 and Proposition 11.6: raw obstruction tests are not counted twice, and an abstract identification that is not an admissible rank quotient does not reduce r_Ω.*

Lemma 3.94 (Boundary degree profiles cannot be quotient-merged). *Let X_1 and X_2 be T-boundaried supports. If*

$$\mathbf{d}_\partial(X_1) \neq \mathbf{d}_\partial(X_2),$$

then no target-complete quotient identifies X_1 and X_2.

Proof. Condition (a) in the definition of a target-complete quotient requires the quotient to preserve the boundary degree profile. If $\mathbf{d}_\partial(X_1) \neq \mathbf{d}_\partial(X_2)$, an identification of X_1 with X_2 would identify two different boundary-degree profiles, so it violates condition (a). Thus two supports lying in different boundary-degree fibres are not eligible for identification, independently of the context-universal target test in condition (b). □

Lemma 3.95 (Context-universality of target-complete identifications). *Let X be a T-boundaried piece, and let $\mathcal{R}(X)$ be any collection of local target-response coordinates of X. Suppose that two coordinates $r_1, r_2 \in \mathcal{R}(X)$ are identified in a target-complete quotient of X. Then r_1 and r_2 have the same target response against every T-boundaried context Y.*

Equivalently, for every $Y \in \mathsf{Ctx}_T$, gluing Y to X cannot distinguish r_1 from r_2 by creating a power-of-two cycle in one case and not in the other. Consequently any identification valid only for the actual outside context $G - X$, but not for all T-boundaried contexts, is target-defective.

Proof. This is precisely the meaning of target-completeness. If some context Y_0 distinguished r_1 and r_2, then one of the two gluings would contain a power-of-two cycle and the other would not. The quotient would fail to preserve the target predicate for Y_0 and therefore would not be target-complete. □

Lemma 3.96 (Replacement lemma for boundaried pieces). *Let $G = X \oplus_T Y$, where X is a proper boundaried piece of G. Suppose there exists a T-boundaried graph X' such that:*

(i) $X' \preceq_T X$*;*

(ii) $\mathbf{d}_\partial(X') = \mathbf{d}_\partial(X)$*;*

(iii) X' contains no internal power-of-two cycle;

(iv) every internal vertex of X' has degree at least 3;

(v) X' is strictly smaller than X in the same lexicographic order used for G.

Then G was not minimal. Consequently no such replacement exists in G.

Proof. Form
$$G' = X' \oplus_T Y.$$
Because $\mathbf{d}_\partial(X') = \mathbf{d}_\partial(X)$, every boundary vertex has the same final degree in G' as it had in G. Vertices of Y are unchanged, and internal vertices of X' have degree at least 3 by hypothesis. Hence $\delta(G') \geq 3$.

We claim that G' has no power-of-two cycle. Let C be a cycle of G', and suppose for contradiction that its length is a power of two. If C is wholly contained in Y, then C is also a cycle of G, contradicting that G is a counterexample. If C is wholly contained in X', then C is an internal power-of-two cycle of X', forbidden by hypothesis (iii). Otherwise C uses internal vertices of both X' and Y; since X' and Y meet only in the boundary ∂, the cycle C lives in $G' = X' \oplus_T Y$ and contains a power-of-two cycle, so by definition of the obstruction profile $Y \in \mathsf{H}_T(X')$. Hypothesis (i) gives $X' \preceq_T X$, that is $\mathsf{H}_T(X') \subseteq \mathsf{H}_T(X)$, whence $Y \in \mathsf{H}_T(X)$. By definition this means $X \oplus_T Y = G$ contains a power-of-two cycle, contradicting that G is a counterexample. Hence G' has no power-of-two cycle. Together with $\delta(G') \geq 3$ and hypothesis (v), G' is a counterexample strictly smaller than G in the lexicographic order, contradicting the minimality of G. □

Definition 3.97 (Target-complete compression)**.** *Let X be a proper boundaried piece. A nontrivial target-complete compression of X is a smaller T-boundaried representative X' satisfying the hypotheses of Lemma 3.96. A quotient or identification of local response coordinates that is not target-complete is called a target-defective quotient.*

Corollary 3.98 (Hereditary target-uncompressibility)**.** *No proper boundaried piece of G admits a nontrivial target-complete compression. Moreover, a non-target-complete identification cannot be used as a replacement that preserves the counterexample predicate: by definition of target completeness and Lemma 3.95, some outside context distinguishes the identified states by producing a power-of-two cycle from one and not from the other.*

Proof. The first assertion is exactly Lemma 3.96. The second assertion is Lemma 3.95. □

Definition 3.99 (Target tester and independent target coordinates)**.** *Let X be a T-boundaried graph. A target tester for a local coordinate r of X is a T-boundaried context Y_r such that the gluing $X \oplus_T Y_r$ contains a power-of-two cycle if and only if the coordinate r takes the obstructing value, while all other coordinates under discussion are held fixed.*

A finite family of local coordinates $\mathcal{I} = \{r_1, \ldots, r_k\}$ is independently target-testable *if its target responses realize 2^k distinct target-complete states; equivalently, for every subset $J \subseteq \{1, \ldots, k\}$ there is a target-complete state of the boundaried piece in which precisely the coordinates indexed by J are obstructing. The* target rank *of a coordinate family is the maximum size of an independently target-testable subfamily.*

Lemma 3.100 (Independent target coordinates consume skeleton entropy)**.** *Let $\mathcal{I}$ be an independently target-testable family of size k arising canonically from graphs in a labelled graph class $\mathcal{G}$. Then*
$$2^k \leq |\mathcal{G}|.$$

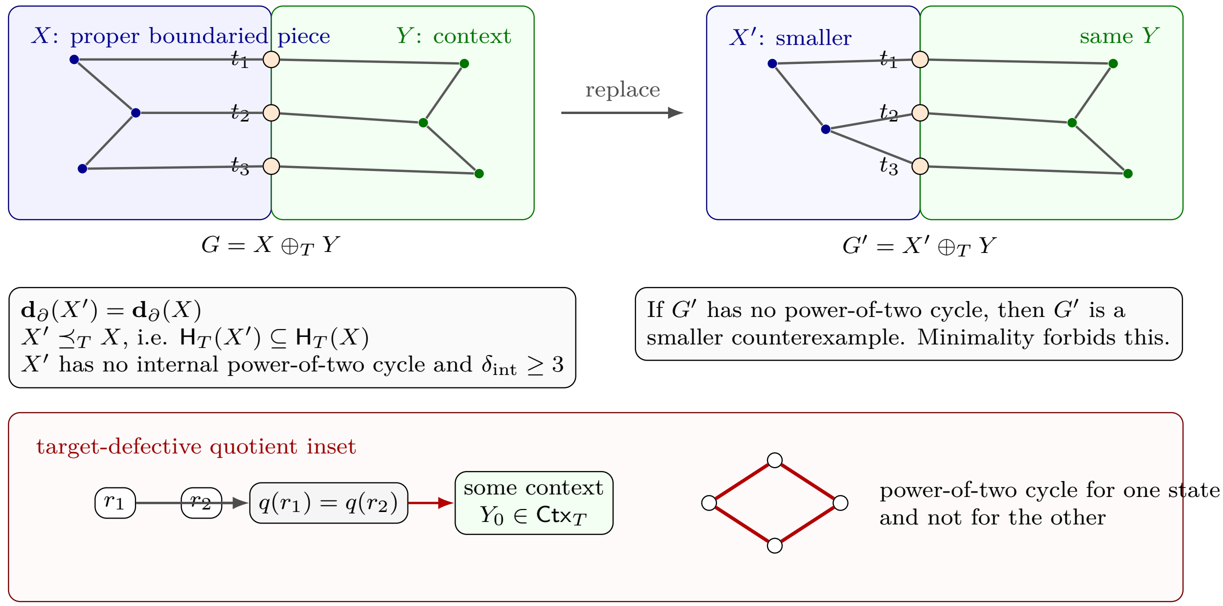

Figure 16: Visual definition of boundaried replacement and target-completeness. The left panel shows a proper boundaried piece X glued to its context Y along the labelled boundary $T = \{t_1, t_2, t_3\}$, whose orange vertices are identified in $G = X \oplus_T Y$. The right panel shows a smaller T-boundaried representative X' glued to the same context. The hypotheses $\mathbf{d}_\partial(X') = \mathbf{d}_\partial(X)$ and $X' \preceq_T X$ mean that the boundary degrees are in the same fibre and every context obstructed by X' is also obstructed by X. If X' has no internal power-of-two cycle and all internal vertices keep degree at least 3, then $X' \oplus_T Y$ would be a smaller counterexample, contradicting minimality. The lower inset records the complementary failure mode: identifying two response coordinates r_1, r_2 is target-defective when some context Y_0 creates a power-of-two cycle for one state and not for the other, so the quotient is not target-complete.

In particular,

$$k \leq \log_2 |\mathcal{G}|.$$

Proof. By independent target-testability, the family $\mathcal{I}$ realizes 2^k distinct target-complete states. If two distinct such states came from the same labelled graph, the canonical target-response map of that graph would be identical in both cases, contradiction. Hence the number of states is at most the number of labelled graphs in the ambient class. □

Framework branch table. The target-algebra and replacement section instantiates the framework branch table as follows.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Target hit	An edge-rooted Mersenne return is present.	No residual demand remains.	The target predicate.	Not applicable.	It is a power-of-two cycle by Lemma 2.2.

Deletion	Deleting an edge would leave a smaller counterexample.	The deleted edge.	Lexicographic minimality.	One edge at a time.	Contradicts Lemma 3.2.
Boundary fibre	A quotient identifies supports with different boundary-degree profiles.	The boundary-degree discrepancy.	The fixed boundary fibre.	One canonical boundary profile per fibre.	Forbidden by Lemma 3.94.
Target defect	An outside context distinguishes two identified coordinates.	A target-defective quotient.	The defect account attached to its declared support.	The finite declared coordinate signature.	Routed by Lemma 3.95.
Compression	A proper boundaried piece has a smaller target-complete representative.	The proper support size.	Minimality.	One replacement per proper boundaried piece.	Contradicts Lemma 3.96 and Corollary 3.98.
Entropy coordinate	Independent target coordinates are realized.	The number of target states.	The labelled skeleton count.	The canonical state map.	Bounded by Lemma 3.100.

4 Sparse rank parameters

Define

$$\beta = m - n + 1, \qquad \lambda = 2n - 3 - m.$$

Then

$$\boxed{\beta + \lambda = n - 2.}$$

Define the surplus above cubic degree

$$\sigma = \sum_{v \in V(G)} (d(v) - 3) = 2m - 3n.$$

Then

$$\boxed{\beta = \frac{n}{2} + 1 + \frac{\sigma}{2}}, \qquad \boxed{\lambda = \frac{n}{2} - 3 - \frac{\sigma}{2}}.$$

The sparse skeleton entropy in the sparse (λ-parametrized) branch is

$$B_{\mathrm{skel}}(n, \lambda) = \log_2 \binom{\binom{n}{2}}{2n - 3 - \lambda}.$$

Definition 4.1 (Near-cubic spine hypothesis)**.** *The* near-cubic spine hypothesis *is the assertion that every lexicographically minimal counterexample surviving to the entropy/net-charge spine satisfies*

$$m = \frac{3}{2}n + O(\sqrt{n}), \qquad \sigma(G) = O(\sqrt{n}).$$

Equivalently, the global surplus is $o(n)$, and the labelled skeleton budget available to the spine is

$$\frac{3}{2}n\log_2 n + o(n\log n).$$

In the final proof, this estimate is supplied by Proposition 3.80 before either the normalized skeleton budget $1.5\,n\log_2 n+o(n\log n)$ or the fan-mass estimate $\sigma(G)=o(|R|)$ is invoked. When a later local lemma states Definition 4.1 among its hypotheses, that line records the branch state needed for the normalized estimate, not an additional global assumption. The local surplus bookkeeping relevant to deriving this hypothesis is recorded in Definitions 3.6 and 3.9. The degree-greater-than-four part is separated by Corollary 3.12 and Remark 3.19; the degree-four local activation is Corollary 3.13; and the non-near-cubic surplus branch is routed by the exact token ledger and the same-token bottleneck closure Lemma 3.77, Theorem 3.78, and Proposition 3.80.

The following lemmas make explicit why this is a safe *upper* bound on every degree of freedom available to a near-cubic graph. The entropy budget is not an encoding assumption: it is merely the number of labelled adjacency matrices in the ambient class.

Lemma 4.2 (Labelled skeleton budget dominates all auxiliary data)**.** *Fix n and m. Let $\mathcal{G}_{n,m}$ be the set of all finite simple labelled graphs on vertex set $[n]$ with exactly m edges. Then*

$$|\mathcal{G}_{n,m}| = \binom{\binom{n}{2}}{m}.$$

Consequently, if a collection of target-complete states is assigned canonically to graphs in $\mathcal{G}_{n,m}$, then the number of realized states is at most

$$\binom{\binom{n}{2}}{m}.$$

In particular, any target-complete response algebra that realizes more than

$$2^{B_{\mathrm{skel}}(n,\lambda)}$$

states among graphs with $m=2n-3-\lambda$ edges cannot be realized by such graphs.

Proof. A labelled simple graph on $[n]$ is uniquely determined by its edge set, a subset of the $\binom{n}{2}$ possible unordered pairs. Thus the number with m edges is exactly $\binom{\binom{n}{2}}{m}$.

All auxiliary objects used in the proof—a maximal induced-P_{13} packing, boundaried-piece decomposition, boundary profiles, spanning-tree choices, and obstruction-response data—are functions of the labelled adjacency matrix once a deterministic tie-breaking rule is fixed. We fix throughout the lexicographically first object among all objects with the required extremal property. Thus no auxiliary structure supplies independent degrees of freedom beyond the adjacency matrix. Hence the number of realized target-complete states is bounded by the number of labelled skeletons. ☐

Lemma 4.3 (Variable edge-count budget)**.** *Let $I \subseteq \{0,1,\dots,\binom{n}{2}\}$ be any interval of possible edge counts. Then the number of labelled graphs on $[n]$ whose edge count lies in I is at most*

$$|I|\max_{m\in I}\binom{\binom{n}{2}}{m}.$$

Consequently the logarithmic budget differs from the largest fixed-m budget by at most $\log_2|I| \le \log_2 n^2 = 2\log_2 n$ whenever I has length at most n^2.

Proof. The graphs with different edge counts are disjoint, so the total number is the sum of $\binom{\binom{n}{2}}{m}$ over $m \in I$. This sum is at most $|I|$ times its largest summand. Taking logarithms gives the claim. $\square$

Lemma 4.4 (Near-cubic specialization of the skeleton budget)**.** *Suppose*

$$m = \frac{3}{2}n + O(\sqrt{n}).$$

Then

$$\log_2 \binom{\binom{n}{2}}{m} = \frac{3}{2}n \log_2 n + O(n) + O(\sqrt{n} \log n).$$

In particular,

$$B_{\mathrm{skel}}(n, \lambda) = \frac{3}{2}n \log_2 n + o(n \log n)$$

in the near-cubic branch.

Proof. Let $N = \binom{n}{2}$. The standard binomial estimate

$$\binom{N}{m} \leq \left(\frac{eN}{m}\right)^m$$

gives

$$\log_2 \binom{N}{m} \leq m \log_2 \left(\frac{eN}{m}\right).$$

If $m = \frac{3}{2}n + O(\sqrt{n})$, then $N = \frac{1}{2}n^2 + O(n)$ and therefore

$$\frac{eN}{m} = \Theta(n).$$

Hence

$$\log_2 \binom{N}{m} \leq \left(\frac{3}{2}n + O(\sqrt{n})\right)(\log_2 n + O(1)) = \frac{3}{2}n \log_2 n + O(n) + O(\sqrt{n} \log n).$$

The matching lower asymptotic is not needed in the proof; the displayed upper bound is what is used. Since $O(n) + O(\sqrt{n} \log n) = o(n \log n)$, the final statement follows. $\square$

Remark 4.5 (Scope of the near-cubic budget)**.** *Lemma 4.4 is the conversion step from $m = \frac{3}{2}n + O(\sqrt{n})$ to the entropy budget used later. The surplus-token routing branch supplies this edge-count estimate before the lemma is invoked in the final proof.*

Remark 4.6 (Robustness of the skeleton budget)**.** *The budget counts all labelled simple graphs with the given edge count. The near-cubic, minimality, boundaried-piece, obstruction, and target constraints only restrict this set. Automorphisms only reduce the number of unlabelled graphs, so labelled counting is an overcount. Degree sequences are not additional data: they are determined by the edge set. Likewise, packings, pieces, trees, and profiles are treated canonically and are deterministic functions of the graph. Thus one cannot obtain extra graph-theoretic degrees of freedom by varying these auxiliary choices; doing so would only count the same labelled graph multiple times.*

Lemma 4.7 (State-count comparison)**.** *Let $\mathcal{S}$ be a family of target-complete states realized by labelled graphs in a class $\mathcal{G}$. If the state map is canonical, then*

$$|\mathcal{S}| \le |\mathcal{G}|.$$

Equivalently, if the proof establishes K independently testable Boolean target coordinates, so that at least 2^K target-complete states are realized, then necessarily

$$K \le \log_2 |\mathcal{G}|.$$

Proof. The canonical state map sends each graph to one state. Therefore the number of realized states cannot exceed the number of graphs. If K independent Boolean target coordinates are realized, they give at least 2^K distinct target-complete states. Hence $2^K \le |\mathcal{G}|$ and the logarithmic inequality follows. □

Framework branch table. The sparse-rank section supplies the global budget branch table.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure
Fixed edge count	States are assigned to a fixed-m labelled graph class.	Target-complete states.	Labelled edge sets.	One canonical state per labelled graph.	Bounded by Lemma 4.2.
Variable edge count	The edge count varies over an interval.	Extra edge-count choices.	The largest fixed-m class.	One edge-count value per graph.	Absorbed by Lemma 4.3.
Near-cubic budget	$m = \frac{3}{2}n + O(\sqrt{n})$.	Skeleton entropy.	$\frac{3}{2}n \log_2 n + o(n \log n)$.	The labelled graph count.	Computed in Lemma 4.4.
State overflow	Independent target states exceed the skeleton count.	Independent Boolean coordinates.	The labelled skeleton budget.	One canonical graph-to-state map.	Contradicts Lemma 4.7.
Non-near-cubic route	The surplus-token branch survives the sparse exits.	Same-token load.	The capacity-token ledger.	The token-fibre partition.	Closed or routed by Proposition 3.80.

5 Induced P_{13} windows

We use the following external theorem.

Theorem 5.1 (Hegde–Sandeep–Shashank)**.** *Every P_{13}-free graph of minimum degree at least 3 contains a cycle whose length is a power of two.*

Reference. This is Theorem 1 of Hegde, Sandeep, and Shashank [1]. □

Corollary 5.2. *The minimal counterexample G contains an induced P_{13}.*

Proof. If G were P_{13}-free, then since $\delta(G) \geq 3$, Theorem 5.1 would give a power-of-two cycle in G, contradicting that G is a counterexample. ☐

Let p_{13} be the maximum size of a vertex-disjoint family of induced copies of P_{13} in G, and put

$$\theta = \frac{p_{13}}{n}.$$

Framework branch table. The induced-window section contributes the external P_{13}-free input and the maximal packing notation consumed by the window entropy branch.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure
External P_{13}-free input	G is P_{13}-free.	Minimum degree at least 3.	The HSS theorem.	The theorem is consumed only through its stated interface.	Closed by Theorem 5.1.
Induced window exists	The minimal counterexample is tested against the external input.	$\delta(G) \geq 3$ and no power-of-two cycle.	The counterexample condition.	If G were P_{13}-free, HSS gives a forbidden target cycle.	Corollary 5.2.
Maximal window packing	A vertex-disjoint induced-P_{13} family is chosen maximal.	The packing size p_{13}.	The complement R.	Every unchosen induced window meets the packing.	Supplies $\theta = p_{13}/n$ for Proposition 7.42 and the remainder analysis.

6 The P_{13} obstruction algebra

Let

$$P = v_0 v_1 \cdots v_{12}$$

be an induced P_{13}. An outside vertex x has attachment label

$$S(x) = \{i : xv_i \in E(G)\} \subseteq \{0, 1, \dots, 12\}.$$

A nonempty label S is *legal* if a single outside vertex x with attachment label S forms no power-of-two cycle of length 4 or 8 together with P. Indeed, if x is adjacent to v_i and v_j with $i < j$, then the subpath $v_i v_{i+1} \cdots v_j$ of P, which has length $j - i$, together with the two edges xv_i and xv_j forms a cycle of length $(j - i) + 2$. As $1 \leq j - i \leq 12$, this length lies in $\{3, \dots, 14\}$ and is a power of two precisely when $(j - i) + 2 \in \{4, 8\}$, that is $j - i \in \{2, 6\}$. Hence S is legal if and only if

$$|i - j| \notin \{2, 6\} \qquad \forall i, j \in S.$$

Let

$$\mathcal{L} = \{S \subseteq \{0, \dots, 12\} : \ S \neq \varnothing \text{ and } |i - j| \notin \{2, 6\} \ \forall i, j \in S\}$$

be the set of legal nonempty labels.

Lemma 6.1 (Legal label count). *The number of legal nonempty labels is*

$$|\mathcal{L}| = 399.$$

The distribution by size is

$$13, 60, 122, 122, 63, 17, 2$$

for sizes $1, 2, 3, 4, 5, 6, 7$, *respectively.*

Proof. This is a direct enumeration of nonempty subsets of $\{0, \dots, 12\}$ avoiding pairwise differences 2 and 6. The verification code in Section A computes the stated values. □

For $s \geq 0$, define a safety relation C_s on legal labels by

$$C_s(S, T) = 1$$

if and only if

$$s + 2 + |i - j| \notin \mathcal{D} \qquad \forall i \in S,\ j \in T.$$

Thus an outside path of length s between vertices carrying labels S, T is safe through the induced path P precisely when $C_s(S, T) = 1$.

Define the two-step obstruction tensor

$$\Omega_2(S, A, T) = C_1(S, A) C_1(A, T)(1 - C_2(S, T)).$$

So $\Omega_2 = 1$ means two individually safe outside edges compose into an unsafe outside path of length 2.

Lemma 6.2 (Two-step obstruction enumeration). *For the* 399 *legal labels,*

$$|\{(S, A, T) : C_1(S, A) = C_1(A, T) = 1\}| = 543958,$$

and

$$|\{(S, A, T) : \Omega_2(S, A, T) = 1\}| = 432672.$$

Hence the number of non-obstructing locally safe wedges is

$$111286.$$

Therefore the entropy cost of one independent two-step obstruction test is

$$\boxed{c_\Omega = \log_2 \frac{543958}{111286} = 2.28922315244\dots.}$$

Proof. The proof is the same direct finite enumeration as in Lemma 6.1, now applied to ordered triples of legal labels. A locally safe wedge is a triple (S, A, T) with $C_1(S, A) = C_1(A, T) = 1$. It is obstructing when $C_2(S, T) = 0$. The code in Section A performs precisely these tests and returns the displayed integer counts. The entropy value is then the elementary calculation

$$\log_2 \frac{543958}{111286} = 2.28922315244071\dots.$$

□

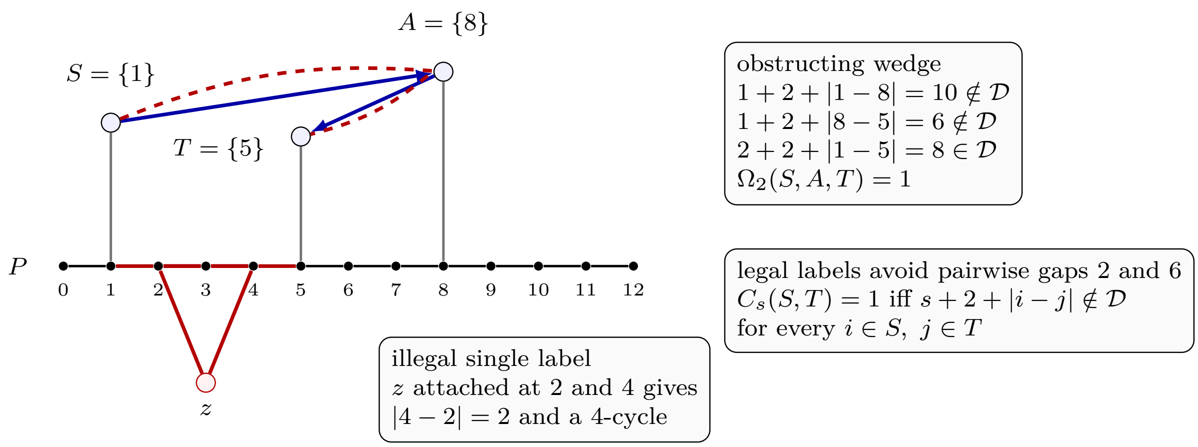

Figure 17: Visual definition of the P_{13} label algebra and the two-step obstruction tensor. The black horizontal path is the induced P_{13}, with indices $0, \dots, 12$ displayed below its vertices. An outside vertex has label equal to the set of path indices to which it is adjacent. The lower red vertex z illustrates an illegal label: attachments at indices 2 and 4 close a 4-cycle through the subpath $v_2v_3v_4$, so legal labels must avoid pairwise gaps 2 and 6. The upper vertices carry the singleton labels $S = \{1\}$, $A = \{8\}$, and $T = \{5\}$. The blue outside edges show the two individually safe length-1 relations $C_1(S, A) = 1$ and $C_1(A, T) = 1$. The dashed red two-step outside path from S to T, together with the red path segment $v_1 \cdots v_5$ and the two attachment edges, has length $2 + 2 + |1 - 5| = 8$; hence $C_2(S, T) = 0$. Thus $\Omega_2(S, A, T) = 1$: two locally safe outside steps compose into an unsafe power-of-two test.

A finite-scale multi-relation obstruction package through length 14 yields the constant

$$\boxed{c_{13} = 118.108581006\dots}$$

bits per independent P_{13}-window package per dyadic scale. This value is the sum of the 91 finite obstruction barriers indexed by ordered pairs of outside path lengths $a, b \geq 1$ with $a + b \leq 14$. The number of such pairs is

$$\left|\{(a, b) : a, b \geq 1,\ a + b \leq 14\}\right| = \sum_{s=2}^{14} (s - 1) = \sum_{k=1}^{13} k = 91,$$

since for each total $s = a + b \in \{2, \dots, 14\}$ there are exactly $s - 1$ admissible pairs $(a, s - a)$ with $1 \leq a \leq s - 1$; the individual barriers are computed in Section A.

Framework branch table. The obstruction-algebra section is a finite computation interface: every quantity it exports is either an integer count or a named entropy rate verified in Section A.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure

Illegal label	An outside vertex attaches at path-index distance 2 or 6.	The attachment label S.	The target predicate.	The induced path segment plus two attachment edges is a 4- or 8-cycle.	Excluded by the counterexample condition.
Legal label enumeration	All nonempty labels avoiding gaps 2 and 6.	The finite set $\mathcal{L}$.	Direct enumeration.	The code tests every subset of $\{0, \dots, 12\}$.	Lemma 6.1.
Two-step obstruction test	$C_1(S, A) = C_1(A, T) = 1$ but $C_2(S, T) = 0$.	The triple (S, A, T).	The finite obstruction tensor.	The same predicates define both the proof and the code check.	Lemma 6.2.
Multi-scale package constant	Ordered path-length pairs $a, b \geq 1$, $a + b \leq 14$.	The 91 barrier rates.	The computational certificate.	The exported value is the displayed sum $c_{13} = 118.108581006\dots$.	Consumed by Lemma 7.1.

7 Entropy bound on independent P_{13} windows

This section carries out the normalized entropy accounting under Definition 4.1 in the high-entropy (window) branch of the routing in Proposition 11.6, in which the remainder contributes its full per-vertex skeleton entropy. Let R be the complement of a maximal disjoint induced-P_{13} packing. The next lemma is the window package consumed by the density bound.

Lemma 7.1 (Independent multi-scale P_{13}-window package)**.** *Assume the sparse surplus exits have been discharged and the near-cubic spine estimate of Definition 4.1 has been supplied. Let $\mathcal{P}$ be a maximal vertex-disjoint induced-P_{13} packing in G. Then the canonical multi-scale package attached to $\mathcal{P}$ contains a family $\mathcal{I}_{\mathrm{win}}$ of independently target-testable coordinates with*

$$|\mathcal{I}_{\mathrm{win}}| \geq (c_{13} - o(1))|\mathcal{P}| \log_2 n, \qquad c_{13} = 118.108581006\dots.$$

The package is the disjoint union of the packages attached to the distinct packed windows. Hence no coordinate is counted for two windows. Moreover, in any entropy comparison in which a baseline spine coordinate family is already present, $\mathcal{I}_{\mathrm{win}}$ is adjoined to that family as independent target coordinates before Lemma 3.100 is applied.

Proof. Fix one packed induced P_{13}, with path vertices $v_0, \dots, v_{12}$. For an outside path of length s, the finite label relation $C_s(S, T)$ records exactly which ordered pairs of legal P_{13}-labels can be joined without creating a power-of-two cycle through the window. For every ordered pair (a, b) with $a, b \geq 1$ and $a + b \leq 14$, let $W_{a,b}$ be the number of locally safe triples (S, A, T) with

$$C_a(S, A) = C_b(A, T) = 1,$$

and let $F_{a,b}$ be the number of those triples that remain safe after the composed outside path is tested, i.e. with $C_{a+b}(S, T) = 1$. The (a, b)-barrier therefore has flatness cost

$$\gamma_{a,b} = \log_2(W_{a,b}/F_{a,b}).$$

The direct finite computation in Section A evaluates these numbers for all

$$\{(a,b) : a, b \geq 1,\ a+b \leq 14\},$$

and gives

$$\sum_{a,b} \gamma_{a,b} = 118.10858100609198\ldots = c_{13}.$$

The multi-scale package uses $\lfloor \log_2 n \rfloor - O(1)$ separated dyadic scales for these finite barriers. The $O(1)$ loss absorbs endpoint collisions with the finitely many reserved boundary and tie-breaking choices inside the canonical packing. At each remaining scale, the target tester is the corresponding edge-rooted Mersenne completion through the window. The relations C_a, C_b, C_{a+b} were defined precisely so that changing the barrier state changes that tester and leaves the other displayed coordinates fixed. Thus, for one window, the package supplies

$$\left(\sum_{a,b} \gamma_{a,b} - o(1)\right) \log_2 n = (c_{13} - o(1)) \log_2 n$$

independently target-testable coordinates.

Distinct packed windows are vertex-disjoint, and the canonical assignment of a coordinate records the unique packed window through which its tester is routed. The product of the target-complete states on the distinct windows is therefore again target-complete; equivalently, the tester for one window does not change the label state assigned to another packed window. The window packages are therefore independent across $\mathcal{P}$, giving the displayed lower bound. The same product argument applies when a baseline spine family has already been declared: the baseline coordinates and the window coordinates are combined into one independently target-testable family, and only then is the labelled skeleton bound of Lemmas 3.100 and 4.2 applied. Hence the window coordinates are not counted twice. □

Definition 7.2 (Hot/cold window ledger)**.** *Set*

$$c_{\text{hot}} := 118.10858100609198\ldots, \qquad \theta_{\text{win}} := \frac{3}{2c_{\text{hot}}} = 0.012700177982179218\ldots.$$

For a packing density $\theta = p_{13}/n$, set

$$\tau(\theta) := \frac{15\theta}{1 - 13\theta}.$$

Then

$$\tau(\theta) < \frac{1}{4} \quad \Longleftrightarrow \quad \theta < \frac{1}{73} = 0.0136986301369863\ldots,$$

and

$$\tau(\theta) < \frac{3}{13} \quad \Longleftrightarrow \quad \theta < \frac{1}{78} = 0.0128205128205128\ldots.$$

The value $1/78$ is the route-8 private-support threshold, and $1/73$ is the negative-net-charge threshold.

Fix a maximal vertex-disjoint induced-P_{13} packing $\mathcal{P}$ and the canonical entropy comparison used on the branch under discussion. A packed window is hot *when that comparison contains, for this window, the full canonical package of independently target-testable coordinates with rate*

$$(c_{\text{hot}} - o(1)) \log_2 n.$$

Equivalently, hotness is witnessed by the injective list of declared window coordinates supplied by Lemma 7.1 for that window and retained in the current comparison. A packed window not carrying such a live package in the current comparison is cold. *We write*

$$\mathcal{P} = \mathcal{P}_{\text{hot}} \sqcup \mathcal{P}_{\text{cold}}, \qquad C := |\mathcal{P}_{\text{cold}}|.$$

Definition 7.3 (Surviving cold branch)**.** *The* surviving cold branch *is the branch on which:*

(i) *no target cycle has been found;*

(ii) *no sparse surplus exit, target-defective quotient, target-complete compression, or support-dependence exit is active;*

(iii) *no unpaid exit-*(4) *peel remains;*

(iv) *no Type B bridge or handoff residual remains outside its ledger;*

(v) *no Type A route-8 residual remains outside Theorem 15.48; and*

(vi) *the sparse surplus branch has already supplied the spine estimate in the form* $m = \frac{3}{2}n + o(n)$ *and* $\sigma(G) = 2m - 3n = o(n)$*.*

On this branch, $|V_{\geq 4}(G)| \leq \sigma(G) = o(n)$*. Hence all but* $o(n)$ *packed* P_{13}*-windows are ambient-cubic.*

Lemma 7.4 (Hot failure forces cold mass)**.** *Assume Definition 7.3. If the live-hot entropy comparison does not close, then*

$$C \geq (\theta - \theta_{\text{win}})n - o(n).$$

In particular, at the route-8 threshold $\theta = 1/78$*,*

$$C \geq 0.000120334838333602\, n - o(n),$$

and at the negative-net-charge threshold $\theta = 1/73$*,*

$$C \geq 0.000998452154807083\, n - o(n).$$

Proof. The hot windows contribute at least

$$(c_{\text{hot}} - o(1))|\mathcal{P}_{\text{hot}}| \log_2 n$$

independently target-testable coordinates. If these coordinates alone exceed the near-cubic skeleton budget $\frac{3}{2}n \log_2 n + o(n \log n)$, then Lemmas 3.100 and 4.2 close the branch. Therefore, on a branch where the hot comparison has not closed,

$$c_{\text{hot}}|\mathcal{P}_{\text{hot}}| \log_2 n \leq \frac{3}{2} n \log_2 n + o(n \log n).$$

Dividing by $c_{\text{hot}} \log_2 n$ gives

$$|\mathcal{P}_{\text{hot}}| \leq \frac{3}{2c_{\text{hot}}} n + o(n) = \theta_{\text{win}} n + o(n).$$

Since $p_{13} = |\mathcal{P}| = \theta n$,

$$C = |\mathcal{P}| - |\mathcal{P}_{\text{hot}}| \geq (\theta - \theta_{\text{win}})n - o(n).$$

The two displayed numerical lower bounds are obtained by substituting $\theta = 1/78$ and $\theta = 1/73$. $\square$

Definition 7.5 (Cold skeleton and branch excess)**.** *Let*

$$\mathcal{P}_{\text{cold}}^{\text{cub}} := \{P \in \mathcal{P}_{\text{cold}} : V(P) \cap V_{\geq 4}(G) = \varnothing\}.$$

By Definition 7.3,

$$|\mathcal{P}_{\text{cold}} \setminus \mathcal{P}_{\text{cold}}^{\text{cub}}| = o(n).$$

For each $P \in \mathcal{P}_{\text{cold}}^{\text{cub}}$, contract the path P to one skeleton vertex p and keep one incident half-edge for every edge of G leaving P. The resulting incidence object is the ambient-cubic cold skeleton $\mathfrak{S}_{\text{cold}}$*. If $s(P)$ is the number of external stubs of P, define the branch-excess contribution of P by*

$$b(P) := \max\{0, s(P) - 2\},$$

and set

$$b(\mathfrak{S}_{\text{cold}}) := \sum_{P \in \mathcal{P}_{\text{cold}}^{\text{cub}}} b(P).$$

The branch-excess units are selected concretely as follows. The global lexicographic order on vertices and edges induces an order on the external stubs of every ambient-cubic cold window. The first two stubs of P are called the transit stubs*; the remaining $s(P) - 2$ stubs, when $s(P) > 2$, are the* selected branch-excess half-edges *of P. Let*

$$\mathcal{E}_{\text{br}}(P)$$

be this selected set and let

$$\mathcal{E}_{\text{br}} := \bigcup_{P} \mathcal{E}_{\text{br}}(P).$$

Thus

$$|\mathcal{E}_{\text{br}}| = b(\mathfrak{S}_{\text{cold}}).$$

All later cold-branch counts are counts of these selected half-edges. Hence a stub can be charged at most once at its cold-window endpoint.

Lemma 7.6 (Cold window stub excess)**.** *On Definition 7.3,*

$$b(\mathfrak{S}_{\text{cold}}) \geq 13C - o(n).$$

Proof. Let P be an ambient-cubic induced P_{13} with vertices $v_0, \ldots, v_{12}$. The internal path edges contribute

$$2 \cdot 12 = 24$$

to the sum of degrees inside P, while the ambient cubic degrees contribute

$$13 \cdot 3 = 39.$$

Thus P has exactly $39 - 24 = 15$ external stubs. In the skeleton, a degree-two corridor through p can absorb at most two of these stubs without branching. Hence $b(P) = 15 - 2 = 13$ for every ambient-cubic cold window. By Definition 7.3, only $o(n)$ cold windows are not ambient-cubic. Summing over the remaining windows gives the claim. $\square$

Lemma 7.7 (Short self-return filter)**.** *Let a cold-window outside self-return have outside length ℓ. Sweeping over the window offsets $\{0, 1, \ldots, 12\}$ tests the whole interval $[\ell, \ell + 12]$. For $1 \le \ell \le 39$, the interval avoids $\{4, 8, 16, 32\}$ only for*

$$\ell \in \{17, 18, 19, 33, 34, 35, 36, 37, 38, 39\}.$$

For $1 \le \ell \le 32$, the only surviving lengths are

$$\ell \in \{17, 18, 19\}.$$

All other short self-returns realize a power-of-two cycle.

Proof. The interval $[\ell, \ell+12]$ contains 4 exactly when $1 \le \ell \le 4$, contains 8 exactly when $1 \le \ell \le 8$, contains 16 exactly when $4 \le \ell \le 16$, and contains 32 exactly when $20 \le \ell \le 32$. Their union inside $\{1, \ldots, 39\}$ is

$$\{1, \ldots, 16\} \cup \{20, \ldots, 32\}.$$

The complement is precisely the displayed set. Intersecting this complement with $\{1, \ldots, 32\}$ gives $\{17, 18, 19\}$. If the interval contains one of $4, 8, 16, 32$, the corresponding offset closes a cycle of that length through the window, contradicting the counterexample condition. □

Definition 7.8 (Cold bounded configuration)**.** *A* cold bounded configuration *is a finite boundaried support with two boundary interfaces x, y and two same-interface x-y representatives $Q[x, y]$ and E. In a terminal corridor these representatives are the two bounded completion strands between the same interfaces; in a repeated-state corridor one representative is the actual corridor segment and the other is the canonical representative determined by the repeated cold corridor state. The configuration also carries the inherited boundary degree profile, P_{13}-window labels, and target-response profile. Its increment is*

$$\delta := |E| - |Q[x, y]|.$$

It is length-changing *if $\delta \neq 0$, and* equal-length *if $\delta = 0$. The word bounded means that the number of internal vertices and all boundary data are bounded by the fixed finite cut-state constant used by the local target algebra; consequently there are only finitely many bounded configuration types.*

Definition 7.9 (Cold corridor states and first failures)**.** *Let X_{cold} be the union of the ambient-cubic cold windows, with their internal path edges retained. Delete the interiors of these windows and look at a connected component K of the remaining outside graph. The* boundary stubs *of K are the edges from K to ambient-cubic cold windows. By Lemma 3.4, no such component has only one boundary stub, since that unique boundary edge would be a bridge of G.*

Order the boundary stubs of K lexicographically as

$$h_1, \ldots, h_m \qquad (m \ge 2)$$

and give them the cyclic successor relation $h_i \mapsto h_{i+1}$, with $h_{m+1} = h_1$. If $h_i = \epsilon \in \mathcal{E}_{\mathrm{br}}(P)$ is a selected branch-excess half-edge, choose inside K the lexicographically first simple path joining the outside endpoint of h_i to the outside endpoint of h_{i+1}. Together with the two boundary stubs h_i, h_{i+1}, this path is the cold return corridor *of ϵ. Thus each selected branch-excess half-edge has exactly one corridor, and each boundary stub is the successor of at most one selected half-edge.*

For an initial segment J of this corridor, let $T(J)$ be its two active boundary interfaces. Its cold corridor state *is the finite two-boundary cut-state obtained from $\rho^{\mathrm{ex}}_{T(J)}(J)$ by retaining exactly the boundary-degree profile, the two active boundary half-edges, the cold-window offsets met at the two* *interfaces, and the declared local coordinates of Definition 3.87 whose support is contained in the bounded active interface. It is not the full labelled prefix. If two prefixes have the same finite cut-state but differ in exact target response against some compatible context, that discrepancy is recorded as a first failure of type* (F2) *below. Thus, after excluding (F2), equality of cold corridor states is equality for every target-response coordinate used by the local replacement. Since the boundary has size two and the retained window and local-coordinate labels are finite, the set of possible cold corridor states is bounded by a constant* Q_{cold} *depending only on the fixed declared signature. Set*

$$M_{\text{cold}} := Q_{\text{cold}} + 30, \qquad B_{\text{cold}} := 15\big(1 + 3(2^{M_{\text{cold}}+2} - 1)\big), \qquad D_{\text{cold}} := M_{\text{cold}}B_{\text{cold}} + 1.$$

The additive 30 *only covers the two* P_{13}*-window interfaces and the two boundary stubs. Thus* M_{cold} *is a uniform upper bound for the number of vertices in a first-failure cold exchange, and* B_{cold} *is a uniform upper bound for the number of selected cold half-edges whose first-failure support can contain a fixed subcubic vertex.*

The first failure *of* ϵ *is the first initial segment, in the lexicographic order along the corridor, at which one of the following happens:*

- *(F1) a compatible completion through a cold-window offset realizes a power-of-two cycle;*
- *(F2) two prefixes with the same displayed boundary data have different target response against some compatible outside context;*
- *(F3) two prefixes have the same exact target response against every outside context and one gives a strictly smaller proper representative;*
- *(F4) the corridor first enters a declared Type B handoff envelope or the route-8 response support already recorded in the branch state;*
- *(F5) no earlier alternative occurs and either the corridor reaches its successor boundary stub before* $Q_{\text{cold}} + 1$ *states have been read, or a cold corridor state repeats.*

In case (F5), the bounded support is the whole terminal corridor in the first subcase, and the support between the two equal states in the second subcase. With its two boundary interfaces, this bounded support is the first-failure cold exchange *of* ϵ*.*

Lemma 7.10 (Cold corridor first-failure routing)**.** *Assume Definition 7.3. For every* $\epsilon \in \mathcal{E}_{\text{br}}$*, the cold return corridor of* ϵ *has a first failure. The first failure is routed as follows:*

- *(i) case* (F1) *is a power-of-two cycle in* G*;*
- *(ii) case* (F2) *is a target-defective quotient, hence belongs to the sparse exit or to the exit-*(4) *ledger;*
- *(iii) case* (F3) *is a target-complete compression of a proper support;*
- *(iv) case* (F4) *is an already named Type B or route-8 handoff;*
- *(v) case* (F5) *is a cold bounded configuration in the sense of Definition 7.8.*

Consequently, on the surviving cold branch, every selected branch-excess half-edge that is not in the already discarded $o(n)$ *handoff or surplus ledgers supplies a canonical bounded-configuration incidence.*

Proof. The return corridor exists by Lemma 3.4. Follow its initial segments until one of the alternatives in Definition 7.9 occurs. If the successor boundary stub is reached before $Q_{\mathrm{cold}} + 1$ states have been read, then the whole corridor, together with the two boundary stubs and the two P_{13}-window interfaces, has at most $M_{\mathrm{cold}} = Q_{\mathrm{cold}} + 30$ vertices and gives the terminal subcase of (F5). Otherwise, before the successor is reached, $Q_{\mathrm{cold}} + 1$ states are read; two states are equal by the definition of Q_{cold}, and the first such equality gives the repeat subcase of (F5). Hence a first failure always exists.

If the first failure is (F1), the displayed completion is literally a cycle of power-of-two length in G, impossible in a counterexample. If it is (F2), the actual quotient is valid only for the current outside context and fails for another compatible context. By Lemma 3.95, this is not target-complete, so it is a target-defective quotient. In the global branch ledger such defects are exactly the sparse exits; in the Type A location they are the exit-(4) peels, and both are excluded in Definition 7.3.

In (F3), context-universality holds and the displayed representative is strictly smaller while preserving the boundary-degree profile and the exact target response. Thus it satisfies Definition 3.89 and is forbidden by Lemma 3.96 and Corollary 3.98. In (F4), the corridor has reached precisely one of the declared interfaces of Definition 14.19 or the route-8 response support, so the charge is transferred to the already existing Type B or route-8 ledger.

It remains to consider (F5). In the terminal subcase, the whole corridor has bounded size and two boundary interfaces. In the repeat subcase, the two repeated states have the same two boundary interfaces, the same inherited boundary-degree profile, the same window labels, and the same exact target-response coordinates; the support between them is bounded by Q_{cold}. In both subcases the support carries two same-interface representatives: the actual corridor representative and the terminal or repeated-state exchange representative with the same retained cut-state. Thus it is a cold bounded configuration in the sense of Definition 7.8. All cases except (F5) are excluded on the surviving branch, up to the $o(n)$ windows and corridors already charged to the non-ambient-cubic, handoff, or surplus ledgers. Hence every remaining selected branch-excess half-edge supplies a canonical bounded-configuration incidence. □

Lemma 7.11 (Cold branch-excess extracts bounded configurations)**.** *Assume Definition 7.3. If the cold skeleton has branch excess $b = b(\mathfrak{S}_{\mathrm{cold}})$, then it contains a family of pairwise vertex-disjoint candidate cold bounded configurations of size*

$$N_{\mathrm{conf}} \geq \frac{b}{D_{\mathrm{cold}}} - o(n).$$

In particular, after hot failure,

$$N_{\mathrm{conf}} \geq \frac{13C}{D_{\mathrm{cold}}} - o(n) \geq \frac{13(\theta - \theta_{\mathrm{win}})}{D_{\mathrm{cold}}} n - o(n).$$

Proof. By Definition 7.5, there are exactly $b = |\mathcal{E}_{\mathrm{br}}|$ selected branch-excess half-edges, and each is selected once. By Lemma 7.10, after deleting the already routed $o(n)$ surplus, non-ambient-cubic, Type B, and route-8 handoff incidences, each remaining selected half-edge supplies one canonical first-failure cold bounded configuration. Let $\mathcal{G}_{\mathrm{cand}}$ be this candidate family. Then

$$|\mathcal{G}_{\mathrm{cand}}| \geq b - o(n).$$

It remains only to extract a vertex-disjoint subfamily without double-counting overlapping local supports. Each member of $\mathcal{G}_{\mathrm{cand}}$ has at most M_{cold} vertices by Definition 7.9. If a candidate support contains a vertex of degree at least 4, then the corresponding corridor first enters the high-degree handoff ledger and was already removed; hence every remaining candidate lies in the subcubic part of the surviving branch. In a subcubic graph the ball of radius $M_{\mathrm{cold}} + 2$ around a vertex has at most

$$1 + 3(2^{M_{\mathrm{cold}}+2} - 1)$$

vertices, and every ambient-cubic cold window contributes at most 15 external stubs. Therefore a fixed vertex belongs to the support of at most

$$B_{\mathrm{cold}} = 15\big(1 + 3(2^{M_{\mathrm{cold}}+2} - 1)\big)$$

candidate configurations. A fixed candidate support has at most M_{cold} vertices, so it meets at most $M_{\mathrm{cold}}B_{\mathrm{cold}}$ other candidate supports. The intersection graph of $\mathcal{G}_{\mathrm{cand}}$ thus has maximum degree at most $M_{\mathrm{cold}}B_{\mathrm{cold}}$.

A greedy maximal independent set in a graph of maximum degree Δ has size at least the number of vertices divided by $\Delta + 1$. With $\Delta = M_{\mathrm{cold}}B_{\mathrm{cold}}$, this gives a vertex-disjoint candidate configuration family of size

$$N_{\mathrm{conf}} \geq \frac{|\mathcal{G}_{\mathrm{cand}}|}{M_{\mathrm{cold}}B_{\mathrm{cold}} + 1} \geq \frac{b}{D_{\mathrm{cold}}} - o(n).$$

The remaining inequalities use Lemmas 7.4 and 7.6. □

Lemma 7.12 (Bounded-configuration trichotomy)**.** *No length-changing cold bounded configuration survives on Definition 7.3. More explicitly, every such configuration falls into one of the following three cases:*

G1. Hit-realized. *Some compatible live completion and window offset close a power-of-two cycle. This contradicts the counterexample condition.*

G2. Hit-distinguished. *Some compatible outside context distinguishes the two representatives by power-of-two truth value without already realizing the cycle in the current graph. The induced quotient is target-defective, so it is routed to the sparse exit or exit-*(4) *ledger.*

G3. Silent. *No compatible outside context and no relevant scale distinguishes the two representatives. Then replacing the longer representative by the shorter one preserves the boundary degree profile and the target response against every context, creates no power-of-two cycle, and strictly decreases the support. This is a nontrivial target-complete compression of a proper support.*

Each case is impossible on the surviving cold branch.

Proof. The cases are exhaustive by whether a compatible completion realizes a power-of-two hit, distinguishes power-of-two truth without realization in G, or never distinguishes the two representatives.

In G1, the completed representative plus the relevant window segment is a cycle whose length is a power of two. This contradicts the definition of a counterexample.

In G2, the two local responses agree in the actual quotient but disagree in a compatible context. By Lemma 3.95, such an identification is not target-complete; equivalently it is a target-defective quotient. This is one of the named surplus exits or, inside Type A, an exit-(4) peel. Both are excluded by Definition 7.3.

In G3, all compatible contexts give the same target response for the two representatives. Because $\delta \neq 0$, one representative is strictly longer. Replacing the longer representative by the shorter one keeps the same boundary vertices, the same boundary degree profile, and the same target response against every context. It cannot create a power-of-two cycle, since any such cycle would distinguish the two responses. Thus the replacement is a smaller target-complete representative of a proper support, forbidden by Lemma 3.96 and Corollary 3.98. Hence no length-changing bounded configuration survives. □

Definition 7.13 (Finite same-interface cold table)**.** *The* same-interface cold table *is the finite table whose rows are equal-length cold bounded configurations and the short self-return exceptions of Lemma 7.7, recorded with:*

(T1) the two boundary vertices and their boundary-degree profile;

(T2) the two terminal cold-window stubs and their offsets in $\{0, \dots, 12\}$*;*

(T3) the exact response profile generated by Definition 3.87;

(T4) the target truth value of every compatible completion represented by that exact profile.

The table is finite because the support size, boundary size, window labels, and declared coordinate labels are bounded by Definition 7.9. A row is realizing *if some compatible completion gives a power-of-two cycle,* distinguishing *if two same-interface representatives have different target truth value in some compatible context, and* neutral *otherwise.*

Lemma 7.14 (Same-interface cold table closure)**.** *Every row of the same-interface cold table is routed to one of the already closed outcomes: a power-of-two cycle, a target-defective quotient, an existing Type B or route-8 handoff, or a target-complete proper-support compression. In particular, an equal-length cold bounded configuration and a short exceptional self-return cannot be a terminal cold residual.*

Proof. Let R be a row of the table. If R is realizing, the corresponding completion is a power-of-two cycle in G, which contradicts the counterexample condition. If R first meets a declared Type B or route-8 interface, then by the definition of the surviving cold branch the charge is transferred to that already closed ledger.

Assume R is not realizing and does not enter a handoff interface. If there is a compatible context distinguishing the two same-interface representatives, then Lemma 3.95 says that their identification is not target-complete. This is a target-defective quotient, hence a sparse exit or an exit-(4) peel, both excluded by Definition 7.3.

It remains to handle a neutral row. Neutrality means that the two representatives have the same boundary-degree profile and the same target response against every compatible context. The row is a first-failure row: otherwise it would not have entered the cold table. Therefore the quotient identifying the two same-interface representatives removes at least one declared branch-excess coordinate while preserving every coordinate needed by the target response. By Definition 3.91, such a proper-support target-complete quotient is admissible only when it has a strictly smaller proper representative in the sense of Definition 3.89. That representative satisfies the hypotheses of Lemma 3.96, and is forbidden by Corollary 3.98. Hence no neutral row survives either. □

Lemma 7.15 (Increment arithmetic for cold configurations)**.** *Let a homogeneous cold configuration have increment* δ*, oriented so* $\delta \geq 0$*. The following cases exhaust the finite arithmetic:*

(a) If $1 \leq \delta \leq 12$*, then the achievable length blocks*

$$[L + j\delta, L + j\delta + 12] \qquad (0 \leq j \leq N)$$

overlap. If their union spans a dyadic scale, the branch is G1; otherwise the whole family lies in one bounded dyadic gap and is a bounded configuration type handled by Lemma 7.12.

(b) *If $\delta \geq 13$ and the doubling orbit modulo δ hits one of the thirteen smear residues $L, L+1, \ldots, L+12$, the branch is G1.*

(c) *If $\delta \geq 13$ and the doubling orbit avoids all thirteen smear residues, the odd part of δ gives a periodic response class. A target-visible response class is G2, and a target-invisible response class is G3.*

(d) *If $\delta = 0$, the equal-length switch belongs to the finite same-interface cold table; each table row is closed by Lemma 7.14.*

For odd δ, the sufficient hit criterion

$$\mathrm{ord}_\delta(2) > \delta - 13$$

forces case (b). For even $\delta = 2^a u$, the same criterion applies to the odd part u after the bounded transient $k < a$.

Proof. For (a), consecutive blocks overlap because the next block begins at $L+(j+1)\delta \leq L+j\delta+12$. Hence their union is an interval. An interval containing a power of two gives G1; if it contains none, all corresponding completions live inside one bounded gap between consecutive powers of two, so only finitely many target-response types occur and Lemma 7.12 applies.

For (b), a congruence $2^k \equiv L + r \pmod{\delta}$ with $0 \leq r \leq 12$ means that, after adding the appropriate number of homogeneous copies, one attainable length is 2^k. This is exactly a hit-realized configuration.

For (c), avoidance of all smear residues means the orbit of powers of two is confined to the complement of the thirteen target residues modulo the odd part of δ. The residue class is then a periodic response class. If some compatible context sees the response class, the two representative responses are target-distinguished and G2 applies. If no compatible context sees it, the response class is target-equivalent and G3 applies.

For (d), equal-length switches cannot be charged as length-changing increments. They are finite because the configuration support and boundary data are bounded, and Lemma 7.14 closes every row of the finite same-interface table.

If δ is odd and $\mathrm{ord}_\delta(2) > \delta - 13$, the doubling orbit has more residues than the complement of the thirteen smear residues. Therefore it must hit a smear residue, giving (b). If $\delta = 2^a u$, the initial powers with $k < a$ form a bounded transient, and for $k \geq a$ the congruence reduces to the odd modulus u. □

Table 22: Cold branch quantitative residual ledger.

Case	**Trigger**	**Quantitative constraint**	**Charged resource**	**Outcome**
0	Sparse non-spine branch	$\sigma(G)$ routes to sparse exit, Type B, or $o(n)$	sparse capacity-token ledger	closed or spine supplied
1	$\theta < 1/78$	$\tau(\theta) < 3/13$	route-8 support collision	closed

Case	Trigger	Quantitative constraint	Charged resource	Outcome
2	$1/78 \le \theta < 1/73$	delicate density interval	hot/cold branch	routed through rows 4–16
3	$\theta \ge 1/73$	$\tau(\theta) \ge 1/4$	hot/cold branch	routed through rows 4–16
4	Hot package live	$(c_{\text{hot}} - o(1))\lvert\mathcal{P}_{\text{hot}}\rvert \log_2 n$ bits	entropy budget	density cap
5	Hot branch fails	$C \ge (\theta - \theta_{\text{win}})n - o(n)$	cold skeleton	linear cold family
6	Non-cubic cold windows	at most $\sigma(G) = o(n)$	surplus ledger	sublinear loss
7	Ambient-cubic cold window	exactly 15 external stubs	stub count	13 branch-excess units
8	Cold skeleton extraction	$N_{\text{conf}} \ge 13C/D_{\text{cold}} - o(n)$	first-failure and bounded-overlap constant D_{cold}	bounded configurations
9	Short self-return	only $17, 18, 19, 33, \dots, 39$ survive for $\ell \le 39$	finite short table	hit or bounded configuration
10	G1	live completion hits a shifted power	target algebra	power-of-two cycle
11	G2	context distinguishes the representatives	target-defect ledger	sparse exit or exit-(4)
12	G3	no context distinguishes the representatives	replacement	compression contradiction
13	$1 \le \delta \le 12$	overlapping length blocks	target algebra or finite table	G1, G2, or G3
14	$\delta \ge 13$ orbit hits smear	$2^k \equiv L + r \pmod{\delta}$, $0 \le r \le 12$	target algebra	G1
15	$\delta \ge 13$ orbit avoids smear	periodic response class on odd part	response-class quotient	G2 or G3
16	$\delta = 0$	finite same-interface cold table	target response table	hit, defect, handoff, or compression

Theorem 7.16 (Cold branch quantitative closure)**.** *No terminal cold branch survives after the near-cubic spine estimate has been supplied. More precisely, on Definition 7.3, either the route-8 response-support inequality closes $\theta < 1/78$, or the live-hot entropy comparison closes, or the hot failure produces a cold family whose branch excess yields a bounded configuration; that configuration is routed by Lemmas 7.10, 7.12, 7.14 and 7.15 to a power-of-two cycle, a target-defective quotient or exit-*(4) *route, a Type B handoff already in its ledger, a route-8 closure, or a target-complete compression. All these outcomes are excluded on the surviving cold branch.*

Proof. If $\theta < 1/78$, then $\tau(\theta) < 3/13$ by Definition 7.2; this is exactly the private-support inequality used in Theorem 15.48, so the route-8 branch closes.

Assume next that $\theta \ge 1/78$. If the live-hot entropy comparison closes, Lemmas 3.100 and 4.2

give the density cap and the branch has no cold residual. If it does not close, then Lemma 7.4 gives

$$C \geq (\theta - \theta_{\mathrm{win}})n - o(n).$$

The route-8 threshold already gives

$$C \geq 0.000120334838333602\, n - o(n),$$

so C is linear on every remaining branch. By Lemma 7.6,

$$b(\mathfrak{S}_{\mathrm{cold}}) \geq 13C - o(n),$$

and by Lemma 7.11,

$$N_{\mathrm{conf}} \geq \frac{13C}{D_{\mathrm{cold}}} - o(n).$$

Thus the remaining branch contains bounded candidate configurations linearly many times unless one of the discarded $o(n)$ surplus or handoff ledgers is already active. Since D_{cold} is a fixed constant and $\theta_{\mathrm{win}} < 1/78$, the displayed lower bound is positive for all sufficiently large n; hence at least one bounded candidate configuration is present on every remaining branch. A length-changing candidate is impossible by Lemma 7.12; an equal-length or exceptional short candidate is routed by Lemmas 7.14 and 7.15. The outcomes are precisely the rows of Table 22: target cycle, target defect or exit-(4), already routed Type B or route-8 handoff, or target-complete compression. Each is forbidden in Definition 7.3. Hence the cold branch has no terminal survivor. □

7.1 The dense-packing residual: nodes [158]–[164]

The exact finite reading of the realization sentence used in Lemma 7.1 and Proposition 7.42 ("all target-complete window states are realized by labelled near-cubic skeletons") is a branch test, not an assertion: it holds or fails on the residual at hand. This subsection records that test and the branch it opens.

Definition 7.17 (Window-package realization test)**.** *Let $\mathcal{P}$ be the fixed maximal packing and let $\mathcal{G}_{n,m}$ be the labelled skeleton class of Lemma 4.2. The joint window package of $\mathcal{P}$ is* realized *if there is an assignment of target-complete states to labelled skeletons whose range has at least $2^{c_{13}p_{13}\log_2 n}$ elements, i.e. if the full package code of $\mathcal{P}$ is realized canonically by $\mathcal{G}_{n,m}$. Node [158] decides this. Its yes-branch is the hot/cold split [22] as written; its no-branch is the* dense-packing residual *[159], on which*

$$2^{c_{13}p_{13}\log_2 n} > |\mathcal{G}_{n,m}|, \qquad \textit{equivalently} \qquad \theta > \theta_{\mathrm{win}} + o(1).$$

Lemma 7.18 (Dense deficiency routing)**.** *On the dense-packing residual, if $\tau(\theta) = 15\theta/(1-13\theta) < 1/4$ in the exact form of node [56] (with the $\sqrt{n}$ allowance of the near-cubic spine), then the negative-net-charge collision Proposition 13.7 and Lemma 13.4 and its localization [57]–[64] apply verbatim, with the deficiency cap of node [161] in place of the density cap [24]; the residual therefore enters node [25] and is closed by the Type A and Type B continuations exactly as the spine is.*

Proof. Nodes [56]–[64] consume the density cap only through the inequality $\partial_+(R) - \sigma(R) < \frac{1}{4}|R|$, i.e. $\tau < 1/4$ after the exact $\sqrt{n}$ allowance; node [161] supplies precisely that inequality from the deficiency test [160], and nothing else in [25]–[64] reads the density cap. The Type A and Type B continuations read the same cap through [56]. □

Lemma 7.19 (Dense hot/cold pass)**.** *On the dense-packing residual with $\tau(\theta) \geq 1/4$, the hot/cold split [22] and the cold branch [145]–[157] can be executed unchanged, and every closure of that pass fires as on the spine: the live-hot overflow [23], the route-8 arm [146]/[147] and the live-hot cap [149], the bounded arm of [153] returning to [24], and, on the linear arm, the first-failure outcomes (F1)/G1 (a power-of-two cycle), (F2)/G2 (a target-defective identification, Lemma 3.95) and (F3)/G3 (a target-complete compression of a proper support, Lemma 3.96 and Corollary 3.98). The residual outcomes of the pass are exactly the two nodes [163] and [164] below.*

Proof. The pass uses only facts of the current residual: the packing, the label algebra, bridgelessness (Lemma 3.4), the return corridors, and the first-failure routing of Lemma 7.10; the density sentence is not among them. The three named refutations are the ones of Lemma 7.12. Since the boundaried pieces of R are induced-P_{13}-free and subcubic, they have bounded diameter, so every return corridor is terminal in the sense of the (F5) terminal subcase, and the only outcome not refuted by a ledger fact is the neutral equal-length terminal configuration; the only entropy comparison not closed by the realized hot family is the one with $\mathcal{P}_{\mathrm{hot}} = \varnothing$. ☐

Definition 7.20 (Neutral equal-length terminal configuration; node [163])**.** *] On the dense residual, a terminal (F5) configuration whose two same-interface strands Q (the return corridor) and E (the terminal exchange representative of its retained cut-state) between the window attachment points have equal length ℓ and are context-equivalent: no compatible outside context and no scale distinguishes them. Exchanging Q and E therefore preserves the vertex and edge counts, the boundary-degree profile, the baseline, and the target response in every context. A neutral configuration is a* symmetry *of the residual, and it carries no target information: neither the trichotomy of Lemma 7.12 nor the label relations C_s of node [18] see it, and by Definition 3.91 its identification of the two strands is an abstract identification of labels that reduces no graph-realizable rank.*

Lemma 7.21 (Symmetry split of the neutral configuration; node [163])**.** *] A neutral equal-length terminal configuration is of exactly one of two kinds: either its second representative E is a canonical replacement piece different from the corridor Q as a boundaried piece (node [165]), or E is a genuine second strand of G between the same attachment vertices, so that Q and E form a symmetric strand pair (node [167]).*

Proof. By Definition 7.8, the second representative is either the canonical representative determined by the retained cut-state or the second bounded completion strand of the terminal corridor; in the first case it is a piece to be exchanged for Q, in the second it is a subgraph of G. The two cases are exclusive because a canonical representative equal to Q as a piece is the trivial configuration, which is not a proper configuration. ☐

Lemma 7.22 (Refined minimality closes the canonical-replacement case; nodes [165]**.** *–[166]] Refine the minimality of node [4] to the lexicographic order on $(|V|, |E|, \Phi)$, where $\Phi(G)$ is the multiset of the boundaried pieces of the canonical decomposition of G in the fixed canonical order on bounded boundaried pieces (the order in which canonical representatives are chosen). If a neutral bounded configuration has a canonical replacement piece $E \neq Q$, then the graph G' obtained by exchanging Q for E is a counterexample with the same $(|V|, |E|)$ and $\Phi(G') < \Phi(G)$, contradicting minimality. Hence on a minimal counterexample every neutral configuration has $Q = E$ and node [165] is empty.*

Proof. G' has the same vertex and edge counts because $|E| = |Q|$; it has minimum degree at least three because E has the same boundary-degree profile as Q and internal degree at least three by Definition 3.89(d); and it has no power-of-two cycle because Q and E have the same target response against every compatible context, in particular against the actual complement of the configuration. The exchange takes place inside one boundaried piece of the canonical decomposition, whose stubs and window labels are unchanged (the retained cut-state records them), so $\Phi(G')$ is $\Phi(G)$ with that piece replaced by a piece that precedes it in the canonical order; the multiset order therefore strictly decreases. Every earlier use of minimality in the proof compares graphs of strictly smaller $(|V|, |E|)$, which remain smaller in the refined order. $\square$

Lemma 7.23 (Two-strand finite check; node [167]**.** *] Let Q and E be a symmetric strand pair: two internally disjoint outside strands of the same length ℓ (edges, both stubs included) between two attachment coordinates of a packed window at distance d. The bounded support of the pair closes exactly the cycles $\ell + d$ (each strand with the window segment) and 2ℓ (the two strands). Hence the pair is a power-of-two hit, node [155], whenever $\ell + d \in \mathcal{D}$ or $2\ell \in \mathcal{D}$; in particular every symmetric strand pair with $\ell \in \{2, 4, 8, 16, \dots\}$ is closed by its pair cycle at every gap. The finite enumeration at the window order* 13 *and strand lengths $\ell \le 40$ (formalized as `Graph/TwoStrandEnumeration.lean`, `survivors 13 40`) closes* 96 *of the* 533 *configurations; the surviving configurations are exactly the pairs with $\ell \notin \mathcal{D}$ and $\ell + d \notin \mathcal{D}$, and their strand lengths are all lengths except* $2, 4, 8, 16, 32$.

Proof. The two strands share their attachment vertices and are internally disjoint, so their union is a cycle of length 2ℓ, and each strand together with the window subpath of length d between the attachment coordinates is a cycle of length $\ell + d$; no other cycle lies in the bounded support. The target accepts a length exactly when it is a power of two, which gives the criterion; the enumeration is a finite computation over $\ell \le 40$, $0 \le d \le 12$. $\square$

Lemma 7.24 (Symmetric pairs attach only at endpoints; node [168]**.** *] Let P be an ambient-cubic packed window. Every interior vertex of P has exactly two window neighbours and hence exactly $\delta - 2 = 1$ external stub; only the two endpoints have $\delta - 1 = 2$. A genuine symmetric strand pair needs two distinct stubs at each of its attachment vertices, so its attachment vertices are endpoints of P (and $d = 12$ unless the pair returns to one endpoint), a window carries at most one such pair, on its four endpoint stubs, and the* 11 *interior stubs are single-stub attachments. Consequently, if the selected branch-excess half-edges of Definition 7.5 are taken among the interior stubs —* 11 *per window,* 9 *after the two corridor ends are absorbed — no selected half-edge is the strand of a genuine symmetric pair; the surviving configurations of Lemma 7.23 ($\ell \notin \mathcal{D}$, $\ell + d \notin \mathcal{D}$) never occur among the selected half-edges, and the extraction of Lemmas 7.6 and 7.11 survives with $9C$ in place of $13C$.*

Proof. An interior path vertex is adjacent to its two path neighbours and, the window being ambient-cubic and induced, to no other window vertex; its remaining $\delta - 2$ incidences are stubs. The endpoints have one path neighbour and $\delta - 1$ stubs. Two internally disjoint strands from a common attachment vertex use two distinct stubs there. The stub count of Lemma 7.6 splits as $11 \cdot 1 + 2 \cdot 2 = 15$, so choosing the selected half-edges among the 11 interior stubs leaves 9 branch-excess units per window and excludes every genuine symmetric pair from the selection. (Formalized: `Graph/WindowStubStructure.lean`, ledger fact `coldWindowStubStructure` on the dense linear residual.) $\square$

Definition 7.25 (All-cold entropy comparison; node [164]**.** *] On the dense residual, the entropy comparison [53] with $\mathcal{P}_{\text{hot}} = \varnothing$: the remainder code $|\mathcal{G}(R)|$ alone (the window package contributing nothing) exceeds every realizable skeleton state count. It is closed by Lemma 7.26.*

Lemma 7.26 (Remainder glue injection)**.** *Let $\mathcal{G}(R)$ be the remainder class of Definition 11.1 read with its glue constraints: subcubicity on the boundaried-piece part, componentwise P_{13}-freeness, no* *internal 3-core, positive net-deficiency density at most the inherited cap, and the inherited edge count $e(H) = e(G[R])$ (equivalently, the inherited net-deficiency numerator). Then the map*

$$H \longmapsto G[R := H],$$

which keeps the packed windows and all window–remainder incidences of G and replaces $G[R]$ by H, is an injection of $\mathcal{G}(R)$ into the labelled skeleton class $\mathcal{G}_{n,m}$. Consequently $|\mathcal{G}(R)| \leq |\mathcal{G}_{n,m}|$, the remainder code is realized by construction, and the all-cold comparison [164] cannot be active.

Proof. For $H \in \mathcal{G}(R)$ the glued graph has vertex set $V(G)$, the outer edges of G and the edges of H, hence m edges by the inherited edge count, and it is a labelled simple graph on n vertices, i.e. a member of $\mathcal{G}_{n,m}$. Two candidates that glue to the same graph have the same edges inside $V(R)$, so the map is injective. On the all-cold arm the comparison [53] demands $|\mathcal{G}(R)|$ states (the obstruction sharpening of Definition 12.1 is realized inside $\mathcal{G}(R)$ and is not charged again, cf. Remark 12.4); the injection realizes each of them by a distinct labelled skeleton, so the demand is at most the budget and [53] cannot be active. □

Remark 7.27 (Where the density bound is proved)**.** *With [158]–[164] in place, Proposition 7.42 is proved on the yes-branch of [158] (the realization sentence) and, on its no-branch, for $\tau(\theta) < 1/4$ by Lemma 7.18; on the dense arm the neutral configuration [163] is split by Lemma 7.21 into the canonical-replacement case, closed by the refined minimality of Lemma 7.22, and the symmetric strand pair, closed by the two-strand check of Lemma 7.23 whenever a closing length is a power of two, and, for the surviving pairs, by the endpoint structure of Lemma 7.24 (they are never selected interior half-edges). The all-cold comparison [164] is closed by the remainder glue injection of Lemma 7.26. The dense regime therefore has no open node; what remains for the formal ledger is bookkeeping (the exchange representative E as a piece, the refined minimality order, and the restriction of the extraction to interior half-edges), recorded in the audit.*

7.2 The small-order repair: nodes [173]–[177]

The collision of node [56] was written with asymptotic allowances: the surplus bound $\sigma(G) \leq C_{\mathrm{sp}}\sqrt{n}$ of node [19], the configuration-extraction loss $B_{\mathrm{cold}}\,\sigma(G)$ of the bounded arm of node [153], and the dyadic-scale factor of the density cap. Read that way, the collision needs n larger than a threshold N_0 determined by those allowances, and the range $n < N_0$ would be a separate obligation. The allowances are, however, bounds on quantities that the residual already carries exactly (σ_W, σ_R, $|\mathcal{P}_{\mathrm{hot}}|$, $C = |\mathcal{P}_{\mathrm{cold}}|$, and the skeleton budget $|\mathcal{G}_{n,m}|$), so the collision is decided on the object, and its failure is a structural residual, not a size condition.

Lemma 7.28 (Exact collision test)**.** *On the large-budget residual, node [56]'s collision is the exact inequality*

$$15p_{13} + \sigma_W - \sigma_R < \tfrac{1}{4}\,(n - 13p_{13}), \qquad p_{13} = |\mathcal{P}_{\mathrm{hot}}| + C,$$

in which $|\mathcal{P}_{\mathrm{hot}}|$ is bounded by the exact entropy comparison $2^{c_{13}\log_2 n\,|\mathcal{P}_{\mathrm{hot}}|} \leq |\mathcal{G}_{n,m}|$ of node [22]. Node [173] decides this inequality on the current object. If it holds, nodes [58]–[64] follow verbatim; no condition on n is used. If it fails, then

$$C \;\geq\; \frac{n - 73|\mathcal{P}_{\mathrm{hot}}| - 4(\sigma_W - \sigma_R)}{73},$$

so the residual carries linearly many cold windows, and it lies on the bounded arm of node [153] (the linear arm has already been closed), i.e. its cold mass was charged as the configuration-extraction loss (per-window excess $+\,B_{\mathrm{cold}}$) $\sigma(G)$.

Proof. The stub supply of node [29] is exact, $\partial_+(R) \le 15p_{13} + \sigma_W$, and $|R| = n - 13p_{13}$; the density input of node [24] is exact after node [22] (the hot family's package code is realized and counted against $|\mathcal{G}_{n,m}|$), and the cold count is a fact of the residual. Hence the collision is an integer inequality of the object. Its failure rearranges to the displayed lower bound on C, and node [153]'s linear arm has been closed, so the residual is the bounded arm, whose only content is the configuration-extraction loss. □

Lemma 7.29 (Absorbed configurations are fan data)**.** *Let ϵ be a selected branch-excess half-edge of a cold window on the absorbed-configuration residual, with return corridor and first-failure support J. Exactly one of the following holds.*

(i) J contains no vertex of degree at least 4. Then J is subcubic, the count of Lemma 7.11 charges ϵ in full, and its first failure is (F1)–(F5) exactly as in Lemma 7.10; the (F5) configuration is closed by Lemmas 7.12, 7.14 and 7.21 to 7.24 (node [176]).

(ii) J contains a vertex z of degree at least 4. Since $V_{\ge 4}(G)$ is independent (node [10]), every neighbour of z has degree 3, and the corridor enters z through one of its incidences and leaves through another: z is a heavy centre and the corridor's two configurations at z are separated connector tails in the sense of Lemma 14.25 and Definition 14.19. The half-edge ϵ is therefore decorated handoff fan data at z, and it enters the Type B branch at node [65] (node [177]), where the heavy-neighbourhood normal form, the fan certificate, and the B1/B2 ledgers of nodes [67]–[85] close it.

In particular the loss $B_{\text{cold}}\,\sigma(G)$ of the bounded arm of node [153] is not a loss: every half-edge it discards is charged to the Type B ledger, and after that charge the bounded arm's cold mass is exactly the mass of the genuine configurations, which node [153]'s linear arm has closed.

Proof. The dichotomy is by whether J meets $V_{\ge 4}(G)$. In case (i) the subcubic ball count behind B_{cold} is not consumed, so the half-edge is one of the candidates of Lemma 7.11 and is routed as there. In case (ii), a first-failure support is a bounded connected piece with two active interfaces, and a degree-≥ 4 vertex on it has all its neighbours of degree 3 by node [10]; the two corridor incidences at z are distinct, so the segments of the corridor on either side of z are two connector tails separated at z, which is the decorated handoff configuration of Lemma 14.25: the charge is transferred to the Type B ledger and node [65] receives z as its heavy centre. Node [153]'s bounded arm bounds the mass of case (ii) by (per-window excess$+B_{\text{cold}}$)$\sigma(G)$; with case (ii) charged to Type B, the remaining cold mass consists of case (i) half-edges, which is what the linear arm extracts and closes. □

Remark 7.30 (No sufficient-order condition)**.** *With nodes [173]–[177] the collision of node [56] is decided exactly on every residual and the absorbed configurations are charged to the ledger that closes them. Consequently the sufficient-order reading of node [57] ($n \ge N_0$) is never used: every asymptotic allowance of the near-cubic spine has been replaced by the corresponding exact quantity of the residual, and the theorem is proved for all orders.*

7.3 The trivial neutral configuration: compression closure, nodes [169]–[172]

After nodes [165]–[168] the dense residual is left with the case $Q = E$: every selected corridor is terminal, neutral, and its own canonical representative. No bounded local test sees anything wrong with such a graph: this residual is exactly the one on which the manuscript's entropy sentence does its work. We make that sentence exact as a compression statement, and we isolate the one step that carries it.

Definition 7.31 (Blocked class)**.** *Fix the vertex set $V(G)$, the edge count m, and the fixed maximal packing $\mathcal{P}$ with its window positions. The* blocked class $\mathcal{B}(\mathcal{P}) \subseteq \mathcal{G}^{\delta \geq 3}_{n,m}$ *is the set of labelled graphs on $V(G)$ with m edges and* minimum degree at least three *that contain every window of $\mathcal{P}$ as an induced P_{13} at its position and in which no cycle of length a power of two passes through any window. The minimum-degree clause is essential and is the structure the plain skeleton budget $\binom{\binom{n}{2}}{m}$ does not charge: see Remark 7.40. A window is* blocked at scale 2^j *in H if no cycle of length 2^j of H passes through it. On the trivial neutral-configuration residual, $G \in \mathcal{B}(\mathcal{P})$, and every window of G is blocked at every scale.*

Definition 7.32 (Fixed-scale barrier overlap system)**.** *Fix a separated scale 2^j, a barrier (a, b), the edges outside the window interiors, and all barrier states exposed before (j, a, b) in the canonical encoding order. For a packed window P, a* completion support *is the canonical simple edge-rooted completion used to test its (a, b)-barrier state at scale 2^j, including the two outside arms, their first and last window incidences, and every window segment used by the completion. Two window coordinates* overlap *when their completion supports meet outside their root-window interiors.*

For a finite set $\mathcal{U} \subseteq \mathcal{P}$, let $\mathcal{S}(\mathcal{U})$ be the set of joint (a, b)-barrier states on $\mathcal{U}$ realized by graphs in the current conditional fibre. A barrier overlap obstruction *is a nonempty $\mathcal{U}$ for which no ordering $P_1, \ldots, P_t$ has*

$$\left|\mathcal{S}(P_i \mid P_1, \ldots, P_{i-1})\right| \leq F_{a,b} \qquad (1 \leq i \leq t)$$

in every nonempty conditional fibre. It is minimal *when every proper nonempty subfamily has such an ordering. Its* overlap support *is the union of its completion supports and the intervening window segments; a disconnected union is treated componentwise.*

Lemma 7.33 (Failure of additivity produces a local overlap obstruction)**.** *If the conditional saving $\gamma_{a,b}$ fails to add across the windows at the fixed scale 2^j, then the current conditional fibre contains a minimal barrier overlap obstruction. Its overlap support is connected.*

Proof. For a family whose canonical completion supports are pairwise disjoint outside their root windows, changing the state at one root changes no label, outside edge, or completion path occurring in another root state. The finite calculation of Section A therefore applies after conditioning on all the other root states and leaves at most $F_{a,b}$ of the $W_{a,b}$ local states. Exposing such coordinates in any order gives the product bound.

Consequently, failure of the product bound is witnessed by a finite family for which no exposure order works. Choose one minimal by cardinality. If its overlap support were disconnected, expose the connected components one at a time. States in distinct components have disjoint declared supports, while minimality supplies an admissible order inside each proper component. Concatenating these orders would give an admissible order for the whole family, a contradiction. ☐

Definition 7.34 (Serial window system and system increments)**.** *A* serial window system *in a fixed-scale overlap support consists of ordered interfaces*

$$x_0, x_1, \ldots, x_s$$

and, for each i, a finite nonempty family of mutually internally disjoint x_{i-1}–x_i corridor pieces. Pieces belonging to distinct indices have disjoint interiors. Two fixed closing pieces outside the system meet it only at x_0, x_s. Thus choosing one corridor piece at every index and one available window offset produces a simple cycle.

Choose the shortest corridor at the i-th interface as its base corridor. The positive differences between the other corridor lengths and the base length are the system increments*, retained with* *multiplicity. The system is* scale-spanning *when, after deleting any bounded number of cells at its two ends, its shortest and longest realizable cycles still lie on opposite sides of* 2^j*, after the available offsets* $0, \ldots, 12$ *are included.*

Lemma 7.35 (Local uncrossing and serial realizability)**.** *Let* $\mathcal{U}$ *be a minimal barrier overlap obstruction on the trivial neutral-configuration residual. Exactly one of the following occurs.*

(i) an uncrossing realizes a power-of-two cycle;

(ii) the uncrossed responses are distinguished by a compatible context, so the support is G2;

(iii) the responses are context-equivalent and one is a smaller proper representative, so the support is G3;

(iv) the first nonserial intersection is a declared Type B handoff or a route-8 response-support interface;

(v) after deleting common subpaths, the overlap support is a scale-spanning serial window system. Every nonzero system increment is bounded by

$$D_{\mathrm{sys}} := 2M_{\mathrm{cold}} + 26,$$

and every choice counted in its spectrum is an actual simple cycle of G*.*

On node [169], alternatives (i)–(iv) *are already closed.*

Proof. Choose the obstruction first with the fewest completion edges and then with the fewest intersections. Orient all completions from their root stubs. For two intersecting completions, take their first and last common vertices. Between these vertices their union contains two internally disjoint strands. Replacing the strands by the uncrossed pairing preserves the boundary-degree profile and every previously exposed barrier state.

If one pairing closes a cycle of length 2^k, case (i) holds. If an outside context distinguishes the pairings, Lemma 3.95 gives case (ii). If no context distinguishes them and the pieces have different size, the shorter pairing is a target-complete replacement, giving case (iii) by Lemma 3.96 and Corollary 3.98. If their first separation is a high-degree or declared response-support interface, the first-failure routing of Lemma 7.10 gives case (iv). Otherwise the strands are equal-length neutral pieces. Node [166] gives $Q = E$, so they may be identified in the overlap description. This removes an intersection without changing the conditional fibre, contrary to secondary minimality unless the intersection is one common subpath.

Apply this uncrossing successively. A branching intersection has three completion strands at the same first-separation interface. At an ambient-cubic interface this is the cubic-switch configuration of Lemma 14.24; at a high-degree interface it is the handoff configuration of Lemma 14.25 and Definition 14.19. These are case (iv). Otherwise the intersection graph has maximum degree two. A cyclic intersection graph has no distinguished first member; removing any member leaves the same conditioned obstruction, contrary to minimality. The intersection graph is a path. Cutting at its successive common subpaths gives the ordered interfaces and disjoint corridor families of Definition 7.34.

If a corridor cell has more than $2M_{\mathrm{cold}} + 26$ edges, read the cold cut-state from both ends. Before the readings meet, one side reads $Q_{\mathrm{cold}} + 1$ states, so two repeat. The intervening piece is a first-failure exchange of Definition 7.9. Outcomes (F1)–(F4) give cases (i)–(iv). In (F5), a nonzero difference is closed by Lemmas 7.12 and 7.15; a zero difference is $Q = E$ and shortens the overlap description. Hence every cell has the stated bound.

Distinct cells have disjoint interiors. At every interface, the chosen incoming and outgoing strands use the two recorded distinct stubs. Concatenating one strand from each cell with the closing pieces therefore repeats no vertex and gives a simple cycle. If all realizable cycles lay on one side of 2^j, the composed barrier truth value would be fixed by the previously exposed states and the local $F_{a,b}$ bound would hold. If deleting a bounded end segment destroyed this property, the dependence would be supported on that bounded segment and would be an (F5) cold-table row. Neither alternative survives in the minimal obstruction, so the system is scale-spanning. □

Lemma 7.36 (Finite sumset interval for a serial system)**.** *Let $\mathfrak{S}$ be supplied by Lemma 7.35, and let $d_1, \dots, d_q \in \{1, \dots, D_{\mathrm{sys}}\}$ be its nonzero increments, with multiplicity. Call a value frequent when it occurs at least $2D_{\mathrm{sys}}^2$ times, and let g be the gcd of the frequent values. The scale-spanning case has at least one frequent value. Let $\mathcal{R} \subseteq \mathbb{Z}/g\mathbb{Z}$ be the residues obtained by adding an arbitrary submultiset of the remaining increments and one offset in $\{0, \dots, 12\}$. There is a constant C_{sys}, depending only on D_{sys}, such that the realizable cycle-length spectrum contains*

$$L + r + tg \qquad (r \in \mathcal{R}, \quad C_{\mathrm{sys}} \le t \le T_r - C_{\mathrm{sys}}),$$

where L is the shortest system length and T_r is the available number of g-steps after the rare submultiset representing r is fixed. Moreover $|\mathcal{R}| \ge \min\{13, g\}$. Empty end ranges belong to the bounded cold table.

Proof. There are at most D_{sys} increment values. If none were frequent, the system would have boundedly many nonzero cells; after neutral cells are removed by $Q = E$, it would be a bounded cold-table row, contrary to scale-spanning. Hence g is defined.

Fix a choice of the rare increments and divide the frequent values by g. Their gcd is one. The residue proof of the Frobenius lemma gives, using generators at most D_{sys}, a representative of every residue modulo the least generator with value bounded in terms of D_{sys}. The frequency threshold supplies enough further copies of the least generator to fill the central interval. Applying the same argument to complementary corridor choices fills the interval downwards from its maximum. Taking the maximum of the finitely many conductor bounds gives C_{sys}.

The thirteen offsets give thirteen consecutive residues before reduction modulo g, hence at least $\min\{13, g\}$ members of $\mathcal{R}$. Rare increments can only enlarge this set. Every represented sum is an actual simple cycle by Lemma 7.35. □

Lemma 7.37 (System increment arithmetic and periodic response class; node [172)**.** *] On the trivial neutral-configuration residual, a scale-spanning serial window system is impossible. Let g and $\mathcal{R}$ be as in Lemma 7.36.*

(a) If the doubling orbit modulo the odd part of g meets one of the realizable residues in $\mathcal{R}$, the central spectrum contains 2^k.

(b) If the orbit avoids those residues, the residue map on the ordered interfaces is a periodic response class. A compatible context that sees the response class gives G2. If no context sees it, quotienting equal residue states gives G3. If its minimal support reaches a declared trace interface, it is the periodic trace response class of Definition 14.48 and is closed by Proposition 14.59 and Theorem 14.61.

For odd g, the sufficient hit criterion is

$$\mathrm{ord}_g(2) > g - |\mathcal{R}|,$$

and in particular $\mathrm{ord}_g(2) > g - 13$ suffices for $g \ge 13$. For $g = 2^a u$, the same assertion applies to u after $k < a$.

Proof. By Lemma 7.36, away from a bounded end segment every length in each displayed coset is realized by a simple cycle. The system spans the tested scale, so a congruence $2^k \equiv L+r \pmod{g}$, $0 \le r \le 12$, has its quotient in the central coefficient range after the bounded scales have been assigned to the cold table. This proves (a).

Assume avoidance. Give x_i the residue, modulo the odd part of g, of the length of a system route from x_0 to x_i. Two choices give the same residue because their difference is generated by the system increments. Thus the residue is well-defined and transported through every corridor. If two equal-residue states are distinguished by an outside context, the quotient is target-defective by Lemma 3.95, giving G2. Otherwise it is target-complete. On a proper support the shorter residue representative gives G3 by Lemma 3.96 and Corollary 3.98. At a declared route-8 trace interface the residue coordinate uses the same completion-port and packed-window incidences as Definition 14.48; deleting an essential residue incidence is the canonical response-support deletion quotient, closed by Proposition 14.59 and Theorem 14.61.

For odd g, the orbit contains $\mathrm{ord}_g(2)$ distinct residues. More than $g - |\mathcal{R}|$ residues cannot lie in the complement of $\mathcal{R}$. Since $|\mathcal{R}| \ge 13$ for $g \ge 13$, the thirteen-residue criterion follows. For $g = 2^a u$, powers with $k \ge a$ reduce to the orbit modulo u, while the first a powers form the bounded transient. ☐

Lemma 7.38 (Scale additivity or arithmetic closure; node [170])**.** *] Let $2^{j_1} < 2^{j_2} < \cdots < 2^{j_L}$, $L = \lfloor \log_2 n \rfloor - O(1)$, be the separated dyadic scales of Lemma 7.1. For a graph $H \in \mathcal{B}(\mathcal{P})$, a scale 2^j and a barrier (a,b), the (a,b)-barrier state of a window P at scale 2^j is the triple (S,A,T) of labels of P and of the two windows reached from it by outside paths of lengths a and b, read together with the edge-rooted Mersenne completion of P at that scale. The lemma asserts: conditional on the edges of H outside the window interiors, on all barrier states at the scales $2^{j_1}, \dots, 2^{j-1}$, and on the barrier states of the other windows at scale 2^j, the (a,b)-barrier state of P at scale 2^j lies in a set of relative size at most $F_{a,b}/W_{a,b}$ of its a-priori range; hence the conditional savings $\gamma_{a,b} = \log_2(W_{a,b}/F_{a,b})$ add over the barriers (a,b), over the L scales, and over the p windows, or the current graph is closed by Lemma 7.37.*

Proof. This is the exact form of the two sentences of Lemma 7.1 “the multi-scale package uses $\lfloor \log_2 n \rfloor - O(1)$ separated dyadic scales for these finite barriers” and “the product of the target-complete states on the distinct windows is therefore again target-complete”. Within one window and one scale the relative-size bound is the finite computation of Section A: among the locally safe triples (S,A,T) exactly the fraction $F_{a,b}/W_{a,b}$ survives the composed test, and blockedness at scale 2^j is that test. Across scales the separation of the scales makes the completions at scale 2^j disjoint from those at the smaller scales.

At one scale, expose the window states in the canonical order. If every conditional fibre has relative size at most $F_{a,b}/W_{a,b}$, multiplication of the conditional fibre sizes gives the claimed saving. Otherwise Lemma 7.33 supplies a minimal local overlap obstruction. On node [169], Lemma 7.35 turns it into a scale-spanning serial window system, and Lemma 7.37 closes that system. Thus every surviving graph lies on the additive branch. Multiplying over the separated scales and the finite barrier list proves the assertion. ☐

Lemma 7.39 (Blocked graphs compress; node [171])**.** *] Assume Lemma 7.38. Then*

$$|\mathcal{B}(\mathcal{P})| \;\le\; |\mathcal{G}_{n,m}| \cdot 2^{-(c_{13}-o(1))\, p_{13} \log_2 n}, \qquad c_{13} = \sum_{a,b} \gamma_{a,b}.$$

In particular, if $\theta = p_{13}/n > \theta_{\mathrm{win}} + o(1)$, then $|\mathcal{B}(\mathcal{P})| < 1$, so $\mathcal{B}(\mathcal{P}) = \varnothing$ and the trivial neutral-configuration residual is empty.

Proof. Encode $H \in \mathcal{B}(\mathcal{P})$ as follows. First record the edges of H outside the window interiors and the window positions; the number of choices is at most $|\mathcal{G}_{n,m}|$ times the number of ways to place the fixed windows, which is absorbed in the $o(1)$. Then, scale by scale from 2^{j_1} to 2^{j_L} and window by window, record the barrier states of H. By Lemma 7.38 the state of window P at scale 2^j for the barrier (a, b), given everything already recorded, ranges over at most a fraction $F_{a,b}/W_{a,b}$ of its a-priori range, so it costs $\log_2 W_{a,b} - \gamma_{a,b}$ bits instead of $\log_2 W_{a,b}$. The map $H \mapsto$ (outside edges, window positions, all barrier states) is injective, because the barrier states of all windows at all scales together with the outside edges determine the window interiors and their incidences. Summing the savings gives $\sum_{a,b} \gamma_{a,b} \cdot L \cdot p_{13} = (c_{13} - o(1))p_{13} \log_2 n$ bits below $\log_2 |\mathcal{G}_{n,m}| + o(n \log n)$. Since $\log_2 |\mathcal{G}_{n,m}| = \frac{3}{2} n \log_2 n + o(n \log n)$ by Lemma 4.4, the bound is < 0 exactly when $c_{13}\,\theta > \frac{3}{2}$, i.e. $\theta > \theta_{\text{win}}$, and a class with fewer than one element is empty; $G \in \mathcal{B}(\mathcal{P})$ is then impossible. □

Remark 7.40 (Two checks on the blocked class)**.** (a) The class must be near-cubic. *If $\mathcal{B}(\mathcal{P})$ were taken inside all labelled graphs with m edges, it would contain every graph in which the p_{13} windows are isolated induced paths and the remaining $m - 12p_{13}$ edges are arbitrary on the other $n - 13p_{13}$ vertices; that family alone has at least $\binom{\binom{n}{2}}{m} 2^{-12p_{13} \log_2 n - O(n)}$ members, so no compression by $c_{13}p_{13} \log_2 n$ bits with $c_{13} > 12$ is possible there. The entropy step therefore has to be read in the class of labelled graphs with minimum degree at least three, where the fifteen external stubs of every ambient-cubic window are forced; this is the structure not charged by $\binom{\binom{n}{2}}{m}$, and it is why Definition 7.31 carries the minimum-degree clause.* (b) A consistency check of the constant. *In the near-cubic class one can build blocked graphs by making the incidence structure between windows and the components of $G - \bigcup V(P)$ a forest (then no cycle passes through a window); the degree constraints force components with s stubs to have at least $1, 4, 5$ vertices for $s = 3, 2, 1$, and the count shows such graphs exist only for $\theta \le 1/81$. Since $1/81 < \theta_{\text{win}} = 1/78.7\ldots$, this family does not contradict Lemma 7.39; it shows that the threshold cannot be lowered below $1/81$ by this route. Above that threshold a nonforest window-incidence system can survive only through overlapping long completions. The fixed-scale overlap and system-increment lemmas charge precisely that case, without assigning the same completion saving independently to every window that it traverses.*

Remark 7.41 (Where the entropy sentence lives)**.** *Nodes [169]–[172] are the exact location of the manuscript's final window-system step. Disjoint conditional fibres close by the compression of Lemma 7.39. A failure of disjointness is charged once to its minimal connected completion support and closes by Lemmas 7.35 and 7.37; no global completion is charged separately at each window that sees it.*

Proposition 7.42 (Near-cubic P_{13} window density bounds)**.** *Under Definition 4.1, every maximal independent induced-P_{13} packing satisfies the window-only bound*

$$\theta \le \theta_{\text{win}} + o(1), \qquad \theta_{\text{win}} := \frac{1.5}{118.108581006} = 0.01270017798\ldots.$$

In the high-entropy branch of Proposition 11.6, where the remainder also contributes $\frac{1}{10} \log_2 n$ bits per remainder vertex, the sharper bound is

$$\theta \le 0.01198542083\ldots + o(1).$$

Consequently, under Definition 4.1,

$$|R| \ge (1 - 13\theta_{\text{win}})n - o(n) = 0.83489768623\ldots n - o(n),$$

and

$$\frac{15\theta}{1 - 13\theta} \le \tau_{\text{win}} + o(1), \qquad \tau_{\text{win}} := \frac{15\theta_{\text{win}}}{1 - 13\theta_{\text{win}}} = 0.22817486846\ldots < \frac{1}{4}.$$

Proof. By Lemma 7.1, the window coordinates alone contribute

$$(118.108581006 - o(1))\, p_{13} \log_2 n$$

bits. Since all target-complete window states are realized by labelled near-cubic skeletons under Definition 4.1, Lemmas 3.100 and 4.2 give

$$(118.108581006 - o(1))\, p_{13} \log_2 n \le \frac{3}{2} n \log_2 n + o(n \log n).$$

After division by $n \log_2 n$, this is exactly $\theta \le \theta_{\mathrm{win}} + o(1)$.

In the high-entropy branch of Proposition 11.6, the P_{13}-free remainder contributes at least

$$\frac{n - 13p_{13}}{10} \log_2 n$$

additional independently target-testable bits. Adding this contribution to the window package and comparing with the same skeleton budget gives

$$\frac{n - 13p_{13}}{10} \log_2 n + (118.108581006 - o(1)) p_{13} \log_2 n \le \frac{3}{2} n \log_2 n + o(n \log n).$$

After division by $n \log_2 n$, with $\theta = p_{13}/n$, this is

$$\frac{1 - 13\theta}{10} + 118.108581006\, \theta = 0.1 + 116.808581006\, \theta \le 1.5 + o(1). \tag{1}$$

Solving (1) gives

$$\theta \le \frac{1.4}{116.808581006} + o(1) = 0.01198542083\ldots + o(1).$$

The lower bound on $|R|$ is $|R| = (1 - 13\theta)n$, and the bound on $15\theta/(1 - 13\theta)$ follows by monotonicity of this expression on $0 \le \theta < 1/13$. $\square$

Framework branch table. The P_{13}-window entropy and hot/cold closure uses the following branch table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Window package	Packed windows are active after the sparse surplus exits.	Independent window coordinates.	The near-cubic skeleton budget.	The vertex-disjoint window packing.	Lemma 7.1 and Proposition 7.42.
Route-8 threshold	$\theta < 1/78$.	Private-incidence load.	The route-8 response-support inequality.	The response-support ledger.	Closed by Theorem 15.48.
Hot branch	The live-hot entropy comparison applies.	Hot window bits.	Skeleton entropy.	Packed windows.	Gives the density cap of Proposition 7.42.

Cold branch	Hot failure leaves linear cold mass.	Branch-excess stubs.	First-failure extraction.	The bounded constant D_{cold}.	Closed by Theorem 7.16.
High entropy	The remainder pays $\lvert R\rvert \log_2 n/10$ bits.	Window plus remainder bits.	The same skeleton entropy.	One target-coordinate family.	The sharper cap in Proposition 7.42.

8 The P_{13}-free remainder and positive deficiency

Let $\mathcal{P}$ be a maximal vertex-disjoint family of induced copies of P_{13} in G. Let

$$W = \bigcup_{P\in\mathcal{P}} V(P), \qquad R = G - W.$$

By the near-cubic window-only part of Proposition 7.42,

$$|R| \geq 0.83489768623\ldots n - o(n).$$

Definition 8.1 (Internal 3-core)**.** *For a graph or support X, the* internal 3-core *of X is the maximal induced subgraph of X with minimum degree at least* 3*, obtained equivalently by repeatedly deleting vertices of internal degree at most* 2*. The internal* 3*-core is* empty *if no nonempty subgraph of X has minimum internal degree at least* 3*.*

Lemma 8.2 (Empty internal 3-core of the remainder)**.** *Every component of R is P_{13}-free and has empty internal* 3*-core.*

Proof. By maximality of $\mathcal{P}$, the graph R contains no induced P_{13}, since any such copy would extend $\mathcal{P}$. Hence every component of R is P_{13}-free.

Suppose that a component C of R had a nonempty internal 3-core. Then C would contain a subgraph H with $\delta(H) \geq 3$. Passing to the induced subgraph $G[V(H)] \supseteq H$ only adds edges, so $\delta(G[V(H)]) \geq 3$. Moreover, $G[V(H)]$ is an induced subgraph of the P_{13}-free component C, and is therefore P_{13}-free. By Theorem 5.1, $G[V(H)]$ contains a power-of-two cycle. This is also a cycle of G, contrary to the counterexample assumption. □

Definition 8.3 (Positive deficiency and surplus)**.** *For a subgraph $X \subseteq G$, define its positive deficiency by*

$$\partial_+(X) = \sum_{v\in V(X)} \max\{0, 3 - d_X(v)\}.$$

Define the ambient surplus carried by vertices of X by

$$\sigma(X) = \sum_{v\in V(X)} \max\{0, d_G(v) - 3\}.$$

For the whole graph, write

$$\sigma(G) = \sum_{v\in V(G)} (d_G(v) - 3).$$

Positive deficiency, not signed cubic charge, measures external degree demand: high-degree vertices inside X cannot cancel the fact that vertices of internal degree below 3 *need outside incidences.*

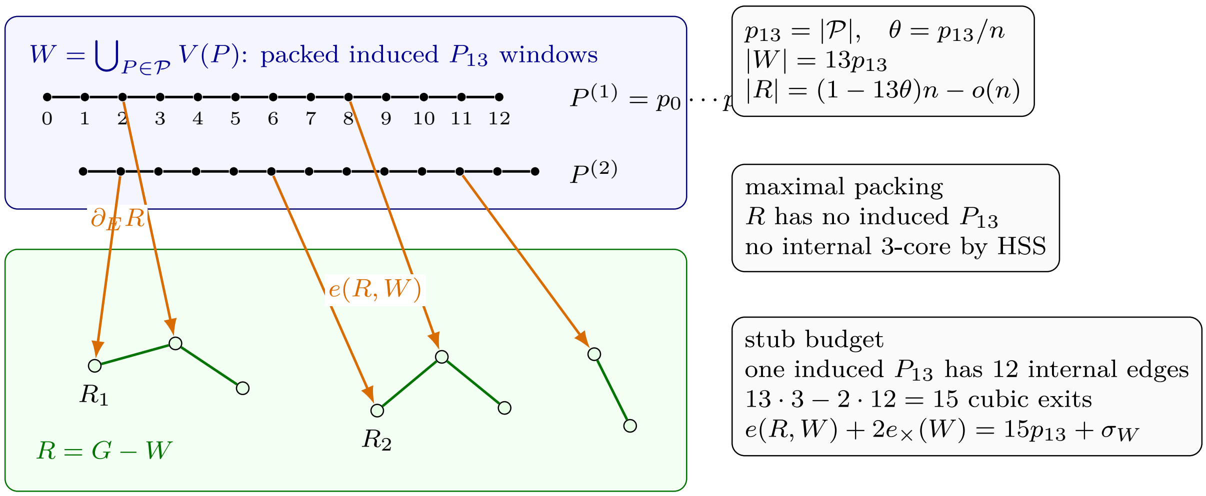

Figure 18: Visual definition of the packed-window and remainder ledger. The blue region is $W = \bigcup_{P\in\mathcal{P}} V(P)$, where $\mathcal{P}$ is a maximal vertex-disjoint family of induced P_{13} windows. Black vertices and black edges are window vertices and their internal path edges; the displayed indices $0, \dots, 12$ name the vertices of $P^{(1)} = p_0 \cdots p_{12}$. The green region is the remainder $R = G - W$; white vertices are remainder vertices, green edges are edges internal to R, and R_1, R_2 mark components of R. The packing size is $p_{13} = |\mathcal{P}|$, its density is $\theta = p_{13}/n$, $|W| = 13p_{13}$, and $|R| = (1 - 13\theta)n - o(n)$. Maximality of $\mathcal{P}$ makes every component of R induced-P_{13}-free; the HSS theorem then forbids any internal subgraph of minimum degree at least 3 inside a component of R. Orange arrows are window–remainder incidences, counted by $e(R, W)$ and by the oriented boundary ledger $\partial_E R$. Since each packed window is induced, $G[W]$ has exactly $12p_{13}$ path edges plus the cross-window edges counted by $e_\times(W)$. The exact degree-sum identity is $e(R, W) + 2e_\times(W) = 15p_{13} + \sigma_W$, so every surplus unit on a window must be spent either as an additional window–remainder incidence or as two half-incidences in a cross-window join.

Definition 8.4 (Window–remainder surplus split)**.** *Let*

$$\sigma_W := \sum_{w\in W} (d_G(w) - 3), \qquad \sigma_R := \sum_{v\in R} (d_G(v) - 3).$$

Since $\delta(G) \geq 3$, these sums are nonnegative,

$$\sigma_R = \sigma(R), \qquad \sigma(G) = \sigma_W + \sigma_R.$$

Let $e_\times(W)$ be the number of edges of G whose endpoints lie in two distinct packed P_{13}-windows. The packed windows are induced, so the edges of $G[W]$ are exactly the $12p_{13}$ path edges inside the windows together with these $e_\times(W)$ cross-window edges.

Lemma 8.5 (External-incidence supply for the P_{13}-free remainder)**.** *Assume Definition 4.1. The P_{13}-free remainder satisfies*

$$\partial_+(R) \leq e(R, W) \leq 15p_{13} + o(n)$$

and hence

$$\partial_+(R) - \sigma_R \leq 15p_{13} + o(n).$$

Equivalently,

$$\frac{\partial_+(R) - \sigma_R}{|R|} \leq \frac{15\theta}{1 - 13\theta} + o(1).$$

Proof. Since $\delta(G) \geq 3$, every vertex $v \in R$ has $d_G(v) \geq 3$, so v is deficient in R only because some of its incidences leave R. Writing $d_G(v) = d_R(v) + e_v$, where e_v is the number of edges from v to $W = G - R$, and using $\max\{0, 3 - d_G(v)\} = 0$,

$$\max\{0, 3 - d_R(v)\} \leq \max\{0, 3 - d_G(v)\} + e_v = e_v.$$

Summing over $v \in R$,

$$\partial_+(R) = \sum_{v \in R} \max\{0, 3 - d_R(v)\} \leq \sum_{v \in R} e_v = e(R, W),$$

the number of edges between R and W.

It remains to bound $e(R, W)$ by the outside-incidence capacity of the windows. Each $P \in \mathcal{P}$ is an *induced* P_{13}, so the subgraph of G on $V(P)$ has exactly its 12 path edges. The number of edges leaving $V(P)$ (to R or to another window) equals the degree sum on $V(P)$ minus twice the internal edges:

$$\sum_{w \in V(P)} d_G(w) - 2 \cdot 12 = \Big(39 + \sum_{w \in V(P)} (d_G(w) - 3)\Big) - 24 = 15 + \sigma(P),$$

where $\sigma(P) = \sum_{w \in V(P)} (d_G(w) - 3)$ is the surplus carried by P and $39 = 13 \cdot 3$ is the cubic degree sum. Every R–W edge leaves some window, so summing over the p_{13} windows,

$$e(R, W) \leq \sum_{P \in \mathcal{P}} (15 + \sigma(P)) = 15p_{13} + \sigma_W \leq 15p_{13} + \sigma(G),$$

where $\sigma_W = \sum_{P \in \mathcal{P}} \sigma(P) \leq \sigma(G)$ since the windows are vertex-disjoint and surplus is nonnegative. Combining the two displays gives

$$\partial_+(R) \leq e(R, W) \leq 15p_{13} + \sigma(G).$$

By Definition 4.1, $\sigma(G) = O(\sqrt{n}) = o(n)$, so

$$\partial_+(R) \leq e(R, W) \leq 15p_{13} + o(n).$$

Since $\sigma_R \geq 0$, this implies

$$\partial_+(R) - \sigma_R \leq 15p_{13} + o(n).$$

Dividing by $|R| = (1 - 13\theta)n + o(n)$ and using $p_{13} = \theta n$ gives the normalized form. □

Lemma 8.6 (Exact window-join identity)**.** *With notation as in Definition 8.4,*

$$e(R, W) + 2e_\times(W) = 15p_{13} + \sigma_W.$$

Proof. Sum degrees over the window vertices. Since $|W| = 13p_{13}$,

$$\sum_{w \in W} d_G(w) = 3|W| + \sigma_W = 39p_{13} + \sigma_W.$$

The same degree sum counts twice the edges internal to W and once the edges from W to R. By Definition 8.4, the internal edges of $G[W]$ are the $12p_{13}$ path edges inside the packed windows and the $e_\times(W)$ cross-window edges. Hence

$$39p_{13} + \sigma_W = 2(12p_{13} + e_\times(W)) + e(R, W).$$

Rearranging gives the identity. □

Lemma 8.7 (Surplus-aware window-stub inequality)**.** *No near-cubic hypothesis is required for the following estimates:*

$$\partial_+(R) \le e(R, W) \le 15p_{13} + \sigma_W$$

and

$$\partial_+(R) - \sigma_R \le 15p_{13} + \sigma_W - \sigma_R.$$

Proof. The proof of Lemma 8.5 gives the first inequality

$$\partial_+(R) \le e(R, W)$$

using only $\delta(G) \ge 3$. By Lemma 8.6,

$$e(R, W) = 15p_{13} + \sigma_W - 2e_\times(W) \le 15p_{13} + \sigma_W,$$

because $e_\times(W) \ge 0$. Subtracting σ_R from the first displayed upper bound gives

$$\partial_+(R) - \sigma_R \le 15p_{13} + \sigma_W - \sigma_R.$$

□

Corollary 8.8 (Boundary-incidence supply)**.** *Assume Definition 4.1. The total positive deficiency of the P_{13}-free remainder satisfies*

$$\partial_+(R) \le 15p_{13} + o(n).$$

Equivalently,

$$\frac{\partial_+(R)}{|R|} \le \tau_{\text{win}} + o(1), \qquad \tau_{\text{win}} = 0.22817486846\ldots.$$

Proof. The proof of Lemma 8.5 gives

$$\partial_+(R) \le e(R, W) \le 15p_{13} + \sigma(G).$$

By Definition 4.1, $\sigma(G) = O(\sqrt{n}) = o(n)$. Hence $\partial_+(R) \le 15p_{13} + o(n)$. Dividing by $|R| = (1 - 13\theta)n + o(n)$ and using $\theta \le \theta_{\text{win}} + o(1)$ from Proposition 7.42 gives the normalized form. □

Framework branch table. The packed-window remainder and deficiency supply close or route residuals as follows.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Forced remainder	The maximal P_{13}-packing leaves R.	Unpacked vertices.	Maximality of the packing.	Componentwise in R.	R is P_{13}-free.
Internal 3-core	A component of R has a subgraph of minimum degree at least 3.	No budget is available.	Theorem 5.1.	One component at a time.	Gives a power-of-two cycle.

Window incidence	Positive deficiency occurs in R.	$\partial_+(R)$.	Window–remainder stubs.	At most 15 cubic exits per packed window, plus surplus.	$\partial_+(R) \le 15p_{13} + o(n)$ by Lemma 8.5 and Corollary 8.8.
Surplus-adjusted supply	Remainder surplus is tracked separately.	$\partial_+(R) - \sigma_R$.	The exact window-join identity.	The stub and cross-window partition.	Lemmas 8.6 and 8.7.

9 Obstruction supply in the piece remainder

For a component C of R, define the number of internal length-two wedges by

$$W_2(C) = \sum_{v \in V(C)} \binom{d_C(v)}{2}.$$

Let

$$W_2(R) = \sum_{C \subseteq R} W_2(C).$$

Lemma 9.1 (Wedge lower bound)**.** *For every component C,*

$$W_2(C) \ge 3|V(C)| - 2\partial_+(C).$$

In particular, under Definition 4.1, in the window-only low-deficiency branch, where $\partial_+(R) \le (\tau_{\mathrm{win}} + o(1))|R|$ with $\tau_{\mathrm{win}} = 0.22817486846\ldots$,

$$W_2(R) \ge \omega_{\mathrm{win}}|R| - o(|R|), \qquad \omega_{\mathrm{win}} = 2.54365026308\ldots.$$

In the sharper high-entropy branch, where $\partial_+(R) \le (0.21296321056\ldots + o(1))|R|$, this improves to

$$W_2(R) \ge \omega|R| - o(|R|), \qquad \omega = 2.57407357888\ldots.$$

Proof. For a component C, let

$$n_i = |\{v \in V(C) : d_C(v) = i\}| \qquad (i \ge 0).$$

Then

$$|V(C)| = \sum_{i \ge 0} n_i,$$

$$\partial_+(C) = 3n_0 + 2n_1 + n_2,$$

and

$$W_2(C) = \sum_{i \ge 0} \binom{i}{2} n_i.$$

A direct calculation gives

$$W_2(C) - (3|V(C)| - 2\partial_+(C)) = 3n_0 + n_1 + \sum_{i \ge 3} \left(\binom{i}{2} - 3 \right) n_i \ge 0.$$

Thus

$$W_2(C) \geq 3|V(C)| - 2\partial_+(C).$$

Summing over the connected components C of R, and using $\sum_C |V(C)| = |R|$ and $\sum_C \partial_+(C) = \partial_+(R)$ (within a component $d_C = d_R$, so the per-component deficiencies add up to $\partial_+(R)$),

$$W_2(R) = \sum_{C \subseteq R} W_2(C) \geq 3|R| - 2\partial_+(R).$$

In the near-cubic window-only low-deficiency branch, Corollary 8.8 gives

$$\partial_+(R) \leq (\tau_{\text{win}} + o(1))|R|,$$

with $\tau_{\text{win}} = 0.22817486846\ldots$. Therefore

$$W_2(R) \geq (3 - 2 \cdot 0.22817486846\ldots - o(1))\,|R| = (\omega_{\text{win}} - o(1))|R|,$$

where

$$\omega_{\text{win}} = 3 - 2 \cdot 0.22817486846\ldots = 2.54365026308\ldots.$$

If the sharper high-entropy cap is available, then $\partial_+(R)/|R| \leq 0.21296321056\ldots + o(1)$. Substituting this value into $W_2(R) \geq 3|R| - 2\partial_+(R)$ gives

$$W_2(R) \geq (3 - 2 \cdot 0.21296321056\ldots - o(1))\,|R| = (\omega - o(1))|R|,$$

where the constant is

$$\omega = 3 - 2 \cdot 0.21296321056\ldots = 2.57407357888\ldots.$$

□

Framework branch table. The obstruction-supply section instantiates the framework table as follows.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Component wedge count	C is a component of R.	$W_2(C)$.	Positive deficiency.	The degree-count identity by internal degree.	$W_2(C) \geq 3\|V(C)\| - 2\partial_+(C)$ by Lemma 9.1.
Window-only demand	$\partial_+(R) \leq (\tau_{\text{win}} + o(1))\|R\|$.	Linear wedge supply.	The window-stub deficiency cap.	Summation over components.	$W_2(R) \geq \omega_{\text{win}}\|R\| - o(\|R\|)$.
High-entropy demand	The sharper high-entropy deficiency cap is available.	Larger wedge supply.	The high-entropy density cap.	The same degree-count identity.	$W_2(R) \geq \omega\|R\| - o(\|R\|)$.

10 Obstruction rank and rank forcing

Definition 10.1 (Obstruction target-rank). *For a boundaried piece C, let $\mathcal{W}_2(C)$ be the set of raw internal length-two obstruction tests in C. A subfamily $\mathcal{A} \subseteq \mathcal{W}_2(C)$* survives *an admissible rank quotient q when q is label-injective on $\mathcal{A}$ in the sense of Definition 3.88. It survives the admissible quotient system when it survives every functional admissible rank quotient of Definitions 3.91 and 3.92. The obstruction rank $\mathrm{r}_\Omega(C)$ is the maximum size of such a surviving subfamily. For the piece remainder R, let $\mathrm{r}_\Omega(R)$ be the corresponding rank of the union of the raw piece obstruction tests. On the surviving hot residual entering node* [47], *quotient survival and exact-code equality with the full target code are retained for the same declared obstruction family. The exact-code equality realizes all $2^{|\mathcal{A}|}$ target states in the sense of Definition 3.99; its failure is the complementary cold residual routed to the later cold-branch analysis. Hence, on this residual, the quotient-system rank and the target rank agree. Closed whole-graph data are evaluated in the exact profile $\rho^{\mathrm{ex}}_{\varnothing}(G)$, so the empty context by itself creates no obstruction-rank identifications.*

Definition 10.2 (Obstruction target-dependence). *Let $\mathcal{A}$ be a finite family of obstruction tests. A* target-dependence *in $\mathcal{A}$ is a pair*

$$(a, \mathcal{B}), \qquad a \in \mathcal{A}, \quad \mathcal{B} \subseteq \mathcal{A} \setminus \{a\},$$

such that a functional admissible rank quotient of the exact response profile determines the target-response value of a from the target-response vector on $\mathcal{B}$. In this definition, the phrase "adjoining a to $\mathcal{B}$ does not increase the number of target-complete states" means precisely that the projection to the $\mathcal{B}$-coordinates is the graph of the function determining the a-coordinate under the admissible quotient. The dependence is proper *when this determination is not a tautological equality of the local coordinate a with one of the coordinates in $\mathcal{B}$.*

A determination certificate *for $(a, \mathcal{B})$ is a tuple*

$$\mathfrak{c} = (X, T, q, \mathcal{P})$$

with the following data.

(a) $X \subseteq G$ is a connected T-boundaried support carrying all coordinates in $\{a\} \cup \mathcal{B}$.

(b) $q : \rho^{\mathrm{ex}}_T(X) \to Q$ is a functional admissible rank quotient.

(c) For any two realizations of $\rho^{\mathrm{ex}}_T(X)$ with the same q-image on the coordinates of $\mathcal{B}$, the q-image of a is the same. Equivalently, the q-value of a is a function of the q-value vector on $\mathcal{B}$.

(d) $\mathcal{P}$ is the finite set of declared paths, boundary incidences, window segments, or connector incidences recorded by the certificate as the support data for the context-universal determination.

Here a realization means a T-boundaried support whose exact response profile maps to the same quotient data Q under q. The support *of $\mathfrak{c}$ is the lexicographically first smallest connected subgraph of G containing the declared supports of a, of all coordinates in $\mathcal{B}$, and of every member of $\mathcal{P}$. The support of $(a, \mathcal{B})$ is the support of a certificate with minimal support. A target-dependence is* inclusion-minimal *if no proper connected subsupport carries a proper target-dependence with the same determined coordinate.*

Lemma 10.3 (Target-rank circuit extraction). *Let $\mathcal{A}$ be a finite family of declared coordinates evaluated under the functional admissible quotients of a fixed exact response profile. If a subfamily*

$\mathcal{I} \subseteq \mathcal{A}$ is maximal among subfamilies that survive every functional admissible rank quotient and $a \in \mathcal{A} \setminus \mathcal{I}$, then there is a subfamily $\mathcal{B} \subseteq \mathcal{I}$ such that $(a, \mathcal{B})$ is a target-dependence in the sense of Definition 10.2. In particular, if no proper target-dependence exists among the coordinates of $\mathcal{A}$, then the whole family $\mathcal{A}$ survives every functional admissible rank quotient and is independently target-testable in the rank sense of Definition 10.1.

Proof. Since $\mathcal{I}$ is maximal for survival, $\mathcal{I} \cup \{a\}$ fails to survive some functional admissible rank quotient

$$q : \rho_T^{\mathrm{ex}}(X) \to Q.$$

The subfamily $\mathcal{I}$ survives q. Hence the failure of label-injectivity for $\mathcal{I} \cup \{a\}$ involves a. By Definition 3.92, the quotient q is functional on $\mathcal{A}$. Therefore there is a finite subfamily $\mathcal{B} \subseteq \mathcal{I}$ whose q-image vector determines the q-image of a. Choose such a subfamily inclusion-minimal. If a is sent to a single quotient value after the quotient forgets its separate label, then $\mathcal{B} = \varnothing$. If a is identified with a coordinate of $\mathcal{I}$, then the functional certificate may be taken with $\mathcal{B} = \{b\}$ for such a coordinate b. The quotient q, together with the declared supports of a, of $\mathcal{B}$, and of the finitely many coordinates used by q, is a determination certificate.

If the dependence is tautological, then a is the same local coordinate as an element of $\mathcal{B}$. Exact coordinate labels keep distinct declared raw coordinates distinct, so a rank loss among distinct raw obstruction coordinates yields a proper dependence. The contrapositive gives the final assertion. □

If $r_\Omega(C) < |\mathcal{W}_2(C)|$, then some coordinate outside a maximal independently target-testable subfamily is target-dependent on that subfamily in the sense of Definition 10.2 and Lemma 10.3. This is the only meaning of "dependence" used below.

Lemma 10.4 (Localization of obstruction dependence)**.** *Let C be a proper boundaried piece and let $\mathcal{W}_2(C)$ be its raw obstruction tests. If*

$$r_\Omega(C) < |\mathcal{W}_2(C)|,$$

then some target-dependence among raw obstruction tests falls into one of the following cases:

(i) a target-defective quotient;

(ii) an admissible nontrivial target-complete compression of a proper boundaried piece;

(iii) a target-dependence that becomes valid only after adjoining a connected support $Z \supsetneq C$.

Proof. A strict rank drop means that some raw obstruction test is target-dependent on the others in the sense of Definition 10.2 and Lemma 10.3. Choose a determination certificate with inclusion-minimal connected support. The certificate has an admissible quotient q and a finite support set $\mathcal{P}$. Either the determination certified by q fails for some T-boundaried outside context, or it is context-universal. In the context-universal case, the connected support of the certificate either equals C or strictly contains C. These alternatives are mutually exclusive and exhaust all certificates:

(i) if the determination fails for some outside context, it is not preserved under context-universality and hence is a target-defective quotient;

(ii) if it holds for every outside context already with support C, then q is a target-complete rank-reducing quotient of the proper boundaried piece C. Since q is admissible, Definition 3.91 supplies a strictly smaller proper representative. Thus this case is a nontrivial target-complete compression of C;

(iii) otherwise it holds for every outside context but only once additional structure outside C is adjoined; taking an inclusion-minimal connected such support yields a connected $Z \supsetneq C$.

□

Lemma 10.5 (Separated wedges are tested by disjoint contexts). *Let $u, v \in V(R)$ have identical rooted radius-r neighbourhood types, $B_r(u) \cong B_r(v)$, and suppose the closed balls $\overline{B}_r(u)$ and $\overline{B}_r(v)$ are vertex-disjoint. Let $w(u), w(v)$ be the corresponding internal length-two wedge tests at the two roots. Then any target tester separating $w(u)$ from $w(v)$ has its support in the complement of $\overline{B}_r(u) \cup \overline{B}_r(v)$. Consequently, any quotient identifying $w(u)$ with $w(v)$ is either context-universal or target-defective.*

Proof. A target tester for a wedge coordinate is a boundaried context glued to the support of that wedge. The wedge itself is an internal length-two path at the root of its radius-r ball. Any closure witnessing a power-of-two target that is not already internal to the ball must leave the ball through its boundary; if the witness stayed wholly inside the ball, the ball would already contain a target cycle, contrary to internal target-safety of the remainder. Hence every tester that separates the two wedge coordinates is supported outside the corresponding closed balls.

Since the rooted balls are isomorphic and vertex-disjoint, the two local wedge states have identical internal data and identical boundary degree profiles. If an outside context distinguishes them, then the identification fails the context-universality condition of target-completeness and is target-defective. If no outside context distinguishes them, the identification is context-universal. These are the only two possibilities. □

Lemma 10.6 (Proper nonlocal-support dependences are forbidden). *Let $C \subsetneq G$ be a proper boundaried piece. Suppose a target-dependence among obstruction tests of C becomes target-complete only after adjoining an inclusion-minimal connected support $Z \supseteq C$. If*

$$Z \subsetneq G,$$

then the dependence is impossible.

Proof. Regard Z as a boundaried graph with boundary

$$\partial Z = \{v \in V(Z) : v \text{ is incident with an edge of } G - Z\}.$$

The response state of Z includes its boundary degree profile and all target-response data inherited from C. Since $Z \subsetneq G$, it is a proper boundaried support. If the dependence fails against some outside ∂Z-context, it is target-defective. If it succeeds against every outside context, it is a nontrivial target-complete compression of the proper support Z, forbidden by Corollary 3.98. □

Definition 10.7 (Repair-network terminology). *Let Z be the support of a determination certificate. A vertex of Z is a* passive degree-two subdivision vertex *if it has degree 2 in Z, is not a boundary vertex of Z, and is not contained in the declared support of any coordinate, path endpoint, packed-window incidence, trace incidence, fan decoration, or response support used by the certificate. Such a vertex records only an unlabelled subdivision of a response path.*

After deleting the original coordinate supports of the dependence from Z, each remaining connected component that meets the deleted part only in boundary leaves is a delayed compensation component. *If such a component has p boundary leaves and all non-leaf vertices have degree 3 except for the recorded surplus contribution σ_Z, then it is a* 1–3 repair network up to surplus. *Its internal vertex count is denoted s, and its cycle rank is denoted β_Z.*

Lemma 10.8 (Enlarged nonlocal supports are repair networks)**.** *Let Z be a connected support on which a formerly nonlocal dependence becomes internal. Suppose Z is target-safe, contains no proper subgraph of minimum degree at least 3, and has no passive degree-two subdivision vertices. Then each delayed compensation component of Z is a 1–3 repair network up to surplus. More precisely, if such a connected component has p boundary leaves, s internal vertices, cycle rank β_Z, and surplus σ_Z, then*

$$s = p - 2 + 2\beta_Z - \sigma_Z.$$

Proof. The component has s internal vertices and p boundary leaves, hence $s + p$ vertices in all. Let e be the number of edges joining two internal vertices; each boundary leaf is attached by a single edge, so the component has $e + p$ edges in total. Each internal vertex has degree 3 up to its surplus, so the internal vertices contribute $3s + \sigma_Z$ to the degree sum, and each of the p leaves has degree 1. The handshake identity $\sum_v d(v) = 2 \cdot (\text{number of edges})$ therefore reads

$$(3s + \sigma_Z) + p = 2(e + p),$$

which simplifies to

$$3s + \sigma_Z = 2e + p. \tag{†}$$

The component is connected, so its cycle rank is

$$\beta_Z = (e + p) - (s + p) + 1 = e - s + 1, \qquad \text{equivalently} \qquad e = s - 1 + \beta_Z.$$

Substituting $e = s - 1 + \beta_Z$ into (†) gives $3s + \sigma_Z = 2(s - 1 + \beta_Z) + p = 2s - 2 + 2\beta_Z + p$, and solving for s yields

$$s = p - 2 + 2\beta_Z - \sigma_Z. \qquad \square$$

Lemma 10.9 (No silent whole-graph dependence)**.** *Let an obstruction target-dependence have inclusion-minimal connected support equal to the whole graph G. Then any attempted rank-reducing quotient is either target-defective, represented by a strictly smaller admissible closed representative, or exact on the raw obstruction labels. In particular, a target-complete whole-graph dependence cannot silently reduce piece obstruction rank.*

Proof. Let $q : \rho^{\mathrm{ex}}_{\varnothing}(G) \to Q$ be the quotient of the closed exact response profile which witnesses the proposed dependence in the sense of Definition 10.2. Since the support is all of G, the boundary is empty and the only actual outside context is the empty graph. Definition 3.91 is the rule that prevents this empty context from identifying all coordinates vacuously.

If the attempted quotient is not target-complete, then it is target-defective by definition and cannot witness a target-dependence. This is the target-defect exit used elsewhere in the proof, so it is not a surviving global rank reduction.

Assume next that q is target-complete and has a strictly smaller admissible closed representative G_q in the sense of Definition 3.90. Thus G_q is finite and simple, $\delta(G_q) \geq 3$, and $\mathsf{H}_{\varnothing}(G_q) \subseteq \mathsf{H}_{\varnothing}(G)$. If G_q contained a power-of-two cycle, then the empty $\varnothing$-boundaried context would belong to $\mathsf{H}_{\varnothing}(G_q)$. The same empty context does not belong to $\mathsf{H}_{\varnothing}(G)$, because G has no power-of-two cycle. This would contradict $\mathsf{H}_{\varnothing}(G_q) \subseteq \mathsf{H}_{\varnothing}(G)$. Hence G_q has no power-of-two cycle. Thus G_q is a strictly smaller counterexample, contradicting the lexicographic minimality of G.

It remains that q is target-complete and has no smaller closed representative. By the closed clause in Definition 3.91, such a quotient is label-injective on the raw obstruction coordinates of $\rho^{\mathrm{ex}}_{\varnothing}(G)$. Therefore it neither forgets a raw obstruction coordinate nor identifies two distinct raw obstruction coordinates. It can record exact global response data, but it is not obstruction-rank-reducing. This contradicts the assumption that the whole-graph dependence reduced r_Ω. $\square$

Lemma 10.10 (Forcing of full obstruction rank)**.** *In a minimal counterexample,*

$$r_\Omega(R) \geq W_2(R) - o(W_2(R)).$$

Proof. If not, then

$$r_\Omega(R) < W_2(R) - o(W_2(R)).$$

By Lemma 10.3, a rank loss among the raw obstruction coordinates gives a proper target-dependence. Choose such a dependence with a determination certificate of inclusion-minimal connected support, and call that support Z. The minimality of Z gives two possibilities.

First suppose $Z \subsetneq G$. Regard Z as a proper boundaried piece with boundary ∂Z. If the dependence is already visible inside a smaller proper boundaried piece $C \subsetneq Z$, then Lemma 10.4 applies to C. Its three alternatives are respectively: a target-defective quotient, a target-complete compression of a proper boundaried piece, or a dependence that requires an enlarged connected support. The first is excluded by the definition of target-completeness; the second is excluded by Corollary 3.98; and the third, because the enlarged support is still proper in G, is excluded by Lemma 10.6. If no smaller $C \subsetneq Z$ carries the dependence, then Z itself is the first proper support on which it becomes target-complete, and Lemma 10.6 again excludes it.

It remains that the inclusion-minimal support is $Z = G$. Then the dependence becomes context-universal only globally. By Lemma 10.9, such a global dependence either creates a target cycle, creates a replacement, or contributes exact non-identifying global response data. The first two alternatives contradict the counterexample condition and minimality; the third does not identify two distinct local obstruction tests and hence is not a proper target-dependence. Thus no proper target-dependence survives. By Lemma 10.3, the raw obstruction tests have no admissible rank loss. Hence

$$r_\Omega(R) = W_2(R),$$

which implies the stated weaker bound $r_\Omega(R) \geq W_2(R) - o(W_2(R))$. □

Corollary 10.11 (Forced obstruction cost)**.** *Assume Definition 4.1. Any surviving residual pays at least*

$$c_\Omega r_\Omega(R) \geq K_{\text{win}}|R| - o(|R|),$$

where

$$K_{\text{win}} = c_\Omega \omega_{\text{win}} = 5.82298307395\ldots.$$

In the sharper high-entropy low-deficiency branch this improves to

$$c_\Omega r_\Omega(R) \geq K|R| - o(|R|),$$

where

$$K = c_\Omega \omega = 5.89262883286\ldots.$$

Proof. This follows from Lemmas 9.1 and 10.10 and the definitions of K_{win} and K. □

Remark 10.12 (Provenance of the obstruction constant)**.** *The two ingredients of Corollary 10.11 play distinct roles. The* independence *of the remainder obstruction tests—that they realize full target-rank (Lemma 10.10)—is forced by the routing of Lemma 10.4: any dependence among the tests would be a target-defective quotient, a target-complete compression of a proper boundaried piece (excluded by Corollary 3.98), or a support dependence. Proper support dependence is excluded by Lemma 10.6; whole-graph support dependence is excluded by the closed exact-profile argument in Lemma 10.9. Thus each route is unavailable in the main spine. The* value c_Ω*, by contrast, enters*

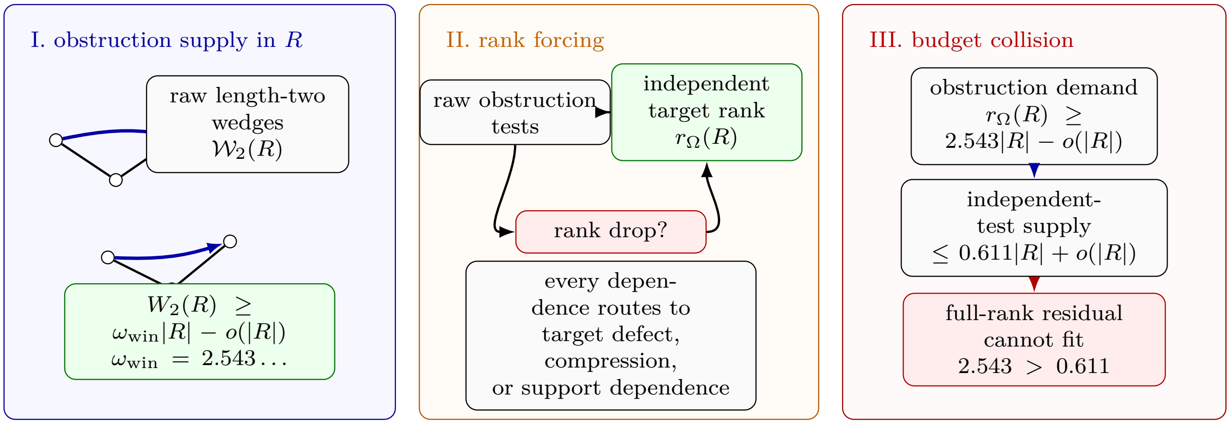

Figure 19: Visual proof of the obstruction supply–budget squeeze. Panel I shows the P_{13}-free remainder R producing raw internal length-two wedge tests; the wedge lower bound gives $W_2(R) \geq \omega_{\rm win}|R| - o(|R|)$, with $\omega_{\rm win} = 2.543\ldots$ on the window-only branch. Panel II shows the rank-forcing step: if the raw tests did not survive as target-rank, a dependence would route to target defect, proper compression, or support dependence, all of which are closed in the minimal counterexample. Thus $r_\Omega(R)$ is linear in $W_2(R)$. Panel III records the resulting collision with the independent-test budget: the full-rank branch demands at least $2.543|R| - o(|R|)$ tests, while the supply side has capacity at most $0.611|R| + o(|R|)$.

through the window-only constant $K_{\rm win} = c_\Omega \omega_{\rm win}$ and through the sharper high-entropy constant $K = c_\Omega \omega$ in the finite-size threshold $\Theta(n)$ of Definition 12.1. As Remark 12.4 records, the closure outside the explicit residuals does not depend on the exact value of c_Ω; what the routing supplies is the independence of the obstruction cost, not its size.

Framework branch table. The obstruction-rank forcing section closes the rank-drop alternatives by the following table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Rank drop	$r_\Omega(R) < W_2(R) - o(W_2(R))$.	A dependent obstruction coordinate.	The target-dependence route.	A finite functional circuit.	Lemmas 10.3 and 10.4.
Proper support	The dependence localizes in $Z \subsetneq G$.	The support profile.	Replacement or target defect.	The proper boundary profile.	Lemma 10.6.
Whole graph	The dependence delocalizes to all of G.	The closed exact profile.	Closed representative minimality.	The empty-boundary exactness rule.	Lemma 10.9.
Full rank	All rank-drop alternatives are excluded.	Raw wedge tests.	Obstruction target rank.	All but $o(W_2)$ tests.	Lemma 10.10 and Corollary 10.11.

11 Per-vertex remainder entropy and two-budget routing

Throughout this section, R is the P_{13}-free remainder of the minimal counterexample, with $|R| = (1-13\theta)n - o(n)$. The aim is to make the entropy branch explicit: the $n \log n$ skeleton budget and the linear forced obstruction cost are used in disjoint regimes.

Definition 11.1 (Per-vertex skeleton entropy of the remainder)**.** *Like every other object in this proof, $\mathcal{G}(R)$ is* local to the residual carried into this node*, not a global family. The remainder R is the one selected by the normalization ledger on the fixed minimal counterexample, and $V(R)$, the boundaried-piece part, and the current net-deficiency cap are all branch-local data inherited with it. Relative to that residual, let $\mathcal{G}(R)$ be the set of labelled simple graphs on $V(R)$ satisfying the remainder constraints already imposed on the branch: subcubicity on the boundaried-piece part, componentwise P_{13}-freeness, no internal 3-core, and positive net-deficiency density at most the current cap. Define*

$$\eta(R) = \frac{\log_2 |\mathcal{G}(R)|}{|R|}.$$

Every candidate carries the same inherited vertex set, so the density comparison is equivalently the comparison of its net-deficiency numerator with the inherited one. The family is used symbolically: no enumeration of labelled graphs is prescribed, and only $|\mathcal{G}(R)|$ is ever consumed, which the inherited data already determines.

Lemma 11.2 (Dominant local type in the low-entropy branch)**.** *Fix $r \geq 1$. Suppose the radius-r local-type coordinate of the remainder is structurally repetitive, in the sense that*

$$\log_2 \big| \{ (B_r(v))_{v \in V(R)} \} \big| = o(|R|),$$

i.e. the number of realizable radius-r type vectors is subexponential in $|R|$. Then for every $\varepsilon > 0$ a single rooted radius-r type covers all but at most $\varepsilon |R|$ vertices of R, for all large n. This hypothesis holds whenever the remainder is in the low-entropy regime $\eta(R) < \frac{1}{10} \log_2 n$ of Definition 11.1 and *the local-type coordinate carries a vanishing fraction of the total skeleton entropy.*

Proof. There are only finitely many rooted subcubic radius-r types; call the number $t = t(r)$, a constant independent of n. Suppose, for contradiction, that no single type covered a $(1-\varepsilon)$-fraction of $V(R)$. Then the empirical type distribution $\pi = (\pi_1, \ldots, \pi_t)$ over the t types satisfies $\max_a \pi_a \leq 1 - \varepsilon$, so its Shannon entropy obeys $H(\pi) \geq h(\varepsilon) > 0$, where $h(\varepsilon) = -\varepsilon \log_2 \varepsilon - (1-\varepsilon) \log_2 (1-\varepsilon)$ is the binary entropy function. The labelled remainder class is closed under relabeling of the fixed vertex set $V(R)$, and the radius-r type vector is a canonical labelled coordinate. Therefore every permutation of one realized type vector with empirical distribution π is again realized by a relabelled graph in the same class. Distinct assignments of types to the labelled vertices give distinct type vectors, so the number of type vectors with empirical distribution π is at least the multinomial coefficient, which by Stirling's approximation satisfies

$$\binom{|R|}{\pi_1 |R|, \ldots, \pi_t |R|} = 2^{\,H(\pi)|R| - o(|R|)} \geq 2^{\,h(\varepsilon)|R| - o(|R|)}.$$

Thus the number of realizable type vectors is $2^{\Theta(|R|)}$, exponential in $|R|$, contradicting the hypothesis that it is subexponential. Hence some rooted radius-r type covers a $(1-\varepsilon)$-fraction of $V(R)$. $\square$

Remark 11.3. *The structural-repetitiveness hypothesis of Lemma 11.2 is stronger than the single-scalar low-entropy regime $\eta(R) < \frac{1}{10} \log_2 n$. The radius-$r$ type coordinate is one canonical coordinate* *of the remainder skeleton, so its bit-content is bounded by $\eta(R)|R|$. When that coordinate is $o(|R|)$ the lemma applies. When it is not $o(|R|)$ but the total remainder entropy is still below $\frac{1}{10}|R|\log_2 n$, no dominance conclusion is asserted here; that nonrepetitive low-entropy case is passed unchanged to the large-budget net-charge analysis, which is independent of the precise obstruction rank fraction by Remark 12.4.*

Lemma 11.4 (Dominant type forces independent obstruction translates)**.** *Suppose a rooted radius-r type with $r \geq 2$ occurs on all but $o(|R|)$ vertices of R and contains an internal length-two wedge at its root. Then*

$$r_\Omega(R) \geq c_r|R| - o(|R|)$$

for a constant $c_r > 0$ depending only on r.

Proof. Let $D \subseteq V(R)$ be the set of vertices carrying the dominant rooted radius-r type, so $|D| \geq |R| - o(|R|)$. Subcubicity bounds each closed radius-r ball by

$$|\overline{B}_r(v)| \leq 1 + 3(2^r - 1),$$

a constant depending only on r. Greedily choose a maximal subset $S \subseteq D$ whose members are pairwise at distance greater than $2r$; then the closed balls $\{\overline{B}_r(u) : u \in S\}$ are pairwise vertex-disjoint. By maximality every vertex of D lies within distance $2r$ of some $u \in S$, and each ball of radius $2r$ has at most

$$b' = 1 + 3(2^{2r} - 1)$$

vertices. Hence

$$|S| \geq |D|/b' \geq |R|/b' - o(|R|).$$

Each ball carries one internal length-two wedge at its root, so these give at least $|S|$ raw obstruction tests. Hence $W_2(R) \geq |S|$. Also subcubicity gives $W_2(R) \leq 3|R|$, because a vertex of degree at most 3 is the center of at most three unordered length-two wedges. By Lemma 10.10,

$$r_\Omega(R) \geq W_2(R) - o(W_2(R)).$$

Since $W_2(R) \geq |S| \geq |R|/b' - o(|R|)$, the error term is $o(|R|)$, and

$$r_\Omega(R) \geq |S| \geq |R|/b' - o(|R|).$$

Take $c_r = 1/b'$. □

Remark 11.5 (A positive fraction of independent tests suffices)**.** *The separated-translate construction already yields independent wedge tests on a positive fraction of the remainder, $|S| \geq |R|/b' - o(|R|)$ with b' a constant. For the only downstream use—the forced cost of Corollary 10.11 feeding the threshold $\Theta(n)$—any positive linear rank $r_\Omega(R) \geq c|R|$ supplies a positive forced cost $c\,c_\Omega|R|$, and by Remark 12.4 the closure outside the explicit residuals holds for every nonnegative value of that cost. The conclusion is therefore insensitive to the precise rank fraction: full rank is the sharpest form, but a positive fraction is all that the remainder of the argument consumes.*

Proposition 11.6 (Two-budget routing)**.** *Assume Definition 4.1. Let R be the P_{13}-free remainder of a minimal counterexample and let $\eta(R)$ be its per-vertex skeleton entropy (Definition 11.1). Exactly one of the following holds.*

(a) ***High-entropy branch:*** $\eta(R) \geq \frac{1}{10}\log_2 n$***.*** *The remainder then contributes at least*

$$\frac{|R|}{10}\log_2 n$$

bits, and Lemma 7.1 together with the remainder accounting gives the packing bound of Proposition 7.42.

(b) ***Structurally repetitive low-entropy branch:*** $\eta(R) < \frac{1}{10}\log_2 n$ *and the radius-2 local-type coordinate has $o(|R|)$ bit-content. Then Lemma 11.2 yields a dominant radius-2 type. If that type contains an internal length-two wedge, then Lemma 11.4 supplies a positive linear obstruction-rank lower bound.*

(c) ***Nonrepetitive low-entropy branch:*** $\eta(R) < \frac{1}{10}\log_2 n$ *and the radius-2 local-type coordinate has nonvanishing linear bit-content. This proposition makes no obstruction-rank claim in this case; the branch is passed to the large-budget net-charge analysis.*

In every case the surviving residual is subsequently passed to the large-budget net-charge analysis of Proposition 13.7, with any additional obstruction cost used only when it has been established by the preceding rank-forcing lemmas.

Proof. The condition $\eta(R) \geq \frac{1}{10}\log_2 n$ and its negation partition all remainders. In the low-entropy case, the radius-2 local-type coordinate either has $o(|R|)$ bit-content or it does not. This gives the three listed cases, which are exhaustive and mutually exclusive.

In branch (a), the realized target-complete remainder states number at least

$$2^{\eta(R)|R|} \geq 2^{(|R|/10)\log_2 n} = n^{|R|/10}.$$

Thus the remainder contributes

$$\frac{|R|}{10}\log_2 n = \frac{n - 13p_{13}}{10}\log_2 n - o(n\log n)$$

bits. Together with the window package of Lemma 7.1 and the near-cubic skeleton budget $\frac{3}{2}n\log_2 n + o(n\log n)$ of Lemma 4.4, the feasibility inequality of Proposition 7.42 follows and bounds θ.

In branch (b), the local-type coordinate is subexponential, so Lemma 11.2 produces a dominant radius-2 type. In the large-budget range the degree-3 bulk has positive density by Lemmas 8.5 and 9.1; if the dominant type contains an internal length-two wedge at its root, then Lemma 11.4 gives a positive linear obstruction-rank lower bound. Branch (c) asserts no such dominance and is simply routed forward. □

Remark 11.7 (Role of the forced obstruction cost)**.** *When Lemma 10.10 is invoked, the window-only forced obstruction cost is $K_{\mathrm{win}}|R| - o(|R|)$. In the sharper high-entropy low-deficiency branch this improves to $K|R| - o(|R|)$, and that sharper value is the numerical constraint used in the entropy cap of Proposition 12.2. The weaker separated-translate input of Lemma 11.4 supplies only a positive linear rank in the structurally repetitive low-entropy branch. The final large-budget closure outside the explicit residuals does not require the exact obstruction constant: Remark 12.4 records that the window-density and net-charge accounting already give the needed normalized deficiency gap.*

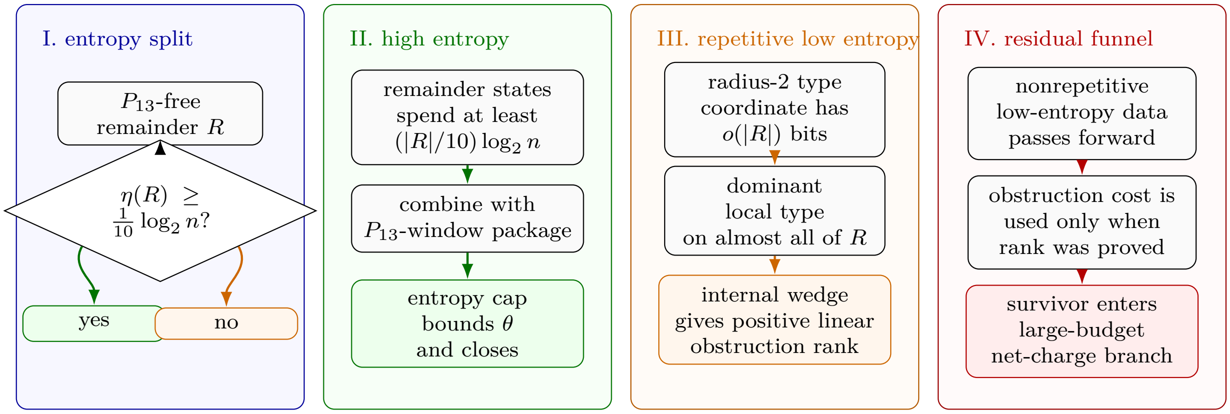

Figure 20: Visual proof of the two-budget routing funnel. Panel I splits the P_{13}-free remainder by its per-vertex skeleton entropy $\eta(R)$. Panel II shows the high-entropy branch: $\eta(R) \geq \frac{1}{10}\log_2 n$ spends $(|R|/10)\log_2 n$ bits, and together with the independent P_{13}-window package gives the packing bound used by the entropy cap. Panel III shows the structurally repetitive low-entropy branch: subexponential radius-2 type data gives a dominant local type, and an internal length-two wedge in that type supplies positive linear obstruction rank. Panel IV records the routing convention of Proposition 11.6: nonrepetitive low-entropy data is passed to the large-budget net-charge branch, and obstruction cost is carried forward only on branches where the preceding rank-forcing lemmas have established it.

Framework branch table. The two-budget routing funnel is summarized by the following framework table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
High entropy	$\eta(R) \geq \frac{1}{10}\log_2 n$.	Remainder bits.	Skeleton entropy.	One target-coordinate family.	Propositions 7.42 and 11.6.
Repetitive low entropy	A radius-2 coordinate has $o(\lvert R\rvert)$ bits.	A dominant local type.	Separated obstruction translates.	Bounded-radius overlap.	Lemmas 11.2 and 11.4.
Nonrepetitive low entropy	The low-entropy data do not give dominance.	No local entropy closure here.	The large-budget route.	Not applicable.	Passed to Section 13.
Cost separation	Obstruction cost is available only after rank forcing.	Independent tests.	The entropy budget.	The rank-forced branch.	Remark 11.7.

12 The admissible entropy cap

In this section assume Definition 4.1. The entropy cap is admissible only when the remaining non-obstruction budget is smaller than the forced obstruction cost. We make this precise. In the high-entropy accounting of Section 7, the window and remainder contributions consume, per factor

$n \log_2 n$, the normalized amount

$$0.1 + 116.808581006\,\theta$$

of the near-cubic labelled skeleton budget

$$1.5\,n \log_2 n + o(n \log n)$$

of Lemma 4.4. Hence the *remaining non-obstruction budget* is

$$(1.5 - (0.1 + 116.808581006\,\theta))n \log_2 n = (1.4 - 116.808581006\,\theta)n \log_2 n.$$

On top of this, Corollary 10.11 imposes the linear forced obstruction cost

$$K|R| - o(|R|) = K(1 - 13\theta)n - o(n), \qquad K = 5.89262883286\ldots.$$

For a residual graph to exist, the window, remainder, and forced-obstruction demands together must fit inside the skeleton budget; equivalently the forced obstruction cost must fit inside the remaining non-obstruction budget,

$$K(1 - 13\theta)n \leq (1.4 - 116.808581006\,\theta)n \log_2 n + o(n \log n).$$

Dividing by n, no residual graph exists once

$$(1.4 - 116.808581006\,\theta) \log_2 n < K(1 - 13\theta). \tag{2}$$

Definition 12.1 (The finite-size packing threshold)**.** *Define*

$$\Theta(n) = \frac{1.4 - \dfrac{K}{\log_2 n}}{116.808581006 - \dfrac{13K}{\log_2 n}},$$

where

$$K = 5.89262883286\ldots.$$

Proposition 12.2 (Entropy cap closes the high-θ branch)**.** *Assume Definition 4.1. In the high-entropy branch of Proposition 11.6, if*

$$\theta > \Theta(n) + o(1),$$

then no minimal residual graph exists.

Proof. Suppose $\theta > \Theta(n) + o(1)$. By the definition of $\Theta(n)$ this is precisely inequality (2), that is, the remaining non-obstruction budget is strictly smaller than the forced full-rank obstruction cost of Corollary 10.11. Then the window package of Lemma 7.1, the remainder bits, and the forced-obstruction bits together strictly exceed the near-cubic skeleton budget $1.5\,n \log_2 n + o(n \log n)$. These bits form one independently target-testable coordinate family, so the number of realized target-complete states would exceed the number of labelled skeletons, contradicting Lemmas 3.100 and 4.2. Hence no residual graph exists in this branch. □

Remark 12.3 (Order of the entropy argument)**.** *The entropy cap is not used to assert an a priori linear upper bound on $r_\Omega(R)$. Instead, the rank-forcing argument first forces full obstruction rank. Entropy is then applied only in the region where the remaining budget is smaller than the forced obstruction cost.*

Remark 12.4 (Robustness of the closure to the forced obstruction cost)**.** *The forced obstruction cost sharpens the high-entropy threshold $\Theta(n)$ of Definition 12.1, but the large-budget closure outside the explicit residuals does not need that sharpening. The near-cubic window-only part of Proposition 7.42 gives*

$$\theta \le \theta_{\mathrm{win}} + o(1), \qquad \theta_{\mathrm{win}} = 0.01270017798\ldots,$$

and therefore

$$\frac{15\theta}{1-13\theta} \le \tau_{\mathrm{win}} + o(1), \qquad \tau_{\mathrm{win}} = 0.22817486846\ldots < \frac{1}{4}.$$

Thus the obstruction-rank and forced-cost machinery is not required for the net-charge closure outside the explicit residuals, which already follows from the window-only density bound together with the local analysis of Sections 13 and 14. Consequently the final closure uses neither the exact value of c_Ω (Remark 10.12) nor the precise rank fraction (Remark 11.5) as an additional hypothesis.

n	$\log_2 n$	$\Theta(n)$	$\tau_\Theta = 15\Theta(n)/(1-13\Theta(n))$
10^3	9.9658	0.007411	0.123019
10^4	13.2877	0.008614	0.145505
10^6	19.9316	0.009776	0.167991
10^9	29.8974	0.010529	0.182982
10^{12}	39.8631	0.010899	0.190477
∞	∞	0.011985	0.212963

Framework branch table. The admissible entropy cap closes only the branch where its numerical inequality is active.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Admissible cap	The remaining non-obstruction budget is smaller than the forced obstruction cost.	Window, remainder, and obstruction bits.	The near-cubic skeleton budget.	One target-complete coordinate family.	Closed by Proposition 12.2.
Below threshold	The entropy inequality does not close.	The large-budget residual.	Net-charge accounting.	Not applicable.	Routed to Section 13.
Robust route	The exact obstruction magnitude is not used.	Window-only deficiency gap.	Net-charge and local discharging.	The same residual ledger.	Remark 12.4.

13 The large-budget residual

Throughout this section assume Definition 4.1. Regardless of whether the high-entropy cap of Proposition 12.2 applies, the near-cubic window-only bound in Proposition 7.42 gives

$$\theta \le \theta_{\mathrm{win}} + o(1), \qquad |R| = n - 13p = (1-13\theta)n \ge (1-13\theta_{\mathrm{win}})n - o(n).$$

By Lemma 8.2, every component of R is P_{13}-free and has empty internal 3-core.

The supply inequality Lemma 8.5 gives

$$\partial_+(R) - \sigma_R \le 15p_{13} + o(n), \qquad \sigma_R = \sigma(R).$$

Dividing by $|R| = (1-13\theta)n + o(n)$ yields

$$\frac{\partial_+(R) - \sigma_R}{|R|} \le \frac{15\theta}{1-13\theta} + o(1).$$

Since $\theta \le \theta_{\mathrm{win}} + o(1)$, every large-budget residual satisfies

$$\boxed{\frac{\partial_+(R) - \sigma_R}{|R|} \le \tau_{\mathrm{win}} + o(1),}$$

where

$$\tau_{\mathrm{win}} = \frac{15\theta_{\mathrm{win}}}{1-13\theta_{\mathrm{win}}} = 0.22817486846\ldots < \frac{1}{4}.$$

Definition 13.1 (Canonical support decomposition)**.** *The* canonical support decomposition *of the residual consists of:*

(i) the connected components $Y_1, \ldots, Y_k$ of R, ordered by the deterministic tie-breaking rule fixed in Lemma 4.2; and

(ii) an assignment of every ambient surplus unit $d_G(h) - 3$ with $h \in V_{\ge 4}(G) \cap V(R)$ to exactly one piece.

All surplus units of $h \in V_{\ge 4}(G) \cap V(R)$ are assigned to the unique piece containing h. Surplus carried by vertices in the packed windows is not assigned to local supports; it has already been absorbed into the $o(n)$ term in Lemma 8.5. For the support associated to Y_i, write

$$|V(X_i)| := |Y_i|, \qquad \sigma(X_i) := \text{the number of surplus units assigned to } Y_i.$$

Thus

$$\sum_i |V(X_i)| = |R|, \qquad \sum_i \sigma(X_i) = \sigma(R).$$

When an assigned surplus unit is represented by an actual high-degree fan vertex in the local Type B analysis, the corresponding fan envelope records that vertex as decoration data; it is not counted a second time in $|V(X_i)|$. Because the pieces are the connected components of R, there are no edges of R between distinct pieces.

Definition 13.2 (Admissible support)**.** *An* admissible support *X is a connected remainder piece Y_i of the canonical support decomposition, together with its assigned surplus units and any local decoration data needed to realize those units in the Type B fan analysis. Its vertex count and positive deficiency are those of the remainder piece:*

$$|V(X)| = |Y_i|, \qquad \partial_+(X) = \sum_{v \in Y_i} \max\{0, 3 - d_{Y_i}(v)\}.$$

Its surplus $\sigma(X)$ is the assigned surplus credit from Definition 13.1. As a boundaried support of the minimal counterexample G, X carries the inherited constraints used in the local analysis: it is P_{13}-free, has empty internal 3-core, is target-safe in the contextual sense (Definition 2.5), has its deficient vertices supplied from the packing W (Lemma 8.5), and is hereditarily target-uncompressible (Corollary 3.98). The quantities $\partial_+(X)$, $\sigma(X)$, and $\mathcal{N}(X)$ below are therefore accounting data for the embedded support, not invariants of an isolated graph.

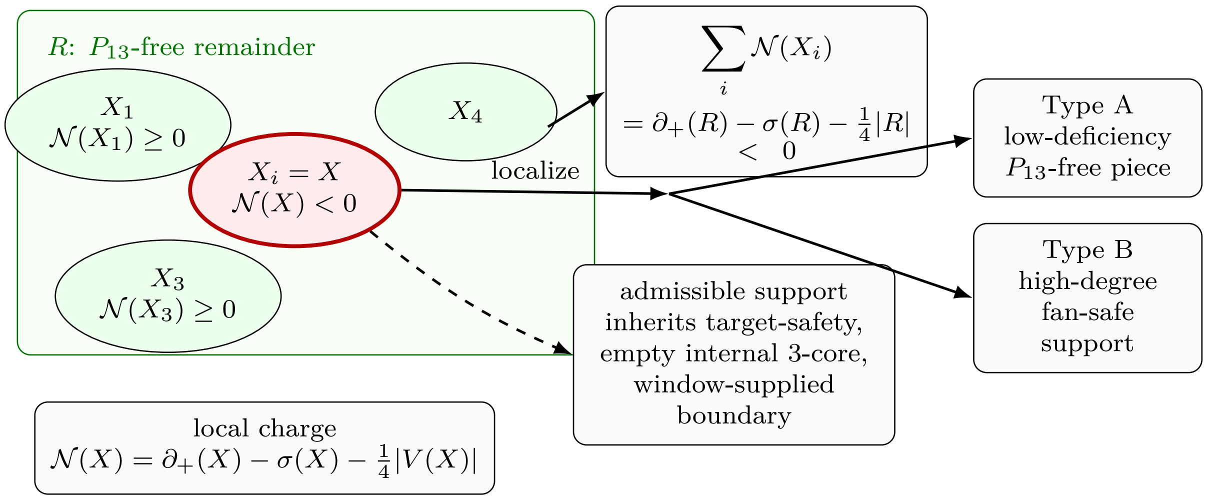

Figure 21: Visual definition of net-charge localization. The green rectangle is the P_{13}-free remainder R, decomposed into connected admissible supports X_i by the canonical support decomposition. Green ellipses show supports with nonnegative net charge; the red ellipse is the localized support $X = X_i$ with $\mathcal{N}(X) < 0$. Each support carries positive deficiency $\partial_+(X_i)$, assigned ambient surplus $\sigma(X_i)$, and net charge $\mathcal{N}(X_i) = \partial_+(X_i) - \sigma(X_i) - \frac{1}{4}|V(X_i)|$. The top budget box records additivity over connected components, $\sum_i \mathcal{N}(X_i) = \partial_+(R) - \sigma(R) - \frac{1}{4}|R| < 0$. The solid arrow labelled "localize" selects a negative support, the dashed arrow records inherited admissibility data, and the two outgoing solid arrows route that support to the Type A low-deficiency boundaried-piece branch or the Type B high-degree fan-safe support branch. The inherited data are contextual target-safety, empty internal 3-core, componentwise P_{13}-freeness, and window-supplied deficient boundary.

Definition 13.3 (Net charge)**.** *For an admissible support X, define*

$$\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \frac{1}{4}|V(X)|.$$

Lemma 13.4 (Net-charge localization under support splitting)**.** *Let R be decomposed canonically into connected admissible supports $X_1, \dots, X_k$. Then*

$$\sum_{i=1}^{k} \mathcal{N}(X_i) \ = \partial_+(R) - \sigma(R) - \frac{1}{4}|R|.$$

Consequently, if

$$\partial_+(R) - \sigma(R) - \frac{1}{4}|R| < -\varepsilon|R|$$

for some fixed $\varepsilon > 0$, then some connected support X_i has $\mathcal{N}(X_i) < 0$.

Proof. The remainder vertex sets Y_i form a partition of $V(R)$, and the ambient surplus units form a disjoint assigned credit partition. By Definition 13.1,

$$\sum_i |V(X_i)| = |R|, \qquad \sum_i \sigma(X_i) = \sigma(R).$$

For each vertex $u \in Y_i$, the component property gives $d_{X_i}(u) = d_R(u)$. Hence the positive deficiencies add exactly:

$$\sum_i \partial_+(X_i) = \partial_+(R).$$

Therefore

$$\sum_i \mathcal{N}(X_i) = \sum_i \partial_+(X_i) - \sigma(R) - \frac{1}{4}|R| = \partial_+(R) - \sigma(R) - \frac{1}{4}|R|,$$

the displayed identity. If the right-hand side is negative by a linear amount and all summands were nonnegative, their sum could not be negative; hence some connected support has negative net charge. $\square$

Corollary 13.5 (Global window-join load)**.** *Let R be decomposed canonically into connected admissible supports $X_1, \ldots, X_k$. Then*

$$\sum_{i=1}^{k} \mathcal{N}(X_i) \leq 15p_{13} + \sigma_W - \sigma_R - \frac{1}{4}|R|.$$

Consequently, if every connected admissible support in this decomposition has nonnegative net charge, then

$$\sigma_W - \sigma_R \geq \frac{1}{4}|R| - 15p_{13} = \frac{n - 73p_{13}}{4}.$$

Equivalently, if $p_{13} \leq \theta n + o(n)$, then the no-negative-support alternative forces

$$\sigma_W - \sigma_R \geq \frac{1 - 73\theta}{4}\, n - o(n).$$

In particular, under the near-cubic window-only cap $\theta \leq \theta_{\text{win}} + o(1)$,

$$\sigma_W - \sigma_R \geq \kappa_{\text{win}} n - o(n), \qquad \kappa_{\text{win}} := \frac{1 - 73\theta_{\text{win}}}{4} = 0.0182217518254\ldots.$$

Proof. By Lemma 13.4,

$$\sum_i \mathcal{N}(X_i) = \partial_+(R) - \sigma_R - \frac{1}{4}|R|.$$

The surplus-aware window-stub inequality Lemma 8.7 gives

$$\partial_+(R) - \sigma_R \leq 15p_{13} + \sigma_W - \sigma_R.$$

Combining the two displays proves the first inequality.

If every X_i has $\mathcal{N}(X_i) \geq 0$, then the left-hand side is nonnegative. Therefore

$$0 \leq 15p_{13} + \sigma_W - \sigma_R - \frac{1}{4}|R|,$$

which is the displayed lower bound on $\sigma_W - \sigma_R$. Since $|R| = n - 13p_{13}$, this lower bound equals

$$\frac{n - 13p_{13}}{4} - 15p_{13} = \frac{n - 73p_{13}}{4}.$$

The density forms follow by substituting $p_{13} \leq \theta n + o(n)$ and then $\theta = \theta_{\text{win}} + o(1)$. $\square$

Remark 13.6 (Meaning of the join-load inequality)**.** *Corollary 13.5 is the exact accounting form of the non-near-cubic load. Surplus carried by R appears with a negative sign in $\mathcal{N}(R)$, so it helps produce a negative admissible support. Avoiding such a support requires a linear excess of window surplus over remainder surplus. By Lemma 8.6, that excess is not abstract: it is realized by additional window–remainder incidences and cross-window joins.*

Proposition 13.7 (Large-budget residual is negative-net-charge)**.** *If the large-budget branch survives and the local lower bound fails by a fixed linear margin, i.e. if for some $\varepsilon > 0$,*

$$\partial_+(R) - \sigma(R) \leq \frac{1}{4}|R| - \varepsilon|R|$$

for all sufficiently large n, then there is a connected admissible support X with

$$\mathcal{N}(X) < 0.$$

Proof. By Lemma 13.4,

$$\sum_i \mathcal{N}(X_i) = \partial_+(R) - \sigma(R) - \frac{1}{4}|R|.$$

By the displayed hypothesis, the right-hand side is negative for all sufficiently large n. Hence at least one connected admissible support X_i has negative net charge. Choose such a support and call it X. □

Framework branch table. The large-budget residual localizes negative mass by the following branch table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Net-charge cap	$\partial_+(R) - \sigma(R) < \frac{1}{4}\lvert R\rvert$.	Negative total charge.	The support decomposition.	The superadditive component sum.	Lemma 13.4.
Localization	The total net charge is negative.	A connected deficit support.	The admissible support class.	The component partition.	Proposition 13.7.
Window load	No negative support appears without surplus load.	$\sigma_W - \sigma_R$.	The window-join identity.	The stub partition.	Corollary 13.5.

14 Local analysis of a negative-net-charge support

Let X be a connected admissible support with

$$\mathcal{N}(X) < 0.$$

We show that X is forced into one of two local obstruction classes.

Remark 14.1 (How to read the local lemmas)**.** *The support X is* admissible *(Definition 13.2): a piece of the minimal counterexample G, so it carries every global constraint, not merely its own induced structure. In particular it is target-safe in the strong, contextual sense (Definition 2.5)—G has no power-of-two cycle meeting $V(X)$, including cycles that leave X through its boundary and return through $G-X$—its deficient vertices are supplied from the packing W (Lemma 8.5), and it is hereditarily target-uncompressible (Corollary 3.98). Consequently the inequalities $\mathcal{N}(X) \geq 0$ proved below are* not *intrinsic properties of subcubic, empty-3-core, P_{13}-free graphs taken in isolation—such graphs may violate them—but properties of supports that admit a completion to a graph of minimum degree three with no power-of-two cycle. Every step uses this admissibility; a lemma read with only its bare graph hypotheses has been read with strictly fewer hypotheses than it states.*

14.1 Type A: low-deficiency P_{13}-free piece

If X carries no high-degree surplus, then $\sigma(X) = 0$ and

$$\mathcal{N}(X) < 0$$

means

$$\partial_+(X) < \frac{1}{4}|V(X)|.$$

If X is subcubic, then $\partial_+(X) = \sum_{v \in V(X)}(3 - d_X(v)) = 3|V(X)| - 2|E(X)|$, so $\partial_+(X) < \frac{1}{4}|V(X)|$ is equivalent to $3|V(X)| - 2|E(X)| < \frac{1}{4}|V(X)|$, that is

$$\frac{2|E(X)|}{|V(X)|} > 2.75.$$

Since X lies in the P_{13}-free remainder, it is P_{13}-free. We first bound its diameter. A shortest path between two vertices is always induced, so if X had two vertices at distance 12, the shortest path between them would be an induced path on 13 vertices, i.e. an induced P_{13}, contradicting P_{13}-freeness. Hence $\operatorname{diam}(X) \leq 11$. Consequently breadth-first layering from any root v_0 has at most $\operatorname{diam}(X) + 1 = 12$ layers; since X is subcubic, the first layer (the neighbours of v_0) has at most 3 vertices, and each vertex in a later layer contributes at most 2 vertices to the next layer, one of its at most three incident edges leading back toward the root. Therefore

$$|V(X)| \leq 1 + 3\sum_{i=0}^{10} 2^i = 1 + 3(2^{11} - 1) = 6142.$$

If the root has degree at most 2, this improves to $1 + 2(2^{11} - 1) = 4095$; in either case the Type A class is finite:

$$|V(X)| \leq 6142, \qquad \partial_+(X) < |X|/4, \qquad X \text{ is } P_{13}\text{-free and target-safe.}$$

The exclusion of this class is established by a contextual density lemma. The lemma is stated separately because the bound is a statement about completions in G, not about isolated P_{13}-free graphs.

Definition 14.2 (Type A support)**.** *A* Type A support *is a connected admissible support X with $\sigma(X) = 0$. Since G has minimum degree at least 3, this condition gives $d_G(v) = 3$ for every $v \in V(X)$. Hence X is subcubic, the deficient vertices of X are exactly the vertices of internal degree at most 2, and the P_{13}-freeness, empty internal 3-core, contextual target-safety, supplied deficiency, and hereditary target-uncompressibility of X are inherited from Definition 13.2. In the negative-net-charge branch a Type A support also satisfies $\partial_+(X) < |V(X)|/4$.*

Definition 14.3 (Receivers, ports, and routed load)**.** *Let X be a Type A support. A* receiver *is a vertex $w \in V(X)$ with $d_X(w) \le 2$, and*

$$q(w) = 3 - d_X(w).$$

Since $\sigma(X) = 0$, every vertex of X has ambient degree exactly 3, so $q(w)$ is also the number of ambient edges from w to $G-X$. An oriented edge $\vec{e} = (w, h)$ with $wh \in E(G)\setminus E(X)$ is a completion port *of w.*

Let X_3 be the subgraph induced by the vertices of X with internal degree 3. The empty internal 3-core condition implies that every component of X_3 has an edge to at least one receiver. For a cubic vertex u, let T_u be the lexicographically first path in X that starts at u, stays in X_3 until its last edge, and ends at a receiver; write this terminal receiver as $r(u)$. The path T_u is the canonical trace *of u. The routed load at w is*

$$L(w) = |\{u \in V(X) : d_X(u) = 3,\ r(u) = w\}|.$$

A receiver is saturated *when $L(w) \ge 4q(w)$ and* unsaturated *when $L(w) \le 4q(w)-1$. For $j = 0, 1, 2$, let*

$$\mathcal{R}_j(X) = \{w \in V(X) : d_X(w) = 2 - j\}.$$

Lemma 14.4 (Canonical receiver loads)**.** *For every Type A support X, the map $u \mapsto r(u)$ is defined for each degree-3 vertex of X, and every degree-3 vertex is assigned to exactly one receiver.*

Proof. The empty internal 3-core hypothesis implies that no component of the induced subgraph X_3 is closed under all three incident edges of its vertices. Therefore each component of X_3 has an edge to a vertex of internal degree at most 2, i.e. to a receiver. The canonical trace T_u is chosen by a fixed lexicographic tie-breaking rule among these receiver-reaching paths, so its terminal receiver $r(u)$ is defined and unique. Thus every degree-3 vertex is assigned to exactly one receiver. □

Definition 14.5 (Receiver-entry returns and visible load)**.** *Let $\vec{e} = (w, h)$ be a completion port. An* anchored return *through $\vec{e}$ is a simple path $P : h \rightsquigarrow w$ in $G - wh$. Its first vertex in X, traversed from h to w, is denoted $\mathrm{ent}_X(P)$.*

If r, w are receivers, a receiver-entry channel *from r to w is a simple path $Q : r \rightsquigarrow w$ in X. An anchored return is a* receiver-entry return *if it decomposes as*

$$P = \Gamma \circ Q,$$

where Γ runs from h to a receiver r with internal vertices outside X, and Q is a receiver-entry channel from r to w.

A receiver-entry return $P = \Gamma \circ Q$ through $\vec{e}$ is visible *for a routed cubic vertex u with $r(u) = w$ if the canonical trace T_u is a subpath of the channel Q, or if Q is the lexicographically first channel assigned to the terminal receiver edge of T_u. The port $\vec{e}$ carries t visible receiver-entry returns when it carries such returns for t distinct routed cubic vertices. Let $L_{\mathrm{vis}}(w, \vec{e})$ be this number. A load routed to w is* visible *if it is visible at some completion port of w; if more than one port sees it, it is assigned to the first such port in the canonical order. The total number of visible loads routed to w is $L_{\mathrm{vis}}(w)$; the remaining loads are* silent *and have count*

$$L_{\mathrm{sil}}(w) = L(w) - L_{\mathrm{vis}}(w).$$

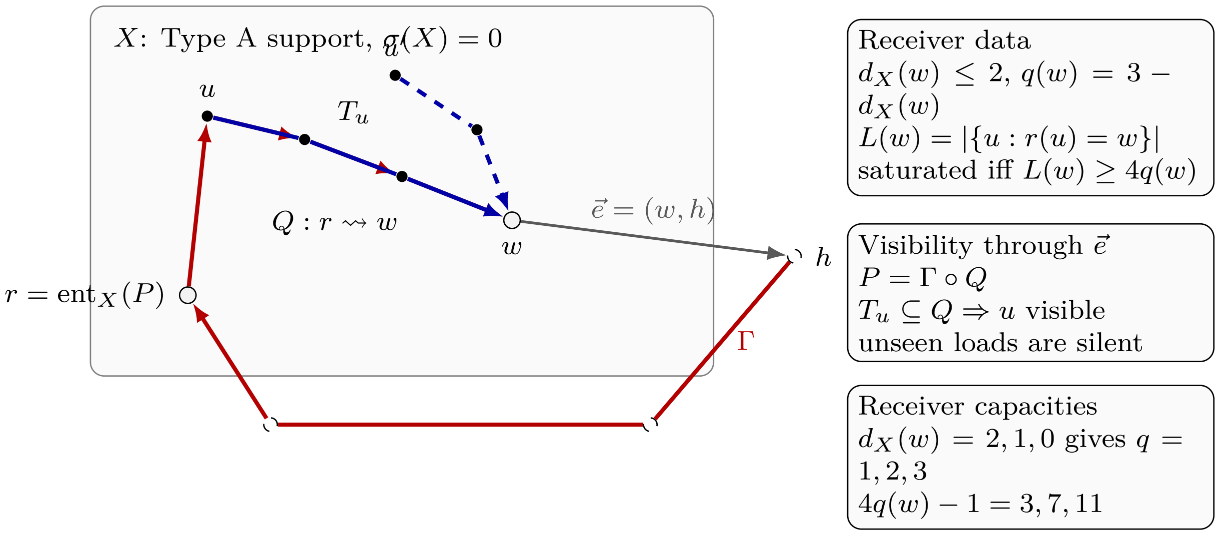

Figure 22: Visual definition of a Type A receiver. The shaded region is the Type A support X, and $\sigma(X) = 0$ records that the support has no ambient surplus. White circled vertices are receivers; the terminal receiver is w, and $r = \mathrm{ent}_X(P)$ is the first vertex of X met by the anchored return P. Black filled vertices are internal vertices of the support or of a displayed trace. The dashed circled vertex h lies outside X, and the grey arrow $\vec{e} = (w, h)$ is the completion port being tested. The red path is the anchored return $P = \Gamma \circ Q$: the outside connector Γ runs from h to r, while the internal receiver-entry channel $Q : r \rightsquigarrow w$ runs inside X. The solid blue path T_u is the canonical trace of the routed cubic vertex u and makes u visible at $\vec{e}$ because $T_u \subseteq Q$; the dashed blue trace from u' illustrates a routed load not seen by this port. The three legend boxes define the receiver data, visibility criterion, and receiver capacities. Here $d_X(w)$ is the degree of w inside X, $q(w) = 3 - d_X(w)$ is the number of missing ambient incidences at w, and $L(w) = |\{u : r(u) = w\}|$ is the number of routed cubic vertices assigned to w. The receiver is saturated when $L(w) \ge 4q(w)$, and the displayed capacities $3, 7, 11$ are the values of $4q(w) - 1$ for $d_X(w) = 2, 1, 0$.

Lemma 14.6 (Common-port return cycle)**.** *Let $\vec{e} = (w, h)$ be a completion port, and let P_1, P_2 be two anchored returns through $\vec{e}$. If P_1 and P_2 are internally vertex-disjoint as paths in $G - wh$, then $P_1 \cup P_2$ is a simple cycle of length $|P_1| + |P_2|$. In particular, if $|P_1| + |P_2| \in \mathcal{D}$, then G contains a power-of-two cycle.*

Proof. By definition, each anchored return through $\vec{e}$ is a simple path from h to w in $G - wh$. Thus the two paths have the same endpoints. The internal-disjointness hypothesis is exactly the hypothesis of Lemma 2.3, so their union is a simple cycle of the stated length. □

Lemma 14.7 (First entry is a receiver)**.** *Let P be an anchored return through a completion port of a Type A support X. Then $\mathrm{ent}_X(P)$ is a receiver.*

Proof. Let $r = \mathrm{ent}_X(P)$. The predecessor of r on P lies outside X, so r has an ambient incidence not counted in $d_X(r)$. Since $\sigma(X) = 0$, every vertex of X has ambient degree 3. Hence $d_X(r) \le 2$, and r is a receiver. □

Lemma 14.8 (Degree-two entry exclusion and entry budget)**.** *Let w be a receiver with $d_X(w) = 2$, and let $\vec{e} = (w, h)$ be its unique completion port. Every anchored return through $\vec{e}$ first enters X at a receiver $r \ne w$. Moreover the number of distinct external edges through which such returns first* *enter X is at most*

$$\sum_{\substack{r \text{ receiver} \\ r\neq w}} q(r).$$

Proof. An anchored return through $\vec{e}$ is a path from h to w in $G - wh$. By Lemma 14.7, its first vertex r in X is a receiver. If $r = w$, then the return enters w from outside X through an ambient edge different from the deleted edge wh. This is impossible because $d_X(w) = 2$ and $\sigma(X) = 0$ give exactly one ambient edge from w to $G - X$, namely wh. Hence $r \neq w$. For each first-entry receiver r, the entering edge is one of the $q(r)$ completion ports of r, giving the displayed bound. ☐

Lemma 14.9 (Ports have anchored returns)**.** *Every completion port of a Type A support has at least one anchored return.*

Proof. A completion port is an oriented edge of G. By Lemma 3.4, the underlying edge lies on a cycle in G. Removing the port edge from that cycle leaves a simple return path in the required orientation. ☐

Definition 14.10 (Connector length and channel spectrum)**.** *Let $\vec{e} = (w, h)$ be a completion port and let*

$$P = \Gamma \circ Q$$

be a receiver-entry return through $\vec{e}$, where Γ runs from h to the first-entry receiver r with internal vertices outside X, and $Q : r \rightsquigarrow w$ lies in X. The path Γ is the connector*, and its length is denoted $g(\Gamma)$.*

For receivers r, w define the internal channel spectrum

$$\Lambda_X(r, w) = \{|Q| : Q \text{ is a simple path in } X \text{ from } r \text{ to } w\}.$$

The interval length *$I_X(r, w)$ is the largest cardinality of an interval of consecutive integers contained in $\Lambda_X(r, w)$.*

Definition 14.11 (Connector configurations and continuation classes)**.** *Let X be a Type A support, let $\vec{e} = (w, h)$ be a completion port, and let*

$$P = \Gamma \circ Q$$

be a receiver-entry return through $\vec{e}$. Write

$$\Gamma = (x_0, x_1, \ldots, x_g), \qquad x_0 = h, \qquad x_g = \mathrm{ent}_X(P),$$

where $x_1, \ldots, x_{g-1} \notin X$. The outside connector configuration *of P is the ordered path Γ together with the first-entry receiver x_g, the channel Q, and the declared response coordinate of Definition 3.87 determined by $(\vec{e}, x_g, g(\Gamma), Q)$.*

Let κ_i, κ_j be two declared response coordinates through the same completion port, with connector configurations Γ_i, Γ_j. They have the same continuation class up to an outside vertex *z when z occurs in both configurations, the two configurations have the same ordered prefix from h to z, and the two coordinates have the same image in the relevant boundary-degree fibre. They* separate at *z when they have the same continuation class up to z and their next incidences after z are distinct. Here the next incidence may be an edge from z to an outside vertex or the first-entry edge from z into X.*

For a finite family $\mathcal{K}$ of declared response coordinates through $\vec{e}$, a vertex $z \notin X$ is a first separator *of $\mathcal{K}$ if some pair of coordinates in $\mathcal{K}$ separates at z and no pair separates at an earlier vertex on their common prefix from h.*

If κ_i, κ_j separate at z, the associated switch support *$S_z(\kappa_i, \kappa_j)$ is the finite connected support consisting of the common prefix from h to z, the two connector tails from z to their first-entry receivers, the two receiver-entry channels in X, the completion port boundary datum, and the declared supports of the two response coordinates. The separator z is* absorbed *when the response identification on this switch support is target-defective, target-complete on a nontrivial response quotient, or target-complete only after adjoining a larger connected support. It is* surviving *when it is not absorbed.*

Lemma 14.12 (Spectral interval constraint)**.** *Let $\vec{e} = (w, h)$ be a completion port and let Γ be a connector from h to a first-entry receiver r. Then*

$$g(\Gamma) + \lambda \notin \mathcal{M} \qquad \text{for every } \lambda \in \Lambda_X(r, w).$$

Consequently, if $\Lambda_X(r, w)$ contains the interval $[a, b] \cap \mathbb{Z}$, then

$$g(\Gamma) \notin \bigcup_{\mu \in \mathcal{M}} ([\mu - b, \mu - a] \cap \mathbb{Z}_{\geq 1}).$$

Proof. For any simple channel $Q : r \rightsquigarrow w$ in X, the concatenation $\Gamma \circ Q$ is a simple return through $\vec{e}$: the internal vertices of Γ avoid X, while Q lies in X. If $g(\Gamma) + |Q| \in \mathcal{M}$, then $\Gamma \circ Q$ together with the port edge gives a power-of-two cycle by Lemma 2.2, contradicting the counterexample condition. Varying Q over all channels gives the first assertion. The displayed band exclusion is the same statement applied to every $\lambda \in [a, b] \cap \mathbb{Z}$. ☐

Remark 14.13 (The receiver-5 witness and the band constraint)**.** *For the explicit 15-vertex near-miss boundaried piece described in the referee material, the receiver-channel spectra from the two other degree-2 receivers to the saturated receiver are both the full interval $[3, 14]$. Thus a connector length g into either entry receiver must avoid*

$$\bigcup_{\mu \in \mathcal{M}} [\mu - 14, \mu - 3].$$

In particular, below 112 the allowed positive lengths are

$$[13, 16] \cup [29, 48] \cup [61, 112],$$

and $[1, 12]$ is forbidden. This is not an additional external theorem; it is the universally quantified target condition $R_e(G) \cap \mathcal{M} = \varnothing$ applied to all channel lengths at once.

Definition 14.14 (Minimal trace-complete supports)**.** *Let u be a routed cubic vertex of a Type A support, and let T_u be its canonical trace ending at $r(u)$. For a connected subgraph $S \subseteq X$ with $T_u \subseteq S$, let*

$$\partial S = \{v \in V(S) : v \text{ is incident with an edge of } G - S\},$$

*and regard S as a ∂S-boundaried graph with the inherited labels. Each local target-response coordinate used in this paper has a declared finite support: for example, a return coordinate is supported on its port and channel, a P_{13} label coordinate on its window, an obstruction coordinate on its length-two wedge, and a boundary-degree coordinate on the corresponding boundary vertex. A coordinate is u-*supported *if its declared support meets T_u, or if it is one of the completion-port labels,* *receiver-entry channel lengths, connector-band constraints, boundary-degree entries, or P_{13} labels whose declared support in Definition 3.87 belongs to a receiver-entry return whose internal channel contains T_u or is canonically assigned to the terminal receiver edge of T_u. Let $\mathcal{R}_u(S)$ be the finite family of u-supported coordinates carried by S. The trace-incidence coordinate recording the labelled path T_u is included whenever $T_u \subseteq S$; hence a trace basin always has at least one internal u-supported coordinate.*

For $S \subseteq X$, let

$$\mathrm{res}_{X,S} : \mathcal{R}_u(X) \dashrightarrow \mathcal{R}_u(S) \cup \{\mathbf{d}_\partial(S)\text{-}entries\}$$

be the partial restriction map which is defined on a coordinate precisely when the coordinate's declared support is carried by S, or when its value is fixed by the boundary degree profile $\mathbf{d}_\partial(S)$. A connected $S \subseteq X$ containing T_u is trace-complete for u *if $\mathrm{res}_{X,S}$ is defined on every coordinate of $\mathcal{R}_u(X)$ and the restricted value equals the value of that coordinate in X. Since X itself is trace-complete for u, the finite set of trace-complete connected subgraphs has lexicographically first inclusion-minimal elements. The* trace basin B_u *is the lexicographically first such element.*

The trace-response state *of B_u is*

$$\rho_u(B_u) = (\mathbf{d}_\partial(B_u), \mathcal{R}_u(B_u), \mathsf{H}_{\partial B_u}(B_u)).$$

The family $\mathcal{R}_u(B_u)$ is the complete declared coordinate family for the u-supported target events used in the route-8 branch: once the boundary degree profile and all entries of $\mathcal{R}_u(B_u)$ are fixed, every compatible outside context has the same truth value for each such declared event. Response-support quotients below are tested only against this declared u-supported target algebra. A response quotient *of $\rho_u(B_u)$ is obtained by identifying or forgetting entries of the finite coordinate family $\mathcal{R}_u(B_u)$ while preserving the boundary degree profile. A realization of such a quotient is a boundaried response state with the same boundary degree profile whose image under the quotient map is the given quotient. The quotient is* target-complete *for $\rho_u(B_u)$ if, for every outside ∂B_u-context compatible with the boundary profile, all realizations of the quotient give the same target predicate as $\rho_u(B_u)$ after gluing. Thus a target-complete quotient identifies only target-indistinguishable response states. The quotient is* nontrivial *if it forgets or identifies a coordinate whose declared support meets $B_u - \partial B_u$, or whose declared support contains an internal edge of B_u, that is, an edge of B_u with both endpoints in $V(B_u)$.*

A response quotient of $\rho_u(B_u)$ is trace-local and support-internal *if it identifies or forgets entries of the fixed coordinate family $\mathcal{R}_u(B_u)$, preserves the full boundary degree profile, and does not delete a boundary incidence from the incidence set used later in Definition 14.48. Thus incidence-restriction and response-support-deletion quotients are excluded from trace-complete-minimality; they are tested only in the two-support clause of Definition 14.49.*

The trace basin B_u is target-complete-minimal *when none of the following four alternatives occurs:*

(a) a trace-local quotient or identification inside $\rho_u(B_u)$ is distinguished by some outside ∂B_u-context, hence is target-defective;

(b) $\rho_u(B_u)$ has a nontrivial target-complete response quotient with the same boundary degree profile; when this quotient is realized by a smaller connected representative, it is a target-complete compression of a proper boundaried piece;

(c) an equality among coordinates of $\rho_u(B_u)$ becomes target-complete only after adjoining a larger connected support $Z \supsetneq B_u$, either with $Z \subsetneq G$ or with $Z = G$;

(d) *two declared outside connector configurations of $\rho_u(B_u)$, with the same boundary-degree image, have a surviving first separator in the sense of Definition 14.11; by Lemma 14.24 this separator has ambient degree at least* 4*.*

Thus a trace basin that is not target-complete-minimal realizes, respectively, one of the quotient, response-compression, support-dependence, or decorated handoff fan alternatives used below in exits (4)–(7).

Definition 14.15 (Silent-core residual profile)**.** *A* silent-core residual profile *is the data*

$$(X, w, A(w), E(w), \mathcal{U}(w), (B_u)_{u \in \mathcal{U}(w)}, \{\Lambda_X(r, w)\}_r, \mathcal{B})$$

obtained from a saturated Type A receiver w when no port carries four visible receiver-entry returns and none of the target, quotient, compression, support-dependence, or decorated handoff fan alternatives applies. Here $A(w)$ is the canonical payable set of Definition 14.29, $E(w)$ is the excess basin, $\mathcal{U}(w)$ is the set of unpaid silent routed vertices in that basin, $(B_u)_{u \in \mathcal{U}(w)}$ is the indexed family of trace basins from Definition 14.14, and $\mathcal{B}$ records the completion-port labels and connector-length bands forced by Lemmas 14.9 and 14.12. Such a profile is admissible *only if every B_u with $u \in \mathcal{U}(w)$ is target-complete-minimal and all connector bands, P_{13} labels, boundary degree profiles, and cross-port theta constraints are compatible with the standing counterexample invariants.*

Definition 14.16 (Saturated receiver exits)**.** *Let w be a saturated receiver in a Type A support. The saturated receiver branch has the following eight exits:*

(1) *an anchored return through a completion port of w has length in $\mathcal{M}$;*

(2) *two anchored receiver-entry returns through one completion port are internally vertex-disjoint as anchored paths and their lengths sum to a power of two;*

(3) *a shared P_{13} window violates the corresponding legal-label relation C_s;*

(4) *a quotient in the canonical exit-(4) family $\mathcal{Q}_4(w)$ of Definition 14.49 is target-defective;*

(5) *the response data give a nontrivial target-complete response compression; when this compression is realized by a smaller proper boundaried piece, it is a target-complete compression in the sense of Lemma 3.96;*

(6) *the equality of responses delocalizes to a proper support or to the whole graph;*

(7) *a high-degree decorated handoff fan envelope is produced;*

(8) *none of the previous alternatives occurs, and the data form the silent-core residual profile of Definition 14.15.*

Exits (1)–(3), (5), and (6) are closed exits*. Exit (4) is the* target-defect peeling *exit: it removes exactly the routed load whose declared coordinate is used by the canonical target-defective quotient, as in Definition 14.33 and Lemma 14.38. Exit (7) is the* Type B handoff *exit. Exit (8) is the* route-8 residual *exit.*

Handoff interface. The next definitions are the Type B objects used by the Type A exit-(7) handoff. They are stated here so that the saturated Type A analysis can invoke the interface without using the later Type B exclusion theorem.

Definition 14.17 (Fan-safe relation)**.** *For a high-degree vertex h, the fan-safe graph $\mathcal{F}_h$ has vertex set $N(h)$. Distinct vertices $u, v \in N(h)$ are adjacent in $\mathcal{F}_h$ when the pair satisfies all of the following five conditions:*

(i) there is no simple u–v return in $G - h$ of length $2^j - 2$;

(ii) the P_{13} labels carried by the pair satisfy the relevant legal-label obstruction relation, in particular $C_2(S(u), S(v)) = 1$ in the length-two fan case;

(iii) the quotient identifying the two boundary-response coordinates is not target-defective;

(iv) no target-complete compression of the fan pair preserves the boundary degree profile in a smaller proper representative;

(v) the equality of the two response coordinates does not delocalize to a proper larger support or to the whole graph.

Definition 14.18 (Marked Type B fan)**.** *A* fan-certificate labelling *of a high-degree center h is a map*

$$S_h : N(h) \longrightarrow \mathcal{L}$$

from the fan neighbours to the legal nonempty P_{13} labels of Lemma 6.1, such that

$$C_2(S_h(u), S_h(v)) = 1 \qquad (u \neq v \in N(h)).$$

A Type B fan is certificate-marked *when this labelling is part of the assigned support data. A high-degree center assigned to a Type B support but not certificate-marked is a* fan-certificate residual center*; it is included in the Type B bridge-residual mass of Definition 14.95 and is not used in the certificate-closed local discharging step.*

A marked Type B fan *centered at a high-degree vertex h is the data of h, the neighbour set $N(h)$, and a fan-certificate labelling S_h. Since $d_G(h) \geq 4$, Lemma 3.2 forces every neighbour of h to have degree three in G. A neighbour $u \in N(h)$ is* cubic-closed *in the marked fan if, after assigning the fan support, the two non-h incidences of u are also assigned to that support. Equivalently, u has internal degree 3 in the assigned fan envelope and contributes charge $-\frac{1}{4}$. Thus*

$$N_G(u) = \{h, x_u, y_u\}.$$

When x_u and y_u lie in the packed P_{13} windows, their incidences are recorded in the window-incidence profile below; when they lie in the same window, the corresponding fan-certificate label is non-singleton. A neighbour that is not cubic-closed has internal degree at most 2 in the assigned fan support and contributes charge at least $\frac{3}{4}$.

Definition 14.19 (Decorated handoff fan envelope)**.** *A* decorated handoff fan envelope *is a pair*

$$\mathfrak{X} = (Y, H),$$

together with the handoff-arm data described below. Here $Y \subseteq R$ is a connected P_{13}-free remainder core and $H \subseteq V_{\geq 4}(G)$ is a set of high-degree decorations assigned to Y.

For each $h \in H$ there is a nonempty assigned first-neighbour set

$$K_h \subseteq N_G(h).$$

For every $a \in K_h$ the envelope records a simple handoff arm

$$A_{h,a} = a = x_0, x_1, \ldots, x_{\ell(a)} = y_{h,a}, \qquad y_{h,a} \in V(Y),$$

whose internal vertices avoid $Y \cup H \cup \{h\}$. The full handoff path from h to Y is the edge ha followed by $A_{h,a}$. The first neighbours in K_h are distinct. The arm union for a fixed h is finite and is recorded as part of the decorated response support; different arms may share a later outside suffix, in which case the shared vertices and edges are counted once in that declared support. The ordinary adjacent fan is the special case $\ell(a) = 0$ for all a, so $K_h \subseteq V(Y)$.

The set K_h is required to be a clique in the fan-safe graph $\mathcal{F}_h$ of Definition 14.17. This condition is imposed on the actual neighbours of h, not on the terminal vertices $y_{h,a}$. Thus any return from a to b in $G - h$ of length $2^j - 2$ would close with the two edges ha, hb to form a cycle of length 2^j; the return is allowed to run through the recorded arms and through Y.

The branch data producing the envelope include the boundary response coordinates incident with h, the handoff arms, their terminal coordinates in Y, and the boundary-degree profile of the decorated support. Its net charge is

$$\mathcal{N}(\mathfrak{X}) = \partial_+(Y) - \sum_{h \in H} (d_G(h) - 3) - \frac{1}{4}|V(Y)|.$$

The P_{13}-free and empty-3-core hypotheses are imposed on the counted core Y; the vertices of H supply surplus and fan-return constraints, while the handoff arms are declared response data used to route the Type A separator into the Type B fan branch.

Definition 14.20 (Decorated Type B envelope supports)**.** *Let $\mathcal{Y}$ be a finite family of pairwise vertex-disjoint Type A cores which produce exit-(7) handoffs. A* decorated Type B envelope support *for $\mathcal{Y}$ is obtained from the bipartite core–center incidence graph $\mathcal{I}(\mathcal{Y})$. The left vertices of $\mathcal{I}(\mathcal{Y})$ are the cores $Y \in \mathcal{Y}$; the right vertices are the ambient high-degree handoff centers; and Y is adjacent to h when one of the exit-(7) handoffs from Y has center h. For a connected component $\mathfrak{C}$ of this incidence graph, write*

$$\mathcal{Y}_{\mathfrak{C}} = \{Y \in \mathcal{Y} : Y \in V(\mathfrak{C})\}, \qquad H_{\mathfrak{C}} = \{h : h \in V(\mathfrak{C})\}.$$

The grouped envelope for $\mathfrak{C}$ has counted core

$$Y^*_{\mathfrak{C}} = \bigcup_{Y \in \mathcal{Y}_{\mathfrak{C}}} Y$$

and center-token sum

$$\omega(\mathfrak{C}) = \sum_{h \in H_{\mathfrak{C}}} \omega(h), \qquad \omega(h) = d_G(h) - 3.$$

This uses the ambient surplus of h whether h lies in the remainder or in a packed P_{13}-window. The grouped envelope uses $\omega(h)$ once inside its incidence component. If the same surplus unit is also canonically assigned to the ordinary support containing h, that ordinary use is counted separately in the at-most-twice global fan-mass bound of Definition 14.95.

The net charge of the grouped envelope is

$$\mathcal{N}(\mathfrak{X}^*_{\mathfrak{C}}) = \sum_{Y \in \mathcal{Y}_{\mathfrak{C}}} \partial_+(Y) - \omega(\mathfrak{C}) - \frac{1}{4} \sum_{Y \in \mathcal{Y}_{\mathfrak{C}}} |V(Y)|.$$

The handoff arms and fan-safe neighbour sets are the union of the corresponding decorated handoff fan envelopes in $\mathfrak{C}$, with shared outside vertices and edges counted once.

Lemma 14.21 (Decorated envelope transfer has no double counting)**.** *Let $\mathcal{Y}$ be a family of pairwise disjoint Type A cores which produce exit-(7) handoffs, and form the grouped decorated Type B envelope supports of Definition 14.20. Then the Type A core deficiencies are counted exactly once in the grouped envelope family, each ambient handoff-center token is counted exactly once in that family, and*

$$\sum_{\mathfrak{C}} \mathcal{N}(\mathfrak{X}^*_{\mathfrak{C}}) = \sum_{Y\in\mathcal{Y}} \left(\partial_+(Y) - \frac{1}{4}|V(Y)|\right) - \sum_{\mathfrak{C}} \omega(\mathfrak{C}).$$

Consequently a negative Type A handoff contribution is not discarded when the branch leaves the Type A calculation; it is transferred to the grouped Type B envelope ledger.

Proof. The cores in $\mathcal{Y}$ are components of the canonical decomposition, so their vertex sets and positive deficiencies are disjoint summands. The connected components of the core–center incidence graph partition the set of cores and the set of handoff centers. Thus a core with handoffs to several centers is counted once, in the single component containing all those centers, and a center serving several cores contributes its token once to that same component. Summing the displayed definition of $\mathcal{N}(\mathfrak{X}^*_{\mathfrak{C}})$ over the incidence components gives the identity. The identity is an equality of the transferred ledger entries, so no core deficiency or grouped-envelope center token is counted twice inside the handoff envelope family. $\square$

Lemma 14.22 (Window handoff center accounting)**.** *Let h be an exit-(7) handoff center lying in a packed P_{13}-window. Then either exit (3) occurs at the saturated receiver that produced the handoff, or the grouped envelope containing h is charged to the surplus token $d_G(h)-3$ of the window vertex h.*

Proof. The handoff arms through a packed window carry the legal P_{13}-labels of Lemma 6.1. If two such arms violate the relevant relation C_s, then the saturated receiver realizes the label-collision exit (3). Otherwise the labels are legal and the handoff survives as a decorated Type B envelope. By Lemma 14.24, the handoff center has $d_G(h) \geq 4$, so it supplies the positive surplus token $d_G(h) - 3$. That token is one of the center tokens of the incidence component containing h in Definition 14.20. $\square$

Lemma 14.23 (Continuation routing at a completion port)**.** *Let X be a Type A support, let $\vec{e}$ be a completion port, and let $\mathcal{K}$ be a finite family of declared response coordinates through $\vec{e}$ lying in one boundary-degree fibre. Suppose the coordinates in $\mathcal{K}$ are pairwise distinct in every target-complete quotient and are distinguished by the actual outside context. Then either one of exits (4)–(6) of Definition 14.16 occurs, or $\mathcal{K}$ has a surviving first separator in the sense of Definition 14.11.*

Proof. Each coordinate in $\mathcal{K}$ has a finite outside connector configuration by Definition 14.11. If two coordinates have the same connector configuration through their first entry into X and the same image in the boundary-degree fibre, then identifying them is an identification of declared coordinates on a finite response state. By Lemma 3.95, either some outside context distinguishes that identification, which is exit (4), or the identification is target-complete. In the latter case it is nontrivial because the two declared coordinates are pairwise distinct in every surviving target-complete quotient; hence it is exit (5) unless its validity requires adjoining a larger connected support, which is exit (6).

Thus, if exits (4)–(6) do not occur, some pair of configurations must separate before their first-entry response data are identified. Since $\mathcal{K}$ is finite and each configuration is finite, there is a first such separator in the prefix order. If that separator were absorbed, the definition of absorbed gives exactly one of exits (4)–(6). Therefore, outside exits (4)–(6), the first separator is surviving. □

Lemma 14.24 (Cubic switch absorption)**.** *Let z be a surviving first separator for two declared response coordinates through one completion port of a Type A support. Then*

$$d_G(z) \geq 4.$$

Proof. Let the two connector configurations have common prefix ending at z. If z is the initial outside vertex h of the completion port, the port edge wh is the root incidence at z; otherwise the last edge of the common prefix is the root incidence. Since the two configurations separate at z, they use two distinct next incidences after z. Hence $d_G(z) \geq 3$, and $d_G(z) \leq 2$ is impossible.

Assume $d_G(z) = 3$. Then the root incidence and the two next incidences used by the separated configurations are all incidences of z. Consequently the switch support S_z of Definition 14.11 has no unused ambient incidence at z. The two separated responses therefore form a finite declared boundaried response state with the same boundary-degree profile: the boundary records the root incidence, the two connector tails to their first entries in X, and the two receiver-entry channels. Consider the quotient that identifies the two separated response coordinates on this finite state. If some compatible outside context distinguishes the two responses, the quotient is target-defective by Lemma 3.95, which is exit (4). Otherwise the identification is target-complete. If all declared support needed for this target-complete identification is already contained in S_z, then the quotient is nontrivial, because the two response coordinates are distinct, and this is exit (5). If some declared support needed for the identification lies outside S_z, choose the lexicographically first smallest connected support $Z \supsetneq S_z$ carrying it. When $Z \subsetneq G$ this is proper support dependence, and when $Z = G$ it is whole-graph support dependence; in both cases the branch is exit (6).

Each of these alternatives is exactly the absorbed case in Definition 14.11. This contradicts that z is surviving. Thus $d_G(z) \neq 3$, and the already-proved inequality $d_G(z) \geq 3$ gives $d_G(z) \geq 4$. □

Lemma 14.25 (High-degree separator handoff)**.** *Let X be a Type A support, and let z be a surviving first separator for a finite family of declared response coordinates through one completion port. Then z, together with the separated connector tails from z to their first-entry data in X, produces a decorated handoff fan envelope in the sense of Definition 14.19.*

Proof. By Lemma 14.24, $d_G(z) \geq 4$, so z is an eligible high-degree decoration. Take $Y = X$ as the counted remainder core and take $H = \{z\}$. For each continuation class that leaves z through a next neighbour a, let $A_{z,a}$ be the remaining connector tail from a to its first-entry receiver in X; if $a \in X$, this tail has length 0. These are simple paths because the receiver-entry returns are simple. Distinct first neighbours give distinct first-neighbour data. If two tails later meet before their first entry into X, the common suffix is part of the finite arm union and is recorded once. Thus the required handoff-arm data are well-defined.

It remains to verify the fan-safe condition on K_z. Suppose that two assigned first neighbours $a, b \in K_z$ are not adjacent in $\mathcal{F}_z$. The five defining failures of Definition 14.17 give, respectively, a return in $G - z$ of length $2^j - 2$, a P_{13} label collision, a target-defective quotient, a target-complete compression, or proper/whole-graph support dependence. In the first case the return together with the two edges za, zb is a power-of-two cycle in G. In the remaining cases the branch is one of exits (3)–(6) of Definition 14.16, so the separator is absorbed rather than surviving. Hence every pair in K_z is fan-safe, and K_z is a clique in $\mathcal{F}_z$. The data just constructed are exactly a decorated handoff fan envelope. □

Lemma 14.26 (Decorated fan admissibility)**.** *If exit (7) of Definition 14.16 occurs in a saturated Type A branch, the resulting decorated handoff fan envelope carries exactly the data required by the Type B fan calculation: contextual target-safety, a P_{13}-free empty-3-core remainder core, hereditary target-uncompressibility of the decorated boundaried profile, and fan-return-safety at every decoration $h \in H$.*

Proof. For a Type A exit (7), the counted core Y is the Type A support X used in Lemma 14.25. Hence Y inherits contextual target-safety, P_{13}-freeness, the empty internal 3-core condition, and hereditary target-uncompressibility from Definition 13.2. The only new vertices counted outside Y are the decorations $h \in H$, and their surplus appears only through the displayed term $d_G(h) - 3$ in Definition 14.19.

For a decoration h, the assigned set K_h consists of actual neighbours of h. If two vertices $a, b \in K_h$ failed the first fan-safe condition, then there would be a simple a–b return in $G - h$ of length $2^j - 2$; adding the two fan edges ha, hb would give a cycle of length 2^j in G. Failures of the other four fan-safe conditions are exactly the label, target-defect, target-compression, and support-dependence exits already removed before exit (7). Thus K_h is a clique in $\mathcal{F}_h$. The handoff arms are declared coordinates in the signature of Definition 3.87, so they do not add undeclared response data. The decorated envelope therefore satisfies the Type B hypotheses later consumed by Lemma 14.93. This lemma verifies those hypotheses; it does not use the conclusion of Lemma 14.93. □

Lemma 14.27 (Raw Type A thresholds)**.** *Let $w \in \mathcal{R}_j(X)$, so $d_X(w) = 2 - j$ and $q(w) = j + 1$. The first routed load value at which w leaves the unsaturated Type A discharging branch is*

$$H_j = 4q(w) = 4(j + 1).$$

Consequently

$$H_0 \le 4, \qquad H_1 \le 8, \qquad H_2 \le 12.$$

If the saturated branch has been eliminated, then every receiver in $\mathcal{R}_j(X)$ satisfies

$$L(w) \le H_j - 1 = 4(j + 1) - 1.$$

Proof. The charge of a receiver is

$$\mathrm{ch}(w) = q(w) - \frac{1}{4}.$$

Each routed cubic vertex assigned to w costs $\frac{1}{4}$. The largest load that can be paid while keeping nonnegative final charge is the largest integer L with

$$q(w) - \frac{1}{4} - \frac{1}{4}L \ge 0,$$

namely $L \le 4q(w) - 1$. The next value, $4q(w)$, is exactly the saturated threshold. Substituting $q(w) = 1, 2, 3$ for internal degrees $2, 1, 0$ gives $4, 8, 12$. After the saturated cases are removed, the surviving capacities are $3, 7, 11$. □

Lemma 14.28 (Visible saturated receiver: exits 1–7)**.** *Let w be a receiver in a Type A support. If some completion port of w carries four visible receiver-entry returns, then the branch realizes one of exits (1)–(7) of Definition 14.16.*

Proof. Fix a port $\vec{e} = (w, h)$ carrying four visible receiver-entry returns. Each return has the form $\Gamma \circ Q$, with the same completion port and with Q carrying a distinct canonical trace T_u. If one return has length in $\mathcal{M}$, Lemma 2.2 gives a power-of-two cycle; this is exit (1). If two returns are internally disjoint as anchored h-to-w paths and their full anchored-return lengths sum to a power of two, the union is a forbidden cycle by Lemma 14.6; this is exit (2). If two traces pass through a common P_{13} window, their labels are governed by the relations C_s of Lemma 6.1; failure of the corresponding C_s test is the stated label collision, which is exit (3).

Assume exits (1)–(3) do not occur. For each of the four visible returns record the local target-response coordinate consisting of its completion port, its first-entry receiver, its connector label, and its receiver-entry channel. These four coordinates lie in one boundary-degree fibre in the sense of Definition 3.81. If identifying two of them fails the context-universal target test of Lemma 3.95, the quotient is target-defective; this is exit (4). If the identification is target-complete on a nontrivial response state, this is exit (5); when it is realized on a proper boundaried piece, it contradicts Corollary 3.98. If the identification becomes target-complete only after adjoining a larger support, it is proper support dependence when that support is proper and whole-graph support dependence when the support is all of G; these are excluded by Lemmas 10.6 and 10.9 and give exit (6).

It remains that the four response coordinates are pairwise distinct in every target-complete quotient and are separated in the actual outside context. Lemma 14.23, applied to this four-coordinate family, gives a surviving first separator for their outside connector configurations. By Lemma 14.24, this separator has ambient degree at least 4. By Lemma 14.25, the separator and its separated connector tails produce a decorated handoff fan envelope. This is exit (7). Lemma 14.26 is used here only to record the handoff interface data; the Type B exclusion conclusion is not used in this Type A lemma. ☐

Definition 14.29 (Excess trace support)**.** *Let w be a receiver with $L(w) \geq 4q(w)$, and assume no completion port of w carries four visible receiver-entry returns. The canonical payable set $A(w)$ is formed in the visible-first elimination order: first list the visible routed loads of w in canonical port order, breaking ties by the canonical trace order, and then list the remaining routed loads by the canonical trace order. Let $A(w)$ be the first $4q(w) - 1$ routed loads in this order, and*

$$E(w) = \{u : r(u) = w\} \setminus A(w)$$

is the excess basin*. Let*

$$\mathcal{U}(w) = \{u \in E(w) : u \text{ is silent}\}.$$

The set $E(w)$ is nonempty because

$$L(w) \geq 4q(w) > 4q(w) - 1 = |A(w)|.$$

Lemma 14.30 (Visible-first excess is silent and quantitative)**.** *Let X be a Type A support. Suppose that no saturated receiver of X has a completion port carrying four visible receiver-entry returns, and form the visible-first excess basins of Definition 14.29. Define*

$$S_{\text{sil}}^{\text{exc}}(X) = \sum_{w} |\mathcal{U}(w)|,$$

where the sum is over the receivers of X and $\mathcal{U}(w) = \varnothing$ for unsaturated receivers. Put

$$D_A(X) = \frac{1}{4}|V(X)| - \partial_+(X).$$

Then

$$S_{\text{sil}}^{\text{exc}}(X) \geq n_3 - 3n_2 - 7n_1 - 11n_0 = 4D_A(X),$$

where $n_i = |\{v \in V(X) : d_X(v) = i\}|$.

Proof. For a receiver w, put

$$c(w) = 4q(w) - 1.$$

If $L(w) \le c(w)$, then w contributes no unpaid routed vertex. If $L(w) > c(w)$, then w is saturated. By hypothesis no completion port of w carries four visible receiver-entry returns. Since w has exactly $q(w)$ completion ports, each port carries at most three visible loads, so

$$L_{\mathrm{vis}}(w) \le 3q(w).$$

For $q(w) \in \{1, 2, 3\}$ one has

$$3q(w) \le 4q(w) - 1 = c(w).$$

The visible-first order therefore pays every visible routed load before the payable set is exhausted. Hence every unpaid routed vertex at w is silent, and

$$|\mathcal{U}(w)| = L(w) - c(w) \qquad \text{whenever } L(w) > c(w).$$

For receivers with $L(w) \le c(w)$ the equality is replaced by the inequality $|\mathcal{U}(w)| = 0 \ge L(w) - c(w)$. Summing over all receivers gives

$$S^{\mathrm{exc}}_{\mathrm{sil}}(X) \ge \sum_w \big(L(w) - c(w)\big).$$

By Lemma 14.4, every internal degree-3 vertex is routed to exactly one receiver, so $\sum_w L(w) = n_3$. The receivers of internal degree $2, 1, 0$ have $q = 1, 2, 3$, and hence capacities $3, 7, 11$, respectively. Thus

$$\sum_w c(w) = 3n_2 + 7n_1 + 11n_0.$$

This proves the first displayed inequality.

Finally,

$$\partial_+(X) = 3n_0 + 2n_1 + n_2, \qquad |V(X)| = n_0 + n_1 + n_2 + n_3,$$

so

$$4D_A(X) = 4\left(\frac{1}{4}|V(X)| - \partial_+(X)\right) = n_3 - 11n_0 - 7n_1 - 3n_2.$$

This is the same expression as above. □

Lemma 14.31 (Reduced silent-excess residual)**.** *Let w be a saturated receiver in a Type A support, and assume no completion port of w carries four visible receiver-entry returns. Then either one of exits (4)–(7) of Definition 14.16 occurs, or every $u \in \mathcal{U}(w)$ has a target-complete-minimal trace basin B_u.*

Proof. Fix $u \in \mathcal{U}(w)$ and consider its trace basin B_u from Definition 14.14. The definition of target-complete-minimality lists four possible ways for B_u to fail to be residual-minimal. If a trace-local quotient or identification inside $\rho_u(B_u)$ is distinguished by an outside context, then the attempted quotient is target-defective, which is exit (4). If $\rho_u(B_u)$ has a nontrivial target-complete response quotient with the same boundary degree profile, then this is exit (5). When the quotient is realized by a smaller proper boundaried piece, it is forbidden by hereditary target-uncompressibility; when it is only a trace-response quotient, it is already failure alternative (b) in Definition 14.14. If an equality among coordinates of $\rho_u(B_u)$ becomes target-complete only after adjoining a larger connected support, then the dependence is proper support dependence when the support is proper and whole-graph support dependence when the support is all of G; this is exit (6). If two declared connector configurations in $\rho_u(B_u)$ have a surviving first separator, then Lemma 14.24 gives ambient degree at least 4, and Lemma 14.25 records that separator as a decorated handoff fan envelope. This is exit (7). Therefore, if none of exits (4)–(7) occurs, none of the four failure alternatives occurs for any $u \in \mathcal{U}(w)$, and each such trace basin is target-complete-minimal. □

Lemma 14.32 (Silent and mixed excess: exits 4–8)**.** *Let w be a saturated receiver in a Type A support, and assume no completion port of w carries four visible receiver-entry returns. Then the excess basin $E(w)$ realizes one of exits (4)–(8) of Definition 14.16.*

Proof. The vertices in $E(w)$ are precisely the routed cubic vertices that cannot be paid by the nonnegative receiver charge. Put $c(w) = 4q(w) - 1$. Since no completion port of w carries four visible receiver-entry returns and w has exactly $q(w)$ completion ports,

$$L_{\text{vis}}(w) \leq 3q(w) \leq 4q(w) - 1 = c(w).$$

The payable set $A(w)$ is ordered visible-first, so all visible routed loads are included in $A(w)$. Since

$$E(w) = \{u : r(u) = w\} \setminus A(w),$$

no vertex of $E(w)$ is visible at any completion port of w; equivalently, every excess vertex is silent. Let $B(w)$ be the boundaried subgraph consisting of the excess basin, the canonical traces from its vertices to w, and the completion-port boundary data through which those traces can be completed. Its boundary response is a coordinate in the target-response collection of Definition 3.81.

If replacing $B(w)$ by its boundary response changes the target predicate in some context, the induced quotient is target-defective; this is exit (4). If the replacement is a nontrivial target-complete response compression, this is exit (5); when supported on a proper boundaried piece it is forbidden by Corollary 3.98, and when it is only a trace-response quotient it violates target-complete-minimality. If the equality of responses becomes target-complete only after adjoining a larger support, the dependence is either proper support dependence, excluded by Lemma 10.6, or whole-graph support dependence, excluded by Lemma 10.9; this is exit (6). If the response of the basin is separated in the outside context, then Lemma 14.23 either has already routed it to exits (4)–(6), or supplies a surviving first separator. By Lemmas 14.24 and 14.25, that separator produces a decorated handoff fan envelope. The admissibility lemma Lemma 14.26 records the Type B interface data for that envelope; it does not use the Type B exclusion theorem. This is exit (7).

If none of exits (4)–(7) occurs, then Lemma 14.31 applies: every unpaid silent routed vertex in $\mathcal{U}(w)$ has a target-complete-minimal trace basin. The only remaining data are exactly those listed in the silent-core residual profile of Definition 14.15: the excess loads are silent after the visible-first payment, their indexed trace basins are target-complete-minimal, no target-defective quotient occurs, no target-complete compression occurs, no support dependence occurs, and no high-degree fan decoration is created. In that residual case, Lemmas 14.9 and 14.12 still impose nonempty anchored-return and connector-band constraints on every completion port. This is exit (8). □

Definition 14.33 (Exit-(4) peeling data)**.** *Let w be a Type A receiver and put*

$$\mathcal{L}(w) = \{u : r(u) = w\}.$$

An exit-(4) witness for a routed load $u \in \mathcal{L}(w)$ *consists of a quotient $q \in \mathcal{Q}_4(w)$, two realizations in the same boundary-degree fibre, and a compatible outside context distinguishing their target* *predicates, such that the declared support of q contains the canonical coordinate of u generated by Definition 3.87. A* peeling set $P_4(w) \subseteq \mathcal{L}(w)$ *is a set of routed loads, each equipped with one exit-(4) witness, and with no routed load listed twice. The witness serves only as a routing record for its selected load. The receiver charge changes because the selected routed load is removed from the pure Type A receiver calculation. The residual load after peeling is*

$$L_4(w) = L(w) - |P_4(w)|.$$

Lemma 14.34 (Exact charge after exit-(4) peeling)**.** *Let w be a Type A receiver with peeling set $P_4(w)$. After the loads in $P_4(w)$ have left the Type A receiver calculation through exit (4), the remaining receiver charge is*

$$q(w) - \frac{1}{4} - \frac{1}{4}L_4(w).$$

In particular, if $L_4(w) \leq 4q(w) - 1$, then the remaining charge at w is nonnegative.

Proof. The receiver w has initial charge $q(w) - \frac{1}{4}$, and every routed cubic vertex assigned to w costs exactly $\frac{1}{4}$ in Lemma 14.43. The peeling set removes exactly $|P_4(w)|$ routed loads from this receiver calculation, with no routed load removed twice. The target-defective witnesses justify routing those selected loads out of the pure Type A branch; the displayed charge counts the selected removed loads. Therefore the number of routed loads still charged to w is $L(w) - |P_4(w)| = L_4(w)$, giving the displayed formula. If $L_4(w) \leq 4q(w) - 1$, then

$$q(w) - \frac{1}{4} - \frac{1}{4}L_4(w) \geq q(w) - \frac{1}{4} - \frac{1}{4}(4q(w) - 1) = 0.$$

□

Lemma 14.35 (Unpeeled visible-return routing)**.** *Let w be a saturated Type A receiver with a peeling set $P_4(w)$. Suppose that a completion port of w carries four visible receiver-entry returns whose routed loads lie in $\mathcal{L}(w) \setminus P_4(w)$. Then the unpeeled data realize one of exits (1)–(7) of Definition 14.16. If the realized exit is (4), the target-defective quotient has declared routed-load support containing at least one of those four unpeeled loads.*

Proof. Let $u_1, u_2, u_3, u_4 \in \mathcal{L}(w) \setminus P_4(w)$ be the four visible loads, and let the corresponding receiver-entry returns pass through the same completion port $\vec{e} = (w, h)$. Each return has the form $\Gamma_i \circ Q_i$, where Q_i contains the canonical trace T_{u_i}. The construction uses only these four returns and their declared response coordinates, so previously peeled loads are outside the data used below.

If one return has length in $\mathcal{M}$, Lemma 2.2 gives exit (1). If two of the four anchored returns are internally vertex-disjoint and their lengths sum to a power of two, Lemma 14.6 gives exit (2). If two traces pass through a common P_{13}-window and fail the legal C_s-relation of Lemma 6.1, this is exit (3).

Assume exits (1)–(3) do not occur. For each i, form the declared target-response coordinate determined by the completion port, the first-entry receiver, the connector label, and the receiver-entry channel of the return for u_i. These four coordinates lie in one boundary-degree fibre. If a quotient identifying two of them is distinguished by a compatible outside context, then it is a target-defective quotient in the canonical family $\mathcal{Q}_4(w)$, of type (Q1) in Definition 14.49. Its declared routed-load support contains the corresponding u_i's, so exit (4) supports an unpeeled load.

If the identification is target-complete on a nontrivial response state, the branch realizes exit (5). If target-completeness requires adjoining a larger support, the branch realizes exit (6), proper or global according to whether the support is proper or all of G. Finally, if the four coordinates remain pairwise distinct in every target-complete quotient and are separated by the actual outside context, Lemma 14.23 supplies either one of exits (4)–(6) or a surviving first separator. In the exit-(4) subcase, the quotient is of type (Q4) in Definition 14.49, and its declared routed-load support is contained in $\{u_1, u_2, u_3, u_4\}$. In the separator subcase, Lemmas 14.24 and 14.25 produce the decorated handoff fan envelope, which is exit (7). ☐

Lemma 14.36 (Unpeeled silent-excess routing)**.** *Let w be a saturated Type A receiver with peeling set $P_4(w)$. Suppose*

$$L_4(w) \geq 4q(w)$$

and no completion port of w carries four visible receiver-entry returns from unpeeled loads. Order $\mathcal{L}(w) \setminus P_4(w)$ by the canonical visible-first order, let $A_4(w)$ be the first $4q(w) - 1$ unpeeled loads, and put

$$E_4(w) = (\mathcal{L}(w) \setminus P_4(w)) \setminus A_4(w).$$

Then $E_4(w)$ is nonempty and realizes one of exits (4)–(8) of Definition 14.16. If the realized exit is (4), the target-defective quotient has declared routed-load support containing a load of $E_4(w)$.

Proof. The inequality $L_4(w) \geq 4q(w)$ gives

$$|E_4(w)| = L_4(w) - (4q(w) - 1) \geq 1.$$

Since no completion port has four visible unpeeled returns and w has exactly $q(w)$ completion ports, the number of visible unpeeled loads is at most $3q(w)$. For $q(w) \in \{1, 2, 3\}$,

$$3q(w) \leq 4q(w) - 1.$$

The visible-first order therefore places every visible unpeeled load in $A_4(w)$. Hence every load in $E_4(w)$ is silent at every completion port of w.

Let $B_4(w)$ be the boundaried subgraph consisting of the loads in $E_4(w)$, their canonical traces to w, and the completion-port boundary data through which those traces can be completed. If the boundary response of $B_4(w)$ is target-defective in some compatible outside context, the corresponding quotient is of type (Q2) in Definition 14.49; its declared routed-load support is $E_4(w)$. This is exit (4), and it supports an unpeeled residual excess load.

If the boundary response of $B_4(w)$ has a nontrivial target-complete compression, the branch realizes exit (5). If equality of the relevant responses becomes target-complete only after adjoining a larger support, the branch realizes exit (6), proper or global according to that support. If the declared connector response data in $B_4(w)$ have a surviving first separator, then Lemmas 14.24 and 14.25 produce a decorated handoff fan envelope, giving exit (7).

Assume none of exits (4)–(7) occurs. For each $u \in E_4(w)$, consider its trace basin B_u. The four failure alternatives in Definition 14.14 are precisely: a trace-local target-defective quotient, a nontrivial target-complete response quotient, proper/whole-graph support dependence of an equality, or a surviving first separator. In the first case the quotient is canonical of type (Q3) and has declared routed-load support $\{u\}$, so it is exit (4). In the second case the branch realizes exit (5). In the third case it realizes exit (6). In the fourth case Lemmas 14.24 and 14.25 produce the exit-(7) decorated handoff fan envelope. Since all four exits are absent, every $u \in E_4(w)$ has a target-complete-minimal trace basin. The loads in $E_4(w)$ are silent, no target-defective quotient occurs, no target-complete compression occurs, no support dependence occurs, and no high-degree fan decoration is created. This is exactly the silent-core residual profile of Definition 14.15, namely exit (8). ☐

Lemma 14.37 (Residual saturated routing after peeling)**.** *Let w be a saturated Type A receiver with a peeling set $P_4(w)$. If*

$$L_4(w) \geq 4q(w),$$

then the unpeeled routed loads at w realize one of exits (1)–(8) of Definition 14.16. If the realized exit is (4), the peeling set can be enlarged by one additional unpeeled routed load.

Proof. If a completion port contains four visible receiver-entry returns from unpeeled loads, Lemma 14.35 gives one of exits (1)–(7). If the exit is (4), the same lemma gives an unpeeled supported load; add the lexicographically first such load to $P_4(w)$.

If no completion port contains four visible receiver-entry returns from unpeeled loads, Lemma 14.36 gives one of exits (4)–(8). If the exit is (4), that lemma gives a supported load in the residual unpeeled excess set; add the lexicographically first such load to $P_4(w)$. □

Lemma 14.38 (Exit-(4) peels one routed load)**.** *Let w be a saturated Type A receiver with peeling set $P_4(w)$. If exit (4) occurs through a quotient $q \in \mathcal{Q}_4(w)$ whose declared support contains the canonical coordinate of an unpeeled routed load $u \in \mathcal{L}(w) \setminus P_4(w)$, then $P_4(w) \cup \{u\}$ is a valid peeling set and the remaining receiver deficit is reduced by exactly* $1/4$.

Proof. By definition of exit (4), q is boundary-degree-preserving and target-defective. Hence there are two realizations in the same boundary-degree fibre and a compatible outside context for which the target predicates differ. The quotient family $\mathcal{Q}_4(w)$ is canonical: its members are exactly the visible-coordinate, silent-basin, trace-basin, continuation/switch, and two-support deletion quotients listed in Definition 14.49. The hypothesis says that the witness uses the declared coordinate of the unpeeled load u. Since $u \notin P_4(w)$, adjoining u preserves the no-duplicate condition in Definition 14.33. Thus

$$P_4(w) \cup \{u\}$$

is a peeling set.

By Lemma 14.34, replacing $P_4(w)$ by $P_4(w) \cup \{u\}$ changes $L_4(w)$ to $L_4(w) - 1$. The remaining receiver charge therefore increases by exactly $\frac{1}{4}$, equivalently the remaining receiver deficit decreases by exactly $\frac{1}{4}$. The bookkeeping records only the removal of the selected load u from the pure Type A receiver sum. □

Lemma 14.39 (Closed saturated exits, exit-(4) peeling, and Type B handoff)**.** *Let a saturated Type A receiver realize one of exits (1)–(7) of Definition 14.16. If the realized exit is one of (1)–(3), (5), or (6), then that branch cannot remain as a negative-charge Type A discharging case. If the realized exit is (4), one routed load is peeled in the sense of Lemma 14.38. If the realized exit is (7), the branch is reclassified as a decorated handoff fan envelope and leaves the Type A charge calculation.*

Proof. Exit (1) gives an edge-rooted Mersenne return, hence a power-of-two cycle by Lemma 2.2. Exit (2) is a power-of-two common-port return cycle by Lemma 14.6. Exit (3) is precisely failure of the legal P_{13} label relation from Lemma 6.1; by definition of the relation, it creates a target event or a target-defective local state. Exit (4) has the one-load role proved in Lemma 14.38: it peels exactly one routed load whose declared coordinate is used by the canonical target-defective quotient, and it reduces the remaining receiver deficit by exactly 1/4. Exit (5) is a nontrivial target-complete response compression. If the compression is realized by a smaller proper boundaried piece, it contradicts hereditary target-uncompressibility (Corollary 3.98); if it occurs only at the trace-basin response level, it is exactly failure alternative (b) in Definition 14.14 and therefore is not an admissible route-8 residual. Exit (6) is excluded by Lemma 10.6 in the proper-support case and by Lemma 10.9 in the whole-graph case. Exit (7) is not a Type A charge case: it is the decorated

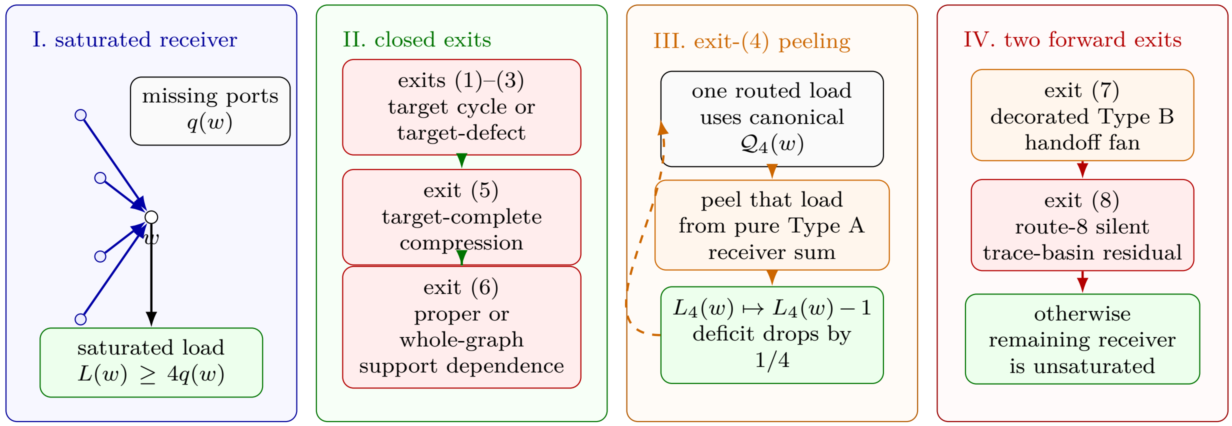

Figure 23: Visual proof of the Type A saturated-exit drain. Panel I shows a Type A receiver w with missing-port count $q(w) = 3 - d_X(w)$ and saturated routed load $L(w) \geq 4q(w)$. Panel II records the exits that close inside the Type A analysis: exits (1)–(3) create a target cycle or target-defective state, exit (5) gives target-complete compression, and exit (6) gives proper or whole-graph support dependence. Panel III isolates exit (4): one routed load using the canonical family $\mathcal{Q}_4(w)$ is peeled out of the pure Type A receiver sum, changing $L_4(w)$ to $L_4(w) - 1$ and reducing the remaining receiver deficit by 1/4; the dashed loop indicates that the receiver is then tested again with smaller residual load. Panel IV shows the two forward exits: exit (7) is reclassified as decorated Type B handoff fan data, while exit (8) is the route-8 trace-basin residual treated by the later response-support reduction and exclusion figures.

handoff fan envelope of Definition 14.19, with admissibility supplied by Lemma 14.26. Thus exits (1)–(3), (5), and (6) are closed inside the Type A analysis, exit (4) is an exact one-load peeling step, and exit (7) is transferred to the Type B branch without using Lemma 14.93. □

Lemma 14.40 (Saturated receiver handoff)**.** *Let X be a Type A support. If a receiver w satisfies $L(w) \geq 4q(w)$, then repeated application of the saturated receiver routing has one of the following outcomes: a closed exit among (1)–(3), (5), and (6) occurs; the Type B handoff exit (7) occurs; the route-8 residual profile of Definition 14.15 occurs; or exit (4) peels finitely many routed loads and the remaining receiver load becomes unsaturated. Consequently every receiver that remains in the pure Type A discharging branch after the closed exits, exit-(4) peeling, route 8, and the Type B handoff have been removed satisfies*

$$L(w) \leq 4q(w) - 1.$$

Proof. Start with the empty peeling set $P_4(w) = \varnothing$. If the residual load $L_4(w)$ is at most $4q(w) - 1$, the receiver is already unsaturated in the remaining Type A charge calculation. Otherwise Lemma 14.37 applies to the unpeeled loads. Exits (1)–(3), (5), and (6) close by Lemma 14.39; exit (7) is transferred to the decorated handoff fan branch by the same lemma; and exit (8) is the route-8 residual profile. If exit (4) occurs, then Lemma 14.38 enlarges $P_4(w)$ by one routed load, so $L_4(w)$ decreases by one. Since $\mathcal{L}(w)$ is finite, this exit-(4) peeling step cannot occur indefinitely. Therefore the iteration ends in one of the listed outcomes. A receiver that remains in the pure Type A discharging branch is precisely one for which the iteration ended by reaching $L_4(w) \leq 4q(w) - 1$; with all peeled loads removed from that branch, this is the displayed unsaturated bound. □

Remark 14.41 (Use of exit-(4) peeling)**.** *The peeling set is a routing record. A load in $P_4(w)$ exits through the target-defect branch before the pure Type A charge is summed. The charge calculation* *therefore applies only to the unpeeled loads counted by $L_4(w)$. Consequently Lemma 14.45 is invoked only after the target-defect alternative has been removed; a support with a remaining exit-(4) witness is routed by alternative (iii) of Lemma 14.44.*

Remark 14.42 (Type A–Type B stratification)**.** *The saturated Type A analysis*

$$\textit{Lemmas } 14.28, 14.32, 14.39, 14.40 \textit{and } 14.43$$

uses no conclusion of Lemma 14.93. The only Type B reference in that block is the handoff interface: Lemmas 14.25 and 14.26 construct and validate the decorated envelope data. The logical direction then reverses: Lemma 14.93 may apply the completed Type A saturated and unsaturated conclusions to the post-ledger core.

The possible alternation terminates. Type A handoffs are assigned to grouped decorated envelopes by the connected components of the core–center incidence graph in Definition 14.20. Each such component uses its finite set of center tokens once in the grouped-envelope ledger. After the Type B ledger removes the corresponding fan/envelope entries, the remaining core is a proper subcore of the previous support or has fewer unused incidence component tokens. Since the support and center-token sets are finite, an $A \to B \to A$ alternation cannot continue indefinitely.

Lemma 14.43 (Unsaturated Type A discharging)**.** *Let X be a Type A support in which every receiver satisfies $L(w) \le 4q(w) - 1$. Then*

$$\partial_+(X) \ge \frac{1}{4}|V(X)|.$$

Equivalently, if $n_i = |\{v \in V(X) : d_X(v) = i\}|$, then

$$n_3 \le 3n_2 + 7n_1 + 11n_0.$$

Proof. Set

$$\mathrm{ch}(v) = (3 - d_X(v)) - \frac{1}{4}.$$

Then $\sum_v \mathrm{ch}(v) = \partial_+(X) - |V(X)|/4$. A cubic vertex has charge $-1/4$, and every cubic vertex is assigned to exactly one receiver by Lemma 14.4. A receiver w has initial charge $q(w) - 1/4$ and pays $1/4$ for each assigned cubic vertex. The unsaturated bound gives

$$q(w) - \frac{1}{4} - \frac{1}{4}L(w) \ge q(w) - \frac{1}{4} - \frac{1}{4}(4q(w) - 1) = 0.$$

All other non-cubic vertices have nonnegative charge and pay nothing. Hence the final total charge is nonnegative, so $\partial_+(X) - |V(X)|/4 \ge 0$. Written by degree classes, the receiver capacities are $3, 7, 11$ for internal degrees $2, 1, 0$, which is exactly $n_3 \le 3n_2 + 7n_1 + 11n_0$. □

Lemma 14.44 (Density forces discharging, target defect, Type B handoff, or route 8)**.** *Let X be a connected admissible support (Definition 13.2) with $\sigma(X) = 0$. Thus X is contextually target-safe (Definition 2.5), P_{13}-free, has empty internal 3-core, has every deficient vertex supplied from the packing W (Lemma 8.5), and is hereditarily target-uncompressible (Corollary 3.98). Then one of the following holds:*

(i)

$$\partial_+(X) \ge \tfrac{1}{4}|V(X)|, \qquad \textit{equivalently} \qquad \frac{2|E(X)|}{|V(X)|} \le 2.75.$$

(ii) X produces a decorated handoff fan envelope in the sense of Definition 14.19.

(iii) X has an exit-(4) witness for a routed load in the sense of Definition 14.33; this branch is routed to the target-defect exit and that routed load is removed from the Type A receiver calculation by Lemma 14.38.

(iv) X contains an admissible silent-core residual profile in the sense of Definition 14.15.

Proof. If a receiver is saturated, Lemma 14.40 applies. Exits (1)–(3), (5), and (6) contradict the standing target, replacement, or support-dependence invariants. Exit (4) gives alternative (iii) and peels one routed load by Lemma 14.38. Exit (7) gives alternative (ii), and exit (8) gives alternative (iv). If no saturated receiver remains after excluding alternatives (ii)–(iv), every receiver in the pure Type A support is unsaturated. The charge calculation of Lemma 14.43 gives

$$\partial_+(X) \geq \frac{1}{4}|V(X)|.$$

Since $\sigma(X) = 0$, this is equivalent to $2|E(X)|/|V(X)| \leq 2.75$. □

Lemma 14.45 (Type A split and charge bound)**.** *Let X be a connected admissible support with $\sigma(X) = 0$, empty internal 3-core, P_{13}-free and target-safe. If X contains no exit-(4) witness for a routed load and contains neither an admissible silent-core residual profile nor a decorated handoff fan envelope, then*

$$\partial_+(X) \geq \frac{1}{4}|V(X)|, \qquad \textit{equivalently} \qquad \frac{2|E(X)|}{|V(X)|} \leq 2.75,$$

so that $\mathcal{N}(X) \geq 0$. Consequently a Type A support with $\mathcal{N}(X) < 0$ must carry an admissible route-8 residual profile, produce the Type B handoff, or contain an exit-(4) witness for a routed load.

Proof. The hypotheses are the admissibility hypotheses used in Lemma 14.44. Since the route-8 residual and Type B handoff alternatives and the exit-(4) witness alternative are excluded by hypothesis, that lemma gives $\partial_+(X) \geq \frac{1}{4}|V(X)|$. Since $\sigma(X) = 0$,

$$\mathcal{N}(X) = \partial_+(X) - \frac{1}{4}|V(X)| \geq 0.$$

Thus a Type A support outside the route-8 residual class, outside the Type B handoff, and outside the exit-(4) target-defect exit cannot have negative net charge. □

Definition 14.46 (Large-budget Type A deficit)**.** *Let $\mathcal{X}$ be a collection of Type A supports whose saturated receivers survive only through the route-8 residual profile of Definition 14.15. Put*

$$D_A(\mathcal{X}) = \sum_{X \in \mathcal{X}} \left(\frac{1}{4}|V(X)| - \partial_+(X) \right).$$

The collection $\mathcal{X}$ carries the large-budget Type A deficit *if*

$$D_A(\mathcal{X}) \geq \left(\frac{1}{4} - \tau_{\mathrm{win}} \right) |R| - o(|R|).$$

Lemma 14.47 (Quantitative burden of the reduced route-8 residual)**.** *Let $\mathcal{X}$ be a collection of Type A supports whose saturated receivers survive only through the route-8 residual profile of Definition 14.15; in particular, no receiver in any $X \in \mathcal{X}$ realizes exits (1)–(7). Let $D_A(\mathcal{X})$ be as in Definition 14.46. Let*

$$N_{\text{basin}}(\mathcal{X}) = \sum_{X\in\mathcal{X}} \sum_{w} |\mathcal{U}_X(w)|,$$

where w ranges over the saturated route-8 receivers of X and $\mathcal{U}_X(w)$ denotes the unpaid silent routed vertices at w. Thus N_{basin} counts the indexed trace-basin entries (u, B_u); two different vertices may contribute two entries even if their underlying boundaried subgraphs coincide. Then

$$N_{\text{basin}}(\mathcal{X}) \geq 4D_A(\mathcal{X}).$$

Consequently, if $\mathcal{X}$ carries the large-budget Type A deficit, then it contains at least

$$4\left(\frac{1}{4} - \tau_{\text{win}}\right)|R| - o(|R|)$$

indexed trace-basin entries. In the limiting window-only cap this coefficient is

$$4\left(\frac{1}{4} - 0.22817486846\ldots\right) = 0.08730052616\ldots.$$

Proof. For each $X \in \mathcal{X}$, the absence of exits (1)–(7) implies in particular that no saturated receiver has a completion port carrying four visible receiver-entry returns. Hence Lemma 14.30 applies to X and gives

$$S^{\text{exc}}_{\text{sil}}(X) \geq 4\left(\frac{1}{4}|V(X)| - \partial_+(X)\right).$$

By Lemma 14.31, every unpaid silent excess vertex counted by $S^{\text{exc}}_{\text{sil}}(X)$ has a target-complete-minimal trace basin unless one of exits (4)–(7) occurs. Those exits are absent by hypothesis. Counting the resulting indexed pairs (u, B_u) and summing the displayed inequality over $X \in \mathcal{X}$ proves $N_{\text{basin}}(\mathcal{X}) \geq 4D_A(\mathcal{X})$. No distinctness of the underlying subgraphs B_u is used.

If $\mathcal{X}$ carries the large-budget Type A deficit, then

$$D_A(\mathcal{X}) \geq \left(\frac{1}{4} - \tau_{\text{win}}\right)|R| - o(|R|)$$

by Definition 14.46. Multiplying by the factor 4 from the first part gives the stated lower bound. Substituting $\tau_{\text{win}} = 0.22817486846\ldots$ gives the displayed coefficient. □

Definition 14.48 (Route-8 response supports and two-support entries)**.** *Let $\mathcal{X}$ be a collection of Type A supports whose saturated receivers survive only through route 8. For a Type A support X, put*

$$\partial_E X = \{(x, xy) : x \in V(X),\ y \notin V(X),\ xy \in E(G)\},$$

the set of oriented boundary incidences leaving X. Since $\sigma(X) = 0$, every vertex of X has ambient degree 3, and hence

$$|\partial_E X| = \partial_+(X).$$

For an indexed entry $\xi = (X, w, u, B_u)$, every declared u-supported coordinate of $\rho_u(B_u)$ that uses the ambient exterior of X records the oriented boundary incidence of X through which it leaves X. More precisely, the declared support of such a coordinate contains a completion-port incidence, first-entry incidence, connector-band endpoint, boundary-degree entry, or packed-window *interface incidence; when that incidence is represented by an edge $xy \in E(G)$ with $x \in V(X)$ and $y \notin V(X)$, its* boundary incidence *is the element $c = (x, xy) \in \partial_E X$. Incidences of ∂B_u internal to X have no boundary incidence unless the declared coordinate canonically assigns them to the terminal completion port of T_u, in which case the boundary incidence is that completion-port incidence. Thus boundary incidences are not extra data attached to B_u; they are the $\partial_E X$-labels already recorded by the declared coordinate signature.*

An indexed route-8 entry is a tuple

$$\xi = (X, w, u, B_u),$$

where $u \in \mathcal{U}_X(w)$ and B_u is the target-complete-minimal trace basin assigned to u. The set of all indexed route-8 entries of $\mathcal{X}$ is denoted

$$\Xi(\mathcal{X}).$$

A boundary incidence $c \in \partial_E X$ is active for ξ *if the declared support of a u-supported coordinate in $\rho_u(B_u)$ contains c as a completion-port incidence, first-entry incidence, connector-band endpoint, boundary-degree entry, or packed-window interface incidence under Definition 3.87.*

For a coordinate $r \in \mathcal{R}_u(B_u)$, let

$$\operatorname{car}_\xi(r) \subseteq \partial_E X$$

be the set of active boundary incidences used by the declared support of r. If $D \subseteq \partial_E X$, the D-restriction *$\rho_u(B_u)|_D$ is the response quotient obtained from $\rho_u(B_u)$ by retaining the full boundary degree profile $\mathbf{d}_\partial(B_u)$ and retaining, apart from this boundary-degree profile, exactly those u-supported coordinates r with $\operatorname{car}_\xi(r) \subseteq D$; all other u-supported coordinates are forgotten. Thus every incidence restriction is taken inside the original boundary-degree fibre.*

A set $D \subseteq \partial_E X$ is a target-complete incidence set for ξ *if the quotient*

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_D$$

is target-complete in the sense of Definition 14.14: for every outside context compatible with the boundary profile, all realizations of $\rho_u(B_u)|_D$ give the same target predicate as $\rho_u(B_u)$ after gluing. Such sets exist: for $D = \partial_E X$, all declared u-supported coordinates are retained, so the completeness clause in Definition 14.14 applies. Let

$$\mathcal{C}_{\mathrm{ess}}(\xi)$$

be the lexicographically first inclusion-minimal target-complete incidence set for ξ. Its elements are the essential incidences *of ξ, and the incidences in $\partial_E X \setminus \mathcal{C}_{\mathrm{ess}}(\xi)$ are* noncore incidences*. Thus*

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)}$$

is target-complete by definition, while for every $c \in \mathcal{C}_{\mathrm{ess}}(\xi)$ the smaller restriction $\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)\setminus\{c\}}$ is not target-complete. Put

$$\alpha_{\mathcal{X}}(\xi) = |\mathcal{C}_{\mathrm{ess}}(\xi)|.$$

An essential incidence for ξ is private for ξ *inside the collection $\mathcal{X}$ if it is not essential for any other indexed route-8 entry in the collection. Let*

$$\pi_{\mathcal{X}}(\xi)$$

be the number of private essential incidences of ξ.

The entry ξ is a two-support route-8 entry *if*

$$\pi_{\mathcal{X}}(\xi) \leq 2.$$

An admissible two-support route-8 residual *is a large-budget route-8 collection containing such an entry.*

Definition 14.49 (Canonical exit-(4) quotient family)**.** *Let w be a saturated receiver in a Type A support X. In the route-8 branch, fix the ambient route-8 Type A collection $\mathcal{X}$ containing X; outside that branch, clauses* (Q1)–(Q4) *below define the same family without using $\mathcal{X}$. The family $\mathcal{Q}_4(w;\mathcal{X})$, abbreviated $\mathcal{Q}_4(w)$ when the ambient collection is fixed, consists of boundary-degree-preserving response quotients generated by the following canonical constructions.*

(Q1) the visible receiver-entry coordinate identifications tested in Lemma 14.28;

(Q2) the silent/excess boundary-response quotient of the basin $B(w)$ in Lemma 14.32;

(Q3) the trace-local quotient or identification of a trace basin in the sense of Definition 14.14, as used in Lemma 14.31;

(Q4) the finite continuation or cubic-switch quotient produced by Lemmas 14.23 and 14.24;

(Q5) for an indexed route-8 entry $\xi = (X, w, u, B_u) \in \Xi(\mathcal{X})$, the response-support deletion quotient

$$\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)} \longrightarrow \rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)\setminus\{c\}}$$

when $c \in \mathcal{C}_{\mathrm{ess}}(\xi)$, the entry is two-support $\pi_{\mathcal{X}}(\xi) \le 2$, and the distinguishing event is recorded by a declared deletion witness in the sense of Definition 14.51.

Every member of $\mathcal{Q}_4(w)$ retains the boundary degree profile of the response state on which it is defined. Exit (4) occurs precisely when one of these listed canonical quotients is target-defective. The declared routed-load support is part of the canonical construction: in (Q1) *it is the set of visible routed loads whose response coordinates are identified; in* (Q2) *it is the routed loads in the basin $B(w)$; in* (Q3) *it is the trace load u whose basin is being tested; in* (Q4) *it is the routed loads whose declared connector coordinates form the finite family $\mathcal{K}$; and in* (Q5) *it is the indexed load u of the route-8 entry $\xi = (X, w, u, B_u)$.*

Definition 14.50 (True route-8 residual entries)**.** *Let $\mathcal{X}$ be a collection of Type A supports whose saturated receivers survive only through route 8, and let*

$$\xi = (X, w, u, B_u) \in \Xi(\mathcal{X})$$

be an indexed route-8 entry. The entry ξ is a true route-8 residual entry *if:*

(R1) w is a saturated receiver of X;

(R2) exits (1)–(7) of Definition 14.16 are absent at w, with exit (4) interpreted through the canonical family $\mathcal{Q}_4(w)$;

(R3) the data at w form an admissible silent-core residual profile of Definition 14.15; and

(R4) B_u is the target-complete-minimal trace basin assigned to u.

A true route-8 residual collection *is a route-8 Type A collection whose indexed entries are true route-8 residual entries. Thus, in a true route-8 residual entry, any occurrence of one of exits (1)–(7) contradicts the definition of the residual.*

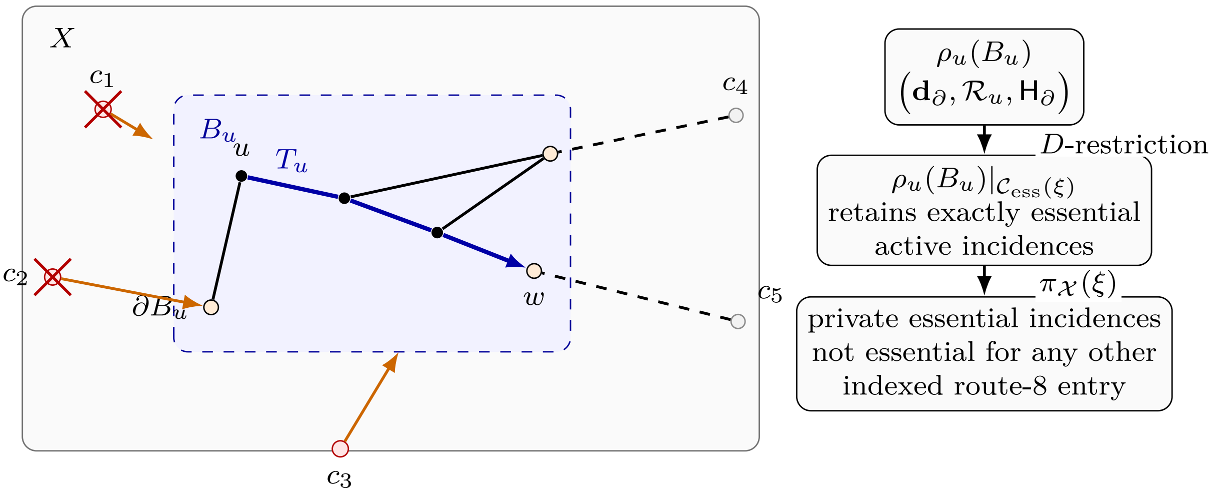

Figure 24: Visual definition of a route-8 trace-basin entry and its response-support core. The grey rectangle is the Type A support X, and the blue dashed shaded region is the target-complete-minimal trace basin $B_u \subseteq X$ assigned to the routed vertex u. Black edges are support edges inside X, the blue arrowed path T_u is the canonical trace from u to the receiver w, and ∂B_u denotes the boundary of the basin inside X. The entry is $\xi = (X, w, u, B_u)$. The labels $c_1, \ldots, c_5$ denote oriented boundary incidences in $\partial_E X$. Orange arrows mark active boundary incidences used by declared u-supported coordinates; dashed black incidences illustrate boundary incidences that remain outside the essential core. Red support markers are elements of the essential response-support set $\mathcal{C}_{\text{ess}}(\xi)$, grey markers are noncore supports, and the red crosses on c_1, c_2 mark supports counted as private for this indexed entry inside the collection $\mathcal{X}$. The response state $\rho_u(B_u)$ records the boundary degree profile $\mathbf{d}_\partial$, the u-supported response coordinates $\mathcal{R}_u$, and the boundary target profile H_∂. A D-restriction keeps only the coordinates whose declared supports lie in the chosen set D; the displayed restriction uses $D = \mathcal{C}_{\text{ess}}(\xi)$. The number $\pi_{\mathcal{X}}(\xi)$ counts the private essential supports of ξ, and a two-support route-8 entry is one with $\pi_{\mathcal{X}}(\xi) \leq 2$.

Definition 14.51 (Response-support deletion witnesses)**.** *Let*

$$\xi = (X, w, u, B_u)$$

be an indexed route-8 entry, and put

$$\mathcal{C} = \mathcal{C}_{\text{ess}}(\xi).$$

For $c \in \mathcal{C}$, the c-deletion quotient *is the quotient*

$$q_{\xi,c} : \rho_u(B_u)|_{\mathcal{C}} \longrightarrow \rho_u(B_u)|_{\mathcal{C}\setminus\{c\}}.$$

A c-deletion witness *is a realization S of $\rho_u(B_u)|_{\mathcal{C}\setminus\{c\}}$, together with a compatible outside context Y, such that*

$$S \oplus Y$$

has a different target predicate from the original core state $\rho_u(B_u)|_{\mathcal{C}}$ glued to the corresponding context. A witnessing target event is chosen as the lexicographically first simple edge-rooted return or simple cycle responsible for this target difference. The declared support *of the witness is the finite set of declared response coordinates from Definition 3.87 used by that event, and its* boundary-incidence support *is the union of the boundary incidences of those coordinates in $\partial_E X$. The witness* *is* internal *when this event uses an edge of $B_u - \partial B_u$. It is* mixed *when, in addition, the event uses an edge outside X.*

Lemma 14.52 (Essential boundary incidences have deletion witnesses)**.** *Let $\xi = (X, w, u, B_u)$ be an indexed route-8 entry. For every*

$$c \in \mathcal{C}_{\mathrm{ess}}(\xi),$$

the c-deletion quotient $q_{\xi,c}$ is not target-complete. Equivalently, there exists a c-deletion witness in the sense of Definition 14.51.

Proof. By definition, $\mathcal{C}_{\mathrm{ess}}(\xi)$ is inclusion-minimal among target-complete incidence sets for ξ. Therefore

$$\mathcal{C}_{\mathrm{ess}}(\xi) \setminus \{c\}$$

is not a target-complete incidence set. The full-to-core quotient

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)}$$

is target-complete. If $q_{\xi,c}$ were target-complete, then Lemma 3.86 would make the composite

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)\setminus\{c\}}$$

target-complete, contradicting the preceding sentence. Hence $q_{\xi,c}$ is not target-complete. Unwinding the definition of target-completeness gives a compatible outside context and a realization of the smaller restricted response state with different target predicate. Choosing the lexicographically first simple edge-rooted return or simple cycle that accounts for the difference gives the stated deletion witness. □

Lemma 14.53 (Deletion witnesses are declared)**.** *Let $\xi = (X, w, u, B_u)$ be an indexed route-8 entry, and let $c \in \mathcal{C}_{\mathrm{ess}}(\xi)$. Every c-deletion witness obtained from Lemma 14.52 has declared support and response support in the sense of Definition 14.51. Moreover its boundary-incidence support contains c.*

Proof. The witness distinguishes the core state $\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)}$ from the smaller restriction $\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)\setminus\{c\}}$ in a compatible outside context. If the witnessing target event used only retained coordinates, then its truth value would be the same in the two glued states, because the two states agree on every retained declared u-supported coordinate and on the full boundary degree profile. Hence the event uses at least one coordinate forgotten by the deletion quotient.

By the definition of D-restriction in Definition 14.48, the coordinates forgotten when passing from $\mathcal{C}_{\mathrm{ess}}(\xi)$ to $\mathcal{C}_{\mathrm{ess}}(\xi)\setminus\{c\}$ are precisely declared u-supported coordinates whose boundary-incidence support contains c. These coordinates are generated by the declared response-coordinate signature Definition 3.87. Taking the finite set of declared coordinates used by the lexicographically first witnessing event gives the declared support, and the union of their boundary incidences gives the boundary-incidence support. This support contains c. □

Lemma 14.54 (Two-incidence deletion quotients are canonical exit-(4) quotients)**.** *Let $\mathcal{X}$ be a route-8 Type A collection, let*

$$\xi = (X, w, u, B_u) \in \Xi(\mathcal{X})$$

be an indexed entry with $\pi_{\mathcal{X}}(\xi) \leq 2$, and let $c \in \mathcal{C}_{\mathrm{ess}}(\xi)$. If c has a declared c-deletion witness, then $q_{\xi,c} \in \mathcal{Q}_4(w)$.

Proof. The quotient $q_{\xi,c}$ retains the full boundary degree profile by the definition of D-restriction in Definition 14.48. The hypothesis $\pi_{\mathcal{X}}(\xi) \leq 2$ says that ξ is a two-support route-8 entry. The declared witness hypothesis supplies the final datum required in clause (Q5) of Definition 14.49. Therefore $q_{\xi,c}$ is one of the canonical exit-(4) quotients at w. $\square$

Lemma 14.55 (Response-support deletion is exit (4))**.** *Let $\mathcal{X}$ be a route-8 Type A collection, let*

$$\xi = (X, w, u, B_u) \in \Xi(\mathcal{X})$$

be a two-support indexed route-8 entry, and put

$$\mathcal{C} = \mathcal{C}_{\mathrm{ess}}(\xi).$$

If $c \in \mathcal{C}$ has a declared c-deletion witness, then exit (4) of Definition 14.16 occurs at the saturated receiver w.

Proof. The response-support core restriction

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_{\mathcal{C}}$$

is target-complete by the definition of $\mathcal{C}_{\mathrm{ess}}(\xi)$. Thus, for every compatible outside context, replacing $\rho_u(B_u)$ by the core state $\rho_u(B_u)|_{\mathcal{C}}$ preserves the target predicate.

Let S and Y be the c-deletion witness. Thus S is a realization of

$$\rho_u(B_u)|_{\mathcal{C}\setminus\{c\}},$$

the context Y is compatible with the same boundary degree profile, and

$$S \oplus Y$$

has a target predicate different from the core state $\rho_u(B_u)|_{\mathcal{C}}$ glued to the corresponding context. Since the full-to-core quotient is target-complete, Lemma 3.86 implies that the same realization S and context Y distinguish the composite quotient

$$\rho_u(B_u) \longrightarrow \rho_u(B_u)|_{\mathcal{C}\setminus\{c\}}.$$

This composite quotient preserves the boundary degree profile and is a response quotient inside $\rho_u(B_u)$: it is obtained by forgetting exactly the declared u-supported coordinates whose boundary-incidence support uses c, after the noncore incidences have already been forgotten by the target-complete core restriction. Hence it is a boundary-degree-preserving response quotient distinguished by an outside context.

By Lemma 14.54, $q_{\xi,c}$ belongs to $\mathcal{Q}_4(w)$. The preceding paragraph proves that this canonical quotient is target-defective. This is precisely exit (4) of Definition 14.16. $\square$

Lemma 14.56 (Response-support cut parity)**.** *Let $\xi = (X, w, u, B_u)$ be an indexed route-8 entry, and replace its response state by the canonical core restriction $\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)}$. If a surviving u-supported target event is realized by a simple edge-rooted return or a simple cycle which uses an internal edge of B_u and also uses an edge outside X, then its declared support contains at least two distinct boundary incidences from $\mathcal{C}_{\mathrm{ess}}(\xi)$.*

Proof. For an edge-rooted return, add the root edge to the return path; for a cycle do nothing. In both cases we obtain a simple cycle C of G. Since the target event uses an internal edge of $B_u \subseteq X$ and also an edge outside X, the set

$$E(C) \cap \delta_G(V(X)), \qquad \delta_G(V(X)) = \{ab \in E(G) : |\{a,b\} \cap V(X)| = 1\},$$

is nonempty. A cycle crosses every cut an even number of times. Because C is simple, it cannot use the same cut edge twice. Hence C uses at least two distinct boundary edges of X.

Each such crossing is recorded in the declared support of the corresponding u-supported coordinate: it is a completion-port incidence, a first-entry incidence, a connector endpoint, a boundary-degree entry, or a packed-window interface incidence in the signature of Definition 3.87. Since the event survives in the restricted state $\rho_u(B_u)|_{\mathcal{C}_{\mathrm{ess}}(\xi)}$, every boundary incidence in its declared support lies in $\mathcal{C}_{\mathrm{ess}}(\xi)$. The two distinct cut crossings therefore give two distinct boundary incidences from $\mathcal{C}_{\mathrm{ess}}(\xi)$. □

Lemma 14.57 (Internal trace quotient witnesses are mixed)**.** *Let $\xi = (X, w, u, B_u)$ be an indexed route-8 entry, let $\mathcal{C} \subseteq \partial_E X$, and let $\rho^\circ_{\mathcal{C}}$ be the quotient of $\rho_u(B_u)|_{\mathcal{C}}$ that retains the full boundary degree profile and forgets exactly the u-supported coordinates whose declared support contains an internal edge of B_u. If*

$$\rho_u(B_u)|_{\mathcal{C}} \longrightarrow \rho^\circ_{\mathcal{C}}$$

is not target-complete, then some distinguishing realization contains a surviving u-supported target event which is a simple edge-rooted return or a simple cycle, whose declared support contains an internal edge of B_u, and which uses an edge outside X.

Proof. Failure of target-completeness gives a compatible outside context and two realizations of $\rho_u(B_u)|_{\mathcal{C}}$ with the same image in $\rho^\circ_{\mathcal{C}}$, but with different target predicates after gluing. Choose a target event present in one glued realization and absent in the other. If the event is an edge-rooted return, add its root edge; otherwise use the target cycle itself. The event is represented by a simple cycle.

The two realizations have the same image in $\rho^\circ_{\mathcal{C}}$, so a distinguishing event must use at least one coordinate forgotten by $\rho_u(B_u)|_{\mathcal{C}} \to \rho^\circ_{\mathcal{C}}$. The forgotten coordinates are precisely the u-supported coordinates whose declared support contains an internal edge of B_u. Hence the chosen event is u-supported and uses such an internal edge.

If the event were contained entirely in X, it would be a power-of-two cycle in the contextually target-safe support X, impossible in the counterexample. Thus the event also uses an edge outside X. Since the event is present in a realization of the restricted state, it survives in $\rho_u(B_u)|_{\mathcal{C}}$. □

Lemma 14.58 (Zero- and one-incidence silent entries collapse)**.** *Let $\xi = (X, w, u, B_u)$ be an indexed route-8 entry. If*

$$\alpha_{\mathcal{X}}(\xi) \le 1,$$

then one of exits (4)–(7) of Definition 14.16 occurs. Consequently, every indexed entry in a true route-8 residual satisfies

$$\alpha_{\mathcal{X}}(\xi) \ge 2.$$

Proof. Let

$$\mathcal{C} = \mathcal{C}_{\mathrm{ess}}(\xi).$$

Replacing $\rho_u(B_u)$ by the restriction $\rho_u(B_u)|_{\mathcal{C}}$ deletes exactly the noncore incidences. This quotient is target-complete by the definition of the canonical response-support core.

Assume $|\mathcal{C}| \le 1$. Let $\rho_{\mathcal{C}}^{\circ}$ be the further quotient of $\rho_u(B_u)|_{\mathcal{C}}$ that retains the full boundary degree profile and forgets every u-supported coordinate whose declared support contains an internal edge of B_u. This quotient is nontrivial because Definition 14.14 includes the trace-incidence coordinate of $T_u \subseteq B_u$, and the trace contains an internal edge of B_u.

We claim that $\rho_u(B_u)|_{\mathcal{C}} \to \rho_{\mathcal{C}}^{\circ}$ is target-complete. Suppose not. By Lemma 14.57, there is a surviving u-supported target event in the core-restricted response state which uses an internal edge of B_u and an edge outside X. Hence Lemma 14.56 applies and forces its declared support to contain at least two distinct boundary incidences from $\mathcal{C}$, contradicting $|\mathcal{C}| \le 1$. Thus the quotient is target-complete.

The nontrivial target-complete quotient above is precisely failure alternative (b) in the definition of target-complete-minimality (Definition 14.14), unless one of the other alternatives occurs first: a distinguishing outside context is alternative (a), validity only after adjoining a larger support is alternative (c), and first separation at a high-degree outside vertex is alternative (d). These are exits (4), (6), and (7), respectively, while alternative (b) is exit (5). Therefore an indexed route-8 entry with at most one essential incidence cannot survive while exits (4)–(7) are absent. In a true route-8 residual those exits are absent by definition, so every surviving indexed entry has at least two essential incidences. ☐

Proposition 14.59 (Response-support reduction for route 8)**.** *Let $\mathcal{X}$ be a collection of route-8 Type A supports that carries the large-budget Type A deficit in the sense of Definition 14.46. Then $\mathcal{X}$ contains a two-support route-8 entry. Equivalently, if no two-support route-8 entry exists, no route-8 Type A collection carries the large-budget Type A deficit.*

Proof. Suppose that no indexed route-8 entry in $\mathcal{X}$ is two-support. By Definition 14.48, $\Xi(\mathcal{X})$ is the set of indexed trace-basin entries. Then $|\Xi(\mathcal{X})| = N_{\text{basin}}(\mathcal{X})$, and every $\xi \in \Xi(\mathcal{X})$ has at least three private essential incidences:

$$\pi_{\mathcal{X}}(\xi) \ge 3.$$

Private essential incidences belonging to different indexed entries are disjoint by definition. They are boundary incidences of the corresponding Type A supports, so

$$3N_{\text{basin}}(\mathcal{X}) \le \sum_{X \in \mathcal{X}} |\partial_E X| = \sum_{X \in \mathcal{X}} \partial_+(X).$$

The collection $\mathcal{X}$ is a subcollection of the canonical support decomposition. Since the canonical supports are the connected components of R, their positive deficiencies add exactly, and all summands are nonnegative. Hence

$$\sum_{X \in \mathcal{X}} \partial_+(X) \le \partial_+(R).$$

By Corollary 8.8 and the near-cubic spine estimate $\sigma(G) = o(|R|)$ from Definition 4.1,

$$\partial_+(R) \le \tau_{\text{win}}|R| + o(|R|).$$

Hence

$$3N_{\text{basin}}(\mathcal{X}) \le \tau_{\text{win}}|R| + o(|R|).$$

On the other hand, Lemma 14.47 and Definition 14.46 give

$$N_{\text{basin}}(\mathcal{X}) \ge 4\left(\frac{1}{4} - \tau_{\text{win}}\right)|R| - o(|R|).$$

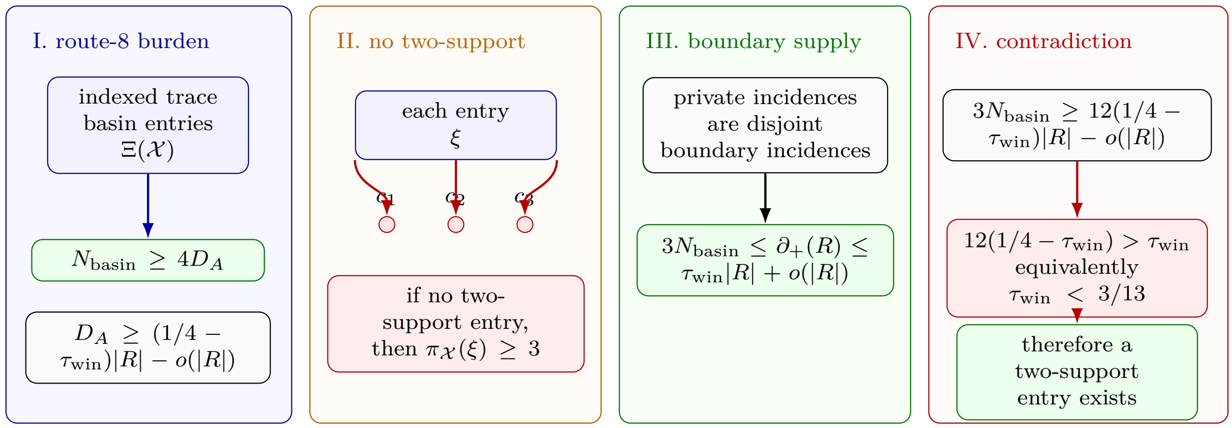

Figure 25: Visual proof of the route-8 response-support reduction squeeze. Panel I records the lower side: a route-8 Type A collection $\mathcal{X}$ carrying the large-budget deficit has $N_{\text{basin}}(\mathcal{X}) \geq 4D_A(\mathcal{X})$, and D_A is linear in $(1/4 - \tau_{\text{win}})|R|$. Panel II shows the negation of the conclusion: if no indexed entry is two-support, then every entry has at least three private essential supports. Panel III records the upper side: private essential supports are disjoint boundary incidences of the canonical supports, so $3N_{\text{basin}}(\mathcal{X}) \leq \partial_+(R) \leq \tau_{\text{win}}|R| + o(|R|)$. Panel IV combines the lower and upper bounds; since $12(1/4 - \tau_{\text{win}}) > \tau_{\text{win}}$, equivalently $\tau_{\text{win}} < 3/13$, the no-two-support alternative is impossible.

Combining the two inequalities yields

$$\tau_{\text{win}}|R| + o(|R|) \geq 12\left(\frac{1}{4} - \tau_{\text{win}}\right)|R| - o(|R|).$$

This is impossible, because

$$12\left(\frac{1}{4} - \tau_{\text{win}}\right) = 0.26190157848\ldots > 0.22817486846\ldots = \tau_{\text{win}}.$$

Equivalently, the numerical condition is

$$\tau_{\text{win}} < \frac{3}{13}.$$

Thus a surviving large-budget route-8 collection must contain a two-support route-8 entry. □

Definition 14.60 (Terminal two-incidence route-8 obstruction)**.** *A terminal two-support route-8 obstruction is a pair*

$$(\mathcal{X}, \xi), \qquad \xi = (X, w, u, B_u) \in \Xi(\mathcal{X}),$$

with the following properties.

(T1) $\mathcal{X}$ is a large-budget route-8 Type A collection carrying the large-budget Type A deficit of Definition 14.46.

(T2) The entry ξ is a true route-8 residual entry in the sense of Definition 14.50: exits (1)–(7) of Definition 14.16 are absent at its saturated receiver, and B_u is target-complete-minimal.

(T3) The canonical response-support core is nontrivial:

$$\alpha_{\mathcal{X}}(\xi) = |\mathcal{C}_{\text{ess}}(\xi)| \geq 2.$$

(T4) The entry is two-support in the private-incidence sense:

$$\pi_{\mathcal{X}}(\xi) \leq 2.$$

(T5) For every $c \in \mathcal{C}_{\text{ess}}(\xi)$, the c-deletion witness of Lemma 14.52 is part of the obstruction data, with its witnessing target event, declared support, and boundary-incidence support recorded as in Lemma 14.53.

The definition allows essential incidences of ξ that are also essential for other indexed route-8 entries. Thus it is exactly the residual produced by Proposition 14.59; it is not restricted to the special case $|\mathcal{C}_{\text{ess}}(\xi)| = 2$.

Theorem 14.61 (Two-incidence route-8 exclusion)**.** *There is no terminal two-support route-8 obstruction in the sense of Definition 14.60.*

Proof. Suppose that $(\mathcal{X}, \xi)$ is a terminal two-support route-8 obstruction, and write

$$\xi = (X, w, u, B_u), \qquad \mathcal{C} = \mathcal{C}_{\text{ess}}(\xi).$$

By condition (T3) in Definition 14.60, the set $\mathcal{C}$ has at least two elements; choose $c \in \mathcal{C}$. Condition (T4) says that ξ is two-support, and condition (T5) records the declared c-deletion witness supplied by Lemmas 14.52 and 14.53. By Lemma 14.54, the corresponding deletion quotient lies in $\mathcal{Q}_4(w)$. Therefore Lemma 14.55 gives exit (4) at the saturated receiver w.

Condition (T2) says that ξ is a true route-8 residual entry, so exits (1)–(7) are absent at w. This contradicts the occurrence of exit (4). Hence no terminal two-support route-8 obstruction exists. □

Proposition 14.62 (Route-8 closure)**.** *No route-8 Type A collection carries the large-budget Type A deficit.*

Proof. Let $\mathcal{X}$ be a route-8 Type A collection carrying the large-budget Type A deficit. By Proposition 14.59, $\mathcal{X}$ contains an indexed two-support entry

$$\xi = (X, w, u, B_u)$$

with $\pi_{\mathcal{X}}(\xi) \leq 2$. Since the collection survives only through route 8, ξ is a true route-8 residual entry in the sense of Definition 14.50; in particular, exits (1)–(7) are absent. By Lemma 14.58, every indexed entry in a true route-8 residual has at least two essential incidences, so

$$\alpha_{\mathcal{X}}(\xi) \geq 2.$$

For each essential incidence, Lemmas 14.52 and 14.53 supplies the required declared deletion witness. Therefore $(\mathcal{X}, \xi)$ is a terminal two-support route-8 obstruction, contradicting Theorem 14.61. Hence no such collection $\mathcal{X}$ exists. □

Remark 14.63 (Type A route-8 closure)**.** *After Proposition 14.59, the only Type A route-8 configuration not eliminated by the private-incidence count is a target-complete-minimal silent trace basin with at least two essential response incidences and at most two private essential incidences. The zero- and one-essential-incidence cases are removed by Lemma 14.58. The remaining two-support entry is excluded by Lemmas 14.54 and 14.55 and Theorem 14.61: an essential incidence has a declared deletion witness, the two-support condition places the corresponding deletion quotient in the canonical family $\mathcal{Q}_4(w)$, and exit (4) is absent in a true route-8 residual.*

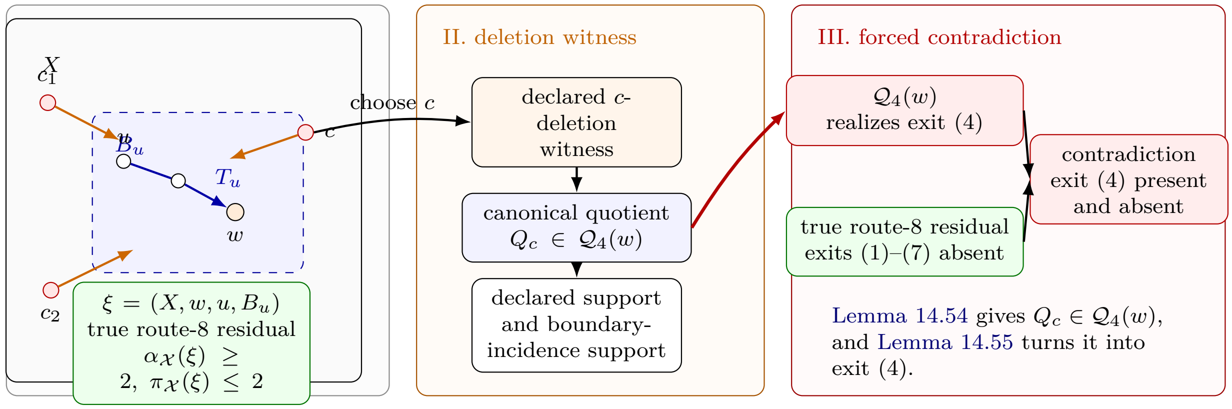

Figure 26: Visual proof of the terminal two-support route-8 exclusion. Panel I shows a terminal entry $\xi = (X, w, u, B_u)$. The grey rectangle is the Type A support X, the blue dashed region is the target-complete-minimal basin B_u, the orange vertex is the saturated receiver w, and the red boundary nodes are essential boundary incidences. The displayed terminal data are $\alpha_{\mathcal{X}}(\xi) \geq 2$, $\pi_{\mathcal{X}}(\xi) \leq 2$, and true route-8 residuality. Panel II isolates one essential incidence c. Its declared deletion witness has declared support and boundary-incidence support, and the two-support deletion lemma places the corresponding quotient in the canonical exit-(4) family $\mathcal{Q}_4(w)$. Panel III records the contradiction: the response-support deletion-exit lemma realizes exit (4) at w, while true route-8 residuality says that exits (1)–(7), including exit (4), are absent. Thus the terminal two-support obstruction of Definition 14.60 cannot occur.

Remark 14.64 (Admissibility is essential; the bound is not intrinsic)**.** *Subcubicity and an empty* 3*-core alone do not yield* 2.75*: a* 2*-degenerate subcubic graph can have average degree approaching* 3*. The point of Lemma 14.44 is therefore not a bare extremal statement about isolated subcubic graphs. It is a discharging statement for* admissible *supports, where every receiver also carries the contextual target algebra, the replacement constraints, the window-stub accounting, and the saturated-exit alternatives of Definition 14.16. The raw receiver thresholds are* 4, 8, 12*, not the discharging capacities. They mark the saturated handoff values* H_0, H_1, H_2 *of Lemma 14.27. Once exits 1–3, 5, and 6 are closed, exit 4 has peeled its target-defective routed loads, and exit 7 is handed to Type B by Lemma 14.40, the remaining Type A charge capacities are* 3, 7, 11 *for internal degrees* 2, 1, 0*. Route 8 is not a bare finite graph class: by Lemma 14.47 it is reduced to indexed target-complete-minimal silent trace-basin entries with a linear burden whenever it carries the large-budget charge deficit, and Proposition 14.59 reduces that branch further to the two-support residual of Definition 14.48. Thus the density bound is a statement about completable boundaried pieces carrying the full target algebra and replacement constraints, not about isolated subcubic graphs.*

Framework branch table. The Type A subsection closes its local residuals by the following framework table.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure
Direct exits	Saturated exits (1)–(3) occur.	A target hit.	The Mersenne-return algebra.	One local receiver.	Closed by Definition 14.16 and Lemma 14.39.

Target defect	Exit (4) supplies a routed-load witness.	One peel token.	The exit-(4) ledger.	Finite descent in the peeling set.	Definition 14.33 and Lemmas 14.34 and 14.38.
Compression or support dependence	Exits (5) and (6) occur.	A proper support.	Replacement and support dependence.	The declared support.	Lemma 14.39.
Type B handoff	Exit (7) occurs.	Decorated fan data.	The Type B ledger.	One grouped handoff envelope.	Lemmas 14.25 and 14.40.
Unsaturated boundaried piece	Receiver loads remain below the saturated thresholds.	3/7/11 discharging capacity.	The deficiency budget.	The receiver partition.	Lemmas 14.43 and 14.45.
Route 8	A silent trace-basin residual remains.	Response-support burden.	Incidence-deletion witnesses.	The terminal two-support package.	Proposition 14.62.

14.2 Type B: high-degree fan-safe surplus obstruction

Definition 14.65 (Assigned Type B charge ledger)**.** *Let X be a connected assigned Type B support. Write Y_X for the counted remainder core of X, and write H_X for the high-degree fan centers whose surplus units are assigned to X by Definition 13.1. Thus*

$$\partial_+(X) = \partial_+(Y_X), \qquad |V(X)| = |V(Y_X)|, \qquad \sigma(X) = \sum_{h \in H_X} (d_G(h) - 3).$$

A high-degree vertex that appears only as boundary context for another support is not an element of that support's set H_X. If the same ambient vertex lies in the local drawing of a support and also supplies a surplus unit, it is entered once in H_X and is not available as a non-center response support.

For $y \in Y_X$ set

$$\delta_X^+(y) = \max\{0, 3 - d_{Y_X}(y)\}, \qquad \mathrm{ch}_X(y) = \delta_X^+(y) - \frac{1}{4},$$

and for $h \in H_X$ set

$$\mathrm{ch}_X(h) = -(d_G(h) - 3) - \frac{1}{4}.$$

The Type B augmented ledger is

$$\widehat{\mathsf{Ch}}_B(X) = \sum_{y \in Y_X} \mathrm{ch}_X(y) + \sum_{h \in H_X} \mathrm{ch}_X(h).$$

It satisfies the exact accounting identity

$$\mathcal{N}(X) = \widehat{\mathsf{Ch}}_B(X) + \frac{1}{4}|H_X|. \tag{B-ledger}$$

Consequently $\widehat{\mathsf{Ch}}_B(X) \geq 0$ *implies* $\mathcal{N}(X) \geq 0$*, hence*

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|.$$

In a fan envelope, a cubic-closed neighbour has $\delta_X^+ = 0$ *and contributes* $-\frac{1}{4}$*; a fan neighbour that is not cubic-closed has internal degree at most* 2 *in the assigned envelope and contributes at least* $\frac{3}{4}$*.*

For a grouped decorated Type B envelope support $\mathfrak{X}_{\mathfrak{C}}^*$ *of Definition 14.20, the same ledger is used with* $Y_X = Y_{\mathfrak{C}}^*$ *and with the component center-token sum* $\omega(\mathfrak{C})$ *in place of the canonically assigned surplus units. The grouped ledger is never summed separately from the canonical Type A pieces that generated it; the transfer identity of Lemma 14.21 is the summation rule. Hence an external handoff center, including a center inside a packed* P_{13}*-window, can enter the Type B bridge only through its grouped envelope component. In the global fan-mass bound, that surplus unit may also appear once through the ordinary canonical support containing the center, and this is the only allowed double use.*

Suppose X carries high-degree surplus, and let $h \in H_X$. By Lemma 3.2, the vertices of H_X are independent. Write

$$d_G(h) = 3 + t_h.$$

The surplus at h is t_h, and every unordered pair of neighbours $u, v \in N(h)$ gives a fan path

$$u - h - v$$

of length 2. If $G - h$ contains a simple u–v path of length

$$2^j - 2 \in \{2, 6, 14, 30, \ldots\},$$

then that path, together with the two edges hu and hv, is a cycle of length $(2^j - 2) + 2 = 2^j$. Thus every neighbour pair of h must be fan-return-safe.

For every $h \in H_X$,

$$N(h) \text{ is a clique in } \mathcal{F}_h,$$

because the failure of one of the five fan-safe conditions is, respectively, a power-of-two cycle, a legal-label contradiction, a target-defective quotient, a forbidden target-complete compression, or proper/whole-graph support dependence. Hence

$$d_G(h) \leq \omega(\mathcal{F}_h).$$

Also,

$$d_G(h) \leq \rho(A(\mathcal{F}_h)) + 1$$

by the spectral clique bound: if a clique has size r, the corresponding principal submatrix is $J_r - I_r$, whose largest eigenvalue is $r - 1$, and eigenvalue interlacing gives $r - 1 \leq \rho(A(\mathcal{F}_h))$. The Type B branch is the case in which a negative-charge support contains at least one assigned high-degree fan center. The fan-safe graph and marked fan data are those of Definitions 14.17 and 14.18; the fan ledger below accounts for the surplus terms at those centers.

Lemma 14.66 (Fan label packing and certificate-closed charge)**.** *In the Type B fan-certificate graph, the* P_{13}*-label projection with adjacency relation* $C_2(S, T) = 1$ *has clique number at most* 8*. Every clique in that projection containing a non-singleton label has size at most* 7*. Consequently every certificate-marked Type B fan has* $d_G(h) \leq 8$*, and every certificate-marked Type B fan with*

$$c - \frac{11 - k}{4} \leq 0, \qquad k = d_G(h), \quad c = |\{u \in N(h) : u \text{ is cubic-closed}\}|,$$

has nonnegative closed-neighbourhood charge in the augmented ledger of Definition 14.65. These are the certificate-closed fans of Definition 14.71.

Proof. Let D be the graph on $\{0,\dots,12\}$ in which two distinct indices are adjacent exactly when their difference is 4 or 12. If $S_1,\dots,S_m$ are pairwise C_2-compatible labels, choose one representative $r_i \in S_i$ from each label. The relation $C_2(S_i,S_j)=1$ says precisely that no cross-difference between S_i and S_j is 0, 4, or 12. Hence the representatives $r_1,\dots,r_m$ are distinct and form an independent set in D.

The graph D is the disjoint union of the component

$$0-4-8-12-0$$

and the three paths

$$1-5-9, \qquad 2-6-10, \qquad 3-7-11.$$

Thus

$$\alpha(D)=2+2+2+2=8,$$

and every C_2-compatible label family has size at most 8. In a certificate-marked fan the labelling $S_h : N(h)\to\mathcal{L}$ sends the neighbour set to such a C_2-compatible label family. Since $C_2(S,S)=0$, the labels of distinct neighbours are distinct. Hence

$$d_G(h)=|N(h)|=|S_h(N(h))|\le 8.$$

Now suppose the clique contains a non-singleton label S. Every other label is contained in

$$A=\{0,\dots,12\}\setminus N_D[S],$$

where $N_D[S]$ is the closed D-neighbourhood of S, because a cross index equal to an index of S or at distance 4 or 12 from it would violate C_2. Choose two distinct indices $a,b\in S$. Deleting the closed D-neighbourhood of $\{a,b\}$ reduces the independence number of D by at least 2. If a and b lie in different components of D, each deletion reduces the corresponding component's independence number by at least 1. If they lie in the same component, that component is either a three-vertex path or the four-cycle above; in either case the closed neighbourhoods of two distinct vertices leave independence number 0 in that component, whereas the original component had independence number 2. Since $N_D[\{a,b\}]\subseteq N_D[S]$, it follows that

$$\alpha(D[A])\le 6.$$

Representatives of all labels other than S form an independent set in $D[A]$, so there are at most 6 of them. Therefore every clique containing a non-singleton label has size at most 7.

For the charge statement, write $k=d_G(h)$ and let c be the number of cubic-closed neighbours. The first packing bound gives $k\le 8$. The certificate-closed condition used below is exactly the nonpositive-deficit condition

$$c-\frac{11-k}{4}\le 0.$$

Equivalently, $c\le 1$ when $k\le 7$, and $c=0$ when $k=8$. With the augmented Type B charge of Definition 14.65, the center contributes

$$\mathrm{ch}_X(h)=3-k-\frac{1}{4}.$$

In the worst certificate-closed case with $k \le 7$, one neighbour is cubic-closed and contributes $-\frac{1}{4}$, while the remaining $k-1$ neighbours contribute at least $\frac{3}{4}$ each. Therefore

$$\mathrm{ch}_X(h) + \sum_{u \in N(h)} \mathrm{ch}_X(u) \ge \left(3 - k - \frac{1}{4}\right) - \frac{1}{4} + \frac{3}{4}(k-1) = \frac{7-k}{4} \ge 0.$$

If $k = 8$, certificate-closed means $c = 0$, so all eight neighbours contribute at least $\frac{3}{4}$, and

$$\mathrm{ch}_X(h) + \sum_{u \in N(h)} \mathrm{ch}_X(u) \ge \left(3 - 8 - \frac{1}{4}\right) + 8 \cdot \frac{3}{4} = \frac{3}{4} > 0.$$

Thus every certificate-closed marked fan has nonnegative closed-neighbourhood charge. □

Remark 14.67 (The fan-degree bound is label packing)**.** *The clique bound $d_G(h) \le \omega(\mathcal{F}_h)$ is an explicit structural consequence of the label algebra, not a free parameter. Two neighbours attaching to a common P_{13} window with labels S, T are fan-return-safe through that window exactly when $C_2(S,T) = 1$: the wedge $u - h - v$ is the length-two outside path, closing the window into a cycle of length $4 + |i-j| \notin \mathcal{D}$. Since $C_2(S,S) = 0$, distinct neighbours carry distinct labels. Lemma 14.66 proves that every pairwise C_2-compatible set has size at most 8, and that every such set containing a non-singleton label has size at most 7. The spectral inequality $d_G(h) \le \rho(A(\mathcal{F}_h)) + 1$ is a secondary quantitative form of this same packing constraint. A reader who treats the fan as a bare subcubic graph, ignoring this label algebra, will not see the bound—just as ignoring admissibility hides the Type A bound (Remark 14.1).*

Definition 14.68 (Type B window-incidence profile)**.** *Let W be the union of the fixed maximal packed induced P_{13} windows. A Type B fan-window profile records the assigned fan neighbours that lie in the remainder side of a decorated envelope; thus each such neighbour lies in $R = G - W$. A* window incidence *of a cubic-closed neighbour u is a triple*

$$(u, P, i)$$

where $u \notin W$, $P = p_0 \cdots p_{12}$ is a packed window, and $up_i \in E(G)$. A Type B fan-window profile is fully window-supported *when every cubic-closed neighbour has both non-h neighbours in W. It is* same-window supported *at u when the two non-h neighbours of u lie in one packed window.*

A non-window fan incidence *of a cubic-closed neighbour u is an edge uz with $z \notin W \cup \{h\}$. A cubic-closed neighbour is* window-supported, mixed-supported, *or* internal-supported *according as it has two, one, or zero window incidences among its two non-h incidences. Thus a mixed-supported neighbour has one window incidence and one non-window fan incidence, while an internal-supported neighbour has two non-window fan incidences.*

For a surplus port $p = (h, x)$ whose port vertex x is recorded on the remainder side of the fan-window profile, the neighbour $x \in N(h)$ is treated as a cubic-closed fan neighbour once the two non-h incidences xa_p and xb_p are assigned to the fan envelope. Its support type is read from the positions of a_p and b_p: it is window-supported, mixed-supported, or internal-supported according as two, one, or neither of those endpoints lies in W. This convention does not depend on whether a_pb_p is an edge of G.

Definition 14.69 (Fan-closed surplus port)**.** *Let h be a high-degree fan center, let $p = (h, x)$ be a surplus port at h, and write*

$$N_G(x) = \{h, a_p, b_p\}.$$

In an assigned Type B fan-window profile $\mathfrak{F}_h$, the port p is fan-closed *if the following three conditions hold.*

(a) *The port vertex x is recorded as a remainder-side fan neighbour of h.*

(b) *The two non-h incidences xa_p and xb_p are assigned to the fan envelope of $\mathfrak{F}_h$.*

(c) *Each assigned incidence is recorded as either a window incidence or a non-window fan incidence in the sense of Definition 14.68.*

Thus a fan-closed port contributes one cubic-closed neighbour to the count $c(\mathfrak{F}_h)$ of Definition 14.71. A fan-closed port may be triangular, when $a_pb_p \in E(G)$, or open, when $a_pb_p \notin E(G)$.

Lemma 14.70 (Fan-compatible open pairs give fan-closed local data)**.** *Let h be a high center, and let $p = (h, x)$ and $q = (h, y)$ be fan-compatible open ports at h. In any assigned Type B fan-window profile that records x and y as remainder-side fan neighbours of h and assigns the four incidences*

$$xa_p, \quad xb_p, \quad ya_q, \quad yb_q$$

to the fan envelope, the ports p and q are two distinct fan-closed ports. Moreover, these four incidences are pairwise distinct as local incidences. If no choice of the local B1 entry containing these incidences and the other assigned Type B fan entries is pairwise disjoint, then B2 has a disjoint-incidence failure, which is recorded as a Type B overlap obstruction by Lemma 14.87.

Proof. The ports are distinct, so $x \neq y$. Since both ports have center h, both x and y lie in $N_G(h)$; by Lemma 3.8, they are cubic. Hence

$$N_G(x) = \{h, a_p, b_p\}, \qquad N_G(y) = \{h, a_q, b_q\}.$$

The hypotheses of the assigned profile are exactly the three conditions in Definition 14.69 for p and for q. Thus both ports are fan-closed.

It remains to check that the four displayed incidences are locally distinct. The two incidences at x are distinct because $a_p \neq b_p$, and the two incidences at y are distinct because $a_q \neq b_q$. No incidence incident with x can equal an incidence incident with y, since $x \neq y$. Definition 3.9 gives $x \notin s(q)$, $y \notin s(p)$, and $s(p) \cap s(q) = \varnothing$. Therefore no endpoint of one displayed incidence can equal the endpoint of another displayed incidence except at its own port vertex, and the four incidences are pairwise distinct as local incidences.

The preceding argument is local to the fan at h. In the ambient assigned Type B support, the other fan demands may have several candidate entries. If no selection of those entries together with the local B1 entry for h is pairwise disjoint, then the global disjoint-incidence clause B2 fails. By Lemma 14.87, such a failure is represented by a minimal Type B overlap obstruction. $\square$

Definition 14.71 (Positive-deficit Type B fan-window residual)**.** *Let $\mathfrak{F}$ be a marked Type B fan centered at h, and write*

$$k = d_G(h), \qquad c(\mathfrak{F}) = |\{u \in N(h) : u \textit{ is cubic-closed in } \mathfrak{F}\}|.$$

The closed-neighbour deficit *of $\mathfrak{F}$ is*

$$D_B(\mathfrak{F}) = c(\mathfrak{F}) - \frac{11-k}{4}.$$

The fan is certificate-closed *when*

$$D_B(\mathfrak{F}) \leq 0.$$

Equivalently, certificate-closed means $c(\mathfrak{F}) \leq 1$ for $k \leq 7$ and $c(\mathfrak{F}) = 0$ for $k = 8$.

A positive-deficit Type B fan-window residual *is an admissible marked Type B fan-window profile with $D_B(\mathfrak{F}) > 0$ whose recorded window labels are pairwise C_2-compatible whenever they lie in a common packed window, and whose fan-window data do not realize a Mersenne return, a target-defective quotient, a target-complete compression, or proper/whole-graph support dependence. In this residual every cubic-closed neighbour is recorded with its two non-h incidences, marked as internal-supported, mixed-supported, or window-supported according as zero, one, or two of those incidences lie in W. Thus for $k \le 7$ the residual has at least two cubic-closed neighbours; for $k = 8$ the single cubic-closed case is also residual, because its local charge deficit is $D_B = 1/4$.*

Definition 14.72 (Hybrid Type B incidence accounting)**.** *Let $\mathfrak{F}$ be a positive-deficit Type B fan-window residual. Write*

$$c_W(\mathfrak{F}), \quad c_M(\mathfrak{F}), \quad c_I(\mathfrak{F})$$

for the numbers of window-supported, mixed-supported, and internal-supported cubic-closed neighbours, respectively. Thus

$$c = c_W + c_M + c_I.$$

The window-incidence count *and* non-window incidence count *are*

$$I_W(\mathfrak{F}) = 2c_W + c_M, \qquad I_N(\mathfrak{F}) = c_M + 2c_I.$$

The hybrid non-window demand *is*

$$D_N(\mathfrak{F}) = \max\left\{0,\ D_B(\mathfrak{F}) - \frac{1}{2}I_W(\mathfrak{F})\right\}.$$

Equivalently,

$$D_N(\mathfrak{F}) = \max\left\{0,\ \frac{1}{2}I_N(\mathfrak{F}) - \frac{11-k}{4}\right\}.$$

Definition 14.73 (Closed fan-window pair)**.** *Let $P = p_0p_1\cdots p_{12}$ be a packed induced P_{13} window. Suppose u is a cubic-closed neighbour of a Type B fan center h and is same-window supported at P. Its* closed label

$$S(u) = \{a_u, b_u\} \subseteq \{0,\dots,12\}, \qquad a_u < b_u,$$

meaning that u is adjacent to h, p_{a_u}, p_{b_u} and has no other incident edge in the assigned fan-window envelope. For two cubic-closed neighbours u, v same-window supported at P, put $S(v) = \{a_v, b_v\}$ and define

$$L_h(x,y) = 4 + |x-y| \qquad (x \in S(u),\ y \in S(v)).$$

If the four endpoints interlace, say $a_u < a_v < b_u < b_v$ after possibly swapping u and v, define

$$L_\times(u,v) = 4 + (a_v - a_u) + (b_v - b_u).$$

If the endpoints do not interlace, $L_\times(u,v)$ is undefined.

Definition 14.74 (Direct-cycle-free closed pair)**.** *A closed fan-window pair (u,v) is* direct-cycle-free *when all of the following arithmetic conditions hold:*

(a)

$$b_u - a_u,\ b_v - a_v \notin \{2,6\};$$

(b)

$$|x-y| \notin \{0,4,12\} \qquad (x \in S(u),\ y \in S(v));$$

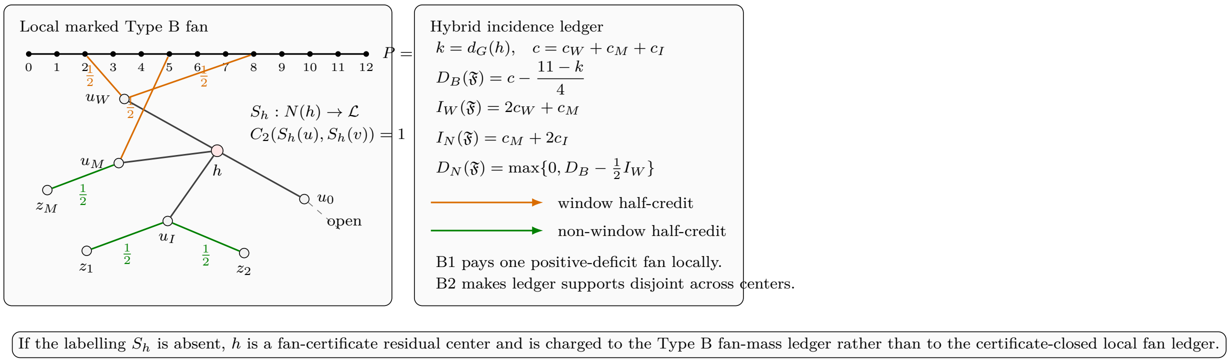

Figure 27: Visual definition of the Type B fan-window and hybrid ledger data. The left panel is the local marked Type B fan, with high-degree center h and neighbours in $N(h)$; the right panel is the hybrid incidence ledger. The horizontal black path $P = p_0 \cdots p_{12}$ is a packed induced P_{13} window, with the displayed indices naming its vertices. The labelling $S_h : N(h) \to \mathcal{L}$ assigns fan certificates, and the condition $C_2(S_h(u), S_h(v)) = 1$ states pairwise fan-safety for distinct neighbours. The neighbour u_W is window-supported because both of its non-h incidences go to window vertices; u_M is mixed-supported because one incidence goes to the window and one to a non-window vertex z_M; u_I is internal-supported because its two non-h incidences go to non-window vertices z_1, z_2; and u_0 is an open neighbour, so it is not counted as cubic-closed in this local ledger. Orange edges are window incidences, green edges are non-window incidences, and each displayed $\frac{1}{2}$ is the half-credit contributed by one such incidence. In the ledger, $k = d_G(h)$, $c = c_W + c_M + c_I$ counts cubic-closed neighbours by support type, $D_B(\mathfrak{F}) = c - (11 - k)/4$ is the closed-neighbour deficit, $I_W = 2c_W + c_M$ is the window-incidence count, $I_N = c_M + 2c_I$ is the non-window-incidence count, and $D_N = \max\{0, D_B - \frac{1}{2}I_W\}$ is the remaining non-window demand. B1 denotes the local payment of one positive-deficit fan, while B2 denotes the cross-center disjointness ledger needed to sum those payments globally. The bottom boxed note records the alternate case: if the labelling S_h is absent, then h is charged as a fan-certificate residual center in the Type B fan-mass ledger.

(c) if $L_\times(u, v)$ is defined, then

$$L_\times(u, v) \notin \{8, 16\}.$$

Lemma 14.75 (Direct fan-window cycle eliminations)**.** *Let u, v be distinct cubic-closed neighbours of a Type B fan center h that are same-window supported at the same packed window P. If the closed fan-window pair (u, v) is not direct-cycle-free, then the envelope contains a power-of-two cycle.*

Proof. Write $S(u) = \{a_u, b_u\}$ and $S(v) = \{a_v, b_v\}$ as in Definition 14.73. If $b_u - a_u \in \{2, 6\}$, then

$$up_{a_u} \, P \, p_{b_u} u$$

is a simple cycle of length $(b_u - a_u) + 2 \in \{4, 8\}$, because $u \notin W$ and P is a simple induced path. If $b_v - a_v \in \{2, 6\}$, then

$$vp_{a_v} \, P \, p_{b_v} v$$

is a simple cycle of length $(b_v - a_v) + 2 \in \{4, 8\}$ for the same reason.

If $|x - y| \in \{0, 4, 12\}$ for some $x \in S(u)$ and $y \in S(v)$, then

$$uhvp_y \, P \, p_x u$$

is a simple cycle: the vertices u and v are distinct and lie outside W, whereas the segment p_yPp_x lies inside the packed window. Its length is $2+1+|x-y|+1=4+|x-y|\in\{4,8,16\}$.

Finally suppose the endpoints interlace. After possibly swapping u and v, write

$$a_u<a_v<b_u<b_v.$$

The two path segments $p_{a_u}Pp_{a_v}$ and $p_{b_u}Pp_{b_v}$ are internally vertex-disjoint, and $u,v\notin W$. Hence

$$up_{a_u}\,P\,p_{a_v}vp_{b_v}\,P\,p_{b_u}u$$

is a simple cycle of length

$$4+(a_v-a_u)+(b_v-b_u)=L_\times(u,v).$$

If this length is 8 or 16, the cycle is a power-of-two cycle. These cases are exactly the negations of the direct-cycle-free conditions. $\square$

Lemma 14.76 (Two-window direct cycle eliminations)**.** *Let u,v be distinct cubic-closed neighbours of a Type B fan center h. Suppose u and v have incidences to two distinct packed windows*

$$P=p_0\cdots p_{12},\qquad Q=q_0\cdots q_{12},$$

with

$$up_i,\ vp_j,\ uq_a,\ vq_b\in E(G).$$

If

$$|i-j|+|a-b|\in\{0,4,12\},$$

then the fan-window envelope contains a power-of-two cycle.

Proof. By the definition of a window incidence, u and v lie outside W. The packed windows are vertex-disjoint, so the path segments p_iPp_j and q_aQq_b are vertex-disjoint and avoid u and v. Hence

$$up_i\,P\,p_jvq_b\,Q\,q_au$$

is a simple cycle. Its length is

$$1+|i-j|+1+1+|a-b|+1=4+|i-j|+|a-b|.$$

If $|i-j|+|a-b|\in\{0,4,12\}$, this length is 4, 8, or 16. $\square$

Lemma 14.77 (Positive-deficit fan-window budget)**.** *Let $\mathfrak{F}$ be an admissible positive-deficit Type B fan-window residual centered at h, with $k=d_G(h)$ and $c=c(\mathfrak{F})$. Then*

$$4\le k\le 8,\qquad D_B(\mathfrak{F})>0,$$

and more explicitly either $k\le 7$ and $c\ge 2$, or $k=8$ and $c\ge 1$. If $k=8$, no cubic-closed neighbour is same-window supported with a non-singleton label. For every k, every same-window supported cubic-closed neighbour has internal label gap outside $\{2,6\}$, every same-window closed fan-window pair in $\mathfrak{F}$ is direct-cycle-free, and in any assigned Type B support containing $\mathfrak{F}$ the closed fan neighbourhood has augmented charge lower bound

$$\mathrm{ch}_X(h)+\sum_{u\in N(h)}\mathrm{ch}_X(u)\ge\frac{11-k}{4}-c.$$

Equivalently, the remaining Type B charge deficit is exactly

$$D_B(\mathfrak{F}) = c - \frac{11-k}{4} > 0.$$

Assume in addition that $\mathfrak{F}$ is fully window-supported and contains no direct cycle of the forms in Lemmas 14.75 and 14.76. Then its c cubic-closed neighbours use exactly $2c$ distinct packed-window incidences, and

$$D_B(\mathfrak{F}) \leq \frac{1}{2} \cdot 2c - \frac{3}{4}.$$

Proof. The residual is a marked Type B fan, so its certificate labelling $S_h : N(h) \to \mathcal{L}$ is part of the data. Therefore the fan degree bound gives $k \leq 8$ by Lemma 14.66. If $k = 8$, then a same-window supported cubic-closed neighbour would carry a non-singleton label, and the non-singleton clique cap in Lemma 14.66 would force $k \leq 7$, a contradiction. Hence no cubic-closed neighbour of an actual 8-fan residual is same-window supported with a non-singleton label. Since the fan is Type B, $k \geq 4$, so $4 \leq k \leq 8$. The positive-deficit hypothesis is $D_B(\mathfrak{F}) > 0$, i.e.

$$c > \frac{11-k}{4}.$$

As c is integral, this gives $c \geq 2$ for $k \leq 7$ and $c \geq 1$ for $k = 8$.

If a cubic-closed neighbour u is same-window supported at $P = p_0 \cdots p_{12}$ and has closed label $S(u) = \{a_u, b_u\}$ with $b_u - a_u \in \{2, 6\}$, then

$$up_{a_u} \, P \, p_{b_u} u$$

is a simple cycle of length 4 or 8, contradicting admissibility. If some same-window closed pair were not direct-cycle-free, Lemma 14.75 would give a power-of-two cycle, contradicting admissibility. Hence every single same-window label has gap outside $\{2, 6\}$, and every same-window closed pair is direct-cycle-free.

For the augmented Type B charge, the fan center contributes

$$\mathrm{ch}_X(h) = (3-k) - \frac{1}{4}.$$

Each cubic-closed neighbour contributes $-\frac{1}{4}$, and each of the remaining $k-c$ neighbours has internal degree at most 2 in the assigned fan support and therefore contributes at least $\frac{3}{4}$. Hence

$$\mathrm{ch}_X(h) + \sum_{u \in N(h)} \mathrm{ch}_X(u) \geq \left(3 - k - \frac{1}{4}\right) - \frac{c}{4} + \frac{3}{4}(k-c) = \frac{11-k}{4} - c.$$

Because $D_B(\mathfrak{F}) > 0$, this lower bound is strictly negative, and the deficit is $D_B(\mathfrak{F}) = c - \frac{11-k}{4}$.

Now assume $\mathfrak{F}$ is fully window-supported and contains no direct cycle of the two displayed forms. Each cubic-closed neighbour has exactly two non-h neighbours, both in packed windows, so the number of packed-window incidences counted with multiplicity is $2c$. These incidences are distinct. Indeed, two different closed neighbours cannot use the same packed-window vertex: if up_i and vp_i were both present, then the fan-cross cycle

$$u - h - v - p_i - u$$

has length 4. One closed neighbour cannot use the same packed-window vertex twice because the graph is simple. Incidences in different packed windows are distinct because the packing is vertex-disjoint. Thus the residual uses exactly $2c$ distinct packed-window incidences.

Since $k \le 8$,

$$\frac{11-k}{4} \ge \frac{3}{4}.$$

Therefore

$$D_B(\mathfrak{F}) = c - \frac{11-k}{4} \le c - \frac{3}{4} = \frac{1}{2} \cdot 2c - \frac{3}{4}.$$

Equivalently, half a unit of reserved credit per distinct packed-window incidence pays the local Type B deficit with at least three quarters of a unit of slack. □

Lemma 14.78 (Hybrid incidence budget for positive-deficit fans)**.** *Let $\mathfrak{F}$ be an admissible positive-deficit Type B fan-window residual centered at h. Suppose that none of the direct-cycle conclusions of Lemmas 14.75 and 14.76 occurs. Then the $2c(\mathfrak{F})$ non-h incidences of the cubic-closed neighbours of h are pairwise disjoint as incidences. The window incidences supply half-credit $\frac{1}{2}I_W(\mathfrak{F})$, and the non-window incidences supply a disjoint local reserve of size at least*

$$D_N(\mathfrak{F}) = \max\left\{0,\ D_B(\mathfrak{F}) - \frac{1}{2}I_W(\mathfrak{F})\right\}.$$

Consequently the hybrid fan entry supplies at least $D_B(\mathfrak{F})$.

Proof. Let u and v be distinct cubic-closed neighbours of h. If a non-h neighbour z were incident with both u and v, then

$$u - h - v - z - u$$

would be a simple 4-cycle, because u and v are distinct, $z \ne h$, and the fan neighbours have ambient degree 3. This is forbidden by target-safety. One cubic-closed neighbour cannot use the same non-h endpoint twice, since G is simple. Hence the two non-h incidences of all cubic-closed neighbours are pairwise disjoint as incidences.

The window incidences among them are counted by $I_W(\mathfrak{F})$ and the non-window incidences by $I_N(\mathfrak{F})$. Each such incidence carries half-credit in the hybrid fan ledger: packed-window incidences use the window-incidence item of Definition 14.83, and non-window incidences use the local internal/mixed reserve item in the same definition. Thus the total local incidence capacity is

$$\frac{1}{2}I_W(\mathfrak{F}) + \frac{1}{2}I_N(\mathfrak{F}) = \frac{1}{2} \cdot 2c(\mathfrak{F}) = c(\mathfrak{F}).$$

Since

$$D_B(\mathfrak{F}) = c(\mathfrak{F}) - \frac{11-k}{4}$$

and $k \le 8$, this total capacity pays $D_B(\mathfrak{F})$ with slack at least $\frac{11-k}{4} \ge \frac{3}{4}$.

After the window half-credit has been applied, the remaining demand is

$$\max\left\{0,\ D_B(\mathfrak{F}) - \frac{1}{2}I_W(\mathfrak{F})\right\} = D_N(\mathfrak{F}).$$

The available non-window half-credit is

$$\frac{1}{2}I_N(\mathfrak{F}) = c(\mathfrak{F}) - \frac{1}{2}I_W(\mathfrak{F}) = D_B(\mathfrak{F}) + \frac{11-k}{4} - \frac{1}{2}I_W(\mathfrak{F}),$$

which is at least $D_N(\mathfrak{F})$. The disjointness proved in the first paragraph permits these non-window incidences to be chosen disjointly inside the single fan entry. □

Lemma 14.79 (Hybrid B1 fan ledger)**.** *Every admissible positive-deficit Type B fan-window residual satisfies the hybrid B1 alternative of Definition 14.90: after the direct cycle, target-defect, compression, and support-dependence alternatives are excluded, the packed-window incidences and the local internal/mixed incidence reserve form a fan ledger entry whose total capacity is at least $D_B(\mathfrak{F})$.*

Proof. The excluded target-defect, compression, and support-dependence alternatives are part of the definition of a positive-deficit residual. If one of the direct fan-window cycle conclusions occurs, the first alternative in B1 holds. In the remaining case, Lemma 14.78 supplies a disjoint hybrid incidence entry of capacity at least $D_B(\mathfrak{F})$. This is exactly the hybrid B1 ledger entry. □

Proposition 14.80 (Fan-closed ports route to the Type B fan ledger)**.** *Let h be a high-degree fan center of degree $k \geq 4$, and let $\mathfrak{F}_h$ be an admissible Type B fan-window profile that records h as a fan center. Suppose that $\mathfrak{F}_h$ contains a set $\mathcal{Q}$ of $r \geq 2$ fan-closed surplus ports at h. Then the following hold.*

(a) The ports in $\mathcal{Q}$ contribute at least r cubic-closed neighbours of h.

(b) If the fan at h is certificate-marked and survives the direct-cycle, target-defect, compression, and support-dependence alternatives in B1, then it is a positive-deficit Type B fan-window residual with

$$D_B(\mathfrak{F}_h) \geq r - \frac{11-k}{4} \geq \frac{k-3}{4} > 0.$$

(c) The hybrid B1 ledger supplies local incidence capacity at least $D_B(\mathfrak{F}_h)$. The paying incidences include the two non-h incidences xa_p and xb_p of every fan-closed port $p = (h, x) \in \mathcal{Q}$, classified by Definition 14.68 as window, mixed, or non-window incidences according to the locations of a_p and b_p.

(d) If the fan at h is not certificate-marked, then h is a fan-certificate residual center. If the local B1 entries cannot be made disjoint from the other Type B entries in the assigned support, then the support contains a minimal Type B overlap obstruction. These are exactly the two Type B bridge residual alternatives of Definition 14.90.

Proof. Let $p = (h, x) \in \mathcal{Q}$. Since p is fan-closed, Definition 14.69 records x as a remainder-side fan neighbour of h and assigns the two non-h incidences xa_p and xb_p to the fan envelope. By Lemma 3.8, x is cubic. Hence x has internal fan degree 3 in the assigned envelope, so it is cubic-closed in the sense of Definition 14.18. Distinct ports at h have distinct port vertices, and therefore the r ports in $\mathcal{Q}$ give at least r cubic-closed neighbours. This proves (a).

Assume now that the fan is certificate-marked and survives the direct-cycle, target-defect, compression, and support-dependence alternatives in the B1 statement. Let $c = c(\mathfrak{F}_h)$ be the number of cubic-closed neighbours recorded in the fan envelope. By (a), $c \geq r$. The closed-neighbour deficit is

$$D_B(\mathfrak{F}_h) = c - \frac{11-k}{4}.$$

Therefore

$$D_B(\mathfrak{F}_h) \geq r - \frac{11-k}{4} \geq 2 - \frac{11-k}{4} = \frac{k-3}{4}.$$

Since $k \geq 4$, the last quantity is positive. Thus the fan satisfies the positive-deficit condition of Definition 14.71. The profile is admissible by hypothesis; the certificate marking gives the required C_2-compatibility of the recorded labels; and the absence of the direct-cycle, target-defect, compression, and support-dependence alternatives is exactly the surviving B1 hypothesis. Hence all clauses of Definition 14.71 are satisfied. This proves (b).

Under the hypotheses of (b), Lemma 14.79 applies. It invokes Lemma 14.78, which gives disjoint local half-credit on the $2c$ non-h incidences of the cubic-closed fan neighbours and pays at least $D_B(\mathfrak{F}_h)$. For every fan-closed port $p = (h, x) \in \mathcal{Q}$, the two relevant incidences are exactly xa_p and xb_p. Definition 14.68 classifies them according to whether their non-h endpoints lie in the packed-window union W. This proves (c).

If the fan is not certificate-marked, then the definition of Definition 14.18 makes h a fan-certificate residual center. If the fan is certificate-marked and the local B1 entries exist but cannot be made globally disjoint inside the assigned support, then the disjoint-incidence clause B2 in Definition 14.90 fails. By Lemma 14.87, that failure contains a minimal Type B overlap obstruction. These two alternatives are exactly the two cases in the definition of a Type B bridge residual, proving (d). □

Corollary 14.81 (Fan-compatible open pairs route to the Type B fan ledger)**.** *Let h be a high center of degree $k \geq 4$, and let $p = (h, x)$ and $q = (h, y)$ be fan-compatible open ports at h. In any admissible Type B fan-window profile that records h as a fan center, records x and y as remainder-side fan neighbours, and assigns the four non-h incidences of p and q, the fan at h has two fan-closed ports. Consequently it routes through Proposition 14.80. If the direct-cycle, target-defect, compression, or support-dependence alternative occurs, the branch is already closed. If the fan is certificate-marked and those closed alternatives are absent, then*

$$D_B(\mathfrak{F}_h) \geq \frac{k-3}{4} > 0,$$

so the B1 ledger pays the local positive-deficit fan. If the fan is not certificate-marked, or if the B1 entries cannot be made globally disjoint, the remaining Type B outcome is a fan-certificate residual center or a B2 overlap obstruction.

Proof. Lemma 14.70 shows that p and q are two distinct fan-closed ports in the assigned profile, with locally distinct non-h incidences. Applying Proposition 14.80 with $\mathcal{Q} = \{p, q\}$ gives the displayed deficit bound and the listed Type B alternatives. □

Corollary 14.82 (Triangular ports route to the Type B fan ledger)**.** *Let h be a heavy center of degree $k \geq 5$, and suppose that the triangular alternative in Corollary 3.12 holds at h. Let $\mathcal{T}_h$ be a set of $k-2$ triangular ports at h. In any admissible Type B fan-window profile that records h as a fan center, records the port vertices x_i of $\mathcal{T}_h$ as remainder-side fan neighbours, and assigns the two triangle incidences of every port in $\mathcal{T}_h$, the fan at h has at least $k-2$ fan-closed ports. Hence it routes through Proposition 14.80; in the certificate-marked B1 survivor,*

$$D_B(\mathfrak{F}_h) \geq (k-2) - \frac{11-k}{4} = \frac{5k-19}{4} > 0.$$

Proof. For a triangular port $p_i = (h, x_i)$, the vertex x_i is a neighbour of h and is cubic by Lemma 3.8. Its two non-h neighbours are precisely a_i and b_i, and both incidences $x_i a_i$, $x_i b_i$ are present in G. Once those two incidences are assigned to the fan envelope, Definition 14.69 makes p_i fan-closed. The $k-2$ ports in $\mathcal{T}_h$ have distinct port vertices, so they contribute at least $k-2$ fan-closed, hence cubic-closed, neighbours. Applying Proposition 14.80 with $\mathcal{Q} = \mathcal{T}_h$ gives the stated alternatives and the stronger displayed deficit bound. □

Definition 14.83 (Type B ledger incidences)**.** *Let X be a connected assigned Type B support. A* Type B ledger item *in X is one of the following pieces of support-charge data.*

(a) a non-center vertex $v \in Y_X$ which is not entered as an assigned high-degree center, carrying the charge

$$\mathrm{ch}_X(v) = \delta_X^+(v) - \frac{1}{4}$$

from Definition 14.65;

(b) an assigned high-degree center $h \in H_X$, carrying the augmented center charge

$$\mathrm{ch}_X(h) = -(d_G(h) - 3) - \frac{1}{4};$$

(c) a packed-window incidence (u, P, i) of Definition 14.68, carrying capacity $1/2$ when it is used as the half-incidence credit of Lemma 14.77;

(d) a non-window fan incidence uz of Definition 14.68, carrying capacity $1/2$ when it is used as the local internal/mixed half-credit of Lemma 14.78;

(e) a local internal/mixed reserve block Q, contained in the assigned non-window part of a positive-deficit fan, whose capacity is the sum of the support charges of its vertices and incidences in the same support-charge ledger.

The support of a block or incidence is its underlying vertex/incidence set. Two ledger items are disjoint *when their underlying supports are disjoint and neither is part of the ordinary deficiency reserve. The* ordinary deficiency reserve *is the unrefined pool of positive-deficiency units $\delta_X^+(y)$ and packed-window boundary-incidence units counted by $\partial_+(X)$ and by the stub supply in Lemma 8.5. A Type B ledger item is disjoint from this reserve precisely when the same vertex or incidence unit is not also used as one of those unrefined reserve units.*

Definition 14.84 (Candidate Type B ledger entries). *Let X be a connected assigned Type B support. A* Type B demand *is attached to a marked fan center $h \in H_X$.*

(a) If the marked fan at h is certificate-closed, a candidate ledger entry is a set $A_h \subseteq N(h) \setminus H_X$ such that

$$\mathrm{ch}_X(h) + \sum_{v \in A_h} \mathrm{ch}_X(v) \geq 0.$$

Its ledger items are the center h together with the vertices of A_h.

(b) If the marked fan at h is positive-deficit, a candidate ledger entry is the fan envelope at h, together with all packed-window incidences and enough non-window fan incidences used by its cubic-closed neighbours to pay the remaining demand $D_N(\mathfrak{F}_h)$ of Definition 14.72. Each chosen incidence has capacity $1/2$. By Lemma 14.78, this entry pays $D_B(\mathfrak{F}_h)$ with slack at least $(11 - k)/4$ before unused incidence credits are discarded. Its ledger items are the center h, the cubic-closed fan neighbours whose closed-neighbour deficit is being paid, and the chosen incidences.

A disjoint refined Type B ledger *is a choice of one candidate entry for every Type B demand in X such that all chosen ledger items are pairwise disjoint. It is* maximal *when, after the chosen entries are removed, the remaining core has no high-degree center and no decorated Type B handoff whose center belongs to X.*

Definition 14.85 (Minimal Type B overlap obstruction). *Let X be a connected assigned Type B support. A* Type B overlap obstruction *is a pair*

$$\mathcal{O} = (\mathcal{D}, \{\mathcal{E}(d)\}_{d\in\mathcal{D}})$$

where $\mathcal{D}$ is a nonempty finite set of Type B demands in X and $\mathcal{E}(d)$ is the set of candidate ledger entries for d, such that no choice of entries

$$E_d \in \mathcal{E}(d) \qquad (d \in \mathcal{D})$$

has pairwise disjoint ledger supports. The obstruction is minimal *when every proper nonempty subfamily of $\mathcal{D}$ does admit such a disjoint choice.*

The overlap support $Z(\mathcal{O})$ *is the connected support induced by the fan centers in $\mathcal{D}$, all vertices and incidences appearing in their candidate entries, and the paths or window segments that witness the corresponding fan-safe relations. If this set is disconnected, each connected component is treated as a separate overlap obstruction.*

Lemma 14.86 (Finite completion of the Type B support assignment). *Let X be a connected assigned Type B support with no fan-certificate residual center. Suppose that no finite family of Type B demands in X forms a Type B overlap obstruction in the sense of Definition 14.85. Then the Type B demands of X admit a maximal disjoint refined Type B ledger.*

Proof. Let $\mathcal{M}$ be the finite set of vertices of X that are either high-degree centers or centers of decorated Type B handoffs in the residual core determined by the canonical support assignment. Since X is finite, $\mathcal{M}$ is finite. Start with the currently declared Type B demand family. If its disjoint refined ledger is not maximal, then some vertex of $\mathcal{M}$ has not yet been entered as a demand center. Add the lexicographically first such vertex to the demand family, using the candidate entries supplied by the Type B assignment rules and by Lemma 14.40 for decorated handoff centers.

After adding finitely many demands, absence of a Type B overlap obstruction means precisely that the enlarged finite demand family has a choice of candidate entries with pairwise disjoint ledger supports. Rechoose the whole ledger for the enlarged family if necessary. Each step marks one previously unprocessed vertex of $\mathcal{M}$, and no step creates a new vertex of $\mathcal{M}$. Hence after at most $|\mathcal{M}|$ steps no such vertex remains. The resulting disjoint refined ledger is maximal by the definition of Definition 14.84. ☐

Lemma 14.87 (Bridge failure produces a minimal overlap obstruction). *Let X be a connected assigned Type B support with no fan-certificate residual center. If the disjoint-incidence part of the B2 refined support ledger of Definition 14.90 fails for X, then X contains a minimal Type B overlap obstruction in the sense of Definition 14.85.*

Proof. For each high-degree fan center of X, form the finite set of candidate ledger entries in Definition 14.84. The sets are finite because X is finite, every fan neighbour has degree 3 by Lemma 3.2, and a packed-window incidence uses one of the finitely many labels $0, \ldots, 12$ in a fixed packed window.

Assume that no choice of one candidate entry for every high-degree fan demand has pairwise disjoint ledger supports. Among all nonempty subfamilies of demands for which the disjoint choice fails, choose one minimal by cardinality; this gives $\mathcal{O}$. If the union of the chosen fan centers, candidate ledger items, and witnessing path/window segments is disconnected, one connected component already contains a subfamily with no disjoint choice, contradicting minimality unless the component is taken as the obstruction. Hence $Z(\mathcal{O})$ is connected after passing to that component.

The hybrid B1 fan entry is supplied locally by Lemma 14.79. Thus a disjoint-incidence failure after B1 is exactly the overlap obstruction constructed above. A failure of the maximal residual core condition is a different alternative: the remaining core still contains a high-degree center or a decorated Type B handoff, and is routed again through Lemma 14.40 and the Type B support assignment rather than through the overlap-obstruction construction. □

Lemma 14.88 (Global-to-local reflection of Type B overlap)**.** *Let $\mathcal{O}$ be a minimal Type B overlap obstruction contained in a connected admissible support X. Then its overlap support $Z = Z(\mathcal{O})$ inherits the following global constraints.*

- *(a)* Contextual target safety: *no cycle of G meeting $V(Z)$ has power-of-two length. Equivalently, no oriented edge in the overlap support has a Mersenne return through the actual outside context.*
- *(b)* High-degree separation: *the high-degree fan centers in Z are independent, and every neighbour of such a center has ambient degree* 3*.*
- *(c)* Window compatibility: *every packed-window incidence in Z is one of the $15p_{13} + o(n)$ outside incidences counted by Lemma 8.5; same-window and two-window incidences satisfy the direct-cycle exclusions of Lemmas 14.75 and 14.76.*
- *(d)* Replacement obstruction: *every quotient or identification among the ledger-response coordinates of Z that would remove a support conflict is either target-defective, target-complete with a smaller proper representative and therefore forbidden by Corollary 3.98, or valid only after adjoining a larger support and therefore routed to Lemmas 10.6 and 10.9. If none of these alternatives occurs, the coordinate remains part of the minimal overlap obstruction.*
- *(e)* Minimal overlap: *every proper connected sub-obstruction of $\mathcal{O}$ has a refined disjoint ledger.*

Proof. The support X is admissible by construction of the local analysis. Hence it is embedded in the fixed minimal counterexample G and carries all global constraints listed in Definition 13.2 and Remark 14.1. Since $Z \subseteq X$, any power-of-two cycle of G meeting $V(Z)$ would also meet $V(X)$, contradicting contextual target safety of X; this proves (a).

For (b), apply Lemma 3.2. No edge joins two vertices of ambient degree at least 4, so the fan centers are independent. If $u \in N(h)$ for a high-degree center h, then the edge hu has one endpoint of degree 3; since $d_G(h) \geq 4$, the endpoint forced to have degree 3 is u.

For (c), each packed-window incidence is by definition an edge between $R = G - W$ and the packed window union W. Such edges are exactly the incidence supply counted in Lemma 8.5. If same-window or two-window labels violated the displayed arithmetic exclusions, the corresponding direct-cycle lemma would produce a power-of-two cycle meeting Z, contradicting (a).

For (d), regard the candidate ledger entries and their ledger items as local target-response coordinates of the boundaried support Z. If an attempted identification of two such coordinates is distinguished by some outside context, it is target-defective by Lemma 3.95. If the identification is target-complete and supplies a smaller proper boundaried representative with the same boundary degree profile, it is a target-complete compression and is forbidden by Corollary 3.98. If the identification becomes valid only after adjoining a larger support, the proper case is forbidden by Lemma 10.6, and the whole-graph case is forbidden by Lemma 10.9. When none of these alternatives applies, no quotient, compression, or support dependence removes the support conflict, so the coordinate stays in the overlap obstruction. These are exactly the replacement and support-dependence alternatives already used in the Type A saturated exits and in the Type B fan-safe relation.

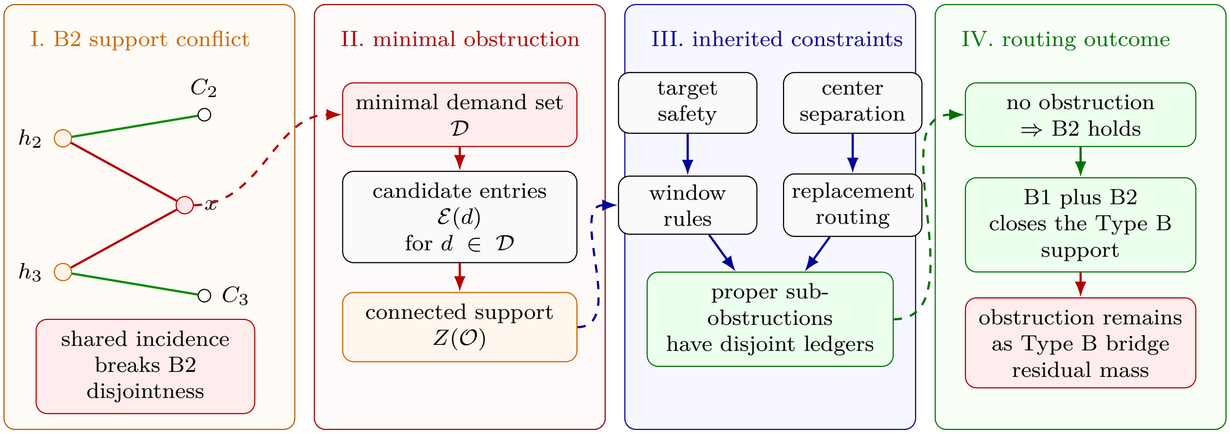

Figure 28: Visual proof of the Type B overlap-obstruction routing. Panel I shows the B2 failure mechanism: two Type B fan demands may have local B1 entries, but their candidate support sets can compete for a shared support x, so the chosen entries are not disjoint. Panel II records the finite minimality step of Lemma 14.87: among all failed demand families, a minimal one gives a Type B overlap obstruction $\mathcal{O} = (\mathcal{D}, \{\mathcal{E}(d)\})$, and its involved fan centers, supports, paths, and window segments form the connected overlap support $Z(\mathcal{O})$. Panel III displays the global constraints inherited by $Z(\mathcal{O})$ in Lemma 14.88: contextual target safety, high-degree separation, window compatibility, replacement/support-dependence routing, and minimality of the overlap. Panel IV gives the routing consequence of Proposition 14.89: if no such obstruction remains, B2 holds and B1 closes the Type B support; if the obstruction remains, it is a controlled Type B bridge residual charged later in the fan-mass ledger.

Finally, (e) is the minimality clause in Definition 14.85: every proper nonempty subfamily of demands admits disjoint candidate entries, hence every proper connected sub-obstruction has a refined disjoint ledger. □

Proposition 14.89 (Type B bridge residual as a global-to-local obstruction)**.** *Every disjoint-incidence Type B bridge residual is represented by a minimal Type B overlap obstruction satisfying all constraints of Lemma 14.88. Conversely, if an admissible support outside the route-8 residual has no fan-certificate residual center and no such overlap obstruction, then B2 holds there; together with the hybrid B1 ledger of Lemma 14.79, Proposition 14.92 closes the Type B support.*

Proof. The first assertion is Lemma 14.87 followed by Lemma 14.88. For the converse, Lemma 14.79 gives the local B1 entry for every positive-deficit fan. The absence of fan-certificate residual centers means that every high-degree center in the Type B support is certificate-marked. Since no multi-demand overlap obstruction is present, Lemma 14.86 supplies a maximal disjoint refined Type B ledger. This is exactly B2. With the local B1 entries and B2 available, Proposition 14.92 gives

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|$$

for every connected admissible Type B support outside the route-8 residual. □

Definition 14.90 (Type B bridge statements)**.** *The Type B bridge consists of the local hybrid fan ledger B1 and the global disjoint support ledger B2. The local ledger B1 is proved in Lemma 14.79; the remaining bridge assertion is the global disjointness statement B2 used in Proposition 14.92.*

(B1) Hybrid fan ledger. *Every positive-deficit Type B fan-window residual is one of the following: it contains a direct power-of-two cycle of the forms in Lemmas 14.75 and 14.76; it realizes a target-defective quotient, a target-complete compression, or proper/whole-graph support dependence; or its packed-window incidences together with its local internal/mixed non-window incidences contain a disjoint hybrid fan entry of capacity at least $D_B(\mathfrak{F})$. The window part contributes $\frac{1}{2}I_W(\mathfrak{F})$, and the non-window part contributes the remaining demand $D_N(\mathfrak{F})$ of Definition 14.72.*

(B2) Refined support ledger. *For any connected assigned Type B support X with no fan-certificate residual center, the high-degree fan centers of X admit a canonical refined ledger in the sense of Definition 14.84. Clauses (a)–(c) are the disjoint-incidence part of B2; clause (d) is the maximal support-assignment part.*

(a) Each certificate-closed fan center is assigned a set of incident non-center vertices whose augmented charge, together with the center charge, is nonnegative. The center and the assigned non-center vertices are ledger items for that entry, and no ledger item is assigned to two different fan centers.

(b) For any maximal canonical family $\mathcal{F}$ of positive-deficit Type B residual fans in X, each fan center has its hybrid B1 ledger entry: packed window incidences carry credit $1/2$, and non-window fan incidences carry the local internal/mixed half-credit needed to pay D_N. These incidences are disjoint from the ordinary deficiency reserve of Definition 14.83.

(c) The certificate-closed entries in (a), the local reserve entries supplied by B1 for positive-deficit fans, and the half-incidence entries in (b) are mutually disjoint refinements of the augmented ledger $\widehat{\mathsf{Ch}}_B(X)$ of Definition 14.65. After the support assignment is forgotten, this ledger gives $\mathcal{N}(X)$ by the identity (B-ledger).

(d) The ledger is maximal for the Type B support assignment: after the ledger entries are removed, the remaining core has no high-degree center, no decorated Type B handoff whose center belongs to X, and no external decorated handoff that is not already covered by a grouped decorated envelope processed in the same canonical step.

A Type B bridge residual *is either:*

(i) a connected assigned Type B support containing a fan-certificate residual center of Definition 14.18; or

(ii) a connected assigned Type B support in which the local B1 entries have been supplied, the support assignment has been made maximal, and the disjoint-incidence part of B2 fails; or

(iii) a grouped decorated Type B envelope support whose envelope ledger cannot be closed by the center tokens of its core–center incidence component.

In cases (ii) and (iii), the support contains a minimal Type B overlap obstruction in the sense of Definition 14.85; by Proposition 14.89, that obstruction carries the global-to-local constraints of Lemma 14.88.

Lemma 14.91 (Post-ledger Type A core admissibility)**.** *Let X be a Type B support or a grouped decorated Type B envelope support. Remove from its counted core the vertices and incidences used by the certificate-closed fan entries, the local B1 fan entries, and the grouped decorated envelope entries. Each connected component Z of the remaining non-window core is an admissible Type A support after its inherited boundary profile is recorded. In particular, Z is P_{13}-free, contextually target-safe, hereditarily target-uncompressible, and has supplied deficiency charged to the unused ordinary deficiency reserve and the window-stub reserve.*

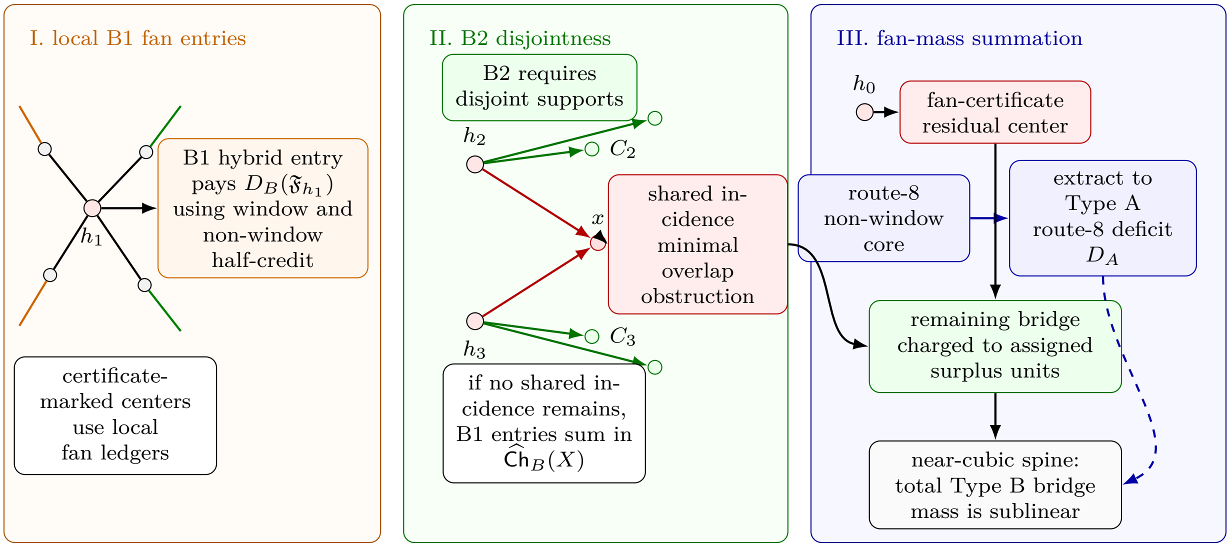

Figure 29: Visual definition of the Type B bridge and fan-mass summation. Panel I shows the local B1 entry for a positive-deficit Type B fan centered at h_1. Window incidences and non-window incidences contribute half-credit, and the B1 ledger pays the local deficit $D_B(\mathfrak{F}_{h_1})$. Panel II shows the global B2 disjointness requirement. Centers h_2 and h_3 may use their own support sets C_2 and C_3, but a shared support x is forbidden by B2; a minimal unresolved sharing pattern is the minimal Type B overlap obstruction of Definition 14.85. Panel III shows the fan-mass summation used for Type B bridge residuals. A center without the required certificate enters the fan-certificate residual class, a route-8 non-window core is extracted into the Type A route-8 deficit ledger D_A, and the remaining Type B bridge entries are charged to assigned surplus units. After this extraction, the Type B bridge mass is sublinear in the near-cubic spine, so it cannot carry the linear large-budget deficit.

Proof. The remaining non-window core is an induced subgraph of the original remainder core after deleting a declared family of ledger items. Hence every connected component Z is P_{13}-free and contextually target-safe by heredity. The target-uncompressibility condition is hereditary by Corollary 3.98: a target-complete compression of Z would extend, with the recorded boundary profile, to a target-complete compression of the original support.

Deleting ledger items can create new deficient boundary vertices in Z. Each such deficit is caused by one of the removed incidences. The ledger item was either part of a fan entry already paid in the Type B ledger, an ordinary deficiency-reserve unit of Definition 14.83, or a packed-window incidence charged by Corollary 8.8. The remaining boundary profile records the unused part of those reserves, so the supplied-deficiency hypothesis in Definition 13.2 transfers to each component. Since components are treated separately, no connectivity assumption is inherited across a deleted ledger item. Therefore every connected component is an admissible Type A support with its own recorded boundary data. □

Proposition 14.92 (Type B reduction under the refined ledger)**.** *Let X be a connected admissible Type B support outside the route-8 Type A residual. Assume that X contains no fan-certificate residual center and that the refined support ledger B2 of Definition 14.90 holds. Then X satisfies*

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|.$$

Proof. Since X contains no fan-certificate residual center, every high-degree center assigned to X is certificate-marked. Use the canonical refined ledger supplied by bridge statement (B2). Its certificate-closed entries are nonnegative by Lemma 14.66, and B2(a) makes the corresponding non-center vertices disjoint across fan centers. A positive-deficit fan that realizes a direct cycle, quotient, compression, or support dependence is forbidden by the standing admissibility and minimality invariants. For every remaining positive-deficit fan $\mathfrak{F}$, the hybrid B1 ledger of Lemma 14.79 supplies a fan entry of capacity at least $D_B(\mathfrak{F})$: packed-window incidences pay their half-credit and non-window incidences pay the residual demand $D_N(\mathfrak{F})$. Bridge statement B2(b) makes these incidences disjoint from the ordinary deficiency reserve of Definition 14.83, and B2(c) makes the resulting fan entries mutually disjoint from the certificate-closed entries and from the other positive-deficit fan entries. Clause (c) records these credits as entries of the same augmented support-charge ledger. Summing over the positive-deficit fans therefore supplies at least the total positive-deficit Type B deficit.

After the fan centers are processed, the remaining core has no high-degree surplus, no decorated Type B handoff whose center belongs to X, and no external decorated handoff left uncovered by the grouped envelope step, by the maximality clause B2(d). Apply Lemma 14.91 to its connected components. Since the route-8 residual is excluded, the Type A saturated exits are closed or handed to the already processed Type B family, and Lemma 14.43 gives nonnegative charge on each remaining component. The certificate-closed fans, the bridge-paid positive-deficit fans, and the residual components have nonnegative total charge in the refined augmented ledger. Clause B2(c) says that these entries are mutually disjoint and refine $\widehat{\mathsf{Ch}}_B(X)$, so

$$\widehat{\mathsf{Ch}}_B(X) \geq 0.$$

The identity (B-ledger) in Definition 14.65 gives

$$\mathcal{N}(X) = \widehat{\mathsf{Ch}}_B(X) + \frac{1}{4}|H_X| \geq 0.$$

Since $\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \frac{1}{4}|V(X)|$, this is exactly

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|.$$

$\square$

Lemma 14.93 (Type B exclusion under the refined ledger)**.** *Let X be a connected admissible support carrying high-degree surplus. Assume that X contains no fan-certificate residual center and admits the refined support ledger B2 of Definition 14.90. If X contains neither an admissible route-8 residual profile nor an admissible positive-deficit Type B fan-window residual, then*

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|,$$

so that $\mathcal{N}(X) \geq 0$. Consequently, a Type B support with $\mathcal{N}(X) < 0$ and with no route-8 or positive-deficit fan-window residual is a Type B bridge residual: it witnesses failure of B2.

Proof. Use the assigned Type B notation of Definition 14.65. The set H_X consists of the high-degree fan centers whose surplus units are assigned to X; it is independent by Lemma 3.2, and

$$\sigma(X) = \sum_{h \in H_X} (d_G(h) - 3).$$

It is enough to prove $\widehat{\mathsf{Ch}}_B(X) \geq 0$, because then the identity (B-ledger) gives $\mathcal{N}(X) \geq 0$.

Step 1 (fan degree is bounded; the closed fan neighbourhood carries nonnegative charge). Fix $h \in H_X$ and write $k = d_G(h) = |N(h)|$. Since X contains no fan-certificate residual center, the fan at h is certificate-marked in the sense of Definition 14.18. By the fan-return-safety analysis above, $N(h)$ is a clique in the fan-safe graph $\mathcal{F}_h$, built from the five local safety relations (i)–(v).

Degree bound. The certificate labelling $S_h : N(h) \to \mathcal{L}$ satisfies $C_2(S_h(u), S_h(v)) = 1$ for every distinct $u, v \in N(h)$. Hence $S_h(N(h))$ is a clique in the fan-certificate graph of Lemma 14.66. That lemma gives

$$k = d_G(h) \leq 8,$$

and the spectral inequality $d_G(h) \leq \rho(A(\mathcal{F}_h)) + 1$ is the quantitative form of this bound.

Closed-neighbourhood charge for the certificate-closed subcase. We use the augmented charge ch_X of Definition 14.65. The center has

$$\mathrm{ch}_X(h) = 3 - k - \frac{1}{4}.$$

Cubic-closed neighbours contribute $-\frac{1}{4}$, while non-cubic-closed fan neighbours have internal degree at most 2 in the assigned fan envelope and contribute at least $\frac{3}{4}$.

Let $\mathfrak{F}_h$ be the marked fan centered at h, let $c = c(\mathfrak{F}_h)$ be its number of cubic-closed neighbours, and put

$$D_B(\mathfrak{F}_h) = c - \frac{11 - k}{4}.$$

If $D_B(\mathfrak{F}_h) > 0$, then, because X is admissible, the fan-window data cannot realize a target cycle, target-defective quotient, target-complete compression, or support dependence. Moreover, if a cubic-closed neighbour u is same-window supported at $P = p_0 \cdots p_{12}$ with closed label $\{a, b\}$ and $b - a \in \{2, 6\}$, then it gives the cycle

$$up_a \, P \, p_b u$$

of length 4 or 8, and any two-neighbour same-window incompatibility gives a direct fan-window cycle by Lemma 14.75. Hence the recorded same-window labels satisfy the compatibility requirements of Definition 14.71. Thus Definition 14.71 would make $\mathfrak{F}_h$ a positive-deficit Type B residual, contrary to the hypothesis of this lemma. Therefore

$$D_B(\mathfrak{F}_h) \leq 0.$$

The charge calculation is then explicit:

$$\mathrm{ch}_X(h) + \sum_{u \in N(h)} \mathrm{ch}_X(u) \geq \left(3 - k - \frac{1}{4}\right) - \frac{c}{4} + \frac{3}{4}(k - c) = \frac{11 - k}{4} - c = -D_B(\mathfrak{F}_h) \geq 0.$$

This proves nonnegative charge for every non-residual fan neighbourhood. In particular, when $k = 8$, the exclusion of the positive-deficit residual forces $c = 0$.

Step 2 (B2 ledger and residual core). Step 1 is a local calculation at a single fan center. It does not identify disjoint paying ledger items for different centers. The required disjointness is exactly the refined support ledger B2. By B2(a), every certificate-closed fan center is assigned a non-center vertex set whose augmented charge together with the center charge is nonnegative, and no ledger item is assigned to two fan centers. Since the hypothesis excludes positive-deficit fan-window residuals, every marked fan centered in H_X is certificate-closed by Step 1. Thus all high-degree centers of X are covered by pairwise disjoint nonnegative ledger entries.

Remove these ledger entries and call the remaining support X_0. By B2(d), X_0 has no high-degree center, no decorated Type B handoff whose center belongs to X, and no external decorated handoff left uncovered by the grouped envelope step. Apply Lemma 14.91 and decompose X_0 into connected admissible Type A components. If one of these components contained a saturated receiver, then Lemma 14.40 would close one of exits (1)–(3), (5), or (6), peel an exit-(4) target-defective routed load, create a decorated Type B handoff, or create the route-8 residual profile. The closed exits contradict the standing target, replacement, or support-dependence invariants, while an exit-(4) peel leaves through the target-defect branch already excluded from the post-ledger core. The decorated handoff is excluded by B2(d), including the clause that every external handoff must already be covered by the grouped envelope step. The route-8 residual is excluded by the hypothesis of this lemma. Hence every receiver in every component of X_0 is unsaturated, and Lemma 14.43 gives nonnegative total charge on those components.

The fan ledger entries have nonnegative augmented charge by Step 1 and B2(a), and the remaining core has nonnegative charge by the unsaturated discharging lemma. Clause B2(c) says that these entries are a disjoint refinement of $\widehat{\mathsf{Ch}}_B(X)$, so summing the pieces gives

$$\widehat{\mathsf{Ch}}_B(X) \geq 0.$$

By (B-ledger), $\mathcal{N}(X) \geq 0$, equivalently $\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|$. □

Remark 14.94. *The fan-certificate labelling bounds the degree of a certificate-marked fan by the label-packing constant* 8. *A high-degree center without such a labelling is not placed in the certificate-closed local discharging calculation; it is a Type B bridge residual center and is charged by the fan-mass estimate below. For certificate-marked fans, the charge proof closes exactly the nonpositive-deficit cases, i.e.* $D_B = c - (11-k)/4 \leq 0$. *Thus a fan with* $k \leq 7$ *becomes residual only when it has at least two cubic-closed neighbours, while an* 8*-fan becomes residual already with one cubic-closed neighbour. If that* 8*-fan neighbour were same-window supported with a non-singleton label, the non-singleton label cap would force* $k \leq 7$; *therefore any actual* $k = 8, c = 1$ *residual must be internal, mixed, or supported on two windows. Outside the positive-deficit residual of Definition 14.71, outside fan-certificate residual centers, and outside route 8, the exclusion requires the refined support ledger B2. That ledger supplies exactly the disjointness needed to sum the certificate-closed fan charges without counting any ledger item twice.*

Degree-four Type B ledger. The degree-four remainder is the no-branch of node [68] in Fig. 6, expanded as node [78] in Fig. 7. Here

$$k = d_G(h) = 4, \qquad d_G(h) - 3 = 1, \qquad D_B(\mathfrak{F}_h) = c - \frac{7}{4},$$

where c is the number of cubic-closed neighbours of the fan center h. Thus the exact local possibilities are

c	0	1	2	3	4
$D_B = c - \frac{7}{4}$	$-\frac{7}{4}$	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{5}{4}$	$\frac{9}{4}$
closed-neighbour charge $-D_B$	$\frac{7}{4}$	$\frac{3}{4}$	$-\frac{1}{4}$	$-\frac{5}{4}$	$-\frac{9}{4}$
local B1 incidence capacity c	0	1	2	3	4

For $c \geq 2$, the hybrid incidence capacity pays the exact local deficit with slack

$$c - D_B = \frac{7}{4}.$$

The table below records what a degree-four Type B support must satisfy if it is still present after all named local exits have been tested. The word *survivor* means a connected assigned Type B support with negative net charge outside route 8 which has not been routed to the fan-mass residual node.

Constraint	**Quantitative consequence when $k = 4$**	**What a survivor must have**	**What a survivor must not have**	**Route**
High-degree bookkeeping	The center supplies exactly one surplus unit: $\sigma(h) = d_G(h) - 3 = 1$. By deletion-criticality, every neighbour of h has ambient degree 3, and high-degree centers are independent.	A single assigned degree-four fan center, with cubic fan neighbours in the assigned Type B support.	An edge from h to another high-degree center, or a non-cubic fan neighbour.	[68]→[78]→[79]
Local closed-neighbour charge	$D_B = c - \frac{7}{4}$. Hence $c = 0, 1$ gives nonnegative local charge, while $c = 2, 3, 4$ gives deficits $\frac{1}{4}, \frac{5}{4}, \frac{9}{4}$, respectively.	At least two cubic-closed neighbours if it is locally positive-deficit.	A negative local fan charge with $c \leq 1$; those cases are certificate-closed.	[79]→[81]
Certificate marking	If the fan has the certificate labelling $S_h : N(h) \to \mathcal{L}$, the degree cap $k \leq 8$ is automatic and the fan is eligible for the certificate/B1 ledger. If the labelling is absent, the center is charged as a bridge residual.	The labelling S_h if it is to proceed to the local fan ledger.	An unlabeled fan center remaining in the certificate-closed local calculation.	[80]→[84]
Direct power-of-two-cycle exclusions	Any same-window single label gap in $\{2, 6\}$ gives a 4- or 8-cycle. For two same-window closed neighbours, endpoint distances $\{0, 4, 12\}$ or interlace length 8 or 16 give a power-of-two cycle. For two windows, $\lvert i-j\rvert+\lvert a-b\rvert \in \{0, 4, 12\}$ gives a 4-, 8-, or 16-cycle.	All recorded same-window and two-window labels must satisfy exactly these arithmetic exclusions.	Any violating label pair; the displayed endpoint arithmetic constructs an explicit power-of-two cycle.	[81]
Target-algebra exits	A positive-deficit fan is tested for target-defect, target-complete compression, and proper/whole-graph support dependence before the B1 ledger is used.	Target-compatible fan-window data after those exits fail.	A target-defective quotient, a target-complete smaller representative, or proper/whole-graph support dependence.	[81]

Constraint	Quantitative consequence when $k = 4$	What a survivor must have	What a survivor must not have	Route
Local B1 incidence budget	The c cubic-closed neighbours use $2c$ pairwise distinct non-h incidences. Half-credit gives total capacity c, pays $D_B = c - \frac{7}{4}$, and leaves slack $\frac{7}{4}$. Window credit is $\frac{1}{2}I_W$; the remaining non-window demand is $D_N = \max\{0, D_B - \frac{1}{2}I_W\}$.	A local hybrid B1 entry for every positive-deficit fan $c = 2, 3, 4$.	A claimed local deficit with no assigned incidences paying D_B.	[81]
Global B2 disjointness	B1 pays one fan at a time. To sum several degree-four fan entries, their candidate entries and half-incidences must be pairwise disjoint and disjoint from the ordinary deficiency reserve.	A refined disjoint Type B ledger for all degree-four fan demands in the support. If it exists, the support has $\mathcal{N}(X) \geq 0$ outside route 8.	Double-use of a vertex, incidence, or ordinary deficiency-reserve unit inside the asserted summed ledger.	[81]→[82]
B2 failure	If B1 exists locally but B2 cannot be chosen globally, the support contains a minimal Type B overlap obstruction. Every proper connected sub-obstruction has a disjoint ledger.	A minimal family of conflicting degree-four fan demands, with all proper subfamilies disjointly payable.	A nonminimal conflict, or a conflict removable by a quotient, compression, or support dependence already excluded by the target algebra.	[81]→[83]
Global-to-local reflection	The overlap support inherits contextual target safety, high-degree separation, window-stub capacity, direct-cycle-free labels, replacement obstruction, and minimal overlap.	All those inherited constraints simultaneously.	A local picture that violates any inherited constraint; such a violation exits through a power-of-two cycle, a target defect, compression, support dependence, or a smaller disjoint ledger.	[83]
Fan-mass bound	For degree-four centers, $\sum_h (d_G(h) - 3)$ is just the number of assigned centers. If $m_4(X)$ is that number, then after any route-8 non-window core is extracted the bridge residual deficit satisfies $\mathcal{N}_-(X) \leq 8m_4(X)$. Summed in the near-cubic spine, allowing one canonical use and one grouped-envelope use of a center surplus, $M_B \leq 16\sigma(G) = O(\sqrt{n}) = o(\|R\|)$.	Only sublinear total Type B bridge mass, unless the support hands a non-window core to the Type A route-8 ledger.	A linear large-budget Type B deficit outside route 8 carried solely by degree-four bridge residuals.	[84]→[85]

Definition 14.95 (Type B residual fan mass)**.** *Let $\mathcal{X}_B$ be a subcollection of the canonical admissible supports which are Type B bridge residuals in the sense of Definition 14.90, together with any*

grouped decorated Type B envelope supports produced from Type A exit-(7) handoffs. Window-center envelopes are included in the same family; by Lemma 14.22 they use the surplus $d_G(h) - 3$ of their high-degree window center, not the packed-window stub reserve. For an ordinary assigned Type B support X, let

$$H_X = \{h : \ h \text{ is a high-degree Type B fan center assigned to } X\}$$

using the assigned-center convention of Definition 14.90. For a grouped decorated envelope support $X = \mathfrak{X}^*_{\mathfrak{C}}$, put $H_X = H_{\mathfrak{C}}$ and $\omega(h) = d_G(h) - 3$ for every $h \in H_{\mathfrak{C}}$. Let

$$\mathcal{N}_-(X) = \max\{0, -\mathcal{N}(X)\}.$$

The Type B residual fan mass *of $\mathcal{X}_B$ is*

$$M_B(\mathcal{X}_B) = \sum_{X \in \mathcal{X}_B} \mathcal{N}_-(X), \qquad H_B(\mathcal{X}_B) = \bigsqcup_{X \in \mathcal{X}_B} H_X, \qquad S_B(\mathcal{X}_B) = \sum_{X \in \mathcal{X}_B} \sum_{h \in H_X} s_X(h),$$

where $s_X(h) = d_G(h) - 3$ for ordinary assigned Type B supports and $s_X(h) = \omega(h) = d_G(h) - 3$ for grouped decorated envelope supports. The symbol $\bigsqcup$ is a tagged union of center occurrences. Within the ordinary role, high-degree centers are disjoint by the canonical assignment in Definition 13.1. Within the grouped-envelope role, centers are disjoint because the core–center incidence components of Definition 14.20 partition the handoff-center set. The same vertex may occur once in each role, and this is the only allowed double occurrence. Hence each surplus unit is counted at most twice in S_B, and

$$S_B(\mathcal{X}_B) \leq 2\sigma(G).$$

Lemma 14.96 (Bounded deficit for grouped decorated envelopes)**.** *Let $X = \mathfrak{X}^*_{\mathfrak{C}}$ be a grouped decorated Type B envelope support with center set $H_{\mathfrak{C}}$. Assume that the non-window core left after deleting the grouped fan envelopes of $\mathfrak{C}$ contains no admissible route-8 Type A residual profile. Then*

$$\mathcal{N}_-(X) \leq 8 \sum_{h \in H_{\mathfrak{C}}} (d_G(h) - 3).$$

Proof. All Type A cores and all centers in the incidence component $\mathfrak{C}$ are grouped before the Type B ledger is evaluated. Thus each center $h \in H_{\mathfrak{C}}$ contributes its negative center term once in the grouped envelope, and the fan neighbours and half-incidences are taken in the same component ledger. The local calculation at a fixed center is the same as the one used for an ordinary Type B bridge center: with $k = d_G(h)$ and $c \leq k$ cubic-closed neighbours, the negative part contributed by that center envelope is at most

$$\left(k - 3 + \frac{1}{4}\right) + \frac{c}{4} \leq \frac{5}{4}k - \frac{11}{4} \leq 8(k-3),$$

the last inequality being equivalent to $27k \geq 85$. Summing this estimate over $h \in H_{\mathfrak{C}}$ gives the displayed center-token bound. Grouping by incidence components prevents either a core with several handoff centers or a center serving several cores from being counted more than once inside the envelope family.

After the grouped fan envelopes of $\mathfrak{C}$ are removed, no handoff incident with a core of $\mathcal{Y}_{\mathfrak{C}}$ remains uncovered; any such handoff would be an edge of the same incidence component. By the hypothesis, the remainder has no route-8 Type A residual profile. Lemma 14.91 decomposes the remainder into admissible Type A components, and Lemmas 14.40 and 14.43 gives nonnegative charge on those components. Therefore only the displayed grouped fan-envelope negative part can remain unpaid, giving the asserted bound. ∎

Lemma 14.97 (Grouped decorated envelope bound with route-8 cores extracted)**.** *Let $X = \mathfrak{X}^*_{\mathfrak{C}}$ be a grouped decorated Type B envelope support with center set $H_{\mathfrak{C}}$. After deleting the grouped fan envelopes of $\mathfrak{C}$, let $\mathcal{A}_X$ be the canonical collection of Type A route-8 residual supports contained in the remaining non-window core. Then*

$$\mathcal{N}(X) \geq -D_A(\mathcal{A}_X) - 8 \sum_{h \in H_{\mathfrak{C}}} (d_G(h) - 3).$$

Proof. The center-envelope estimate in the proof of Lemma 14.96 gives total unpaid center-envelope contribution at least

$$-8 \sum_{h \in H_{\mathfrak{C}}} (d_G(h) - 3).$$

After the grouped fan envelopes of $\mathfrak{C}$ are removed, no decorated handoff incident with a core of the incidence component remains uncovered. Lemma 14.91 decomposes the remaining non-window core into admissible Type A components. The components in $\mathcal{A}_X$ are exactly those carrying route-8 residual profiles. Every other component has no route-8 residual profile and no decorated Type B handoff, so Lemma 14.45 gives nonnegative net charge on it.

For $Y \in \mathcal{A}_X$, one has $\sigma(Y) = 0$, and hence

$$\mathcal{N}(Y) = \partial_+(Y) - \frac{1}{4}|V(Y)| = -\left(\frac{1}{4}|V(Y)| - \partial_+(Y)\right).$$

Summing over $Y \in \mathcal{A}_X$ gives the route-8 core contribution $-D_A(\mathcal{A}_X)$. Combining this with the center-envelope estimate gives the displayed inequality. □

Lemma 14.98 (Bounded deficit per Type B bridge center)**.** *Let X be a connected assigned Type B bridge residual. Assume that the non-window core left after deleting the Type B fan envelopes contains no admissible route-8 Type A residual profile. Then*

$$\mathcal{N}_-(X) \leq 8 \sum_{h \in H_X} (d_G(h) - 3).$$

Proof. The estimate uses the augmented ledger of Definition 14.65. It does not assume that the B2 disjoint-incidence ledger exists. It bounds the unpaid negative part left by a failure of B2.

Fix a center $h \in H_X$, and write

$$k = d_G(h), \qquad t_h = k - 3 \geq 1.$$

Let c be the number of cubic-closed neighbours in the fan envelope at h; if h is not certificate-marked, c is still the number of fan neighbours whose two non-h incidences are assigned to the envelope. In all cases $c \leq k$. Let E_h be the fan envelope consisting of h, its assigned fan neighbours in Y_X, and the non-h incidences of the cubic-closed neighbours that appear in the local B1 candidate entry when such an entry exists. In the augmented ledger the center contributes

$$-(k-3) - \frac{1}{4}.$$

Each cubic-closed neighbour contributes $-\frac{1}{4}$. Every other fan neighbour has internal degree at most 2 in the assigned fan envelope and contributes at least $\frac{3}{4}$, and every incidence in the B1 entry carries nonnegative capacity. Therefore the negative part of E_h is bounded by

$$\left(k - 3 + \frac{1}{4}\right) + \frac{c}{4} \leq \left(k - 3 + \frac{1}{4}\right) + \frac{k}{4} = \frac{5}{4}k - \frac{11}{4} \leq 8(k-3) = 8t_h. \tag{1}$$

The last inequality is equivalent to $27k \geq 85$, and holds for every $k \geq 4$.

If h is not certificate-marked, there is no local certificate-closed entry to choose; by definition it is already a Type B bridge residual center. The bound (1) is the only estimate used for that center. If h is certificate-marked and certificate-closed, the closed-neighbourhood calculation (Lemmas 14.66 and 14.93) supplies a nonnegative candidate entry. If h is certificate-marked and positive-deficit, Lemma 14.79 supplies a local hybrid incidence entry whose capacity is at least $D_B(\mathfrak{F}_h)$. Thus every center is either bounded directly by (1), or has a local candidate envelope E_h whose unpaid negative part is bounded by (1). A bridge residual is a certificate failure or a failure to choose the local candidate entries with disjoint ledger supports after the support assignment is maximal. Omitting a contested ledger item deletes only one of the nonnegative support capacities listed in Definition 14.83; the negative terms that remain unpaid are contained in the corresponding envelopes E_h. Hence, after all overlaps are ignored rather than resolved,

$$\widehat{\mathsf{Ch}}_B(X) \geq -8 \sum_{h \in H_X} (d_G(h) - 3). \tag{2}$$

After the fan-envelope terms are removed, the residual core has no high-degree center, no decorated Type B handoff whose center belongs to X, and no external decorated handoff left uncovered by the grouped envelope step, by the maximality clause in Definition 14.90. By the hypothesis of the lemma it also has no route-8 Type A residual profile. Lemma 14.91 decomposes it into admissible Type A components. Therefore the Type A saturated handoff Lemma 14.40 and the unsaturated discharge Lemma 14.43 give nonnegative net charge on each component. Combining this nonnegative core contribution with (2) and the identity (B-ledger) gives

$$\mathcal{N}(X) \geq -8 \sum_{h \in H_X} (d_G(h) - 3) + \frac{1}{4}|H_X|.$$

Therefore

$$\mathcal{N}_-(X) \leq 8 \sum_{h \in H_X} (d_G(h) - 3),$$

as claimed. ☐

Lemma 14.99 (Type B bridge bound with route-8 cores extracted)**.** *Let X be a connected assigned Type B bridge residual. After deleting the Type B fan envelopes, let $\mathcal{A}_X$ be the canonical collection of Type A route-8 residual supports contained in the remaining non-window core. Then*

$$\mathcal{N}(X) \geq -D_A(\mathcal{A}_X) - 8 \sum_{h \in H_X} (d_G(h) - 3).$$

Proof. Fix a high-degree center $h \in H_X$ and put

$$k = d_G(h), \qquad t_h = k - 3.$$

Then $k \geq 4$ and $t_h \geq 1$. Let c_h be the number of cubic-closed fan neighbours whose two non-h incidences are assigned to the fan envelope at h. Since all such neighbours are neighbours of h, one has $0 \leq c_h \leq k$. In the augmented Type B ledger the center contributes negative charge

$$-(k-3) - \frac{1}{4},$$

and the c_h cubic-closed neighbours contribute total negative charge $-c_h/4$. All other fan neighbours have internal degree at most 2 in the assigned fan envelope and contribute at least $3/4$, while the B1 incidence ledger items have nonnegative capacity. Therefore the negative part of the fan envelope assigned to h is at most

$$\left(k-3+\frac{1}{4}\right)+\frac{c_h}{4}\le\left(k-3+\frac{1}{4}\right)+\frac{k}{4}=\frac{5}{4}k-\frac{11}{4}.$$

For $k\ge 4$,

$$\frac{5}{4}k-\frac{11}{4}\le 8(k-3),$$

because this inequality is equivalent to $27k\ge 85$. Summing over $h\in H_X$ gives the fan-envelope bound

$$\widehat{\mathsf{Ch}}_B(X)\ge -8\sum_{h\in H_X}(d_G(h)-3)$$

before the remaining non-window core is discharged.

The maximality clause in Definition 14.90 leaves the remaining core with no high-degree center, no decorated Type B handoff whose center belongs to X, and no external decorated handoff left uncovered by the grouped envelope step. By Lemma 14.91, decompose the core into admissible Type A components. The pieces in $\mathcal{A}_X$ are exactly the components carrying route-8 residual profiles. Every other Type A component has no route-8 residual and no decorated Type B handoff, so Lemma 14.45 gives nonnegative net charge on it.

For a piece $Y\in\mathcal{A}_X$, $\sigma(Y)=0$, hence

$$\mathcal{N}(Y)=\partial_+(Y)-\frac{1}{4}|V(Y)|=-\left(\frac{1}{4}|V(Y)|-\partial_+(Y)\right).$$

Summing over $Y\in\mathcal{A}_X$ gives total core contribution $-D_A(\mathcal{A}_X)$. Combining this core contribution with the fan-envelope bound and the identity (B-ledger) gives the displayed inequality. □

Proposition 14.100 (Type B bridge residuals are sublinear)**.** *Assume Definition 4.1. Let $\mathcal{X}_B$ be any canonical subcollection of Type B bridge residual supports whose non-window cores contain no admissible route-8 Type A residual profile. Then*

$$M_B(\mathcal{X}_B)\le 8\,S_B(\mathcal{X}_B)\le 16\,\sigma(G)=O(\sqrt{n})=o(|R|).$$

Proof. For an ordinary assigned Type B bridge residual, use Lemma 14.98. For a grouped decorated envelope residual, use Lemma 14.96. Summing these estimates over $X\in\mathcal{X}_B$ gives $M_B(\mathcal{X}_B)\le 8S_B(\mathcal{X}_B)$. The at-most-twice convention in Definition 14.95 gives

$$S_B(\mathcal{X}_B)\le 2\sigma(G)=2\sum_{v\in V(G)}(d_G(v)-3).$$

The near-cubic spine hypothesis gives $\sigma(G)=O(\sqrt{n})$ (Definition 4.1), while Proposition 7.42 gives $|R|\ge 0.83489768623\ldots n-o(n)$. Hence $\sigma(G)=o(|R|)$, and the displayed estimate follows. □

Framework branch table. The Type B fan-safe surplus obstruction is discharged by the following framework branch table.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure

Fan-compatible pair	An open pair closes two ports at a high center.	Positive-deficit fan data.	The Type B local ledger.	One fan center.	Corollary 14.81 and Proposition 14.80.
Triangular ports	The heavy center has triangular alternatives.	Fan-closed ports.	The Type B ledger.	Triangular shoulder cases.	Corollary 14.82.
Certificate-closed fan	The local fan ledger has nonpositive deficit.	Local deficit.	The fan-certificate calculation.	One marked fan.	Lemmas 14.66 and 14.93.
Positive-deficit fan	The certificate-closed calculation does not pay the fan.	Hybrid incidence demand.	B1 half-credit incidences.	Window and non-window incidences.	Lemmas 14.78 and 14.79.
B2 failure	The disjoint ledger cannot be selected globally.	A minimal overlap obstruction.	The fan-mass ledger.	Assigned surplus units.	Definition 14.85 and Proposition 14.89.
Bridge residual	Non-window cores carry no route-8 profile.	Bridge mass.	Global surplus.	At most twice assigned surplus.	Proposition 14.100.

15 The branch-closure theorem

Theorem 15.1 (Entropy-admissible closure outside explicit residuals)**.** *Assume Definition 4.1. The two local obstruction classes outside the explicit residuals isolated in Section 14 are empty:*

(a) ***Type A exclusion*** *(Lemma 14.45). Every admissible subcubic P_{13}-free target-safe boundaried piece C with $\sigma(C) = 0$ and empty internal 3-core, no exit-(4) witness for a routed load, no admissible route-8 residual profile, and no decorated Type B handoff satisfies*

$$\partial_+(C) \geq \frac{1}{4}|V(C)|.$$

(b) ***Type B bridge reduction*** *(Proposition 14.92, with Lemma 14.93 as the certificate-closed specialization). Every connected admissible high-degree fan support X outside the Type B bridge residual and containing no admissible route-8 residual profile satisfies*

$$\partial_+(X) - \sigma(X) \geq \frac{1}{4}|V(X)|.$$

Then every large-budget residual outside the target-defect exit, outside the Type A route-8 residual branch, and outside the Type B bridge residual branch is closed.

Proof. If the high-entropy remaining budget is small, Proposition 12.2 closes that high-entropy branch. In every branch that survives to the large-budget analysis, including the low-entropy branches of Proposition 11.6, the near-cubic window-only cap gives

$$\frac{\partial_+(R) - \sigma_R}{|R|} = \frac{\partial_+(R) - \sigma(R)}{|R|} \leq \tau_{\text{win}} + o(1).$$

Since $\tau_{\mathrm{win}} < 1/4$, the local lower bound

$$\partial_+(R) - \sigma(R) \geq \frac{1}{4}|R| - o(|R|)$$

would contradict the cap. More explicitly, for all sufficiently large n,

$$\partial_+(R) - \sigma(R) \leq \left(\frac{1}{4} - \varepsilon\right)|R|, \qquad \varepsilon = \frac{1}{2}\left(\frac{1}{4} - \tau_{\mathrm{win}}\right) > 0.$$

Thus the hypothesis of Proposition 13.7 holds, and if the branch survives there is a connected admissible support X with $\mathcal{N}(X) < 0$.

By the Type A/Type B analysis, such an X is either a low-deficiency boundaried piece or a high-degree fan-safe surplus support. If X is Type A and contains an exit-(4) witness for a routed load, then it leaves the pure Type A net-charge calculation through the target-defect exit of Definition 14.33 and Lemma 14.38; this exit is excluded in the present branch. If X is Type A and contains the route-8 residual profile, then it lies in the explicit Type A route-8 residual branch, which is excluded by the hypothesis of this theorem. If X is Type A and produces a decorated Type B handoff, the handoff is transferred to the grouped decorated Type B envelope ledger by Lemma 14.21; if the handoff center lies in a packed window, Lemma 14.22 charges it to the center's own surplus token or gives exit (3). Outside the target-defect exit, the Type A route-8 branch, and the Type B bridge residual, Lemma 14.45 excludes the pure Type A case. In the Type B case, the local B1 ledger is Lemma 14.79, and being outside the bridge residual means that B2 holds. Hence Proposition 14.92 gives $\mathcal{N}(X) \geq 0$; in the certificate-closed specialization this is exactly Lemma 14.93. This contradicts $\mathcal{N}(X) < 0$. Hence every large-budget residual outside the target-defect exit and the explicit residual branches is closed. □

Framework branch table. The branch-closure theorem uses the following branch table.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
High entropy	The admissible entropy cap is active.	Window, remainder, and obstruction bits.	The skeleton budget.	One coordinate family.	Proposition 12.2.
Negative support	The large-budget branch survives.	Negative net charge.	Local support analysis.	The canonical support.	Proposition 13.7.
Pure Type A	The support avoids exit (4), route 8, and handoff.	Low-deficiency piece charge.	Type A discharging.	The receiver partition.	Lemma 14.45.
Type B	The support avoids the bridge residual.	Fan-safe surplus support charge.	B1 and B2 ledger entries.	Assigned fan centers.	Proposition 14.92.

Explicit residuals	Target defect, Type A route 8, or Type B bridge mass remains.	Residual mass.	Demand and bridge ledgers.	Global accounting.	Routed to Theorem 15.48 and Proposition 14.100.

15.1 External demand accounting for the target-defect residual

Throughout assume Definition 4.1. The purpose of this section is to account, in the large-budget net-charge ledger, for the supports that leave the pure Type A branch through the target-defect exit of Definition 14.33. Such a support has strictly negative net charge (a peeled routed load is a cubic vertex still carrying $-\frac{1}{4}$ in $\sum_v \mathrm{ch}(v)$), so it is neither a nonnegative support nor, in general, a route-8 residual nor a Type B bridge residual. Any partition that counts only those three classes omits the target-defect negative mass. The large-budget ledger therefore sums over a single collection that contains *every* negative Type A support, and the response-support reduction is applied to that unified collection.

15.1.1 The unified negative Type A collection

Definition 15.2 (Unified negative collection)**.** *Let R be the P_{13}-free remainder of a minimal counterexample satisfying Definition 4.1, decomposed into the canonical connected admissible supports of Definition 13.1. Let*

$$\tilde{\mathcal{X}} = \{\, X : \ \sigma(X) = 0,\ \mathcal{N}(X) < 0,\ X \text{ produces no decorated Type B handoff}\}.$$

For $X \in \tilde{\mathcal{X}}$ write $\delta(X) = -\mathcal{N}(X) = \frac{1}{4}|V(X)| - \partial_+(X) > 0$ and set

$$\tilde{D}_A = \sum_{X\in\tilde{\mathcal{X}}} \delta(X).$$

Remark 15.3. *$\tilde{\mathcal{X}}$ contains, simultaneously, the Type A supports carrying an admissible route-8 residual profile* and *the Type A supports that leave through the target-defect exit; the two are not distinguished at the level of net charge. A support that hands off to Type B is excluded from $\tilde{\mathcal{X}}$ and is treated in the Type B bridge ledger of Definition 14.95.*

Lemma 15.4 (The unified collection carries the whole large-budget deficit)**.** *In the large-budget branch,*

$$\tilde{D}_A \ \geq\ \left(\tfrac{1}{4} - \tau_{\mathrm{win}}\right)|R| - o(|R|).$$

Proof. By Lemma 13.4,

$$\sum_i \mathcal{N}(X_i) = \partial_+(R) - \sigma(R) - \tfrac{1}{4}|R| \leq -\left(\tfrac{1}{4} - \tau_{\mathrm{win}}\right)|R| + o(|R|),$$

the inequality being the large-budget net-deficiency cap obtained from Proposition 7.42 and Lemma 8.5. Split the canonical supports into three groups: the members of $\tilde{\mathcal{X}}$; the Type B bridge residual ledger entries $\mathcal{X}_B$; and all remaining supports. A remaining support has $\sigma(X) = 0$ and $\mathcal{N}(X) \geq 0$, or is a nonnegative Type B support; in either case it contributes nonnegatively. Hence

$$\sum_i \mathcal{N}(X_i) \ \geq\ -\tilde{D}_A - \sum_{X\in\mathcal{X}_B} \mathcal{N}_-(X) \ \geq\ -\tilde{D}_A - 8S_B(\mathcal{X}_B) - o(|R|) \ \geq\ -\tilde{D}_A - o(|R|),$$

using $\mathcal{N}(X) = -\delta(X)$ on $\tilde{\mathcal{X}}$, Proposition 14.100 for the middle inequality, and the at-most-twice bound $S_B(\mathcal{X}_B) \leq 2\sigma(G) = o(|R|)$ of Definition 14.95 for the last. Combining the two displays gives the claim. $\square$

15.1.2 Every unit of deficit is a silent-excess entry with a nonempty response-support core

The next lemma is the point of the reformulation: it shows that the deficit of *every* $X \in \tilde{\mathcal{X}}$ – route-8 or target-defect – is carried by silent-excess routed loads, each of which possesses a trace basin, and hence a canonical essential-incidence core. It uses only counting and the target-safety of admissible supports; it does not use target-complete-minimality.

Definition 15.5 (Unified indexed entries)**.** *For $X \in \tilde{\mathcal{X}}$ and a saturated receiver w of X, let $\mathcal{U}_X(w)$ be its silent unpaid routed loads (Definition 14.29). For $u \in \mathcal{U}_X(w)$ let B_u be its trace basin (Definition 14.14). The set of* unified indexed entries *is*

$$\tilde{\Xi} = \{\, \xi = (X, w, u, B_u) : \ X \in \tilde{\mathcal{X}}, \ u \in \mathcal{U}_X(w) \,\}, \qquad \tilde{N} = |\tilde{\Xi}|.$$

An entry is a route-8 entry *if B_u is target-complete-minimal (Definition 14.14); otherwise it is a* target-defect entry*, and one of the four failure alternatives of Definition 14.14 holds for B_u.*

Lemma 15.6 (Unified burden)**.**

$$\tilde{N} \ \geq \ 4\tilde{D}_A.$$

Proof. Fix $X \in \tilde{\mathcal{X}}$. Because X produces no Type B handoff, no saturated receiver of X realizes exit (7); because X lies in a minimal counterexample, none realizes exits (1)–(3), (5), or (6), since each of those is a standing-invariant contradiction (Lemma 14.39). Thus every saturated receiver of X realizes only exit (4) or exit (8); in particular no completion port carries four visible receiver-entry returns. Hence Lemma 14.30 applies and gives

$$S^{\mathrm{exc}}_{\mathrm{sil}}(X) \ \geq \ 4\Big(\tfrac{1}{4}|V(X)| - \partial_+(X)\Big) = 4\delta(X).$$

By Lemma 14.31, every vertex counted by $S^{\mathrm{exc}}_{\mathrm{sil}}(X)$ lies in some $\mathcal{U}_X(w)$ and thus indexes an entry of $\tilde{\Xi}$ (its basin is target-complete-minimal, giving a route-8 entry, or realizes one of exits (4)–(7); exits (5),(6) are contradictions and exit (7) is excluded, so the only alternative to route 8 is a target-defect entry via exit (4)). Summing over $X \in \tilde{\mathcal{X}}$,

$$\tilde{N} = \sum_{X \in \tilde{\mathcal{X}}} \sum_{w} |\mathcal{U}_X(w)| \ \geq \ \sum_{X \in \tilde{\mathcal{X}}} S^{\mathrm{exc}}_{\mathrm{sil}}(X) \ \geq \ 4 \sum_{X \in \tilde{\mathcal{X}}} \delta(X) = 4\tilde{D}_A. \qquad \square$$

Lemma 15.7 (Minimality-free support lower bound)**.** *Every entry $\xi = (X, w, u, B_u) \in \tilde{\Xi}$, route-8 or target-defect, has*

$$\alpha_{\mathcal{X}}(\xi) = |\mathcal{C}_{\mathrm{ess}}(\xi)| \ \geq \ 2.$$

Proof. For a route-8 entry this is Lemma 14.58. Let ξ be a target-defect entry and put $\mathcal{C} = \mathcal{C}_{\mathrm{ess}}(\xi)$. Suppose for contradiction that $|\mathcal{C}| \leq 1$.

Because ξ is a target-defect entry, one of the four alternatives of Definition 14.14 holds for B_u. Alternatives (b)–(d) are exits (5),(6),(7): the first two are standing-invariant contradictions (Corollary 3.98 and Lemmas 10.6 and 10.9) and the third is excluded because $X \in \tilde{\mathcal{X}}$ produces no handoff. Hence alternative (a) holds: a trace-local quotient q_0 of $\rho_u(B_u)$ is target-defective, i.e. distinguished by a compatible outside context. By definition of a trace-local quotient (Definition 14.14), q_0 preserves the boundary degree profile and forgets an internal coordinate whose declared support contains an internal edge of B_u.

Apply Lemma 14.57 to ξ with the boundary-incidence set $\mathcal{C}$ and the quotient $\rho^{\circ}_{\mathcal{C}}$ that forgets exactly the u-supported coordinates whose declared support contains an internal edge of B_u. Since

q_0 is target-defective and is one such forgetting quotient, the core-to-ρ° quotient $\rho_u(B_u)|_{\mathcal{C}} \to \rho^\circ_{\mathcal{C}}$ is not target-complete. The lemma then supplies a surviving u-supported target event realized by a simple edge-rooted return or simple cycle that uses an internal edge of B_u and an edge outside X. Add the root edge if the event is edge-rooted. The result is a simple cycle which uses an edge of X and an edge outside X. It therefore crosses the cut $\delta_G(V(X))$ a positive even number of times. Because the cycle is simple, two distinct cut edges occur. The event survives in the response-support-restricted state, so both cut crossings are recorded by declared coordinates retained by $\mathcal{C}$. Thus its declared support contains at least two distinct boundary incidences of $\mathcal{C}$, so $|\mathcal{C}| \geq 2$, contradicting $|\mathcal{C}| \leq 1$. Therefore $\alpha_{\mathcal{X}}(\xi) \geq 2$. □

15.1.3 Unified response-support reduction

Proposition 15.8 (Unified two-support reduction)**.** *In the large-budget branch, $\tilde{\Xi}$ contains a two-support entry, that is, an entry ξ with $\pi_{\mathcal{X}}(\xi) \leq 2$.*

Proof. Suppose not; then every $\xi \in \tilde{\Xi}$ has $\pi_{\mathcal{X}}(\xi) \geq 3$. By Lemma 15.7 the essential-incidence cores are well defined and, by definition of privacy (Definition 14.48), private essential incidences of distinct entries are disjoint oriented boundary incidences of their supports. Hence

$$3\tilde{N} \;\leq\; \sum_{X \in \tilde{\mathcal{X}}} |\partial_E X| \;=\; \sum_{X \in \tilde{\mathcal{X}}} \partial_+(X) \;\leq\; \partial_+(R) \;\leq\; \tau_{\mathrm{win}}|R| + o(|R|),$$

using $|\partial_E X| = \partial_+(X)$ for ambient-cubic supports and the near-cubic window-only cap $\partial_+(R) \leq \tau_{\mathrm{win}}|R| + o(|R|)$ of Corollary 8.8. On the other hand, Lemmas 15.4 and 15.6 give

$$\tilde{N} \;\geq\; 4\tilde{D}_A \;\geq\; 4\Big(\tfrac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|).$$

Combining,

$$12\Big(\tfrac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|) \;\leq\; \tau_{\mathrm{win}}|R| + o(|R|),$$

which forces $12(\frac{1}{4} - \tau_{\mathrm{win}}) \leq \tau_{\mathrm{win}}$, i.e. $\tau_{\mathrm{win}} \geq \frac{3}{13}$. This contradicts $\tau_{\mathrm{win}} = 0.22817\ldots < 0.23076\ldots = \frac{3}{13}$ (Remark B.1). Hence a two-support entry exists. □

Remark 15.9 (The margin is unchanged)**.** *Proposition 15.8 spends the boundary-incidence budget $\partial_+(R) \leq \tau_{\mathrm{win}}|R|$ exactly once, at three incidences per entry, and draws no additional boundary or window-stub capacity. The tight inequality is still $\tau_{\mathrm{win}} < \frac{3}{13}$ with the 1.12% relative margin of Remark B.1; no new capacity is consumed, so that margin is preserved verbatim.*

15.1.4 External demand accounting for target-defect entries

The route-8 two-support case is already closed in the manuscript: if B_u is target-complete-minimal and $\pi_{\mathcal{X}}(\xi) \leq 2$, the declared deletion witnesses make $(\tilde{\mathcal{X}}, \xi)$ a terminal two-support route-8 obstruction, which is excluded by Theorem 14.61. The new accounting below treats only the other two-support outcome, namely an entry whose trace basin fails target-complete-minimality by an exit-(4) target-defective quotient.

Definition 15.10 (Exit-(4) demand token)**.** *An* exit-(4) demand token *is a tuple*

$$\mathfrak{t} = (\xi, q, S_0, S_1, Y, E), \qquad \xi = (X, w, u, B_u) \in \tilde{\Xi},$$

with the following data.

(P1) *ξ is a target-defect entry: B_u is not target-complete-minimal, and the surviving failure alternative is alternative* (a) *of Definition 14.14.*

(P2) *q is the corresponding trace-local quotient. It preserves the boundary degree profile of $\rho_u(B_u)$, forgets at least one declared u-supported coordinate whose support contains an internal edge of B_u, and is distinguished by the compatible outside context Y.*

(P3) *S_0 and S_1 are two realizations in the same boundary-degree fibre with $q(S_0) = q(S_1)$, but*

$$S_0 \oplus Y \quad \text{and} \quad S_1 \oplus Y$$

have different target predicates.

(P4) *E is the lexicographically first simple edge-rooted return or simple cycle which witnesses this target difference. Its declared support is the set of declared coordinates of Definition 3.87 used by E.*

The boundary-incidence set *of* $\mathfrak{t}$ *is*

$$K(\mathfrak{t}) = \{\, c \in \partial_E X : \ c \text{ occurs in the declared support of } E \,\}.$$

The token is cheap *if $|K(\mathfrak{t})| = 2$. It is* actual-context *if the context Y is the actual exterior of X in G, with the recorded boundary degree profile. Otherwise it is* profile-only.

Lemma 15.11 (Demand tokens have at least two response supports)**.** *Every exit-(4) demand token $\mathfrak{t}$ satisfies*

$$|K(\mathfrak{t})| \geq 2.$$

Proof. The quotient q forgets a u-supported coordinate whose declared support contains an internal edge of B_u, and the witnessing event E uses one of the forgotten coordinates. Hence E uses an internal edge of B_u. If E were contained in X, adding the root edge in the edge-rooted case would give a power-of-two cycle contained in the contextually target-safe support X, impossible in the counterexample. Thus E also uses an edge outside X.

Add the root edge if E is edge-rooted. We obtain a simple cycle of G which meets both sides of the cut $\delta_G(V(X))$. A cycle crosses a cut an even number of times, and a simple cycle cannot use one cut edge twice. Thus there are at least two distinct boundary edges of X in E. Each such crossing is recorded in the declared support of E as a completion-port incidence, first-entry incidence, connector endpoint, boundary-degree entry, or packed-window interface incidence. Therefore the set $K(\mathfrak{t})$ of boundary incidences in the declared support has size at least two. ☐

Definition 15.12 (Exit-(4) demand ledger)**.** *The ledger is used after exits* (5)–(7) *have been removed or transferred. Thus every surviving target-defect entry fails target-complete-minimality through alternative* (a) *and has a demand token. Let $\mathcal{T}_4$ be the finite set of those tokens, obtained from the target-defect entries of $\tilde{\Xi}$ by choosing the lexicographically first witness data in Definition 15.10. A* 2/3-demand ledger *on $\tilde{\Xi}$ consists of the following partition:*

$$\tilde{\Xi} = \Xi_3 \sqcup \Xi_2 \sqcup \Xi_{\mathrm{res}},$$

together with an assignment $A(\xi) \subseteq \partial_E X$ for every $\xi = (X, w, u, B_u) \in \Xi_3 \cup \Xi_2$, subject to:

(L1) *if ξ is a route-8 entry, then either the branch is already closed by Theorem 14.61, or $\xi \in \Xi_3$ and $A(\xi)$ consists of three private essential incidences of ξ. Thus no surviving route-8 entry is placed in Ξ_2 or Ξ_{res};*

(L2) if ξ is a target-defect entry and its token is t_ξ, then $A(\xi) \subseteq K(\mathsf{t}_\xi)$;

(L3) $|A(\xi)| = 3$ for $\xi \in \Xi_3$, and $|A(\xi)| = 2$ for $\xi \in \Xi_2$;

(L4) the incidence sets $A(\xi)$ are pairwise disjoint over $\Xi_3 \cup \Xi_2$;

(L5) Ξ_{res} contains exactly the entries for which no such assignment was made.

Among all such ledgers, choose once and for all the lexicographically first ledger maximizing first $|\Xi_3|$, then $|\Xi_2|$. Define

$$N_3 = |\Xi_3|, \qquad N_2 = |\Xi_2|, \qquad N_{\mathrm{res}} = |\Xi_{\mathrm{res}}|.$$

The external-demand defect *of the ledger is*

$$\mathsf{P}_{\mathrm{ext}} = N_2 + 3N_{\mathrm{res}}.$$

Lemma 15.13 (Demand ledger does not overcount)**.** *For the demand ledger of Definition 15.12,*

$$3N_3 + 2N_2 \le \sum_{X \in \tilde{\mathcal{X}}} |\partial_E X| = \sum_{X \in \tilde{\mathcal{X}}} \partial_+(X) \le \partial_+(R).$$

Equivalently,

$$3\tilde{N} - \mathsf{P}_{\mathrm{ext}} \le \partial_+(R).$$

Proof. Every assigned incidence is an oriented boundary incidence of the Type A support which contains the corresponding entry. Clause (L4) says that no assigned incidence is used by two entries. Hence the total number of assigned incidences is at most $\sum_{X \in \tilde{\mathcal{X}}} |\partial_E X|$, while clauses (L3) and (L4) make this total exactly $3N_3 + 2N_2$.

For $X \in \tilde{\mathcal{X}}$ one has $\sigma(X) = 0$; hence every vertex of X has ambient degree 3, and the identity in Definition 14.48 gives $|\partial_E X| = \partial_+(X)$. The canonical supports are disjoint components of the remainder decomposition, so their positive deficiencies add into $\partial_+(R)$, with all omitted terms nonnegative. This proves the first display. Since

$$\tilde{N} = N_3 + N_2 + N_{\mathrm{res}},$$

we have

$$3\tilde{N} - \mathsf{P}_{\mathrm{ext}} = 3(N_3 + N_2 + N_{\mathrm{res}}) - (N_2 + 3N_{\mathrm{res}}) = 3N_3 + 2N_2,$$

which gives the equivalent form. ☐

Definition 15.14 (Two-terminal demand records)**.** *Let $\xi \in \Xi_2 \cup \Xi_{\mathrm{res}}$ be a target-defect entry, and let t_ξ be its token. If t_ξ is actual-context, choose the lexicographically first ordered pair (c_1, c_2) of distinct boundary incidences in $K(\mathsf{t}_\xi)$ that occur as consecutive crossings of $\partial_E X$ along the witnessing event E. The corresponding* actual outside corridor *is the subpath of E between those crossings whose internal vertices lie outside X. The record*

$$\mathsf{a}(\xi) = (X, w, u, B_u, c_1, c_2, E, \textit{outside corridor})$$

is the actual two-terminal demand record *of ξ.*

If t_ξ is profile-only, the two realizations S_0, S_1 differ in at least one declared coordinate forgotten by q. Let $r(\xi)$ be the lexicographically first such coordinate. The record

$$\mathsf{p}(\xi) = (X, w, u, B_u, q, r(\xi), S_0, S_1, Y)$$

is the profile demand record *of ξ.*

Lemma 15.15 (Demand records are canonical)**.** *Every target-defect entry in* $\Xi_2 \cup \Xi_{\mathrm{res}}$ *has exactly one of the two records in Definition 15.14. Actual records and profile records are disjoint classes of data.*

Proof. The token is either actual-context or profile-only by definition, and not both. In the actual-context case, the mixed event E uses an edge outside X. A simple cycle crosses the cut $\delta_G(V(X))$ an even positive number of times; hence there is at least one pair of consecutive boundary crossings bounding an outside subpath. The imposed lexicographic order chooses one such pair and one outside corridor.

In the profile-only case, S_0 and S_1 have the same image under q but different target predicates after gluing to Y. They cannot be the same exact response state. Since q preserves the boundary degree profile, their difference is recorded by at least one declared coordinate forgotten by q. The lexicographic rule chooses $r(\xi)$. Thus exactly one record is assigned in either case, and the record type is determined by whether the context is actual or profile-only. $\square$

Proposition 15.16 (External-demand reduction)**.** *After the route-8 two-support exclusion has been applied, every surviving unified entry is accounted for in exactly one of the following ways:*

class	*charge used now*	*remaining data*
Ξ_3	3 *disjoint boundary incidences*	*none*
Ξ_2	2 *disjoint boundary incidences*	*actual or profile demand record*
Ξ_{res}	0 *assigned incidences*	*actual or profile demand record*

No boundary incidence is counted twice in the first two rows, and no entry is discarded.

Proof. The route-8 case with at most two private essential incidences is closed by Theorem 14.61; otherwise the ledger definition puts the route-8 entry in Ξ_3 with three private essential incidences. A target-defect entry has the demand token of Definition 15.10, and Lemma 15.11 gives at least two available boundary incidences for that token. The demand ledger chooses the maximal disjoint assignment into Ξ_3 and Ξ_2, leaving the unassigned entries in Ξ_{res}. Clause (L4) gives the no-overcount assertion for assigned boundary incidences. Finally, Lemma 15.15 attaches an actual or profile demand record to every target-defect entry not paid at the three-incidence level. Since surviving route-8 entries never lie in $\Xi_2 \cup \Xi_{\mathrm{res}}$, the displayed table is exactly this partition. $\square$

15.1.5 Demand-defect absorption ledger

Definition 15.17 (Actual and profile demand defects)**.** *For* $\xi \in \Xi_2 \cup \Xi_{\mathrm{res}}$*, define its demand weight by*

$$\omega(\xi) = \begin{cases} 1, & \xi \in \Xi_2, \\ 3, & \xi \in \Xi_{\mathrm{res}}. \end{cases}$$

Let $\Xi_{\mathrm{act}} \subseteq \Xi_2 \cup \Xi_{\mathrm{res}}$ *be the entries whose canonical record in Definition 15.14 is actual-context, and let* Ξ_{prof} *be the entries whose canonical record is profile-only. Put*

$$\mathsf{P}_{\mathrm{act}} = \sum_{\xi \in \Xi_{\mathrm{act}}} \omega(\xi), \qquad \mathsf{P}_{\mathrm{prof}} = \sum_{\xi \in \Xi_{\mathrm{prof}}} \omega(\xi).$$

Lemma 15.18 (Exact demand-defect split)**.** *The demand defect splits without overlap:*

$$\mathsf{P}_{\mathrm{ext}} = \mathsf{P}_{\mathrm{act}} + \mathsf{P}_{\mathrm{prof}}.$$

Proof. By Lemma 15.15, each target-defect entry in $\Xi_2 \cup \Xi_{\text{res}}$ has exactly one canonical record, and that record is actual-context or profile-only, but not both. Since surviving route-8 entries do not lie in $\Xi_2 \cup \Xi_{\text{res}}$ by Definition 15.12, the sets Ξ_{act} and Ξ_{prof} form a partition of $\Xi_2 \cup \Xi_{\text{res}}$. Therefore

$$\mathsf{P}_{\text{act}} + \mathsf{P}_{\text{prof}} = \sum_{\xi \in \Xi_2 \cup \Xi_{\text{res}}} \omega(\xi) = |\Xi_2| + 3|\Xi_{\text{res}}| = N_2 + 3N_{\text{res}} = \mathsf{P}_{\text{ext}}.$$

□

Lemma 15.19 (Profile-demand dependence routing)**.** *Let*

$$\mathcal{A}_{\text{prof}} = \{\, r(\xi) : \xi \in \Xi_{\text{prof}} \,\}$$

be the family of profile demand coordinates of Definition 15.14, with repeated coordinates listed once. If a functional admissible rank quotient is rank-reducing on $\mathcal{A}_{\text{prof}}$, then one of the following occurs:

(i) a target-defective quotient;

(ii) an admissible nontrivial target-complete compression of a proper support;

(iii) a target-dependence whose validity requires adjoining a connected support strictly larger than the support carrying the dependent coordinates.

Proof. Let $\mathcal{I} \subseteq \mathcal{A}_{\text{prof}}$ be maximal among subfamilies which survive every functional admissible rank quotient, and choose $a \in \mathcal{A}_{\text{prof}} \setminus \mathcal{I}$. By Definition 3.92, the quotient that kills a's label gives a finite subfamily $\mathcal{B} \subseteq \mathcal{I}$ whose quotient-image vector determines the quotient-image of a. Choose such a pair $(a, \mathcal{B})$ with inclusion-minimal connected declared support. This is the target-dependence extracted in Lemma 10.3, but the argument uses only the definitions of exact response profile, functional quotient, and declared support.

Let C be the connected support of this minimal determination certificate. The determination is tested against T-boundaried outside contexts, where T is the boundary of C. If some compatible context distinguishes two realizations with the same quotient image, the quotient is target-defective, giving (i). If no compatible context distinguishes them and the support is already the proper support C, then the quotient is target-complete on a proper support. Since it is rank-reducing and admissible, it has a strictly smaller proper representative by Definition 3.91; this is the target-complete compression in (ii). The only remaining case is that the determination becomes context-universal only after adjoining further connected support. Minimality of the certificate gives a connected support strictly larger than C, which is (iii). □

Definition 15.20 (Demand absorbers)**.** *A* demand unit *is a pair (ξ, j), where $\xi \in \Xi_2 \cup \Xi_{\text{res}}$ and $1 \le j \le \omega(\xi)$. Let $\mathcal{U}_{\text{press}}$ be the set of all demand units, so that*

$$|\mathcal{U}_{\text{press}}| = \mathsf{P}_{\text{ext}}.$$

A demand unit is absorbed *if it is assigned one of the following single-use certificates.*

(A1) an oriented boundary incidence $c \in \partial_E X$, for the same support X as ξ, which is not already used in any set $A(\xi')$ of the 2/3-demand ledger and is not assigned to another demand unit;

(A2) a profile-dependence certificate of Lemma 15.19 whose conclusion is an admissible proper-support compression, or whose larger-support dependence is one of the support-dependence alternatives closed by the branch split. A certificate whose conclusion is another target-defective quotient is not a type (A2) *absorber; it re-enters the target-defect demand ledger and is counted there unless it is absorbed by type* (A1)*.*

Among all assignments satisfying the single-use conditions in (A1)–(A2)*, choose the lexicographically first assignment which maximizes, in order, the number of absorbed units and then the number of type* (A1) *boundary-incidence absorptions. Let $\mathcal{U}_{\mathrm{abs}} \subseteq \mathcal{U}_{\mathrm{press}}$ be its absorbed units, and put*

$$\mathsf{P}_{\mathrm{open}} = |\mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}|.$$

Lemma 15.21 (Absorbed units improve the boundary count)**.** *Let B_{abs} be the number of demand units absorbed by boundary incidences of type* (A1) *in Definition 15.20. Then*

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} \leq \partial_+(R) + B_{\mathrm{dep}},$$

where B_{dep} is the number of units absorbed by dependence certificates of type (A2)*. On any branch where the type* (A2) *dependence certificates have routed to their proper-compression or support-dependence alternatives, $B_{\mathrm{dep}} = 0$, and hence*

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} \leq \partial_+(R).$$

Proof. The original 2/3-demand ledger assigns $3N_3 + 2N_2$ boundary incidences, and Lemma 15.13 proves that these incidences are pairwise disjoint. Each type (A1) absorber is, by definition, an additional oriented boundary incidence not used by the 2/3-demand ledger and not assigned to another absorbed unit. Therefore the 2/3-ledger together with all type (A1) absorbers uses

$$3N_3 + 2N_2 + B_{\mathrm{abs}}$$

distinct boundary incidences of the supports in $\tilde{\mathcal{X}}$. Since these supports are ambient-cubic and disjoint in the remainder decomposition, each counted external incidence is incident with exactly one deficient boundary slot of R, and no deficient boundary slot is used twice. Hence the number of counted incidences is bounded by the total positive deficiency of R:

$$3N_3 + 2N_2 + B_{\mathrm{abs}} \leq \partial_+(R).$$

The total demand defect decomposes as

$$\mathsf{P}_{\mathrm{ext}} = B_{\mathrm{abs}} + B_{\mathrm{dep}} + \mathsf{P}_{\mathrm{open}}.$$

Using

$$3\tilde{N} - \mathsf{P}_{\mathrm{ext}} = 3N_3 + 2N_2$$

from Lemma 15.13, we get

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} - B_{\mathrm{dep}} = 3N_3 + 2N_2 + B_{\mathrm{abs}} \leq \partial_+(R).$$

This is the first displayed inequality. If the type (A2) absorbers route to their proper-compression or support-dependence alternatives, no surviving demand residual contains those units; equivalently the surviving residual has $B_{\mathrm{dep}} = 0$, giving the second display. $\square$

Proposition 15.22 (Exit-(4) closure from open demand)**.** *Assume Definition 4.1. On a branch where all dependence absorbers of type* (A2) *have routed to their proper-compression or support-dependence alternatives, the target-defect exit-(4) residual is impossible whenever*

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}_{\mathrm{open}}}{|R|} < \varepsilon_{\mathrm{press}}, \qquad \varepsilon_{\mathrm{press}} = 12\left(\frac{1}{4} - \tau_{\mathrm{win}}\right) - \tau_{\mathrm{win}}.$$

Proof. By Lemmas 15.4 and 15.6,

$$\tilde{N} \geq 4\left(\frac{1}{4} - \tau_{\mathrm{win}}\right)|R| - o(|R|).$$

By Lemma 15.21 and the hypothesis that the type (A2) units have routed to their proper-compression or support-dependence alternatives,

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} \leq \partial_+(R).$$

The window-only deficiency cap gives

$$\partial_+(R) \leq \tau_{\mathrm{win}}|R| + o(|R|).$$

Combining the three inequalities gives

$$\left[12\left(\frac{1}{4} - \tau_{\mathrm{win}}\right) - \frac{\mathsf{P}_{\mathrm{open}}}{|R|}\right]|R| \leq \tau_{\mathrm{win}}|R| + o(|R|).$$

The displayed limsup assumption makes the coefficient on the left eventually strictly larger than τ_{win}, a contradiction. $\square$

15.1.6 Window-blocker accounting for the open demand

Definition 15.23 (Canonical window blocker of an open unit)**.** *Let $\upsilon = (\xi, j) \in \mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}$ be an open demand unit, with $\xi = (X, w, u, B_u)$, and let $\mathfrak{t}_\xi$ be the demand token of ξ. Since X is a component of the P_{13}-free remainder R, every ambient boundary incidence of X leaves X through an edge from R to the packed-window set W. Choose the lexicographically first boundary incidence $c(\upsilon) \in K(\mathfrak{t}_\xi)$. If*

$$c(\upsilon) = (x, xy), \qquad x \in V(X), \quad y \in W,$$

let $P(\upsilon) \in \mathcal{P}$ be the unique packed induced P_{13} window containing y. The pair

$$\mathfrak{b}(\upsilon) = (P(\upsilon), c(\upsilon))$$

is the canonical window blocker *of υ. For a packed window P, put*

$$B_{\mathrm{open}}(P) = |\{\upsilon \in \mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}} : P(\upsilon) = P\}|.$$

Lemma 15.24 (Open demand is exactly the window-blocker load)**.** *The canonical window-blocker loads satisfy*

$$\mathsf{P}_{\mathrm{open}} = \sum_{P\in\mathcal{P}} B_{\mathrm{open}}(P).$$

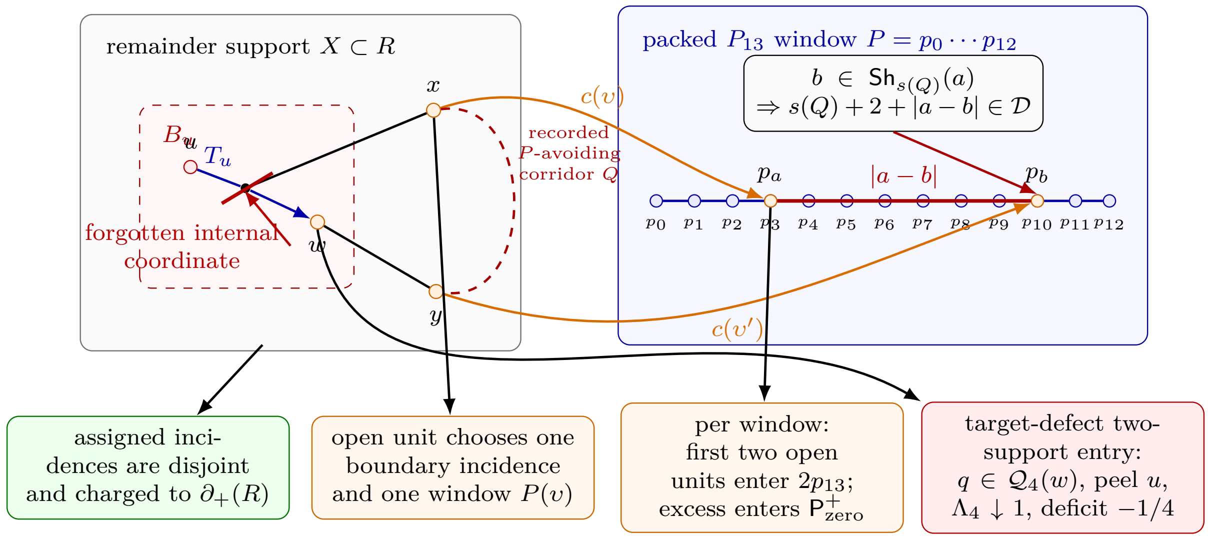

Figure 30: Exit-(4) attachment to a packed P_{13} window and the associated accounting. A target-defect demand token for the entry $\xi = (X, w, u, B_u)$ forgets a declared u-supported coordinate using an internal edge of B_u. The witnessing event must also use the exterior of X, so cut parity gives at least two boundary incidences. Incidences assigned by the 2/3-demand ledger are pairwise disjoint and are counted once in $\partial_+(R)$. An unabsorbed demand unit chooses the lexicographically first incidence $c(v)$; because X is a component of the P_{13}-free remainder, the other endpoint lies in the packed-window set, and the disjointness of the packing gives a unique window $P(v)$. Thus $\mathsf{P}_{\mathrm{open}} = \sum_P B_{\mathrm{open}}(P)$. On each window, the first two open units form the base term $2p_{13}$; any third open unit forms an overload triple. Routed triples leave through the direct power-of-two certificate, an unused boundary absorber, a proper-compression or support-dependence absorber, or the Type B fan-window handoff. If two open units have a recorded P-avoiding corridor Q with $b \in \mathsf{Sh}_{s(Q)}(a)$, the displayed cycle has length $s(Q)+2+|a-b| \in \mathcal{D}$, so the triple is routed. The only surviving excess is the zero-signature term in $\mathsf{P}_{\mathrm{open}} \le 2p_{13} + \mathsf{P}^+_{\mathrm{zero}}$. Independently of that auxiliary window audit, every surviving target-defect two-support entry is a canonical (Q3) exit-(4) quotient, so Lemmas 15.44 and 15.47 peel the routed load u, decrease the remaining deficit by exactly 1/4, and strictly decrease Λ_4.

Proof. Every open demand unit has a demand token. By Lemma 15.11, its boundary-incidence set is nonempty. The lexicographic rule in Definition 15.23 therefore assigns exactly one boundary incidence $c(v)$. Because X is a connected component of R, the other endpoint of the corresponding boundary edge lies in W, and because the packed windows are vertex-disjoint, that endpoint lies in a unique window $P(v)$. Thus the sets counted by $B_{\mathrm{open}}(P)$, $P \in \mathcal{P}$, partition $\mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}$. Their total size is $\mathsf{P}_{\mathrm{open}}$ by the definition of $\mathsf{P}_{\mathrm{open}}$. ☐

Definition 15.25 (Same-window overload triples)**.** *For a packed window $P \in \mathcal{P}$, put*

$$\mathcal{U}(P) = \{v \in \mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}} : P(v) = P\}.$$

A same-window overload triple *at P is a three-element subset $\mathcal{T} \subseteq \mathcal{U}(P)$. If $v \in \mathcal{T}$, write*

$$c(v) = (x_v, x_v p_{a(v)}), \qquad P = p_0 p_1 \cdots p_{12},$$

for its canonical window boundary incidence and its window index.

The triple $\mathcal{T}$ is routed *if at least one of the following certificates is present.*

(O1) Direct power-of-two window certificate. *The demand records of one or two units in $\mathcal{T}$, or an explicitly recorded P-avoiding corridor between their canonical window boundary incidences, together with one or two subpaths of P, contain a simple cycle of length in $\mathcal{D}$. This includes the direct same-window and two-window cycle configurations covered by Lemmas 14.75 and 14.76.*

(O2) Unused boundary-incidence certificate. *Some $v = (\xi, j) \in \mathcal{T}$, with $\xi = (X, w, u, B_u)$, has an oriented boundary incidence $c \in \partial_E X$ which is not used by any set $A(\xi')$ in the 2/3-demand ledger and is not assigned to any absorbed demand unit.*

(O3) Profile-dependence certificate. *A subfamily of profile coordinates belonging to units of $\mathcal{T}$ is rank-reducing under a functional admissible quotient, and the routing supplied by Lemma 15.19 is a proper-support compression or one of the support-dependence alternatives already removed by the branch split.*

(O4) Type B fan-window certificate. *Two units of $\mathcal{T}$ determine fan-closed ports or cubic-closed fan neighbours in an assigned Type B fan-window profile, with the window incidences and non-window incidences required by Lemma 14.70 and Definition 14.90. Thus the local B1 entry is available, and any failure of cross-center disjointness is a B2 overlap obstruction in the sense of Lemma 14.87.*

A same-window overload triple is primitive *if it is not routed.*

Lemma 15.26 (Routed overload triples do not remain open)**.** *Work on a branch where the direct power-of-two cycle alternatives, the type* (A1) *boundary absorbers, the type* (A2) *support-dependence absorbers, and the Type B fan-window handoff have all been applied. Then every surviving same-window overload triple is primitive.*

Proof. Let $\mathcal{T}$ be a same-window overload triple. If (O1) holds, the displayed certificate is a power-of-two cycle in G, contradicting the standing counterexample hypothesis. If (O2) holds, the indicated boundary incidence is an admissible absorber of type (A1) in Definition 15.20. Since the absorber assignment is maximal, the corresponding demand unit cannot still be open. If (O3) holds, Lemma 15.19 supplies an absorber of type (A2) whose conclusion is proper compression or support dependence; on the stated branch such a unit has routed away from the open demand ledger. If (O4) holds, the data are exactly Type B fan-window data. The local B1 entry is supplied by Definition 14.90; if disjointness fails, Lemma 14.87 records the B2 residual. In either case the triple is no longer an open Type A demand overload. Therefore a surviving same-window overload triple has none of (O1)–(O4), and is primitive by definition. ☐

Definition 15.27 (Primitive same-window overload excess)**.** *Assume the branch of Lemma 15.26. A packed window P is* primitive-overloaded *if*

$$B_{\mathrm{open}}(P) = |\mathcal{U}(P)| \geq 3$$

and the lexicographically first three-element subset of $\mathcal{U}(P)$ is a primitive same-window overload triple. The primitive overload excess *is*

$$\mathsf{P}^{+}_{\mathrm{prim}} = \sum_{\substack{P \in \mathcal{P} \\ P \text{ primitive-overloaded}}} (B_{\mathrm{open}}(P) - 2).$$

Definition 15.28 (Window attachment signatures)**.** *Let $P = p_0p_1 \cdots p_{12}$ be a packed window. For an integer $s \geq 0$, define the forbidden distance set*

$$D_s = \{d \in \{0, \ldots, 12\} : s + 2 + d \in \mathcal{D}\}.$$

*For an attachment index $a \in \{0, \ldots, 12\}$, define its s-*signature *on P by*

$$\mathsf{Sh}_s(a) = \{b \in \{0, \ldots, 12\} : |a - b| \in D_s\}.$$

Equivalently,

$$b \in \mathsf{Sh}_s(a) \iff C_s(\{a\}, \{b\}) = 0$$

for the P_{13}-label relation C_s of the main window algebra.

Lemma 15.29 (Exact singleton-signature table)**.** *For every $s \geq 0$,*

$$D_s = \{2^q - s - 2 : q \geq 1,\ 0 \leq 2^q - s - 2 \leq 12\}.$$

In particular,

s	D_s
0	$\{0, 2, 6\}$
1	$\{1, 5\}$
2	$\{0, 4, 12\}$
3	$\{3, 11\}$
4	$\{2, 10\}$
5	$\{1, 9\}$
6	$\{0, 8\}$
7	$\{7\}$
8	$\{6\}$
9	$\{5\}$
10	$\{4\}$
11	$\{3\}$
12	$\{2\}$
13	$\{1\}$
14	$\{0\}$
15, 16, 17	$\varnothing$
18	$\{12\}$
19	$\{11\}$
20	$\{10\}$
21	$\{9\}$
22	$\{8\}$
23	$\{7\}$
24	$\{6\}$
25	$\{5\}$
26	$\{4\}$
27	$\{3\}$
28	$\{2\}$
29	$\{1\}$
30	$\{0\}$.

For $s \geq 18$, the set D_s has at most one element.

Proof. By Definition 15.28, $d \in D_s$ exactly when $s + 2 + d = 2^q$ for some $q \geq 1$. This is equivalent to

$$d = 2^q - s - 2$$

with $0 \leq d \leq 12$, giving the displayed formula. The table is obtained by substituting $s = 0, \dots, 30$ into this formula. If $s \geq 18$, then $[s + 2, s + 14] \subseteq [20, \infty)$. Consecutive powers of two in $[16, \infty)$ differ by at least $16 > 12$, so the interval $[s + 2, s + 14]$ contains at most one power of two. Hence D_s has at most one element. □

Definition 15.30 (Recorded window-signature hit)**.** *Let $\upsilon, \upsilon' \in \mathcal{U}(P)$ be distinct open demand units on the same packed window $P = p_0 \cdots p_{12}$, with canonical boundary incidences*

$$c(\upsilon) = (x, xp_a), \qquad c(\upsilon') = (y, yp_b).$$

A recorded P-avoiding corridor *from υ to υ' is a simple x-y path Q in $G - V(P)$, recorded in the actual demand data of one or both units or in the Type B fan-window handoff data, whose internal vertices avoid P and whose first and last edges are not the canonical boundary edges xp_a, yp_b. Its length is denoted $s(Q) = |E(Q)|$.*

The ordered pair (υ, υ') has a recorded window-signature hit *if the two canonical boundary edges xp_a and yp_b are distinct, such a corridor Q exists, and*

$$b \in \mathsf{Sh}_{s(Q)}(a).$$

When $x = y$, the corridor Q may be the trivial length-zero path; the distinct-incidence condition then says $a \neq b$.

Lemma 15.31 (Window-signature hits are direct power-of-two certificates)**.** *If two open demand units on the same packed window have a recorded window-signature hit, then every same-window overload triple containing them is routed by certificate* (O1) *of Definition 15.25.*

Proof. Use the notation of Definition 15.30. Since $b \in \mathsf{Sh}_{s(Q)}(a)$, the definition of the signature gives

$$s(Q) + 2 + |a - b| \in \mathcal{D}.$$

The path

$$x\,Q\,y\,p_b\,P\,p_a\,x$$

is a simple cycle. Indeed, Q lies in $G - V(P)$, the segment $p_b P p_a$ lies in the induced path P, and the only edges between the two pieces used in the display are the two distinct canonical boundary edges yp_b and xp_a. If $x = y$, then Q is the trivial path and distinctness of the boundary-edge distinctness gives $a \neq b$, so the displayed cycle is $xp_b P p_a x$. In all cases the length is exactly $s(Q) + 2 + |a - b|$, hence is a power of two. This is a direct power-of-two window certificate, which is certificate (O1). □

Definition 15.32 (Zero-signature primitive excess)**.** *Assume the branch of Lemma 15.26. A primitive-overloaded packed window P is* zero-signature *if the set $\mathcal{U}(P)$ has no pair of distinct open units with a recorded window-signature hit. The* zero-signature primitive excess *is*

$$\mathsf{P}^{+}_{\text{zero}} = \sum_{\substack{P \in \mathcal{P} \\ P \text{ zero-signature}}} (B_{\text{open}}(P) - 2).$$

Lemma 15.33 (Primitive excess equals zero-signature excess after window-signature routing)**.** *Work on a branch where the routed overload triples of Definition 15.25 have been removed. Then*

$$\mathsf{P}^{+}_{\mathrm{prim}} = \mathsf{P}^{+}_{\mathrm{zero}}.$$

Proof. Let P be primitive-overloaded. Its lexicographically first three open units form a surviving same-window overload triple. If some pair in that triple had a recorded window-signature hit, then Lemma 15.31 would route the triple by certificate (O1), contrary to the branch hypothesis. More generally, if any distinct pair $v, v' \in \mathcal{U}(P)$ had a recorded window-signature hit, then, since $B_{\mathrm{open}}(P) \geq 3$, choosing any third unit of $\mathcal{U}(P)$ would give a same-window overload triple routed by (O1). Hence no pair in $\mathcal{U}(P)$ has a recorded window-signature hit, so P is zero-signature. Therefore every summand counted by $\mathsf{P}^{+}_{\mathrm{prim}}$ is counted by $\mathsf{P}^{+}_{\mathrm{zero}}$. Conversely, a zero-signature window is primitive-overloaded by definition, so every summand counted by $\mathsf{P}^{+}_{\mathrm{zero}}$ is counted by $\mathsf{P}^{+}_{\mathrm{prim}}$. □

Lemma 15.34 (Open demand after routed overloads)**.** *On a branch where the routed overload triples of Definition 15.25 have been removed,*

$$\mathsf{P}_{\mathrm{open}} \leq 2p_{13} + \mathsf{P}^{+}_{\mathrm{zero}}.$$

Proof. By Lemma 15.24,

$$\mathsf{P}_{\mathrm{open}} = \sum_{P \in \mathcal{P}} B_{\mathrm{open}}(P).$$

If $B_{\mathrm{open}}(P) \leq 2$, then its contribution is at most 2. If $B_{\mathrm{open}}(P) \geq 3$, then the lexicographically first three-element subset of $\mathcal{U}(P)$ is a surviving same-window overload triple. By Lemma 15.26, it is primitive, so P is primitive-overloaded. If its first triple had a recorded window-signature hit, Lemma 15.31 would route it by certificate (O1), contrary to the branch hypothesis. If any other pair in $\mathcal{U}(P)$ had a recorded window-signature hit, that pair together with any third open unit would give a routed same-window overload triple, again contrary to the branch hypothesis. Hence P is zero-signature, and its excess $B_{\mathrm{open}}(P) - 2$ is one summand of $\mathsf{P}^{+}_{\mathrm{zero}}$. Therefore, for every packed window P,

$$B_{\mathrm{open}}(P) \leq 2 + \mathbf{1}_{\{P \text{ zero-signature}\}}(B_{\mathrm{open}}(P) - 2)$$

with no overlap between windows. Summing this pointwise inequality over $P \in \mathcal{P}$ gives the result. □

Lemma 15.35 (Window-blocker accounting audit)**.** *On a branch where the routed overload triples of Definition 15.25 have been removed, every open demand unit is counted in exactly one of the following two places:*

(a) among the first two open units assigned to its packed window, contributing to the base term $2p_{13}$;

(b) among the excess units of a zero-signature primitive-overloaded packed window, contributing to $\mathsf{P}^{+}_{\mathrm{zero}}$.

No open unit is counted for two packed windows, and every same-window overload not counted in $\mathsf{P}^{+}_{\mathrm{zero}}$ has routed through one of (O1)–(O4).

Proof. Lemma 15.24 partitions the open demand units by the unique packed window $P(v)$. Inside a fixed packed window, choose the first two units in the global lexicographic order as the base units; if fewer than two units are present, all units are base units. Any remaining unit exists only when $B_{\mathrm{open}}(P) \geq 3$. On the stated branch, the first three units then form a surviving same-window overload triple. If that triple were routed by one of (O1)–(O4), the branch would have removed that overload before $\mathsf{P}_{\mathrm{open}}$ is counted. Hence, for an open residual unit, the triple is not routed. Lemma 15.26 makes it primitive, and Lemma 15.31 removes every same-window overload triple containing a pair with a recorded window-signature hit. Since $B_{\mathrm{open}}(P) \geq 3$, any recorded-hit pair in $\mathcal{U}(P)$ would lie in such a triple. Thus the window is zero-signature, and all units beyond the first two are counted in the summand $B_{\mathrm{open}}(P) - 2$ of $\mathsf{P}^{+}_{\mathrm{zero}}$. The packed windows are vertex-disjoint and the assignment $P(v)$ is single-valued, so no open unit can appear in two window classes. These two alternatives exhaust the open units because they are applied after the exact partition of Lemma 15.24. □

Lemma 15.36 (Exhaustion of the final open-demand charge)**.** *Assume the type* (A2) *absorbers of Definition 15.20 have routed to their proper-compression or support-dependence alternatives, and assume the routed overload triples of Definition 15.25 have been removed. Then the surviving open demand admits a disjoint partition*

$$\mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}} = \mathcal{U}_{\mathrm{base}} \sqcup \mathcal{U}_{\mathrm{zero}}$$

such that

$$|\mathcal{U}_{\mathrm{base}}| \leq 2p_{13}, \qquad |\mathcal{U}_{\mathrm{zero}}| = \mathsf{P}^{+}_{\mathrm{zero}},$$

and hence

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} \leq \partial_{+}(R), \qquad \mathsf{P}_{\mathrm{open}} \leq 2p_{13} + \mathsf{P}^{+}_{\mathrm{zero}}.$$

Proof. The boundary-incidence inequality

$$3\tilde{N} - \mathsf{P}_{\mathrm{open}} \leq \partial_{+}(R)$$

is Lemma 15.21 after the type (A2) alternatives have routed away.

It remains to identify the open units. By Lemma 15.24, every open demand unit has a unique packed window $P(v)$. For a fixed window P, order $\mathcal{U}(P)$ by the global lexicographic order. Put the first two units, if present, in $\mathcal{U}_{\mathrm{base}}$. Since there are p_{13} packed windows, this gives $|\mathcal{U}_{\mathrm{base}}| \leq 2p_{13}$.

Let v be any remaining open unit on P. Then $B_{\mathrm{open}}(P) \geq 3$. The first three units of $\mathcal{U}(P)$ form a same-window overload triple. If that triple were routed by one of (O1)–(O4), it would have been removed before the present open-demand ledger was formed. Hence the triple is primitive and P is primitive-overloaded. If any pair in $\mathcal{U}(P)$ had a recorded window-signature hit, that pair and any third open unit of $\mathcal{U}(P)$ would form a same-window overload triple routed by (O1) via Lemma 15.31; this is again excluded on the present branch. Therefore P is zero-signature. Put every unit of $\mathcal{U}(P)$ after the first two in $\mathcal{U}_{\mathrm{zero}}$.

The preceding paragraph applies to every open unit not in $\mathcal{U}_{\mathrm{base}}$, and the construction is window-by-window, so the two sets are disjoint and cover $\mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}$. For a zero-signature window P, exactly $B_{\mathrm{open}}(P) - 2$ units are placed in $\mathcal{U}_{\mathrm{zero}}$, and no non-zero-signature window contributes to $\mathcal{U}_{\mathrm{zero}}$. Summing over P gives

$$|\mathcal{U}_{\mathrm{zero}}| = \sum_{\substack{P \in \mathcal{P} \\ P \text{ zero-signature}}} (B_{\mathrm{open}}(P) - 2) = \mathsf{P}^{+}_{\mathrm{zero}}.$$

The last displayed inequality follows from

$$\mathsf{P}_{\text{open}} = |\mathcal{U}_{\text{press}} \setminus \mathcal{U}_{\text{abs}}| = |\mathcal{U}_{\text{base}}| + |\mathcal{U}_{\text{zero}}|.$$

$\square$

Proposition 15.37 (Exit-(4) closure from zero-signature slack)**.** *Assume Definition 4.1. Work on a branch where the type* (A2) *absorbers of Definition 15.20 have routed to their proper-compression or support-dependence alternatives, and where the routed overload triples of Definition 15.25 have been removed. If*

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}^{+}_{\text{zero}}}{|R|} < \varepsilon_{\text{prim}},$$

where

$$\varepsilon_{\text{press}} = 12\left(\frac{1}{4} - \tau_{\text{win}}\right) - \tau_{\text{win}}, \qquad \varepsilon_{\text{prim}} = \varepsilon_{\text{press}} - \frac{2\theta_{\text{win}}}{1 - 13\theta_{\text{win}}} = 0.00330339423\ldots.$$

then the target-defect exit-(4) *residual is impossible.*

Proof. By Lemma 15.36,

$$\mathsf{P}_{\text{open}} \le 2p_{13} + \mathsf{P}^{+}_{\text{zero}}.$$

The near-cubic P_{13}-density bound gives

$$p_{13} \le \theta_{\text{win}} n + o(n).$$

Writing $\theta = p_{13}/n$, the remainder size is

$$|R| = (1 - 13\theta)n + o(n),$$

and $2\theta/(1 - 13\theta)$ is increasing for $0 \le \theta < 1/13$. Therefore

$$\limsup_{|R|\to\infty} \frac{2p_{13}}{|R|} \le \frac{2\theta_{\text{win}}}{1 - 13\theta_{\text{win}}}.$$

Hence the displayed hypothesis implies

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}_{\text{open}}}{|R|} < \frac{2\theta_{\text{win}}}{1 - 13\theta_{\text{win}}} + \varepsilon_{\text{prim}} = \varepsilon_{\text{press}}.$$

Proposition 15.22 now closes the target-defect exit-(4) residual. $\square$

Remark 15.38 (Numerical size of the final signature residual)**.** *Since*

$$\frac{p_{13}}{|R|} \le \frac{\theta_{\text{win}}}{1 - 13\theta_{\text{win}}} + o(1) = 0.01521165789\ldots + o(1),$$

the available slack

$$\varepsilon_{\text{prim}} = 0.00330339423\ldots$$

is exactly

$$\frac{\varepsilon_{\text{prim}}}{\theta_{\text{win}}/(1 - 13\theta_{\text{win}})} = 0.2171620118\ldots$$

zero-signature excess units per packed window on average. Thus the zero-signature criterion alone leaves only residuals satisfying

$$\mathsf{P}^{+}_{\mathrm{zero}} \geq 0.2171620118\ldots p_{13} - o(|R|).$$

Every packed window contributing to this sum carries at least a third open blocker, no pair of open blockers on that window has a recorded C_s-signature hit, and the corresponding open units have no unused boundary absorber, no support-dependence absorber, and no Type B fan-window handoff.

Definition 15.39 (Same-window open-blocker cap)**.** *The open-demand ledger satisfies the* same-window two-blocker cap *if there is an exceptional set*

$$\mathcal{E}_{\mathrm{win}} \subseteq \mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}$$

with $|\mathcal{E}_{\mathrm{win}}| = o(|R|)$ such that, after deleting $\mathcal{E}_{\mathrm{win}}$, every packed window $P \in \mathcal{P}$ receives at most two canonical open blockers:

$$|\{v \in (\mathcal{U}_{\mathrm{press}} \setminus \mathcal{U}_{\mathrm{abs}}) \setminus \mathcal{E}_{\mathrm{win}} : P(v) = P\}| \leq 2.$$

Lemma 15.40 (The two-blocker cap has sublinear overload excess)**.** *If the same-window two-blocker cap of Definition 15.39 holds, then*

$$\mathsf{P}^{+}_{\mathrm{prim}} = o(|R|) \qquad \text{and} \qquad \mathsf{P}^{+}_{\mathrm{zero}} = o(|R|).$$

Proof. Let $\mathcal{E}_{\mathrm{win}}$ be the exceptional set in Definition 15.39. For a fixed packed window P, write $e(P) = |\mathcal{E}_{\mathrm{win}} \cap \mathcal{U}(P)|$. After deleting $\mathcal{E}_{\mathrm{win}}$, at most two units remain on P, so

$$\max\{0, B_{\mathrm{open}}(P) - 2\} \leq e(P).$$

The primitive overload excess is bounded by the total overload excess:

$$\mathsf{P}^{+}_{\mathrm{prim}} \leq \sum_{P \in \mathcal{P}} \max\{0, B_{\mathrm{open}}(P) - 2\} \leq \sum_{P \in \mathcal{P}} e(P) = |\mathcal{E}_{\mathrm{win}}| = o(|R|).$$

The same displayed total overload excess also bounds $\mathsf{P}^{+}_{\mathrm{zero}}$, since zero-signature windows are a subfamily of the packed windows. ∎

Lemma 15.41 (Numerical window-blocker squeeze)**.** *Assume Definition 4.1. If the same-window two-blocker cap of Definition 15.39 holds, then*

$$\limsup_{|R| \to \infty} \frac{\mathsf{P}_{\mathrm{open}}}{|R|} \leq \frac{2\theta_{\mathrm{win}}}{1 - 13\theta_{\mathrm{win}}} = 0.03042331579\ldots < 0.03372671002\ldots = \varepsilon_{\mathrm{press}}.$$

Proof. By Lemma 15.24 and the same-window cap,

$$\mathsf{P}_{\mathrm{open}} \leq 2p_{13} + o(|R|).$$

The near-cubic P_{13}-density bound gives

$$p_{13} \leq \theta_{\mathrm{win}} n + o(n).$$

Writing $\theta = p_{13}/n$, the remainder size is

$$|R| = (1 - 13\theta)n + o(n),$$

and the function $2\theta/(1-13\theta)$ is increasing on the relevant interval $0 \le \theta < 1/13$. Therefore

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}_{\text{open}}}{|R|} \le \frac{2\theta_{\text{win}}}{1-13\theta_{\text{win}}}.$$

Substituting $\theta_{\text{win}} = 0.01270017798\ldots$ gives

$$\frac{2\theta_{\text{win}}}{1-13\theta_{\text{win}}} = 0.03042331579\ldots.$$

The open-demand threshold from Proposition 15.22 is

$$\varepsilon_{\text{press}} = 12\left(\frac{1}{4} - \tau_{\text{win}}\right) - \tau_{\text{win}} = 0.03372671002\ldots,$$

so the displayed bound lies strictly below the threshold. □

Proposition 15.42 (Exit-(4) closure from the same-window cap)**.** *Assume Definition 4.1. On a branch where all dependence absorbers of type* (A2) *have routed to their support-dependence alternatives, the target-defect exit-(4) residual is impossible whenever the same-window two-blocker cap of Definition 15.39 holds.*

Proof. By Lemma 15.41, the same-window cap gives

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}_{\text{open}}}{|R|} < \varepsilon_{\text{press}}.$$

The conclusion is exactly Proposition 15.22. □

Proposition 15.43 (Sharp external-demand closure criterion)**.** *Assume Definition 4.1. In the large-budget branch, suppose the demand ledger of Definition 15.12 satisfies*

$$\limsup_{|R|\to\infty} \frac{\mathsf{P}_{\text{ext}}}{|R|} < \varepsilon_{\text{press}}, \qquad \varepsilon_{\text{press}} = 12\left(\frac{1}{4} - \tau_{\text{win}}\right) - \tau_{\text{win}}.$$

Then no minimal counterexample survives. Numerically,

$$\varepsilon_{\text{press}} = 12(0.02182513154\ldots) - 0.22817486846\ldots = 0.03372671002\ldots.$$

In particular, a sublinear external-demand defect $\mathsf{P}_{\text{ext}} = o(|R|)$ *closes the branch.*

Proof. By Lemmas 15.4 and 15.6,

$$\tilde{N} \ge 4\left(\frac{1}{4} - \tau_{\text{win}}\right)|R| - o(|R|).$$

By Lemma 15.13 and the window-only deficiency cap,

$$3\tilde{N} - \mathsf{P}_{\text{ext}} \le \partial_+(R) \le \tau_{\text{win}}|R| + o(|R|).$$

Combining the two displays gives

$$\left[12\left(\frac{1}{4} - \tau_{\text{win}}\right) - \frac{\mathsf{P}_{\text{ext}}}{|R|}\right]|R| \le \tau_{\text{win}}|R| + o(|R|).$$

If the normalized limsup of P_{ext} is strictly below $\varepsilon_{\text{press}}$, the coefficient on the left is eventually larger than τ_{win} by a fixed positive amount, contradicting the displayed inequality. The numerical value follows from the stated value of τ_{win}. The sublinear case has normalized limsup 0, so it is a special case. □

Lemma 15.44 (Target-defect demand is canonical exit-(4) peel data)**.** *Let $\xi = (X, w, u, B_u) \in \tilde{\Xi}$ be a target-defect entry, and let $\mathfrak{t}_\xi = (\xi, q, S_0, S_1, Y, E)$ be its lexicographically chosen constraint token. Then $q \in \mathcal{Q}_4(w)$, the declared routed-load support of q contains the load u, and q is an exit-(4) witness for u in the sense of Definition 14.33. Consequently, if $u \notin P_4(w)$, then Lemma 14.38 enlarges the peeling set by u and reduces the remaining receiver deficit by exactly $1/4$.*

Proof. Because ξ is target-defect, the surviving failure alternative in Definition 14.14 is alternative (a): a trace-local trace-local quotient or identification inside $\rho_u(B_u)$ is target-defective. The token definition chooses exactly this quotient q, two same-boundary-degree realizations S_0, S_1, and a compatible context Y which distinguishes their target predicates.

Clause (Q3) of Definition 14.49 places every such trace-local quotient of the trace basin B_u in the canonical family $\mathcal{Q}_4(w)$. The same definition records the declared routed-load support of a (Q3) quotient as the trace load u whose basin is being tested. Thus $q \in \mathcal{Q}_4(w)$, it is boundary-degree-preserving, it is target-defective in the compatible context Y, and its declared support contains the canonical coordinate of u. This is precisely an exit-(4) witness for u by Definition 14.33. If u is unpeeled, the last assertion is Lemma 14.38. $\square$

Definition 15.45 (Peeling-reduced unified ledger)**.** *Let $P_4 = (P_4(w))_w$ be a family of valid exit-(4) peeling sets in the sense of Definition 14.33. The* P_4-reduced unified ledger *is obtained from Definitions 15.2 and 15.5 by deleting every routed load listed in its receiver's peeling set and appearing in the unified silent-entry ledger. Put*

$$\mathcal{U}^{\mathrm{un}}(w) = \bigcup_{X \in \tilde{\mathcal{X}}} \mathcal{U}_X(w), \qquad P_4^{\mathrm{un}}(w) = P_4(w) \cap \mathcal{U}^{\mathrm{un}}(w).$$

Thus

$$\mathcal{U}_X^{P_4}(w) = \mathcal{U}_X(w) \setminus P_4^{\mathrm{un}}(w),$$

and

$$\tilde{\Xi}^{P_4} = \{\, (X, w, u, B_u) : X \in \tilde{\mathcal{X}},\ u \in \mathcal{U}_X^{P_4}(w) \,\}, \qquad \tilde{N}^{P_4} = |\tilde{\Xi}^{P_4}|.$$

The remaining Type A deficit represented by this ledger is

$$\tilde{D}_A^{P_4} = \tilde{D}_A - \frac{1}{4} \sum_w |P_4^{\mathrm{un}}(w)|.$$

If $\tilde{D}_A^{P_4} \le 0$, the Type A target-defect/route-8 ledger has no negative mass left to carry.

Lemma 15.46 (Peeling-reduced support reduction)**.** *For every valid family P_4 of exit-(4) peeling sets, the reduced ledger satisfies*

$$\tilde{N}^{P_4} \ge 4\tilde{D}_A^{P_4}.$$

Moreover, if

$$\tilde{D}_A^{P_4} \ge \Big(\frac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|),$$

then $\tilde{\Xi}^{P_4}$ contains a two-support entry.

Proof. Each load in $P_4^{\mathrm{un}}(w)$ is equipped with one exit-(4) witness and is removed exactly once from the unified Type A receiver calculation. By Lemmas 14.34 and 14.38, each such removal decreases the remaining receiver deficit by exactly $1/4$ and does not alter the graph, the support boundary, or any unpeeled load. Hence the silent excess count of Lemma 15.6 loses exactly one indexed possible entry for each load in P_4^{un}, while the deficit side loses exactly $1/4$. The proof of Lemma 15.6 therefore gives

$$\tilde{N}^{P_4} \geq 4\left(\tilde{D}_A - \frac{1}{4}\sum_w |P_4^{\mathrm{un}}(w)|\right) = 4\tilde{D}_A^{P_4}.$$

Assume now that the displayed lower bound on $\tilde{D}_A^{P_4}$ holds and that no entry of $\tilde{\Xi}^{P_4}$ is two-support. Then every remaining entry has at least three private essential incidences. These private incidences are oriented boundary incidences of the supports in $\tilde{\mathcal{X}}$, and private incidences belonging to different entries are disjoint by the definition of $\pi_{\mathcal{X}}$ in Definition 14.48. Deleting peeled loads has not changed G, the supports, or their boundary incidences, so selecting three private incidences for each entry gives

$$3\tilde{N}^{P_4} \leq \sum_{X\in\tilde{\mathcal{X}}} |\partial_E X| = \sum_{X\in\tilde{\mathcal{X}}} \partial_+(X) \leq \partial_+(R) \leq \tau_{\mathrm{win}}|R| + o(|R|),$$

where the equality uses $\sigma(X) = 0$ for $X \in \tilde{\mathcal{X}}$, and the last inequality is Corollary 8.8. The first part of the lemma and the assumed reduced-deficit lower bound give

$$\tilde{N}^{P_4} \geq 4\Big(\frac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|).$$

Combining the two displays yields

$$12\Big(\frac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|) \leq \tau_{\mathrm{win}}|R| + o(|R|),$$

and hence $12(\frac{1}{4} - \tau_{\mathrm{win}}) \leq \tau_{\mathrm{win}}$. Equivalently $\tau_{\mathrm{win}} \geq 3/13$, contradicting $\tau_{\mathrm{win}} = 0.22817\ldots < 3/13$. Therefore $\tilde{\Xi}^{P_4}$ contains a two-support entry. ☐

Lemma 15.47 (Finite exit-(4) descent)**.** *Run the saturated Type A receiver analysis with the peeling sets $P_4(w)$ of Definition 14.33. Whenever the current large-budget residual contains a two-support target-defect entry, one exit-(4) peeling step is performed, the integer*

$$\Lambda_4 = \sum_w \big|\mathcal{L}(w) \setminus P_4(w)\big|$$

decreases by one, and no standing invariant of the current branch is weakened. Hence an infinite target-defect peeling loop is impossible.

Proof. Let $\xi = (X, w, u, B_u)$ be a two-support target-defect entry in the current residual. Entries of the current residual are indexed by unpeeled routed loads; a load already in $P_4(w)$ has left the Type A receiver calculation by Remark 14.41. Hence $u \notin P_4(w)$. By Lemma 15.44, the constraint token of ξ is an exit-(4) witness for u. Applying Lemma 14.38 replaces $P_4(w)$ by $P_4(w) \cup \{u\}$. Thus $L_4(w)$ decreases by one and every other residual load count is unchanged, so Λ_4 decreases by one.

The operation changes only the bookkeeping ledger: the graph G, the packed P_{13}-windows, the boundary degree profiles, the target-safety constraints, and all target-response quotients remain the same. The selected load is removed from the pure Type A receiver sum, and Lemma 14.34 gives the exact $1/4$-unit charge update. Thus the next stage satisfies the same standing invariants; if its large-budget deficit has disappeared, the branch is closed, and otherwise the same large-budget reduction applies to the updated unpeeled ledger. Since Λ_4 is a nonnegative integer, the target-defect peeling step cannot occur indefinitely. ☐

15.1.7 Large-budget closure from demand accounting and finite descent

With the target-defect class accounted for, the large-budget branch closes by a finite exit-(4) descent. The demand ledger proves that no target-defect unit is missing from the incidence count; the descent lemma then identifies each surviving target-defect two-support unit with one of the canonical peelable loads already present in the saturated Type A analysis.

Theorem 15.48 (Large-budget closure with target-defect demand accounting)**.** *Assume Definition 4.1. No minimal counterexample survives in the large-budget branch.*

Proof. Start with the peeling sets $P_4(w) = \varnothing$ and run the following deterministic procedure. At any stage, let $\tilde{D}_A^{P_4}$ be the remaining Type A deficit of Definition 15.45. If

$$\tilde{D}_A^{P_4} < \Big(\frac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|),$$

then the still-unresolved negative mass is too small to realize the large-budget branch: all supports outside the P_4-reduced Type A ledger are nonnegative or belong to the Type B bridge ledger, and the latter has total negative mass $o(|R|)$ by Proposition 14.100. Hence the large-budget net-deficiency cap and Theorem 15.1 close that stage. Thus a surviving stage must satisfy

$$\tilde{D}_A^{P_4} \geq \Big(\frac{1}{4} - \tau_{\mathrm{win}}\Big)|R| - o(|R|).$$

Lemma 15.46 then produces a two-support unified entry in the current unpeeled receiver ledger.

If that two-support entry is a route-8 entry, then Proposition 14.62 applies: the entry gives the terminal two-support route-8 obstruction and Theorem 14.61 excludes it. If instead the entry is target-defect, then Lemma 15.44 makes its demand token a canonical exit-(4) witness for its routed load, and Lemma 15.47 performs one peeling step.

The second alternative cannot occur infinitely many times, because the integer Λ_4 of Lemma 15.47 strictly decreases at each target-defect peel. Hence the procedure terminates. It cannot terminate with a large-budget survivor. Indeed, a terminal surviving stage still satisfies the displayed lower bound on $\tilde{D}_A^{P_4}$, so the reduced support reduction again supplies a two-support entry. The target-defect case would perform another peel and lower Λ_4, contrary to terminality. The two-support entry is therefore route 8, and the previous paragraph excludes it. This contradiction proves that the large-budget branch has no minimal counterexample. ☐

Corollary 15.49 (Large-budget closure with sublinear open demand)**.** *Assume Definition 4.1. After the type* (A2) *support-dependence alternatives of Definition 15.20 have been removed, if*

$$\mathsf{P}_{\mathrm{open}} = o(|R|),$$

then no minimal counterexample survives in the large-budget branch.

Proof. The normalized limsup of $\mathsf{P}_{\mathrm{open}}/|R|$ is 0. The threshold equals

$$12\left(\frac{1}{4} - \tau_{\mathrm{win}}\right) - \tau_{\mathrm{win}} = 0.03372671002\ldots,$$

which is positive. Therefore Proposition 15.22 closes the target-defect exit-(4) residual. The route-8 two-support residual is closed by Theorem 14.61, and the remaining unified entries are already paid by the demand ledger or by the Type B bridge ledger used in Lemma 15.4. Hence no large-budget survivor remains. ☐

Remark 15.50 (Unified demand accounting)**.** *A lower bound for $\sum_i \mathcal{N}(X_i)$ that uses only the route-8 collection and the Type B bridge ledger is not sufficient. Such a bound has the form*

$$\sum_i \mathcal{N}(X_i) \geq -D_A(\mathcal{X}_A^0) - 8S_B,$$

where the right-hand side sums only over the route-8 collection $\mathcal{X}_A^0$ and the Type B bridge ledger. It is insufficient to combine this estimate with Theorem 15.1 by treating every other support as nonnegative: Theorem 15.1(a) proves $\mathcal{N}(X) \geq 0$ only for supports with no *exit-(4) witness, and a support that peels routed loads through the target-defect exit has $\mathcal{N}(X) \geq -\frac{1}{4}|P_4| < 0$. Such a support is neither nonnegative nor (in general) a route-8 residual nor a Type B bridge residual, so its negative mass was omitted from the lower bound, and the deduction "$\mathcal{X}_A^0$ carries the large-budget deficit" did not follow: the deficit could reside entirely in the target-defect class. Definition 15.2 removes the omission by summing over* all *negative Type A supports at once; Lemma 15.7 shows the extra (target-defect) entries obey the same essential-support lower bound as route-8 entries – this is where the minimality-free Lemmas 14.56 and 14.57 are essential. The target-defect mass is assigned to the $2/3$-demand ledger of Definition 15.12. The no-overcount lemma Lemma 15.13 proves the exact raw boundary-incidence inequality*

$$3\tilde{N} - \mathsf{P}_{\text{ext}} \leq \partial_+(R), \qquad \mathsf{P}_{\text{ext}} = N_2 + 3N_{\text{res}},$$

and Proposition 15.43 gives the raw closure threshold

$$\mathsf{P}_{\text{ext}} < \left(12\left(\frac{1}{4} - \tau_{\text{win}}\right) - \tau_{\text{win}}\right)|R| - o(|R|).$$

The absorption ledger of Definition 15.20 refines this by charging additional single-use units before the final squeeze is applied. After those absorptions, the same numerical threshold applies to the explicitly defined open remainder P_{open} by Proposition 15.22. The window-blocker ledger then assigns every open unit to a unique packed P_{13}-window. Same-window overload triples that realize a direct power-of-two cycle, an unused boundary absorber, a support-dependence absorber, or Type B fan-window data are removed by the listed mechanisms in Definition 15.25. After those routed triples are removed, Lemma 15.31 removes every overload triple containing a pair of open blockers with a recorded window-signature hit. The exact residual quantity beyond the two-blocker-per-window base is therefore the zero-signature primitive excess $\mathsf{P}^+_{\text{zero}}$, where no open pair on the overloaded window has such a hit, and

$$\mathsf{P}_{\text{open}} \leq 2p_{13} + \mathsf{P}^+_{\text{zero}}.$$

The remaining numerical slack is

$$\varepsilon_{\text{prim}} = 0.03372671002\ldots - 0.03042331579\ldots = 0.00330339423\ldots.$$

Thus Proposition 15.37 closes exit (4) *whenever $\mathsf{P}^+_{\text{zero}} < \varepsilon_{\text{prim}}|R| - o(|R|)$. The same-window two-blocker cap is the special case $\mathsf{P}^+_{\text{zero}} = o(|R|)$, closed by Lemma 15.41. No target-defect unit is discarded or counted twice: every unit is charged to the original $2/3$-demand ledger, to an unused boundary incidence, to a routed support-dependence certificate, to a Type B handoff, to a recorded power-of-two window-signature route, or to a unique zero-signature window-overload slot.*

Framework branch table. The target-defect demand section closes the remaining large-budget residual by the following table.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure
Unified negative collection	A negative Type A support remains.	$\delta(X) = -\mathcal{N}(X)$.	The unified Type A ledger.	The canonical decomposition.	Definition 15.2 and Lemma 15.4.
Demand ledger	Target-defect or route-8 entries remain.	Demand units.	Boundary incidences.	The no-overcount demand ledger.	Lemma 15.13.
Open demand	Boundary incidences do not pay all target-defect units.	Open demand.	P_{13}-attachment accounting.	Same-window and primitive excess caps.	Propositions 15.22, 15.37 and 15.42.
Finite descent	An exit-(4) peel is available.	One routed load.	The integer Λ_4.	Strict decrease by one.	Lemmas 15.44 and 15.47.
Terminal route 8	No target-defect peel remains.	A two-support route-8 entry.	Deletion witnesses.	The canonical response-support core.	Proposition 14.62 and Theorem 14.61.

A Finite obstruction enumeration code

This appendix reproduces the finite enumerations that certify the constants used above. Each listing is self-contained and was run under CPython 3 (standard library, with `numpy` for the second block); the displayed output immediately below each listing is the certificate. The first block certifies the legal-label count $|\mathcal{L}| = 399$ of Lemma 6.1 together with the two-step obstruction counts 543958, 432672, 111286 of Lemma 6.2, and hence the value of c_Ω.

```
from itertools import combinations
from collections import Counter
import math

labels=[]
for mask in range(1,1<<13):
    S=[i for i in range(13) if (mask>>i)&1]
    ok=True
    for a,b in combinations(S,2):
        if abs(a-b) in (2,6):
            ok=False
            break
    if ok:
        labels.append(tuple(S))

print(len(labels))
print(Counter(map(len,labels)))
```

```
powers={2**k for k in range(2,20)}

def C(s,S,T):
    return all(s+2+abs(i-j) not in powers for i in S for j in T)

W=0
pos=0
flat=0
for S in labels:
    for A in labels:
        if not C(1,S,A):
            continue
        for T in labels:
            if not C(1,A,T):
                continue
            W+=1
            if not C(2,S,T):
                pos+=1
            else:
                flat+=1

print(W,pos,flat)
print(pos/W, flat/W, math.log2(W/flat))
```

The output is:

```
399
Counter({3: 122, 4: 122, 5: 63, 2: 60, 6: 17, 1: 13, 7: 2})
543958 432672 111286
0.7954143518433409 0.20458564815665917 2.28922315244071
```

The second block certifies the constant

$$c_{13} = 118.108581006\ldots$$

used in Lemma 7.1 and Proposition 7.42, as the sum of the 91 finite obstruction barriers; it uses matrix multiplication to avoid an explicit $399^3 \cdot 91$ loop.

```
from itertools import combinations
import numpy as np, math

L=[]
for mask in range(1,1<<13):
    inds=[i for i in range(13) if (mask>>i)&1]
    ok=True
    for a,b in combinations(inds,2):
        if abs(a-b) in (2,6):
            ok=False
            break
    if ok:
        L.append(mask)
assert len(L)==399

def indices(mask):
    return [i for i in range(13) if (mask>>i)&1]
```

```
Ms=[]
for s in range(15):
    forb={2**r-s-2 for r in range(2,20) if 0 <= 2**r-s-2 <= 12}
    M=np.ones((len(L),len(L)), dtype=np.int8)
    for i,S in enumerate(L):
        Si=indices(S)
        for j,T in enumerate(L):
            Tj=indices(T)
            bad=any(abs(a-b) in forb for a in Si for b in Tj)
            if bad:
                M[i,j]=0
    Ms.append(M)

total=0.0
vals=[]
for a in range(1,15):
    for b in range(1,15-a): # a,b >= 1 and a+b <= 14
        middle_counts=Ms[a].astype(np.int32) @ Ms[b].astype(np.int32)
        W=int(middle_counts.sum())
        bad=int((middle_counts*(1-Ms[a+b])).sum())
        flat=W-bad
        val=math.log2(W/flat)
        vals.append((a,b,W,bad,flat,val))
        total += val

print(len(vals))
print(total)
print(min(vals, key=lambda x: x[-1]))
```

The output is:

```
91
118.10858100609198
(4, 4, 6648206, 1190878, 5457328, 0.284770329720...)
```

The third block reproduces the Type B fan constants used in Remark 14.67 and Lemma 14.93. The proof of those constants is the structural label-packing argument in Lemma 14.66. The listing builds the same graph on the 399 labels with adjacency relation $C_2(S,T) = 1$, computes its clique number, and computes the marked variant corresponding to the non-singleton-label cap. The residual class in Definition 14.71 is determined by the positive closed-neighbour deficit $D_B > 0$: for $k \leq 7$ this means at least two cubic-closed neighbours, while for $k = 8$ it also includes the single cubic-closed border case.

```
from fractions import Fraction
from itertools import combinations

labels=[]
for mask in range(1,1<<13):
    S=tuple(i for i in range(13) if (mask>>i)&1)
    if all(abs(a-b) not in (2,6) for a,b in combinations(S,2)):
        labels.append(S)

powers={2**k for k in range(2,20)}
```

```
def C(s,S,T):
    return all(s+2+abs(i-j) not in powers for i in S for j in T)

n=len(labels)
adj=[0]*n
edges=0
for i,S in enumerate(labels):
    for j in range(i+1,n):
        if C(2,S,labels[j]):
            adj[i] |= 1<<j
            adj[j] |= 1<<i
            edges += 1

def maximum_clique(mask):
    best_size=0
    best_bits=0
    def color_sort(P):
        verts=[]
        colors=[]
        U=P
        color=0
        while U:
            color += 1
            Q=U
            while Q:
                bit=Q & -Q
                v=bit.bit_length()-1
                verts.append(v)
                colors.append(color)
                U &= ~bit
                Q &= ~bit
                Q &= ~adj[v]
        return verts, colors
    def expand(size,bits,P):
        nonlocal best_size,best_bits
        if not P:
            if size > best_size:
                best_size=size
                best_bits=bits
            return
        verts,colors=color_sort(P)
        for idx in range(len(verts)-1,-1,-1):
            if size + colors[idx] <= best_size:
                return
            v=verts[idx]
            bit=1<<v
            if P & bit:
                expand(size+1,bits|bit,P & adj[v])
                P &= ~bit
                if size + P.bit_count() <= best_size:
                    return
    expand(0,0,mask)
    return best_size,best_bits
```

```
omega,wit=maximum_clique((1<<n)-1)
omega_ns=0
wit_ns=0
for i,S in enumerate(labels):
    if len(S) >= 2:
        size,bits=maximum_clique(adj[i] | (1<<i))
        if size > omega_ns:
            omega_ns=size
            wit_ns=bits

def bit_labels(bits):
    return [labels[i] for i in range(n) if (bits>>i)&1]

simple_charge_k_le_7=min(
    (Fraction(3-k,1)-Fraction(1,4)-Fraction(1,4)
     + Fraction(3,4)*(k-1), k)
    for k in range(4,8)
)
charge_k8_no_closed=Fraction(3-8,1)-Fraction(1,4)+8*Fraction(3,4)

print('labels', n)
print('fan_safe_edges', edges)
print('omega_C2', omega)
print('omega_C2_witness', bit_labels(wit))
print('omega_with_nonsingleton', omega_ns)
print('nonsingleton_witness', bit_labels(wit_ns))
print('simple_charge_k_le_7', simple_charge_k_le_7[0], 'at_k',
      simple_charge_k_le_7[1])
print('charge_k8_no_closed', charge_k8_no_closed)
```

The output is:

```
labels 399
fan_safe_edges 18301
omega_C2 8
omega_C2_witness [(1,), (2,), (3,), (4,), (9,), (10,), (11,), (12,)]
omega_with_nonsingleton 7
nonsingleton_witness [(0, 1), (2,), (3,), (8,), (9,), (10,), (11,)]
simple_charge_k_le_7 0 at_k 7
charge_k8_no_closed 3/4
```

Remark A.1 (Fan-packing reproducibility check)**.** *The fan-packing bound used in the Type B proof is established structurally in Lemma 14.66. The listing above is retained only as a reproducibility check of the same constants for the* 399*-label graph; it is not an additional premise in the proof.*

Framework branch table. The enumeration appendix is a computational-certificate register. Each row states which finite datum is exported and where the proof consumes it.

Branch	Residual condition	Demand	Payer	Bounded multiplicity	Closure

Legal-label certificate	All subsets of $\{0,\dots,12\}$ are tested for forbidden gaps.	$\|\mathcal{L}\| = 399$ and the size distribution.	The finite subset enumeration.	The predicate in the code is the predicate in Lemma 6.1.	Consumed by Lemma 6.1.
Obstruction certificate	All ordered legal-label triples are tested.	543958, 432672,. 111286	The finite obstruction tensor.	The code applies C_1, C_2 exactly as defined in the text.	Consumed by Lemma 6.2.
Window-package rate	All 91 pairs $a, b \geq 1$, $a + b \leq 14$, are summed.	c_{13}.	The matrix-computed barrier list.	The printed count 91 verifies that every scale pair is included once.	Consumed by Lemma 7.1 and Proposition 7.42.
Type B reproducibility	The 399-label fan graph is checked independently.	Clique and marked-clique constants.	The structural proof of Lemma 14.66.	The code is a reproducibility check, not a separate premise.	Checks Remark 14.67 and Lemma 14.93.

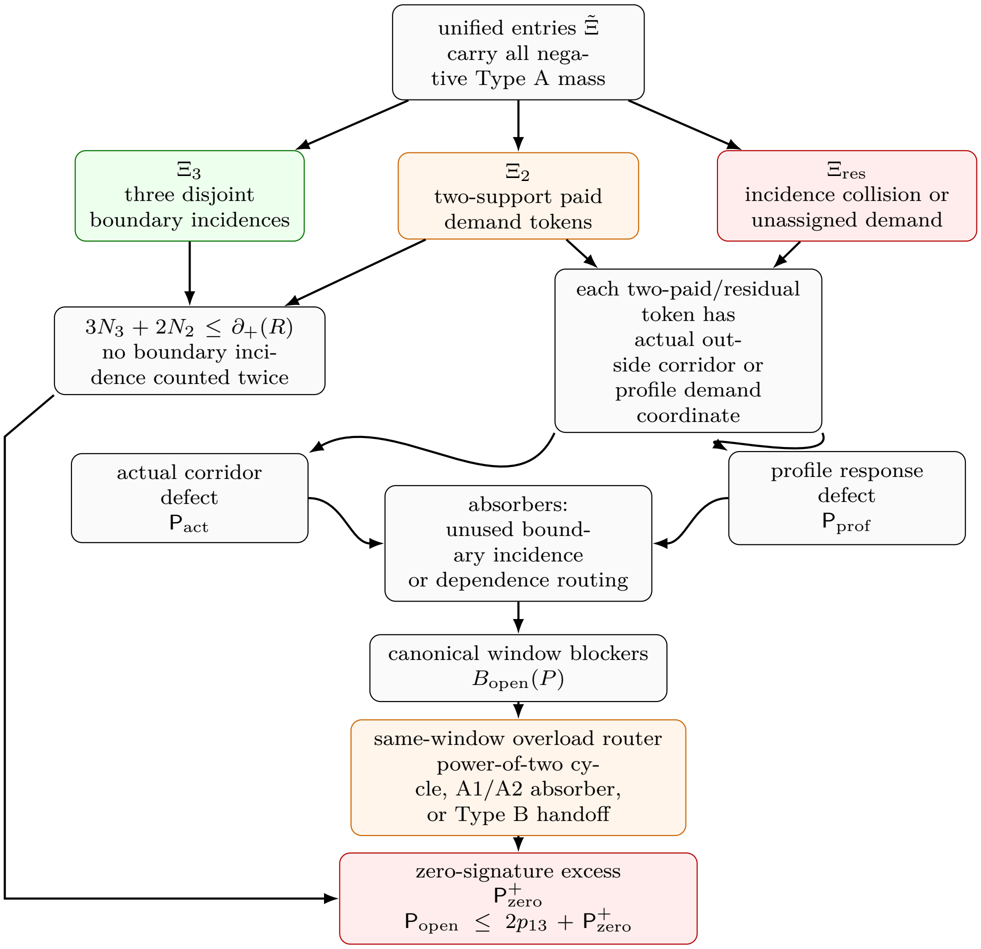

Figure 31: External-demand accounting for exit-(4) target-defect tokens. The three-support class is paid directly by disjoint boundary incidences. The two-support paid class receives two disjoint boundary incidences and leaves one unit of demand defect; this class includes genuinely cheap two-support tokens and tokens whose additional incidences collide with earlier ledger choices. The unassigned class leaves three units of demand defect. Each two-support paid or unassigned target-defect token is represented by either an actual outside corridor in $G - X$ or by a profile-only exact response coordinate. These give the exact split $\mathsf{P}_{\mathrm{ext}} = \mathsf{P}_{\mathrm{act}} + \mathsf{P}_{\mathrm{prof}}$. The absorption ledger then assigns demand units either to unused boundary incidences, which improve the same $\partial_+(R)$ count, or to profile-dependence certificates, which route to support-dependence alternatives. Target-defective dependence outputs re-enter the demand ledger instead of being treated as absorbed. The only demand left in the surviving branch is $\mathsf{P}_{\mathrm{open}}$. The window-blocker ledger assigns every open unit to a unique packed P_{13}-window. A same-window overload triple is routed when it realizes a direct power-of-two cycle, an unused boundary absorber, a support-dependence absorber, or a Type B fan-window handoff. After these routed triples and their recorded window-signature power-of-two hits have been removed, the only extra term beyond two blockers per window is the zero-signature primitive excess $\mathsf{P}^{+}_{\mathrm{zero}}$, with $\mathsf{P}_{\mathrm{open}} \le 2p_{13} + \mathsf{P}^{+}_{\mathrm{zero}}$. Thus Proposition 15.37 closes the branch whenever $\mathsf{P}^{+}_{\mathrm{zero}} < \varepsilon_{\mathrm{prim}}|R| - o(|R|)$. The same-window two-blocker cap is the special case $\mathsf{P}^{+}_{\mathrm{zero}} = o(|R|)$, where the strict numerical inequality is $2\theta_{\mathrm{win}}/(1 - 13\theta_{\mathrm{win}}) = 0.03042331579\ldots < 0.03372671002\ldots = \varepsilon_{\mathrm{press}}$.

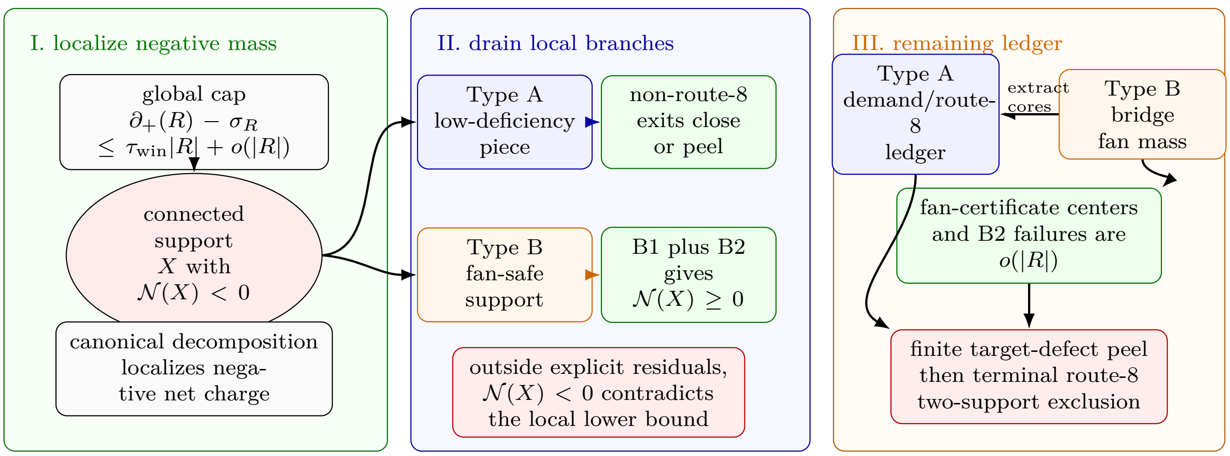

Figure 32: Visual proof of the branch-closure mass ledger. Panel I shows the large-budget cap localizing negative net charge to a connected admissible support X with $\mathcal{N}(X) < 0$. Panel II shows the local branch drain used in Theorem 15.1: Type A supports close through the non-route-8 exits or target-defect peeling, while Type B supports close through the B1 ledger together with B2, giving $\mathcal{N}(X) \ge 0$ outside the explicit residuals. Panel III records the remaining global ledger. Type B bridge and fan-mass residuals are charged to assigned surplus units and become $o(|R|)$ after route-8 non-window cores are extracted into the Type A ledger. The unified demand ledger counts both target-defect and route-8 Type A entries. Target-defect two-support entries are canonical exit-(4) peel data, so finite descent removes them. The terminal survivor would have to be a two-support route-8 obstruction, which is excluded by the support-reduction and exclusion figures.

B Proof resilience: residual counterexamples under lemma failure

For every *closing cell* (terminal node) of the proof-dependency diagram, this appendix records what survives if that cell's closing lemma is *false* and no redundancy covers it. The entry is the forced structure of a hypothetical minimal counterexample confined to that cell—that is, the reduction theorem the proof degrades into at that leaf. The logic is that a closing lemma does not merely delete a possibility when it fails; its negation is a structural hypothesis. A surviving counterexample must satisfy *all standing invariants accumulated up to that node* (it does not lose them) *together with* the negation of the failed lemma's conclusion. Thus every degradation cell is a specified residual problem, not an unconstrained failure mode.

Notation is as in the manuscript: G is the minimal counterexample, $\delta(G) \geq 3$, R is the P_{13}-free remainder, $\theta = p_{13}/n$, $\partial_+(X) = \sum_v \max(0, 3 - d_X(v))$, $\sigma(X) = \sum_v \max(0, d_G(v) - 3)$, $\mathcal{N}(X) = \partial_+(X) - \sigma(X) - \frac{1}{4}|V(X)|$, W_2 is the two-step obstruction supply, c_Ω the flatness entropy cost, and $\beta = m - n + 1$. "Standing invariants" are the invariants established at earlier-or-equal nodes; they are retained in every residual cell below.

Framework branch table. The resilience appendix already gives the detailed degradation rows below; the framework branch table for the appendix is the following compact index.

Branch	**Residual condition**	**Demand**	**Payer**	**Bounded multiplicity**	**Closure**
Covered leaf	A listed closure is removed but a redundant route remains.	The residual cell.	The redundant closure.	The standing invariant set.	Recorded as covered in the detailed tables.
Dense-window auxiliary	The window entropy branch degrades to dense-window blocks.	Bounded-block demand.	The target-compatible external input.	Canonical block boundaries.	Lemma B.2.
Whole-graph support dependence	The rank-drop branch degrades to a whole-graph dependence.	Closed exact-profile data.	Minimality or label-injectivity.	The empty-boundary profile.	Lemma B.3.
Type A failure	The route-8 or exit-(4) closure is weakened.	Terminal response-support package.	The response-support ledger.	Declared witnesses.	Lemma B.4.
Type B failure	The fan ledger or B2 closure is weakened.	Bridge or overlap residual.	The fan-mass ledger.	Assigned surplus.	Remark B.5.

How to read a row.

Column	Meaning
Node	Terminal node id $[k]$ from the diagram, with tikz id.
Closing lemma	The single labeled result that discharges this leaf.
Closing condition	The inequality or identity whose truth closes the leaf.
Slack	Margin by which it currently closes (how wrong the constant can be).
Redundant cover?	Whether another standing route re-closes this cell if the lemma fails.
Residual counterexample	Forced structure of a surviving G if the lemma is false and no cover applies.

A. Trivial / external-input leaves. These do not degrade to nontrivial residual reductions: their negation is either a direct cycle (so G is not a counterexample at all) or a failure of an external theorem, not of this proof.

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false)
[3] `notbad`	def. of counterexample	$\delta(G) \geq 3$ and no C_{2^j}	exact (definitional)	—	None — if it fails, G already has a power-of-two cycle (not a counterexample).
[7] `cycle`	Lemma 2.2	a Mersenne return $\iff$ a 2^k cycle	exact (arithmetic identity)	—	None — failure means the return/cycle dictionary is misstated; G would have a 2^k cycle. Pure bookkeeping.
[16] `hss`	Theorem 5.1 (HSS, external)	P_{13}-free $+ \delta \geq 3 \Rightarrow 2^k$ cycle	external	—	Not this proof's leaf. The proof uses the Hegde–Sandeep–Shashank theorem as its black-box external input; if that theorem is unavailable, the window/remainder spine has no entry point.

B. Density / supply leaves (close by counting; degrade to "dense-window" reductions).

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[23] `overflow`	Proposition 7.12 via Eq. (1)	$\theta \leq 0.011985$ (else window entropy > skeleton budget)	feasibility gap $0.1 + 116.8\theta \leq 1.5$; binds only at $\theta \approx 0.012$	**Target-compatible auxiliary closure.** The dense correlated-window residual is closed in Lemma B.2 by the stated bounded-block input.	G has a *dense induced-P_{13} packing*: $\theta > 0.011985$, so the maximal disjoint packing covers > 15.6% of vertices, yet the window entropy still fits under the skeleton budget. The target-compatible auxiliary closure shows that such entropy-cheap density must either create close cycle lengths in a target-safe boundaried piece or pay quadratic boundary, in both cases restoring a contradiction.
[29] → [28] `stub`	Lemma 8.5	each induced P_{13} exports $\leq 15 + \sigma(P)$ incidences	exact (12-edge identity); only θ feeds the normalized form	**Framework-independent.** Uses only $\delta \geq 3$, "induced P_{13} has 12 edges", and disjointness.	If the constant 15 were miscounted, the residual G would be one whose remainder borrows more than 15 incidences per window, contradicting the induced-path edge count. Thus failure here is a numerical misstatement of the incidence identity, not a new graph-theoretic residual.

C. Obstruction / rank leaves (close by supply-vs-demand; degrade to "flat-remainder" reductions).

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[30] `wedge` (demand floor)	Lemma 9.1	under Definition 4.1, $W_2(R) \geq 2.543 \cdot \|R\|$; high entropy gives $2.574 \cdot \|R\|$	4.1× **over the** 0.611 **supply cap**	**Yes inside the near-cubic spine.** The net-charge collision (diagram node [60], 0.25 vs 0.2282) reaches a contradiction without the obstruction magnitude; Remark 12.4 states closure already holds from the window-only cap.	G has an *obstruction-poor remainder*: $W_2(R) < 2.543 \cdot \|R\|$, i.e. the P_{13}-free remainder admits fewer than 2.543 two-step obstruction tests per vertex. With the standing invariants (empty 3-core, deficiency ≤ 0.2282) this forces an unusually *flat* piece remainder—locally tree-like wedge structure avoiding obstructing triples. *Reduction:* "a counterexample's remainder is obstruction-flat below density 2.543; classify flat P_{13}-free pieces." Recoverable via the net-charge route, which does not consume the wedge magnitude.
[32]/[47] `fullrank`	Lemma 10.10	$r_\Omega(R) \geq W_2 - o(\|R\|)$ (no rank drop)	absorbs $o(\|R\|)$; rank-drop branch D separately closes at [46]	**Yes.** Branch D (Lemma 10.9, inv 34/35) independently closes the rank-drop case; the two-budget split also re-routes.	G has a *rank-deficient obstruction operator*: $r_\Omega(R) < W_2 - \Omega(\|R\|)$, so a linear fraction of obstruction tests are dependent. The proper-support cases are routed to target defect, compression, or proper support enlargement. The whole-graph case is closed by the exact closed response profile: empty context does not identify labelled obstruction coordinates, and a non-injective closed quotient must produce a smaller closed counterexample.
[45] `globalterm`	Lemma 10.9	no silent whole-graph dependence of power-of-two target support	qualitative (closed exactness is label-injective on raw obstruction tests)	**Closed in the main text.** Lemma B.3 records the corresponding degradation-cell closure.	A surviving row [45] residual would require a target-complete whole-graph quotient of $\rho^{\mathrm{ex}}_{\varnothing}(G)$ that is not label-injective on raw obstruction tests and has no smaller closed representative. This contradicts Definition 3.91 and Lemma 10.9.
[50]/[54] `entcap`	Proposition 12.2 / Proposition 11.6	high-θ branch: entropy cap $c_\Omega \cdot$ rank exceeds budget when the rank input is available	routing is exhaustive; only the high-entropy and repetitive low-entropy subcases add new rank/entropy information	**Partial routing.** The nonrepetitive low-entropy subcase is deliberately passed to the large-budget net-charge analysis, not closed here.	G sits in a *nonrepetitive low-entropy regime*: $\eta(R) < \frac{1}{10}\log_2 n$ per vertex, the radius-2 type coordinate has linear bit-content, and the local-type dominance lemma does not apply. *Reduction:* this branch carries no extra obstruction closure claim and is handled only by the later net-charge/local-residual analysis.

D. Net-charge terminus leaves (close by discharging; degrade to "charged-piece" reductions).

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[60] `contrad`	Proposition 13.7 + Lemma 13.4	$\frac{1}{4}\lvert R\rvert \le \partial_+(R) - \sigma(R) = \partial_+(R) - \sigma_R \le \tau_{\mathrm{win}}\lvert R\rvert + o(\lvert R\rvert)$ with $\tau_{\mathrm{win}} < \frac{1}{4}$	$\sim 9.6\%$ (0.25 vs 0.2282 per $\lvert R\rvert$)	**Yes.** The second independent collision form; the obstruction route ([30]) reaches a contradiction on its own with large slack.	G has a *net-charge-negative residual that does not localize*: $\partial_+(R) - \sigma(R) < \frac{1}{4}\lvert R\rvert$ globally, yet net charge fails to concentrate on a connected support (superadditivity localization fails). *Reduction:* "a counterexample distributes a $\frac{1}{4}$-deficit across R with $\mathcal{N}(X) \ge 0$ on every connected piece"—forbids the very cell-wise accounting the discharging needs; isolates a precise superadditivity gap to repair.
[91] **`atomthm`** (Type A, $\sigma = 0$)	Lemma 14.45 via Lemmas 14.27, 14.34 to 14.38, 14.40 and 14.43	outside target-defect peeling, route 8, and the Type B handoff, $\partial_+(X) \ge \frac{1}{4}\lvert V(X)\rvert \iff 2\lvert E(X)\rvert/\lvert V(X)\rvert \le 2.75$, so $\mathcal{N}(X) \ge 0$	raw thresholds $H_0 \le 4$, $H_1 \le 8$, $H_2 \le 12$; exits 1–3, 5, and 6 close; exit 4 peels one target-defective routed load with exact 1/4 receiver-charge accounting; exit 7 hands off to Type B; then unsaturated discharging $11n_0 + 7n_1 + 3n_2 \ge n_3$	**Local split.** The 4/8/12 saturated cases in exits 1–3, 5, and 6 are removed by target, compression, and support enlargement, while exit 4 removes its supported target-defective load before the 3/7/11 charge calculation. Exit 7 is represented by Type B, and route 8 is the reduced silent trace-support branch of Definition 14.15.	G would contain a *dense low-deficiency admissible boundaried piece*: a connected, $\sigma = 0$, empty-3-core, P_{13}-free support with average degree > 2.75. The Type A split sends exits 1–3, 5, and 6 to closed contradictions, exit 4 to target-defect peeling, exit 7 to Type B, and the remaining unsaturated case to discharging; if the route-8 branch carries the large-budget Type A deficit, the declared u-supported response algebra and canonical minimal target-complete response-support cores reduce it to a two-support route-8 residual, with bounded-block boundedness recorded in Lemma B.4.

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[74] `fanthm` (Type B, $\sigma > 0$)	Lemma 14.93 and Proposition 14.100 via Definition 14.18, Definition 14.68, Definition 14.72, Definition 14.73, Definition 14.74, Lemma 14.75, Lemma 14.76, Lemma 14.77, Lemma 14.78, Lemma 14.79, Definition 14.83, Definition 14.84, Definition 14.85, Lemma 14.87, Lemma 14.88, Proposition 14.89, Definition 14.90, Proposition 14.92, Definition 14.71, Definition 14.95, Lemma 14.97, Lemma 14.98, Lemma 14.99, Remark 14.67, and Lemma 14.66	outside route 8, outside fan-certificate residual centers, and outside B2 failure, every certificate-marked fan has $d_G(h) \leq 8$, every nonpositive-deficit marked fan has nonnegative closed-neighbourhood charge, every positive-deficit fan has the local hybrid B1 incidence ledger, and B2 supplies the disjoint ledger needed to sum the fans; in the near-cubic spine, residual centers and B2 failures satisfy $M_B \leq 16\sigma(G) = o(\|R\|)$ after any route-8 non-window core is extracted into D_A and grouped-envelope centers are counted at most twice	certificate-closed iff $D_B \leq 0$; every positive-deficit residual has $D_B = c - \frac{11-k}{4}$ paid by half-credit on the $2c$ disjoint non-h incidences, split into window credit $I_W/2$ and non-window demand D_N; each bridge residual is charged by at most eight times its assigned surplus plus the extracted route-8 Type A deficit	**Structural reduction to B2 plus fan-mass closure.** Direct fan-window and two-window cycles close by explicit path construction; local B1 is the hybrid incidence budget; every disjoint-incidence B2 failure is represented by a minimal overlap obstruction carrying contextual target safety, window-stub capacity, replacement/support-dependence routing, and a sublinear fan-mass budget in the near-cubic spine after route-8 cores are charged to D_A; fan-certificate residual centers and grouped envelopes enter this fan-mass bound directly.	In the near-cubic spine, a surviving Type B bridge obstruction has only sublinear bridge mass outside the Type A demand/route-8 ledger, so it cannot carry the linear large-budget obstruction without entering node [123].

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[76]/[85] `branch` (local closures)	Theorem 15.1	combines [30]+[60]+entropy: outside explicit Type A target-defect/route-8 and Type B bridge residuals, the entropy-admissible branch has no survivor	inherits the available component slacks: about 4.1× in the obstruction form and about 9.6% in the net-charge form; Type B bridge mass is $o(\|R\|)$ after route-8 cores are extracted into the Type A ledger	**Local join.** It leaves only the explicit Type A demand/route-8 ledger for node [123].	G *survives the branch squeeze* only if the negative local support contributes to the explicit Type A demand/route-8 ledger; Type B fan-certificate residual centers and B2 failures are too small in total to carry the required linear deficit.
[123] `closed` (demand descent)	Theorem 15.48	after the branch-kill reduction, the remaining large-budget Type A target-defect/route-8 ledger either peels a target-defect load by exit (4) or forces a two-support route-8 entry	inherits the unified burden lower bound, the demand no-overcount ledger, finite descent of Λ_4, the route-8 burden lower bound, and the private-support upper bound; Type B bridge mass has already been removed at [76]/[85]	**Join node feeding the terminal exclusion.** Target-defect two-support entries decrease Λ_4; the terminal non-peeling case enters node [124], which is discharged by response-support deletion.	G can survive node [123] only if the finite exit-(4) descent has terminated and the remaining entry is the terminal two-support route-8 obstruction through the canonical response-support core ledger.

Node	Closing lemma	Closing condition	Slack	Redundant cover?	Residual counterexample (if false, no cover)
[124] `nogo` (two-support route 8)	Definition 14.50, Lemmas 14.53 to 14.55, and Theorem 14.61, with Definition 14.60 and Proposition 14.62	no terminal two-support route-8 obstruction exists	inherits the route-8 lower bound, the private-incidence upper bound, cut parity, declared deletion witnesses, and all absent-exit constraints	**Terminal local exclusion.** In a two-support entry, declared essential deletion witnesses first give canonical exit-(4) quotients and then realize exit (4), which is absent in a true route-8 residual.	A false instance is exactly a terminal two-support route-8 obstruction: a true route-8 entry with at least two essential incidences, at most two private essential incidences, and declared incidence-deletion witnesses for every essential incidence.

Remark B.1 (Tight margin in the route-8 support reduction)**.** *The private-incidence reduction at node [123] uses the numerical inequality*

$$12\left(\frac{1}{4}-\tau_{\mathrm{win}}\right) > \tau_{\mathrm{win}}, \qquad \textit{equivalently} \qquad \tau_{\mathrm{win}} < \frac{3}{13}.$$

With the printed constants, $\tau_{\mathrm{win}} = 0.22817486846\ldots$*, while* $3/13 = 0.23076923076\ldots$*. The relative margin is therefore about*

$$\frac{3/13-\tau_{\mathrm{win}}}{3/13} = 0.01124\ldots.$$

This is the tight response-support reduction margin. Any later argument that spends additional boundary-incidence or stub capacity before Proposition 14.59 must recheck this inequality.

External-input closures for partial degradation cells. The next three lemmas use target-compatible forms of the auxiliary external inputs listed in the constraint ledger. The named external theorems supply the graph-theoretic alternatives; the target-compatible clauses below state exactly how those alternatives interact with the Mersenne-return algebra and the canonical support ledger of this paper. Thus no closure below depends on an unstated propagation step. For a connected support Y, write $b(Y) = |\partial Y|$ for its boundary size in the ambient counterexample.

Lemma B.2 (Dense-window residual closes by bounded-block input)**.** *Assume the following target-compatible Bondy–Vince/Gao–Ma input for the canonical blocks generated by a row [23] dense-window residual.*

(a) Each canonical block Y is almost cubic, and its sub-cubic vertices are boundary vertices charged to the packed-window incidence ledger of Lemma 8.5.

(b) If Y exports two cycle lengths differing by 1 or 2, then the corresponding two edge-rooted returns lie in one boundary fibre and force either a return length in $\mathcal{M}$ or a target-complete replacement in the sense of Lemma 3.96.

(c) If Y is in the bounded-block regime, then $|Y| < 5b(Y)^2$, and the canonical bounded blocks are boundary-disjoint up to packed-window interfaces; summing their boundary charges restores the feasibility inequality Eq. (1).

Then the dense-window residual in row [23] is empty.

Proof. Suppose row [23] survives. Then the maximal induced-P_{13} packing has $\theta > 0.011985$, while the corresponding window states remain below the skeleton entropy bound. Apply the target-compatible input to the canonical blocks of this residual. In the exported-length alternative, clause (b) gives a Mersenne return or a target-complete replacement. The first contradicts Lemma 2.2; the second contradicts minimality through Lemma 3.96. In the quiet alternative, clause (c) restores Eq. (1), contradicting $\theta > 0.011985$ in the entropy-cheap dense-window branch. The two alternatives exhaust the input, so row [23] is empty. ☐

Lemma B.3 (Global-support closure)**.** *The rank-one global-mode residual in row [45] is empty.*

Proof. Let a row [45] residual survive. By Lemma 10.9, the corresponding global dependence is target-defective, has a strictly smaller admissible closed representative, or contributes exact label-injective global response data. The first alternative cannot witness a target-dependence by Definition 10.2; the second contradicts the lexicographic minimality of G; and the third supplies no rank reduction of $r_\Omega(R)$ by Definitions 3.88 and 3.91. This contradicts the definition of a row [45] rank-one global-mode residual. Hence row [45] is empty. ☐

Lemma B.4 (Route-8 Type A residual is bounded by the bounded-block input)**.** *Assume the following target-compatible Bondy–Vince/Gao–Ma input for route-8 Type A supports.*

(a) Every maximal almost-cubic block of a route-8 Type A support either exports two cycle lengths differing by 1 *or* 2*, or is quiet and satisfies* $|Y| < 5b(Y)^2$*.*

(b) In the exported-length alternative, completion through the ambient boundary gives a Mersenne return or a target-complete replacement.

(c) If all maximal blocks are quiet, gluing them along the canonical boundary of the Type A support gives $|X| < 5b(X)^2$*.*

Then every admissible silent-core residual profile of Definition 14.15 lies in a finite boundary-labelled residual class determined by

$$|X| \leq 6142, \qquad |X| < 5b(X)^2,$$

with all boundary return profiles avoiding $\mathcal{M}$*.*

Proof. Let X be the Type A support underlying an admissible silent-core residual profile. It is connected, subcubic, P_{13}-free, has empty internal 3-core, satisfies $\sigma(X) = 0$, and has $\partial_+(X) < |X|/4$. The P_{13}-free diameter argument in the Type A analysis already gives $|X| \leq 6142$.

Apply the Bondy–Vince/Gao–Ma input to every maximal almost-cubic block of X. If one of them realizes the exported-length alternative, clause (b) creates a Mersenne return or a target-complete replacement, both forbidden in an admissible support. Hence every maximal block is quiet. Clause (c) gives $|X| < 5b(X)^2$. Since $|X| \leq 6142$ and the boundary labels are drawn from the finite 399-label P_{13} algebra, only finitely many boundary-labelled return profiles remain. Thus the auxiliary input bounds route 8 by a finite boundary-labelled residual class with explicit Mersenne-avoiding return profiles; it does not by itself close that residual. ☐

Remark B.5 (Type B residual after the structural fan-window reduction)**.** *The unused external inputs do not close node [74]; the available closure is the structural label-packing lemma recorded in Lemma 14.66. Bondy–Vince/Gao–Ma controls quiet almost-cubic blocks, and Kleitman–West* *controls global tree–cotree profiles; the Type B obstruction is a local fan-safe clique around a high-degree surplus vertex. The packing lemma proves the C_2 clique cap, the marked non-singleton cap, and the spectral bound $d_G(h) \leq \rho(A(\mathcal{F}_h))+1$ for certificate-marked fans, and it discharges certificate-closed marked fans. Fan-certificate residual centers are charged directly to their assigned surplus units. The structural fan-window reduction of Lemmas 14.75 to 14.79 then removes every positive-deficit pair that already contains a direct power-of-two cycle and pays every remaining positive-deficit fan by hybrid half-credit on its window and non-window incidences. The refined support ledger B2 is the local disjointness statement needed to sum those fan entries. By Lemmas 14.87 and 14.88, any disjoint-incidence failure of that bridge ledger is represented by a minimal overlap obstruction carrying contextual target safety, high-degree separation, window compatibility, replacement routing, and minimal-overlap constraints. The fan-mass closure Definition 14.95, Lemmas 14.98 and 14.99, and Proposition 14.100 charges fan-certificate residual centers and disjoint-incidence failures to their assigned surplus units and gives total negative Type B bridge mass $O(\sigma(G)) = o(|R|)$. Thus neither Type B residual form is a terminal large-budget residual.*

Summary: the degradation ladder. Ordered by how much the proof loses if that single leaf fails with no redundant cover:

1. **Definitional or exact-counting leaves:** [29] `stub` — framework-independent counting; [3]/[7] definitional.
2. **Redundantly covered or routed forward:** [30] obstruction demand (covered by [60] net-charge, about 4.1× slack); [60] net-charge (covered by [30], about 9.6% slack); [32]/[47] `fullrank` (covered by Branch D); [50]/[54] entropy routing (high-entropy and repetitive low-entropy subcases add information, while nonrepetitive low entropy is routed to net charge); [123] branch-kill outside explicit residuals.
3. **Closed or bounded by auxiliary bookkeeping:** [23] `overflow` via the bounded-block accounting in Lemma B.2; [45] `globalterm` by the main whole-graph support-dependence lemma, recorded again in Lemma B.3. The route-8 Type A branch is reduced to the terminal two-support obstruction by the canonical response-support core ledger of Proposition 14.59, supplied with declared deletion witnesses by Lemmas 14.52 and 14.53, and closed at node [124] because two-support declared deletion quotients are canonical exit-(4) quotients by Lemma 14.54 and realize exit (4) by Lemma 14.55; it remains bounded-block bounded by Lemma B.4.
4. **Closed by structural Type B bookkeeping:** [74] `fanthm` uses the certificate-marked fan-safe clique bound for the local certificate-closed fan charge, the direct fan-window cycle lemmas for positive-deficit pairs, and the hybrid incidence budget of Lemmas 14.78 and 14.79. The degree-greater-than-four side leaf enters this ledger through fan-closed ports: a fan-compatible open pair routes by Corollary 14.81, and the triangular alternative routes by Corollary 14.82. In both cases Proposition 14.80 supplies $D_B \geq (k-3)/4 > 0$, with the triangular alternative giving the stronger $D_B \geq (5k-19)/4 > 0$. The local closure is reduced to B2 in Definition 14.90; fan-certificate residual centers and disjoint-incidence failures are then bounded globally by the fan-mass estimate Proposition 14.100, while Lemmas 14.97 and 14.99 extract route-8 non-window cores into the Type A deficit ledger. Disjoint-incidence failures are also reflected as minimal overlap obstructions of Definition 14.85.
5. **Reduced by exact surplus-token bookkeeping and geometric routing:** [125]–[144] use the active surplus family, the free/blocked pair split, the exact token-fibre identity, the

classwise token supplies, and the window-join load identity. The exact partition Lemma 3.50 assigns every active-demand pair to exactly one free term or one role fibre. The matching–star bound Lemma 3.47 and the classwise budget theorem Theorem 3.51 give the sharp load alternative, and Lemma 3.77 and Theorem 3.78 discharge any large homogeneous same-token fibre: it realizes a sparse surplus exit, produces ordinary Type B fan data, or is bounded by fixed homogeneous caps. In the bounded case Corollary 3.44 gives $\sigma(G) = O(\sqrt{n})$.

No leaf degrades to nothing. Every failed leaf, stripped of redundancy, leaves a hypothetical counterexample carrying all standing invariants plus one explicit extra constraint. The bounded-block auxiliary input closes the dense-window reduction, and the main whole-graph support-dependence lemma closes the global-mode reduction. In Type A, exits 1–3, 5, and 6 close, exit 4 peels one supported target-defective routed load, exit 7 hands off to Type B, the unsaturated branch is discharged, and route 8 is reduced to the terminal two-support trace-basin obstruction by Proposition 14.59: each entry first passes to its canonical minimal target-complete response-support core inside the declared u-supported response algebra, zero/one-essential-core entries realize exits (4)–(7), each essential response support has a deletion witness, each witness has declared support by Lemma 14.53, and in a two-support entry each such witness gives a canonical exit-(4) quotient by Lemma 14.54 and realizes exit (4) by Lemma 14.55. The indexed linear burden is Lemma 14.47. In Type B, certificate-closed fans have nonnegative local charge and are globally summed through B2, direct fan-window and two-window cycles are structurally removed, and every degree-greater-than-four side leaf supplies two fan-closed ports by Corollaries 14.81 and 14.82; the hybrid B1 incidence budget pays every resulting positive-deficit fan locally. Disjoint-incidence failure of B2 is the minimal overlap obstruction of Definition 14.85 with the global constraints of Lemma 14.88; fan-certificate residual centers, grouped handoff envelopes, and disjoint-incidence failures are bounded by Proposition 14.100 after route-8 non-window cores are extracted by Lemmas 14.97 and 14.99. In the non-near-cubic branch, the exact token ledger and Theorem 3.75 give the explicit lower bound $\psi(N_*(G)/(36(4n + 15p_{13} + 3\sigma(G))))$ for a homogeneous subtype load whenever bounded caps fail; Lemma 3.77 then turns that load into a sparse surplus exit, ordinary Type B fan data, or fixed caps implying the near-cubic estimate. Hence, inside the near-cubic spine or after the surplus-token bottleneck discharge, no linear large-budget deficit remains in the Type A route-8 response-support ledger.

Cross-check against the demand > supply collision. Outside the explicit residuals, the proof's contradiction is reached in two arithmetically independent forms, so no *single* constant kills both.

Form	**Demand**	**Supply**	**Slack**	**Feeder leaves**
Obstruction	$W_2 \geq 2.543 \cdot \lvert R \rvert$	$\leq 0.611 \cdot \lvert R \rvert$ independent tests	4.1×	[30], [47], [48]
Net charge	$\partial_+(R) - \sigma(R) \geq 0.25 \cdot \lvert R \rvert$	$\partial_+(R) - \sigma_R = \partial_+(R) - \sigma(R) \leq 0.2282 \cdot \lvert R \rvert$ (τ_{win})	$\sim 9.6\%$	[60], [91], [74]

The shared ancestor of both forms is invariant 24 (Lemma 8.5, diagram node [29]), the exact incidence-counting leaf.

A note on human–AI collaboration

This paper grew out of a collaboration between the author and several AI systems, principally Claude Fable 5, GPT-5.4, GPT-5.5, and GPT-5.6-Sol.

This project began as a personal effort by the author to learn graph theory and to study how large language models fail when reasoning about mathematics. Because the subject was initially unfamiliar to him, part of the aim was to develop a working protocol for detecting hallucinations and avoiding reliance on arguments that he could not independently follow. The author asked GPT-5.4 to identify difficult problems together with recent progress on them, and selected the Erdős–Gyárfás conjecture and the Hegde–Sandeep–Shashank theorem for P_{13}-free graphs as the starting point.

The first stage proceeded through an iterative dialogue. The author proposed possible approaches in natural language and asked the model to explain why they were insufficient and how they related to methods in the literature. He then asked what a counterexample to each proposed approach would have to look like. The resulting examples became the input to the next round of questions. In particular, the author repeatedly requested explanations at an elementary level or in terms familiar to an AI and machine-learning engineer, with emphasis on the purpose of each technique and on the structural properties of the graph that the current argument had not yet controlled. This process continued for approximately two weeks, until the models no longer produced further counterexamples to the developing strategy.

The next stage converted the dialogue into a mathematical manuscript. The author asked GPT-5.5 to translate the accumulated discussion into LaTeX and carried out more than fifty rounds of revision. In each round, he asked the model to replace informal descriptions with mathematical statements, expand steps that he did not understand, and construct proof-flow diagrams and other figures that made the dependencies of the argument visible. He then continued the process for approximately one hundred rounds with GPT-5.5 and Claude Opus 4.8. Over these iterations, the proof strategy acquired a coherent global organization, and the models became able to track the argument using the same tables and diagrams that the author used to follow it.

The author subsequently developed a red-teaming protocol in which GPT-5.5 and Claude Opus 4.8 were asked to search systematically for missing hypotheses, unsupported transitions, and other errors. When a possible gap was identified, the repair protocol introduced a separate branch for the missing case and attempted to close it using the same cycle of counterexamples, explanations, and structural refinements. After Claude Fable 5 and GPT-5.6-Sol became available, the author repeated this red-teaming process roughly fifty more times. The additional gaps found in these reviews were addressed through further manual steering of the models. The point at which the reviews stopped producing new objections was treated as a reason to move to a stricter form of checking, rather than as a certificate of correctness.

For that purpose, the author designed a Lean framework and an interactive proof explorer. The framework represented the proof as a data-processing pipeline whose nodes record the mathematical facts available at each point in the proof tree. Its purpose was to make the transfer of hypotheses and conclusions explicit and to allow individual proof steps to be inspected independently. The framework was rewritten four times to mitigate recurring failure modes encountered when GPT-5.6-Sol attempted to port the argument to Lean.

Once this infrastructure was in place, the author asked Claude Fable 5 to translate the proof into Lean so that remaining inconsistencies could be exposed and repaired. This process increased the proof architecture from 157 to 180 nodes as gaps were found and new branches were added to close them by the same general strategy. The node-by-node translation is ongoing: each node is being formalized separately in order to force the models to account for its hypotheses, dependencies, and conclusion within a modular proof state.

The author's role throughout was to select the problem and research direction, propose and revise ideas, request explanations and counterexamples, design the review and repair protocols, construct the proof-tracking framework, and decide how identified gaps should be routed through the argument. The AI systems supplied literature-oriented explanations, counterexamples to proposed approaches, mathematical formulations, proof drafts, diagrams, reviews, repairs, and Lean translations. This division of labor reflects the actual development of the manuscript: the author did not begin as an expert in graph theory, and the proof emerged through repeated attempts to make every step explicit enough for him to question, organize, and check. The author also abstracted this process into a replicable protocol for decomposing difficult problems into sequences of standard moves drawn from the literature. The protocol is designed to use the models' strengths in searching, reformulating, and iterating over possible arguments while mitigating recurrent failures in hypothesis tracking, global consistency, and the verification of individual steps.

References

[1] A. S. Hegde, R. Sandeep, and S. Shashank, *Power of two cycles in P_{13}-free graphs*, arXiv:2410.22842, 2024/2025. https://arxiv.org/pdf/2410.22842

[2] J. A. Bondy and A. Vince, *Cycles in a graph whose lengths differ by one or two*, J. Graph Theory 27 (1998), 11–15. https://people.clas.ufl.edu/avince/files/Cycles.pdf

[3] J. Gao and J. Ma, *On a conjecture of Bondy and Vince*, arXiv:1902.05701, 2019. https://arxiv.org/abs/1902.05701

[4] D. J. Kleitman and D. B. West, *Spanning trees with many leaves*, SIAM J. Discrete Math. 4 (1991), 99–106. https://dwest.web.illinois.edu/pubs/manyleaf.pdf